Contents

Contents i

1 Concordance with primary literature 1

1.1 Principal contributions . . . 4
1.2 Relationship to Beilinson–Drinfeld . . . 6
1.3 Relationship to Francis–Gaitsgory . . . 6
1.4 Relationship to Gui–Li–Zeng . . . 6
1.5 Relationship to Ayala–Francis . . . 8
1.6 Relationship to Feigin–Frenkel . . . 9
1.7 Relationship to Costello–Gwilliam . . . 9
1.8 Relationship to Nishinaka–Vicedo . . . 10
1.9 Relationship to Loday–Vallette . . . 13
1.10 Relationship to Malikov–Schechtman . . . 14
1.11 Three-pillar foundational architecture . . . 15
1.12 The three concentric rings . . . 22
1.13 Intrinsic Koszulness: the characterization programme . . . 26
The meta-theorem: twelve characterizations . . . 27
Sufficient conditions for chiral Koszulness . . . 28
Operadic complexity . . . 28
Shadow depth classification . . . 29
Geometric, physics, and categorical characterizations . . . 32
Bifunctor obstruction . . . 33
Computational diagnostics . . . 33
Cross-reference table . . . 34
1.14 Rectification: the organizing meta-theorem . . . 35
1.15 Non-principal $\mathcal{W}$-algebra duality: the transport frontier . . . 37
Proved core. . . . 37
Active frontier. . . . 37
Hook-type quantitative data. . . . 38
$\mathcal{W}_3$ multi-variable shadow archetype. . . . 38
Key constraint. . . . 38
Recent developments. . . . 38
Geometric localization programme. . . . 38
1.16 E_1 modular theory: the ordered bar complex at all genera . . . 39
Status of the E_1 five main theorems. . . . 39

Resolved conjectures. 40
Remaining conjectural frontier. 40
The three bar complexes 40
Relationships. 41
Which is used where. 41
E_1 primacy 41
The averaging map. 41
The thesis. 41
1.17 Structural upgrades 42
Shadow archetype classification and single-line dichotomy. 42
Non-simply-laced Kac–Moody shadow verification. 42
Minimal model shadow obstruction towers. 42
Ran-coherent bar-cobar equivalence. 42
Ambient complementarity routing. 42
Chiral Deligne–Tamarkin principle and the dimensional jump. . . . 42
Modular trace principle. 43
Quartic logarithmic contact class. 43
Shadow separation (Theorem **??**). 43
Shadow tautological classes (Theorem **??**). 43
Chriss–Ginzburg tautological programme. 43
Non-perturbative shadow partition function. 44
Chiral Hochschild descent lemma. 46
Closure properties of chiral Koszulness. 46
Kac–Moody

$-modulepurityequivalence.$

. 46
Three-lens identification of the shadow tower. 46
Generating depth versus algebraic depth. 46
MC3 after the type-A reduction. 47
Holographic central charge identification (conjectural). 47
Analytic shadow realization (Corollary **??**). 47
1.18 The Polyakov programme 47
Dictionary. 47
Five main results. 48
Three frontier items. 48
Bibliographic pointers. 49
Cross-references. 49
1.19 Bridges to Volume II 49
1.20 Towards modular homotopy theory 50
The coderived Ran-space formalism 53
The Lagrangian form of complementarity 54
Completion kinematics and the modular cumulant transform 55
The analytic completion programme 57
The universal Maurer–Cartan class 57

The primitive kernel programme . . . 60
The discriminant as spectral invariant . . . 61
The index theorem for genus expansions . . . 64
The spectral discriminant as characteristic class . . . 67
Chain-level / evaluation-locus derived Drinfeld–Kohno and the Bridge Theorem . . . 67
Summary . . . 69
The holographic completion . . . 69
The open/closed realization and the chiral derived center . . . 72
The four-stage architecture. . . . 73
Five negative principles (permanent). . . . 74
1.21 Conjecture promotions and the true frontier . . . 74
Promotions . . . 75
Reductions . . . 77
The true frontier . . . 77
The four-test boundary of control . . . 82
Two orthogonal axes . . . 83
Three Langlands gaps . . . 85
Spectral continuation programme . . . 86
1.22 Three falsification tests . . . 94
April 2026 session: falsification test results. . . . 96
1.23 Programme status ledger . . . 97
The descent fan . . . 99
1.24 Open problems in the Θ^{oc} framework . . . 101
Closed-sector projections . . . 101
Koszulness programme . . . 103
Categorical and completion targets . . . 104
Bridge and factorization targets . . . 105
Analytic and holographic targets . . . 106
Arithmetic targets . . . 107
Open-sector and globalisation targets . . . 109
1.25 Forbidden slogans . . . 111

2 Editorial constitution and research directions 115
2.1 Constitutional declaration . . . 115
2.2 Conjecture stratification . . . 115
Resolved Entry Theorem and Dependency-Ordered Open Problems . . . 116
Homotopy templates for conjectures . . . 129
2.3 Research directions . . . 130
Higher-dimensional Koszul duality . . . 130
Vassiliev invariants from Feynman transform . . . 132
Anomaly cancellation and Koszul structure . . . 132
AdS_3/CFT_2 as curved Koszul duality . . . 134
3d mirror symmetry from E_1-chiral Koszul duality . . . 135
Noncommutative Hodge theory from genus tower . . . 135
Analytic completion and the sewing envelope . . . 136
The nine directions: overview and interdependence . . . 137

2.4 Programme status assessment 138
Status: nine futures 138
Index of conjectures and related results 142

Bibliography **145**

Chapter 1

Concordance with primary literature

This chapter is the constitution of the monograph. When chapters disagree, this chapter governs. It records the five main theorems with their precise status and locates every result against the prior literature: what is proved, what is conditional, what remains conjectural.

The governing principle is falsification: every claim in these three volumes is presumed false until it appears below with an explicit status tag and a proof reference or independent verification trail. A theorem that contradicts the concordance is not a theorem; it is an error in the chapter that states it. Omission from this chapter is itself a verdict: what is not recorded here has not survived scrutiny. The concordance does not summarize the monograph; the monograph approximates the concordance.

The five main theorems. The algebraic engine rests on five theorem packages. Their proved core and conditional refinements are recorded explicitly below.

	Theorem	Statement	Status
A	Bar-cobar adjunction	$\bar{B}^{\mathrm{ch}} \dashv \Omega^{\mathrm{ch}}$ on $\mathrm{Ran}(X)$, with the algebra-level Verdier half only after $\mathbb{D}_{\mathrm{Ran}}$: $\mathbb{D}_{\mathrm{Ran}}\overline{B}_X(\mathcal{A}_i) \simeq \Omega_X(_j) \simeq \mathcal{A}_j$, (Thm **??**)	[proved here]
B	Bar-cobar inversion	Strict quasi-isomorphism on the Koszul locus $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Thms **??**, **??**); off the locus, the counit is an unconditional coderived coacyclic-equivalence, promoted back to an ordinary quasi-isomorphism on collapse loci	[proved here]
C	Complementarity	C0: coderived fiber-center identification unconditional, ordinary-derived only on the flat perfect locus (Thm **??**). C1: homotopy eigenspace decomposition for all g, with duality/Lagrangian only for $g \geq 1$ and genus 0 treated separately (Thm **??**). C2: shifted-symplectic/BV upgrade conditional on the uniform-weight perfectness package (Thm **??**)	C0/C1: [proved here]; C2: [conditional]
D	Modular characteristic	$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g$ on the proved uniform-weight lane at all genera (Definition **??**, Thm **??**); genus 1 is universal for all families; multi-weight $g \geq 2$ acquires the cross-channel term $\delta F_g^{\mathrm{cross}}$ (Thm **??**); the routing $\Theta_{\mathcal{A}} \to \mathrm{obs}_g \to$ family index is non-circular	[proved here]
H	Chiral Hochschild	On the Koszul locus at generic level: $CH^*(\mathcal{A})$ is concentrated in $\{0, 1, 2\}$, satisfies the duality shift $[2]$, has degree-$\leq$ 2 Hilbert polynomial, and obeys $CH^1(V_k()) \cong$ for affine KM; the collision-depth spectral sequence collapses by local Fulton–MacPherson formality (Thms **??**, **??**; Prop. **??**)	[proved here]

Theorem **??** (the flag theorem) in the Introduction unifies them.

All five descend from a single object. The modular L_∞ convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ carries its L_∞ structure from the Feynman transform of the modular operad; the dg Lie algebra of Definition **??** is its strict model $\mathrm{Conv}_{\mathrm{str}}$, and the homotopy-invariant object is $\mathrm{Def}_\infty^{\mathrm{mod}}(\mathcal{A})$ (Theorem **??**). The universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ exists by the bar-intrinsic construction (Theorem **??**: $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is MC because $D_{\mathcal{A}}^2 = 0$). This is the coinvariant carrier for the main theorem. The ordered lift $\Theta_{\mathcal{A}}^{E_1} \in \mathrm{MC}((\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})^{E_1})$ is constructed separately in the E_1-modular chapter (Theorem **??**), and averaging recovers the concordance-level class $\Theta_{\mathcal{A}}$. The shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq 2} \to \Theta_{\mathcal{A}}^{\leq 3} \to \Theta_{\mathcal{A}}^{\leq 4} \to \cdots$ consists of its finite-order projections. Theorems A–C are consequences of the MC package; Theorems D and H are scalar and Hochschild outputs. The governing equation is $D_{\mathcal{A}}(\Theta_{\mathcal{A}}) + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$.

Convention 1.0.1 *(Three Hochschild theories).* Three distinct Hochschild theories appear in this programme. They differ in input, output, and geometric origin; conflating them is a category error.

(i) *Topological Hochschild cohomology* $^*(A, A)$. Input: an E_1-algebra A (associative up to homotopy). Output: an E_2-algebra (Deligne conjecture, proved by many authors). Geometric origin: the real line and its interval compactifications. This is the classical theory of deformation complexes.

(ii) *Chiral Hochschild cohomology* $CH^*(\mathcal{A}) = CH^*(\mathcal{A}, \mathcal{A})$. Input: a chiral algebra $\mathcal{A}$ on a smooth curve X (equivalently, a vertex algebra on the formal disk D). Output: a graded vertex algebra, concentrated in cohomological degrees $\{0, 1, 2\}$ for modular Koszul algebras (Theorem H). Geometric origin: the curve X and its Fulton–MacPherson compactifications. This is the theory developed in the present monograph.

(iii) *Categorical Hochschild homology* $_*()$. Input: a dg or A_∞-category (e.g. $\mathrm{Rep}_q()$). Output: a chain complex carrying an E_2-structure with Calabi–Yau shifted Poisson structure when is proper Calabi–Yau. Geometric origin: the circle S^1 and its cyclic bar construction.

The geometry determines which theory applies: curve $X \to$ chiral, real line $\to$ topological, Calabi–Yau category $\to$ categorical. When a comparison relates two theories (e.g. equation (**??**) in Chapter **??**), the qualifier on each side is mandatory. Bare "Hochschild" without qualifier is permitted only in historical remarks about the classical (topological) theory; in all other contexts, one of the three adjectives *topological*, *chiral*, or *categorical* is required.

The three structure theorems and the discriminant

Three *structure theorems* govern the representation-theoretic content: Riccati algebraicity, the formality identification, and complementarity (Theorem C). Theorems D and H are *projectional outputs*: κ is the scalar trace of $\Theta_{\mathcal{A}}$, and the chiral Hochschild ring is a polynomial consequence of Koszul functoriality.

- **Algebraicity** (Theorem **??**). On any one-dimensional primary slice L of the cyclic deformation complex, the weighted shadow generating function $H(t) = \sum r\, S_r\, t^r$ satisfies $H(t)^2 = t^4 Q_L(t)$ for an explicit quadratic polynomial Q_L. The shadow obstruction tower is therefore algebraic of degree 2 over $k(c)[t]$, determined by three invariants: the modular characteristic κ, the cubic coefficient α, and the quartic shadow S_4.

- **Formality identification** (Theorem **??**). The degree-r shadow $\mathrm{Sh}_r(\mathcal{A})$ has genus-0 projection equal to the transferred r-ary L_∞ bracket on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ evaluated on the truncated MC element. The genus-0 shadow obstruction tower is the L_∞ formality obstruction tower; the positive-genus corrections lie in the quantum L_∞ layer.

- **Complementarity** (Theorem **??**). For a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$, the genus-$g$ partition functions decompose as $Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$. The Verdier involution sends $\Theta_{\mathcal{A}}$ to $\Theta_{\mathcal{A}^!}$, and the two summands are complementary Lagrangians for the Verdier-side shifted pairing; the ambient formal-moduli upgrade is conditional on perfectness and nondegeneracy.

The organizing invariant is the *critical discriminant*

$$\Delta := 8\,\kappa\, S_4.$$

On each primary line it controls shadow depth (Theorem **??**: $\Delta = 0$ forces $r_{\max} \leq 3$; $\Delta \neq 0$ forces $r_{\max} = \infty$; class C escapes via stratum separation), the growth rate ρ of shadow coefficients (Definition **??**), and the duality constraint: within each family, $\Delta(\mathcal{A}) + \Delta(\mathcal{A}^!)$ is rational with universal numerator (eq. (**??**)).

Relation to classical representation theory. Classical Koszul duality (Priddy; Beilinson–Ginzburg–Soergel) is a theory of graded algebras over a point. It embeds into the genus-0, degree-2, $\Delta = 0$ stratum of

the chiral theory (class G, $r_{\max} = 2$) via formal-disk restriction, but the embedding is not an equivalence: vertex algebras live on the formal disk D, not on a bare point; the deformation retract $\mathbb{A}^1 \to \mathrm{pt}$ is additional data (specifying it and the attendant homotopy transfer is part of the comparison); and on $\mathbb{P}^1$ the global topology of FM compactifications introduces Arnold relations absent over a point. The Drinfeld–Kohno bridge recovers the classical R-matrix as the genus-0 binary shadow projection (Construction **??**; DK-0/1 proved, DK-2/3 proved on the evaluation-generated core, DK-4/5 conjectural). The Feigin–Frenkel center $H^0(\overline{B}(\mathcal{C}_{-h^\vee})) \cong \mathrm{Fun}(\mathrm{Op}_\vee(D))$ (Theorem **??**) is the bar *cohomology* at the critical level $k = -h^\vee$, where $\kappa = 0$ and the scalar class vanishes; it is orthogonal to the full shadow obstruction tower, not a projection of it. Algebraicity, the formality identification, and Lagrangian complementarity have no classical analogue.

Shadow depth classifies complexity within the Koszul world. Within the chirally Koszul standard examples treated in Part **??** (Heisenberg, affine Kac–Moody, bc-$\beta\gamma$, Virasoro, universal principal $\mathcal{W}_N$), shadow depth $r_{\max}$ and the four-class partition G/L/C/M measure L_∞ non-formality, not Koszulness status: class G (formal, $r_{\max} = 2$), class L (single Massey product, $r_{\max} = 3$), class C (quartic then rigid by stratum separation, $r_{\max} = 4$), class M (intrinsically non-formal, $r_{\max} = \infty$).

1.1 Principal contributions

*(1) Configuration space bar-cobar adjunction (Theorems **??** and **??**).* Bar and cobar are realized via explicit integrals over Fulton–MacPherson compactifications $\overline{C}_n(X)$: bar differential by residues at collision divisors, cobar differential by inclusions of strata. The adjunction holds for all Koszul chiral algebras at all genera. Inversion is genus 0 unconditional; at $g \geq 1$ it is unconditional on the CFT-type standard landscape except integer-spin $\beta\gamma$ by PBW propagation, and otherwise conditional on axiom **??**. Three layers (Remark **??**): chiral twisting morphisms (A_0, Theorem **??**), bar concentration (A_1, Theorem **??**), Verdier-geometric duality (A_2, Theorem **??**). *Quillen upgrade*: $B_\kappa \dashv \Omega_\kappa$ is a Quillen equivalence in the Vallette model structure [389, Theorem 2.1] (Theorem **??**); every Ch_∞-algebra rectifies to a strict chiral algebra (Corollary **??**).

*(2) Genus universality (Theorem **??**).* For a Koszul chiral algebra $\mathcal{A}$ with uniform-weight generators, the genus-g obstruction factors as $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ in $H^{2g}(\overline{\mathcal{M}}_g)$. For multi-weight families, only the genus-1 identity is unconditional. Rigidity proves level-direction concentration $\Theta^{\min} = \eta \otimes \Gamma_{\mathcal{A}}$; the stronger all-genera scalar formula $\Theta^{\min} = \kappa \cdot \eta \otimes \Lambda$ and hence $\mathrm{obs}_g = \kappa \cdot \lambda_g$ *fails* for multi-weight families: the full genus-g free energy receives a cross-channel correction $\delta F_g^{\mathrm{cross}}$ (Theorem **??**, Open Problem **??** resolved negatively). Here κ is an algebra-dependent constant and λ_g is the Faber–Pandharipande tautological class on $\overline{\mathcal{M}}_g$. The genus-independent coefficient $\kappa(\mathcal{A})$ is determined by the genus-1 curvature. This is the *scalar* genus universality (Level 0 of Remark 1.20.1): the first Chern class of the factorization homology bundle. The full genus-g invariant, the cohomology $H^*(\overline{B}^{(g)}(\mathcal{A}), D_g)$ and its variation over $\mathcal{M}_g$, is a chain-level categorical invariant (Theorem **??**).

*(3) Deformation-obstruction complementarity (Theorem **??**).* The package splits into three layers. C0 identifies the fiber object with the derived center in the coderived category unconditionally, and in the ordinary derived category on the flat perfect locus (Theorem **??**). C1 gives the homotopy eigenspace decomposition for all genera, with duality and the Lagrangian package for $g \geq 1$, while genus 0 is treated separately (Theorem **??**). C2 is the shifted-symplectic/BV upgrade, conditional on the uniform-weight perfectness package (Theorem **??**). The Kodaira–Spencer map is constructed for all Koszul pairs and all genera (Theorem **??**).

(4) E_1*-chiral Koszul duality (Theorem* **??***).* The bar-cobar adjunction extends to E_1-chiral algebras (nonlocal vertex algebras with R-matrix braiding), covering Yangians and toroidal algebras, beyond the E_∞ (commutative/local) setting of BD and FG.

(5) *Explicit computations.* Complete Koszul duality computations for all standard families: Heisenberg, free fermion, bc-$\beta\gamma$, affine Kac–Moody ($\widehat{\mathfrak{g}}_k \leftrightarrow \mathrm{CE}^{\mathrm{ch}}(\widehat{\mathfrak{g}}_{-k-2h^\vee})$), Virasoro (same-family complementarity partner Vir_{26-c} at M/S-level, with the $\mathcal{W}_\infty$ realization retained as project), $\mathcal{W}_N$ algebras, Yangians, the $\mathcal{N} = 2$ superconformal algebra (Chapter **??**: $\kappa = (k+4)/4$, complementarity $\kappa + \kappa' = 1$, Koszul despite nonvanishing H^2_{CE}), and the moonshine module $V^\natural$ (Chapter **??**: $\kappa = 12$, class M, Niemeier discrimination at $c = 24$). Genus-g free energies through $g = 10$ with Bernoulli asymptotics; bar cohomology dimensions through degree 10.

(6) *Oper differential-form package from the bar complex (Theorem* **??***).* At the critical level $k = -h^\vee$, the bar complex is uncurved and its cohomology recovers the formal-disc oper differential-form package: $H^n(\overline{B}(\widehat{\mathfrak{g}}_{-h^\vee})) \cong \Omega^n(\mathrm{Op}_\vee(D))$. The H^0 identification gives the Feigin–Frenkel center [160]; the H^1 identification gives Kähler differentials of the oper space; and the full cohomological identification follows from the bar-Ext identification combined with the Frenkel–Teleman theorem [179]. This gives a vertex-algebraic chain model interfacing with the oper side of the geometric Langlands programme.

(7) *Chiral Hochschild cohomology (Theorems* **??**, **??***; Proposition* **??***).* On the Koszul locus at generic level, the collision-depth spectral sequence for the Fulton–MacPherson stratification collapses at $E_2 = E_\infty$ because each local $\mathrm{FM}_m(\mathbb{C})$ is formal. Hence $CH^*(\mathcal{A})$ is concentrated in cohomological degrees $\{0, 1, 2\}$, carries the derived duality shift $[2]$, and has Hilbert polynomial of degree at most 2. For the principal $\mathcal{W}$ and Virasoro lanes this polynomial is $1 + t^2$ at generic level. For affine Kac–Moody at generic level: $CH^1(V_k()) \cong$ (Proposition **??**). Critical level is separate: Theorem H does not apply there, and the oper/Feigin–Frenkel package is recorded instead by Theorem **??**. Functoriality under Koszul duality passes through the Connes operator S (Theorem **??**, Corollary **??**).

(8) *Chain-level modular functor (Theorem* **??***).* The Feynman transform of the Getzler–Kapranov modular operad assembles the genus-graded bar complex into a homotopy-coherent modular functor: surface $\mapsto$ cochain complex, degeneration $\mapsto$ chain map, consistency relation $\mapsto$ chain homotopy. H^0 at integrable level recovers conformal blocks (Definition **??**). For uniform-weight algebras, the scalar genus tower $F_g = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ is its first Chern class; $\Delta_{\mathcal{A}}$ its Chern character; the flat connection on its cohomology sheaf furnishes the bar-side comparison surface for the KZ/Hitchin monodromy package on the integrable affine lane (Remark **??**).

(9) *Chiral center theorem (Theorems* **??** *and* **??***).* The chiral Hochschild cochain complex $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ is the operadic center of a logarithmic chiral algebra $\mathcal{A}$ in the holomorphic–topological Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$. The pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\ \mathcal{A})$ is terminal in the category of local chiral Swiss-cheese pairs over $\mathcal{A}$: for every such pair $(B, \mathcal{A}, \mu)$, there exists a unique morphism $\Phi\colon B \to C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ intertwining the brace structures and the mixed operations. Two independent proofs: (i) operadic, via the recognition theorem and homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$ (§**??**); (ii) direct, via the explicit brace morphism with Koszul signs (§**??**).

Remark 1.1.1 *(Geometric Langlands connection).* The bar complex $\overline{B}(\widehat{\mathfrak{g}}_{-h^\vee})$ at critical level is a *chain-level enhancement* of the Feigin–Frenkel center: it encodes functions on the oper space (in H^0), Kähler differen-

tials (in H^1), and higher exterior forms (in $H^{\geq 2}$), reproducing the oper differential-form package central to the Langlands programme (Theorem **??**).

Remark 1.1.2 *(The twisting morphism as organizing datum).* The canonical twisting morphism $\tau\colon \overline{B}_X(\mathcal{A}) \to \mathcal{A}$ underlies the four main theorems (Convention **??**): Theorem A establishes τ; Theorem B inverts it; Theorem C identifies its genus-g curvature; Theorem D computes its leading coefficient. At higher genus, τ extends to a curved twisting cochain $\tau_{\mathrm{mod}} = \tau + \Theta_{\mathcal{A}}$ whose curvature $\kappa(\mathcal{A}) \cdot \omega_g$ governs the entire modular deformation tower (Remark **??**).

1.2 Relationship to Beilinson–Drinfeld

Our Terminology	**BD Terminology**
Chiral algebra	Chiral algebra (same)
E_∞-chiral algebra	Vertex algebra / Chiral algebra
E_1-chiral algebra	No direct analog (generalization)
Chiral bracket μ	Chiral operation $\mu : j_* j^*(A \boxtimes A) \to \Delta_! A$
Bar complex ($\mathcal{A}$)	Chiral chain complex (cf. Chevalley–Eilenberg for Lie case)
Factorization structure	Factorization algebra structure
Ran space $\mathrm{Ran}(X)$	$\mathfrak{R}(X)$ in BD notation

Remark 1.2.1 *(Key extension).* Our E_1-chiral algebras extend BD's chiral algebras by dropping skew-symmetry. BD's chiral algebras are our E_∞-chiral algebras.

1.3 Relationship to Francis–Gaitsgory

Our Terminology	**FG Terminology**
Chiral Koszul duality	Chiral Koszul duality (same)
$\mathsf{Com}^{\mathrm{ch}}$-$\mathsf{Lie}^{\mathrm{ch}}$ duality	Main theorem of FG
$\mathsf{Ass}^{\mathrm{ch}}$-$\mathsf{Ass}^{\mathrm{ch}}$ duality	Not explicitly treated (implicit)
Pro-nilpotence	Pro-nilpotent tensor ∞-category
Bar-cobar equivalence	Koszul duality equivalence

Remark 1.3.1 *(Key extension).* FG establish $\mathsf{Com}^{\mathrm{ch}}$-$\mathsf{Lie}^{\mathrm{ch}}$ duality. We show this is derived from the more fundamental $\mathsf{Ass}^{\mathrm{ch}}$-$\mathsf{Ass}^{\mathrm{ch}}$ self-duality via the deformation Pois $\to$ Ass.

1.4 Relationship to Gui–Li–Zeng

Our Terminology	**GLZ Terminology**
Quadratic chiral algebra	Quadratic chiral algebra (same)
Koszul dual $\mathcal{A}^!$	Quadratic dual $A^!$
Bar complex	Not used (direct quadratic dual)
Non-quadratic duality	Not treated
E_1-chiral algebras	Not treated

PROPOSITION 1.4.1 *(GLZ as special case;* [proved here]*).* The Gui–Li–Zeng quadratic duality is a special case of the $\mathsf{Ass}^{\mathrm{ch}}$-duality of this monograph:

$$\text{GLZ duality} = \mathsf{Ass}^{\mathrm{ch}}\text{-duality}|_{\mathsf{E}_\infty\text{-chiral}}|_{\text{quadratic}}$$

Restricting to E_∞-chiral algebras with quadratic presentations recovers their theory.

Proof. An E_∞-chiral algebra $\mathcal{A}$ with quadratic presentation has a quadratic relation ideal $R \subset j_*j^*(\mathcal{A}^{\boxtimes 2})$. Its $\mathsf{Ass}^{\mathrm{ch}}$ Koszul dual is determined by bar degrees ≤ 2: the generators V^* in $\bar{B}^1$ and the orthogonal complement of relations $R^\perp$ in $\bar{B}^2$ (the higher bar degrees, while nonzero, contribute no independent data for quadratic algebras). The Koszul dual coalgebra $\bar{B}(\mathcal{A})$ is then determined by $R^\perp$, and the Koszul dual algebra $\mathcal{A}^! = \bar{B}(\mathcal{A})^\vee$ is recovered by linear duality under the chiral pairing $j_*j^*(\mathcal{A}^{\boxtimes 2}) \otimes j_*j^*((\mathcal{A}^!)^{\boxtimes 2}) \to \omega_X$. This is the quadratic dual construction of GLZ.

For the E_∞ restriction: our $\mathsf{Ass}^{\mathrm{ch}}$ bar complex uses the full associative chiral product $\mu\colon j_*j^*(\mathcal{A} \boxtimes \mathcal{A}) \to \Delta_!\mathcal{A}$. When $\mathcal{A}$ is E_∞-chiral (i.e., the chiral bracket is skew-symmetric), the bar complex acquires a commutative coalgebra structure, and the Koszul dual is a Lie chiral algebra. This recovers the FG framework as a special case (see Theorem 1.4.2). □

THEOREM 1.4.2 *(FG duality from* $\mathsf{Ass}^{\mathrm{ch}}$ *self-duality;* [proved here]*).* The Francis–Gaitsgory $\mathsf{Com}^{\mathrm{ch}}$-$\mathsf{Lie}^{\mathrm{ch}}$ duality is the associated graded of the $\mathsf{Ass}^{\mathrm{ch}}$ self-duality under the PBW filtration. Precisely: for a quadratic E_∞-chiral algebra $\mathcal{A}$, the $\mathsf{Ass}^{\mathrm{ch}}$ bar complex $\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A})$ carries a filtration $F_\bullet$ (induced by the symmetrization degree of the Σ-action on $C_n(X)$) such that:

$$\mathrm{gr}^F \bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A}) \;\cong\; \bar{B}_{\mathsf{Com}^{\mathrm{ch}}}(\mathrm{Sym}^{\mathrm{ch}}(\mathcal{A})) \;\cong\; \mathrm{CE}^{\mathsf{Lie}^{\mathrm{ch}}}(\mathcal{A})$$

and the associated spectral sequence degenerates at E_1 for Koszul algebras.

Proof. Step 1: PBW filtration. The chiral bar complex $\bar{B}^n_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A})$ consists of sections on $\overline{C}_{n+1}(X)$ with coefficients in $\mathcal{A}^{\boxtimes(n+1)} \otimes \Omega^n_{\log}$. Define $F_p\bar{B}^n$ as the span of elements symmetric under at least $(n-p)$ transpositions of configuration space points. This is the chiral analogue of the degree filtration on the classical associative bar complex (Loday–Vallette [299], §9.3.2).

Step 2: Associated graded identification. On $\overline{C}_n(X)$, the Σ_n-representation decomposes as $j_*j^*(\mathcal{A}^{\boxtimes n}) = \mathrm{Sym}^n_{\mathrm{ch}}(\mathcal{A}) \oplus \bigwedge^2_{\mathrm{ch}}(\mathcal{A}) \otimes \cdots$. The PBW filtration separates these: $F_0/F_1 = \mathrm{Sym}^n_{\mathrm{ch}}(\mathcal{A})$ (the fully symmetric part) is the commutative chiral bar complex, and F_1/F_2 involves the Lie bracket (the antisymmetric part). The associated graded $\mathrm{gr}^F\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}$ is thus the Chevalley–Eilenberg complex for the Lie chiral algebra structure, which is $\bar{B}_{\mathsf{Com}^{\mathrm{ch}}}(\mathrm{Sym}^{\mathrm{ch}}(\mathcal{A}))$ by the classical identification of the commutative bar complex with Chevalley–Eilenberg chains.

Step 3: Spectral sequence degeneration. For Koszul $\mathcal{A}$, the spectral sequence $E_1^{p,q} = H^{p+q}(\mathrm{gr}^F_p\bar{B})$ degenerates at E_1. To see this, note that the associated graded $\mathrm{gr}^F\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A})$ is the Chevalley–Eilenberg complex (Step 2), and the Koszul property of $\mathcal{A}$ as a chiral algebra implies that this commutative bar complex has cohomology concentrated on the diagonal $p+q=0$ (i.e., $E_1^{p,q}=0$ for $p+q\neq 0$). Since $d_1\colon E_1^{p,q} \to E_1^{p+1,q}$ shifts the total degree $p+q$ by 1, it maps from the surviving diagonal to the zero region, hence $d_1 = 0$ and $E_2 = E_1$. All higher differentials vanish for the same degree reason. Therefore $H^*(\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A})) \cong H^*(\mathrm{gr}^F\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A}))$, and the FG duality is recovered as the associated graded of our $\mathsf{Ass}^{\mathrm{ch}}$ duality.

Step 4: Comparison with FG. The FG theorem states: for an E_∞-chiral algebra $\mathcal{A}$, the chiral Koszul dual is a Lie chiral algebra $\mathcal{A}^{!,\mathsf{Lie}^{\mathrm{ch}}}$. In the present framework: the Koszul dual coalgebra $\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A})$ is

an E_1-chiral coalgebra, and the Koszul dual algebra $\mathcal{A}^{!,\mathsf{Ass}^{\mathrm{ch}}} = \bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A})^\vee$ is an E_1-chiral algebra; its associated graded under PBW is $\mathcal{A}^{!,\mathsf{Lie}^{\mathrm{ch}}} = \bar{B}_{\mathsf{Com}^{\mathrm{ch}}}(\mathcal{A})^\vee$. (The cobar $\Omega(\bar{B}_{\mathsf{Ass}^{\mathrm{ch}}}(\mathcal{A}))$ recovers $\mathcal{A}$ itself by bar-cobar inversion, not $\mathcal{A}^!$.) The FG duality is recovered. □

1.5 Relationship to Ayala–Francis

Our Terminology	**AF Terminology**
Non-abelian Poincaré duality	Nonabelian Poincaré duality (same)
FM compactification $\overline{C}_n(X)$	Disk stratification of $\mathrm{Ran}(X)$
Verdier duality on $\overline{C}_n(X)$	Poincaré/Koszul duality for E_2-algebras on oriented surfaces
Bar complex as factorization coalgebra	Factorization homology

Remark 1.5.1 *(Key comparison).* The Ayala–Francis framework [30, 31] establishes non-abelian Poincaré duality as an equivalence between factorization homology and factorization cohomology for framed E_n-algebras on n-manifolds. Our Theorem A (Theorem **??**) is the specialization to dimension $n = 2$ (complex curves viewed as real oriented surfaces) with the additional structure of:

(i) *Holomorphic enrichment.* The AF framework works with topological E_n-algebras and locally constant cosheaves. We work with holomorphic chiral algebras ($\mathcal{D}$-modules on curves), where the Fulton–MacPherson compactification provides the correct holomorphic compactification (not only the topological one). The bar differential involves residues of meromorphic forms, not only topological data.

(ii) *Explicit formulas.* The AF duality is an ∞-categorical equivalence; our realization via configuration space integrals gives explicit chain-level formulas for the bar differential (residues at collision divisors) and the cobar differential (inclusions of FM strata).

(iii) *Higher genus.* For genus $g \geq 1$ curves, the configuration space $C_n(\Sigma_g)$ acquires additional topology ($H^1(\Sigma_g)$), and the bar complex inherits new generators from the handle cycles. The AF framework predicts this (factorization homology on non-simply-connected manifolds), but the explicit identification with moduli space integrals and the Hodge bundle (Theorem C) is new.

Remark 1.5.2 *(Reconstruction versus duality).* Theorem A produces two functors, not one:

$$R_X = \Omega^{\mathrm{ch}}_X \circ \bar{B}^{\mathrm{ch}}_X, \qquad K_X = \Omega^{\mathrm{ch}}_X \circ \mathbb{D}_{\mathrm{Ran}} \circ \bar{B}^{\mathrm{ch}}_X.$$

The first recovers $\mathcal{A}$ (bar-cobar inversion, Theorem B); the second produces $\mathcal{A}^!$ (Koszul duality). The Verdier duality $\mathbb{D}_{\mathrm{Ran}}$ converts the bar *coalgebra* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ into a factorization *algebra*, which cobar then reads as the dual chiral algebra. The four objects $\mathcal{A}$, $\bar{B}^{\mathrm{ch}}(\mathcal{A})$, $\mathcal{A}^{\mathrm{i}} = H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$, $\mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee$ must never be conflated (Remark **??**).

1.6 Relationship to Feigin–Frenkel

Our Terminology	**FF Terminology**
KM Koszul dual $\widehat{\ }_{-k-2h^\vee}$	FF involution $k \mapsto -k - 2h^\vee$
Critical level $k = -h^\vee$	Critical level (same)
Curvature $\kappa \propto (k + h^\vee)$	Center of $\widehat{\ }_{-h^\vee}$
DS reduction $\mathcal{W}^k()$	Drinfeld–Sokolov reduction (same)
$c + c' = 2d$	Not formulated
Genus-g obstruction obs_g	Not formulated

Remark 1.6.1 *(Key comparison).* The Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ is, in this monograph's framework, a consequence of chiral Koszul duality via Verdier duality on Fulton–MacPherson compactifications. The canonical treatment, including the geometric mechanism, the critical-level recovery of the FF center, and the proof of Theorem **??**, appears in Chapter **??**.

The extensions beyond FF are:

(i) The complementarity formula $c(\widehat{\ }_k) + c(\widehat{\ }_{-k-2h^\vee}) = 2d$ (Theorem **??**), which shows the central charge sum is a root datum invariant, independent of the level.

(ii) The genus-g obstruction formula $\mathrm{obs}_g = \kappa \cdot \lambda_g$ for uniform-weight algebras (Theorem **??**), which extends the FF involution to all genera via the Hodge bundle on $\overline{\mathcal{M}}_g$; genus-1 universality is unconditional for all families.

(iii) The Virasoro and $\mathcal{W}_N$ generalizations, where the Drinfeld–Sokolov parametrization introduces nonlinear level shifts (e.g., $c \mapsto 26 - c$ for Virasoro) that do not arise from a linear involution on levels.

1.7 Relationship to Costello–Gwilliam

Our Terminology	**CG Terminology**
Chiral algebra on X	Holomorphic factorization algebra on X
Bar complex $(\mathcal{A})$	Factorization homology $\int_X \mathcal{A}$
Cobar construction $\Omega(C)$	Factorization envelope
OPE residues	Observables in perturbative QFT
Envelope-shadow $\Theta^{\mathrm{env}}(R)$	Shadow obstruction tower of $\mathsf{Fact}(R)$; §1.8

Remark 1.7.1 *(Key comparison).* The Costello–Gwilliam factorization algebra framework [92] relates to ours through the Beilinson–Drinfeld equivalence between chiral algebras and factorization algebras on curves (BD, Chapter 3). The precise comparison:

(i) *Factorization homology.* The CG factorization homology $\int_X \mathcal{A}$ computes the same object as our chiral homology $H_*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$. Our explicit realization via configuration space integrals provides chain-level formulas that the CG framework treats abstractly.

(ii) *Koszul duality.* The CG framework includes Koszul duality for factorization algebras (CG, Vol. 2, Chapter 5), specialized to holomorphic factorization algebras on curves. Their Koszul duality, restricted to this setting, recovers our chiral Koszul duality. Our contribution is the explicit identification of the Koszul dual for all standard families (Kac–Moody, Virasoro, $\mathcal{W}_N$, etc.) via the bar-cobar adjunction on Fulton–MacPherson spaces.

(iii) *Quantum corrections.* The CG perturbative quantization framework produces A_∞ structures from Feynman diagram expansions (CG, Vol. 2, Chapter 3). Our curved A_∞ structure on the bar complex is the chiral-algebraic counterpart: the curvature $m_0 = \kappa \cdot \mathbf{1}$ arises from the highest-order OPE pole (quartic for Virasoro, double for Heisenberg), not from a Feynman diagram sum. The genus universality theorem shows that these two perspectives produce the same genus-g obstruction class.

(iv) *BV-BRST.* The CG BV formalism motivates the BRST/bar comparison package in Chapter **??**. The BV/bar identification is resolved in the coderived category D^{co} for all shadow classes (Theorem **??**); at the chain level, the Heisenberg scalar identification is proved at all genera (Theorem **??**), and classes G and L are proved, class C is conditional on harmonic decoupling, and the chain-level identification fails for class M (Conjecture 2.2.7).

(v) *Monoidal bar-cobar.* Booth–Lazarev [59] construct monoidal model structures on categories of coalgebras, providing the homotopical infrastructure for the bar-cobar adjunction to be a Quillen equivalence in the monoidal setting. On the current record, this is realized by the fusion-compatibility theorem (Theorem **??**): the bar construction is a lax monoidal functor from the relevant complete module category to the bar-comodule category, with quasi-isomorphism on the explicit Koszul/Künneth lanes where the theorem applies.

1.8 Relationship to Nishinaka–Vicedo

Nishinaka [428] constructs an amenably holomorphic prefactorization algebra from a Lie conformal algebra R via a *factorization envelope*, and proves that the associated vertex algebra is the enveloping vertex algebra $U_{\mathrm{vert}}(R)$. The construction explicitly recovers and generalizes the Costello–Gwilliam Kac–Moody factorization algebra [92] and Williams's Virasoro factorization algebra, and extends to super Lie conformal algebras, producing the Neveu–Schwarz and $\mathcal{N} = 2$ vertex superalgebras. Vicedo [429] gives a complementary unified factorization construction for full universal enveloping vertex algebras, treating Kac–Moody, Virasoro, and $\beta\gamma$ in one framework via unital local Lie algebras.

The factorization envelope provides a functorial machine:

$$R \in \mathsf{LCA} \xrightarrow{\text{fact. env.}} \mathsf{Fact}(R) \xrightarrow{\text{assoc. VA}} U_{\mathrm{vert}}(R). \tag{1.1}$$

Combined with the bar-cobar machinery of this monograph, the pipeline extends: the shadow obstruction tower is a functor on Lie conformal input.

Definition 1.8.1 *(Cyclically admissible Lie conformal algebra).* A Lie conformal algebra L is *cyclically admissible* if:

(i) it has a conformal-weight grading $L = \bigoplus_{h\geq 0} L_h$ with $\dim L_h < \infty$;

(ii) the descending filtration $F^m L := \bigoplus_{h\geq m} L_h$ is complete;

(iii) its OPE/λ-bracket has bounded pole order;

(iv) it carries an invariant residue pairing $\langle -, - \rangle \colon L \otimes L \to \omega_X$ compatible with translation and skew-symmetry.

Conditions (i)–(iii) are the Nishinaka admissibility conditions ensuring that the genus-0 factorization envelope $\mathsf{Fact}_X(L)$ exists [428]. Condition (iv) supplies the cyclic structure needed for the cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}(U_{\mathrm{vert}}(L))$ (Definition **??**) and modular completion.

Definition 1.8.2 *(Envelope-shadow functor).* Let R be a cyclically admissible Lie conformal algebra (Definition 1.8.1). Define the *envelope-shadow functor* at degree r by

$$\Theta^{\mathrm{env}}_{\leq r}(R) \;:=\; \Theta_{\leq r}\big(U_{\mathrm{vert}}(R)\big) \;\in\; \mathcal{A}^{\mathrm{sh}}_{r}\big(U_{\mathrm{vert}}(R)\big),$$

where $\Theta_{\leq r}$ is the degree-r truncation of the bar-intrinsic MC element (Theorem **??**). The full envelope-shadow is $\Theta^{\mathrm{env}}(R) := \varprojlim \Theta^{\mathrm{env}}_{\leq r}(R)$ (Theorem **??**).

Definition 1.8.3 *(Envelope-shadow complexity).* The *envelope-shadow complexity* of R is

$$\chi_{\mathrm{env}}(R) \;:=\; r_{\max}\big(U_{\mathrm{vert}}(R)\big) \;=\; \sup\big\{r \geq 2 : \mathcal{A}^{\mathrm{sh}}_{r,0}\big(U_{\mathrm{vert}}(R)\big) \neq 0\big\},$$

the shadow depth (Definition **??**) of the enveloping vertex algebra.

Proposition 1.8.4 *(Finite-jet rigidity).* [proved here] The degree-r envelope-shadow $\Theta^{\mathrm{env}}_{\leq r}(R)$ depends only on the $(r{+}1)$-jet of the enveloping vertex algebra; that is, only on $U_{\mathrm{vert}}(R)/F_{r+1}\,U_{\mathrm{vert}}(R)$, where $F_\bullet$ is the PBW filtration.

Proof. The shadow obstruction tower at degree r is constructed from r-vertex graph sums (Definition **??**). Each vertex carries a transferred operation $\ell^{\mathrm{tr}}_{|v|}$ from the minimal model of $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$, which at degree $\leq r$ sees at most $r{+}1$ OPE poles. These are precisely the data carried by $U_{\mathrm{vert}}(R)/F_{r+1}$. Higher-jet corrections appear only at degree $\geq r{+}1$. □

Proposition 1.8.5 *(Polynomial level dependence).* [proved here] Let R_t be a one-parameter family of Lie conformal algebras with central-extension parameter t (e.g., $t = k$ for affine Kac–Moody, $t = c$ for Virasoro). Then $\Theta^{\mathrm{env}}_{\leq r}(R_t)$ is polynomial in t of degree at most $r{-}1$:

$$\Theta^{\mathrm{env}}_{\leq r}(R_t) \;=\; \sum_{j=0}^{r-1} \theta^{(r)}_j\, t^j.$$

In particular, $\kappa = \theta^{(2)}_1\, t$ is linear (Theorem D), the cubic shadow $\mathfrak{C} = \theta^{(3)}_0 + \theta^{(3)}_1 t + \theta^{(3)}_2 t^2$ is at most quadratic, and the quartic resonance class $\mathfrak{Q}$ is at most cubic in t.

Proof. At degree r, the shadow is a sum over stable graphs with $\leq r$ vertices. Each vertex contributes one factor of the OPE structure constants, which are polynomial in t of degree 1 (for affine/Virasoro type data) or degree 0 (for the structural part). An r-vertex graph thus contributes degree $\leq r$ in t. The leading degree-r coefficient factors through the scalar MC equation and is absorbed into $\Theta_{\leq r-1}$ by the inductive construction (Construction **??**), leaving degree $\leq r{-}1$ for the genuinely new contribution at degree r. □

Proposition 1.8.6 *(Gaussian collapse for abelian input).* [proved here] Let R be an abelian Lie conformal algebra ($[-_\lambda -]_R = 0$). Then:

(i) $U_{\mathrm{vert}}(R)$ is a Heisenberg-type vertex algebra.

(ii) $\chi_{\mathrm{env}}(R) = 2$: the connected shadow obstruction tower terminates at degree 2.

(iii) $\Theta^{\mathrm{env}}(R) = \kappa(R) \cdot \eta \otimes \Lambda$ (scalar saturation).

The Heisenberg algebra is the universal Gaussian fixed point of the envelope-shadow functor: every abelian Lie conformal input produces a class-**G** shadow.

Proof. The abelian bracket gives $U_{\mathrm{vert}}(R) = \mathrm{Sym}^{\mathrm{ch}}(R)$, which is the Heisenberg algebra on the underlying $\mathcal{D}_X$-module with pairing determined by the cocycle of R. By the Heisenberg shadow computation (§**??**), the shadow obstruction tower terminates at degree 2 with the scalar curvature κ. Since $U_{\mathrm{vert}}(R)$ is Heisenberg-type, Part (iii) is the Heisenberg scalar formula (uniform-weight, single-generator), not an application of Theorem **??**. □

Remark 1.8.7 *(Envelope-shadow complexity of standard families).* The envelope-shadow complexity recovers the shadow depth classification (Definition **??**):

R	$U_{\mathrm{vert}}(R)$	$\chi_{\mathrm{env}}(R)$	**Class**
R abelian	Heisenberg	2	**G**
$\hat{}_k$ current	$V_k()$	3	**L**
$\beta\gamma$ current	$\beta\gamma$ system	4	**C**
Virasoro current	Vir_c	∞	**M**

Remark 1.8.8 *(From examples to machine).* The Nishinaka–Vicedo pipeline (1.1) replaces *ad hoc* constructions (free fields for Heisenberg, currents for Kac–Moody, the stress tensor for Virasoro) with a single functorial input: a Lie conformal algebra. The bar-intrinsic construction (Theorem **??**) produces $\Theta^{\mathrm{env}}(R)$ from this input. The super extension (Neveu–Schwarz, $\mathcal{N} = 2$) suggests the same construction for nonlinear chiral families.

Remark 1.8.9 *(Conjectural frontier).* Five directions remain open:

(i) *DS-envelope descent.* If $R \twoheadrightarrow R_{\mathrm{DS}}$ is a Drinfeld–Sokolov reduction, does Θ^{env} descend compatibly? The expectation is $\Theta_{\mathcal{W}_N} = H^0_{Q_{\mathrm{DS}}}(\Theta_{V_k()\otimes\mathcal{F}_{\mathrm{gh}}})$: reduction should commute with the universal MC class, turning DS from an operation on individual examples into a functor on modular Koszul datums (Construction 1.8.10). Reduction should happen *before* shadow-taking: the shadow obstruction tower of $\mathcal{W}_N$ should be the BRST-reduced image of the full current-plus-ghost tower.

(ii) *Plumbing-Taylor theorem.* Identify the degree-r shadow with the r-th Taylor coefficient of factorization homology $\int_{\Sigma_g} U_{\mathrm{vert}}(R)$ near a boundary divisor, using the plumbing-fixture expansion.

(iii) *Lie-conformal complementarity.* Lift the complementarity theorem (Theorem C) to a statement about R and its Lie-conformal dual $R^!$, giving $\Theta^{\mathrm{env}}(R) + \Theta^{\mathrm{env}}(R^!) = H^*({}_g, Z(U_{\mathrm{vert}}(R)))$ at the Lie-conformal level.

(iv) *Critical-level concentration.* At $k = -h^\vee$, the envelope $U_{\mathrm{vert}}(R_{-h^\vee})$ is uncurved; the shadow tower should concentrate on the factorizable Feigin–Frenkel center [430].

(v) *Super shadow machine.* Extend Θ^{env} to super Lie conformal algebras, computing the shadow obstruction tower for Neveu–Schwarz, $\mathcal{N} = 2$, and the superconformal families.

Construction 1.8.10 *(The modular Koszul datum).* Fix a smooth complex curve X and a cyclically admissible Lie conformal algebra L (Definition 1.8.1). The *modular Koszul datum* of L is the six-fold datum

$$\Pi_X(L) \;=\; \big(\mathsf{Fact}_X(L),\;\; \overline{B}_X(L),\;\; \Theta_L,\;\; \mathcal{L}_L,\;\; (V_L^{\mathrm{br}}, T_L^{\mathrm{br}}),\;\; \mathfrak{R}_4^{\mathrm{mod}}(L)\big),$$

where:

(1) $\mathsf{Fact}_X(L)$ is the genus-0 factorization envelope [428];

(2) $\overline{B}_X(L) := \overline{B}(\mathsf{Fact}_X(L))$ is the reduced bar coalgebra;

(3) $\Theta_L \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}(L)\widehat{\otimes}\mathbb{G}_{\mathrm{mod}})$ is the universal modular Maurer–Cartan class (Theorem **??**);

(4) $\mathcal{L}_L$ is the determinant line of the genus family;

(5) $(V_L^{\mathrm{br}}, T_L^{\mathrm{br}})$ is the spectral branch object: a finite-rank complex with its genus-1 transport operator;

(6) $\mathfrak{R}_4^{\mathrm{mod}}(L)$ is the quartic resonance class (Definition **??**).

The slogan: *envelope + bar coalgebra + universal MC class = modular form datum*. Every invariant (scalar anomaly, spectral sheet, nonlinear resonance) is a shadow of Θ_L.

Remark 1.8.11 *(The modular factorization adjunction).* The modular factorization adjunction is proved in Theorem **??**: the modular factorization envelope functor U_X^{mod} is left adjoint to the primitive-current functor $\mathrm{Prim}^{\mathrm{mod}}$ on cyclically admissible Lie conformal algebras. At genus 0, this recovers the Nishinaka–Vicedo envelope [428, 429]. The full modular envelope carries the modular Koszul datum $\Pi_X(L)$ (Construction 1.8.10) functorially.

Remark 1.8.12 *(Execution programme).* The factorization envelope programme decomposes into seven stages:

(1) Pin down the input category $\mathsf{LCA}_{\mathrm{cyc}}(X)$: Nishinaka-admissible plus cyclic pairing plus pronilpotent completion.

(2) Build $\overline{B}_X(L)$ functorially from the genus-0 factorization envelope.

(3) Construct $\mathrm{Def}_{\mathrm{cyc}}(L)$ as cyclic coderivations of $\overline{B}_X(L)$, then transfer stable-graph operations to it.

(4) Solve the Maurer–Cartan equation recursively by total complexity $2g - 2 + n +$ bar weight, using the genus-0 envelope as the base point.

(5) Extract the shadows in order: $\Theta_L \rightsquigarrow \kappa(L),\ \mathcal{L}_L,\ T_L^{\mathrm{br}},\ \Delta_L,\ \mathfrak{R}_4^{\mathrm{mod}}(L)$.

(6) Compute the archetypes: Heisenberg, $\widehat{\mathfrak{sl}}_2$, $\beta\gamma$, Virasoro, principal $\mathcal{W}_N$.

(7) Pass from the E_∞/local case to the ordered E_1 face, where the same package acquires an R-matrix/Yang–Baxter refinement.

Steps (1)–(2) are supplied by the current literature. Steps (3)–(5) are the content of the proved core (Theorem **??**; Definition **??**; Theorem **??**). Steps (6)–(7) are the constructive and spectral layers developed throughout Parts **??**–**??**.

1.9 Relationship to Loday–Vallette

Our Terminology	**LV Terminology**
Operad	Operad (same)
Koszul operad	Koszul operad (same)
Bar construction	Bar construction (same)
Cobar construction Ω	Cobar construction Ω (same)
Twisting morphism	Twisting morphism (same)
Maurer–Cartan equation	Maurer–Cartan equation (same)
$\mathsf{Ass}^! \cong \mathsf{Ass}$	$\mathsf{Ass}^! \cong \mathsf{Ass} \otimes \mathrm{sgn}$ (Thm 7.1.1)
$\mathsf{Com}^! \cong \mathsf{Lie}$	$\mathsf{Com}^! \cong \mathsf{Lie}\{1\}$ (Thm 7.2.1)

Remark 1.9.1 *(Convention for Koszul dual isomorphisms).* In Loday–Vallette [299], §7.1–7.2, the Koszul dual includes the sign representation: $\mathsf{Ass}^! = \mathsf{Ass} \otimes \mathrm{sgn}$ and $\mathsf{Com}^! = \mathsf{Lie}\{1\}$ (with operadic suspension). In this manuscript, we absorb the sign twist into the bar desuspension: our bar construction uses the shifted generators $s^{-1}\mathcal{A}$ whose sign already accounts for the sgn-twist. Concretely, our convention $\mathsf{Ass}^! \cong \mathsf{Ass}$ means that the dual cooperad is isomorphic to Ass as a *graded* cooperad after this shift. All bar-cobar computations in the text use the desuspended convention consistently, so no sign errors propagate. See Appendix **??** for the explicit translation between conventions.

Remark 1.9.2 *(Chiral enhancement).* The LV framework lifts to the chiral setting by replacing vector spaces with D-modules and tensor products with chiral tensor products.

Remark 1.9.3 *(Overview).* Each framework (BD, FG, CG, NV, LV) captures one projection of a single phenomenon: *the duality between commutative and Lie structures, mediated by configuration spaces*. This monograph gives the chain-level realization via configuration space integrals, making all projections simultaneously computable. The bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ is the meeting point: BD's chiral homology, FG's Koszul dual coalgebra, CG's factorization homology on the Ran space, NV's factorization envelope of Lie conformal input, and LV's bar construction, incarnated as a single cochain complex with computable differential. At the homotopy level, Conv_∞ unifies all these frameworks: each is a strictification or truncation of $\mathrm{Def}^{\mathrm{mod}}_\infty(\mathcal{A})$.

1.10 Relationship to Malikov–Schechtman

Malikov–Schechtman [309] introduce *homotopy chiral algebras* (HCA): chiral algebras up to higher homotopies, with the Moore complex of a cosimplicial Lie algebra in a pseudo-tensor category carrying L_∞ operations controlled by the Eilenberg–Zilber operad $\mathcal{Y}(n) = \mathcal{Z}(n) \otimes_k \mathrm{Lie}(n)$. Their main theorem (Theorem 3.1): the Čech complex of a sheaf of chiral algebras is an HCA. In our language (Definition **??**): the HCA operations assemble into a single MC element $\Phi_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\check{C}}_{\mathcal{A}})$ satisfying $D\Phi + \frac{1}{2}[\Phi, \Phi] = 0$.

Our framework	**MS framework**
Convolution algebra $\mathfrak{g}^{\mathrm{bar}}_{\mathcal{A}}$	Convolution algebra $\mathfrak{g}^{\check{C}}_{\mathcal{A}}$
MC element $\Theta_{\mathcal{A}} = (m_1, m_2, \ldots)$	MC element $\Phi_{\mathcal{A}} = (F_1, F_2, \ldots)$
A_∞ operations $\{m_n\}$	HCA operations $\{F_n\}$
Ran space / Fulton–MacPherson	Zariski site / affine cover
Associahedra / moduli $\overline{\mathcal{M}}_{0,n}$	Eilenberg–Zilber operad $\mathcal{Y}(n)$
Modular operad FCom (genus $g \geq 1$)	[Not treated; genus-0 only]
m_3 from $\mathrm{FM}_3(\mathbb{C})$ integration	F_3 from triple intersections
$d^2 = 0$ on $\overline{\mathcal{M}}_{0,n}$	Cosimplicial identities

Remark 1.10.1 *(Complementary resolutions).* The bar and Čech complexes resolve the same cohomology with compatible MC structures. The comparison (Conjecture **??**) is an L_∞ morphism Φ^* sending $\Phi_{\mathcal{A}}$ to $\Theta_{\mathcal{A}}$, generically *non-strict*: the bar side encodes configuration space topology (FM_k strata) while the Čech side encodes cover topology (intersections). The MS framework is purely algebraic; ours works at all genera via FCom and the shadow obstruction tower. The curvature $d^2 = \kappa \cdot \omega_g$ and finite-order shadows have no Čech counterpart in [309].

Remark 1.10.2 *(Genus-*0 *strictness and the asymmetry).* On $\mathbb{P}^1$ with a two-element cover, the Čech HCA is *strict for every chiral algebra* (Proposition **??**): the MC element is $\Phi = (d_M, F_2, 0, 0, \ldots)$ because $\check{C}^p = 0$ for $p \geq 2$ kills all higher operations. The bar A_∞ structure, by contrast, has $m_3 \neq 0$ for non-abelian

algebras (e.g., $\widehat{\mathfrak{sl}}_2$) because the triple collision stratum of $\mathrm{FM}_3(\mathbb{C})$ contributes regardless of the cover. The comparison Φ^* creates m_3 from F_2 via its higher components Φ_2^*. Genuine Čech non-strictness ($F_3 \neq 0$) appears at genus $g \geq 1$, where covers require ≥ 3 opens and triple intersections contribute (Example **??**).

1.11 Three-pillar foundational architecture

Three recent papers force upgrades at three levels of the monograph's foundations. The resulting *three-pillar architecture* is a constitutional amendment.

The three pillars form a triangle centered on the master MC equation $D \cdot \Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$. Each pillar provides one face of $\Theta_{\mathcal{A}}$: Pillar A gives its *local algebraic nature* (a homotopy chiral algebra on the curve), Pillar B gives its *deformation-theoretic home* (the convolution sL_∞-algebra controlling all twisting morphisms), and Pillar C gives its *global geometric skeleton* (the log-FM compactification providing collision geometry). Every theorem reads through all three lenses simultaneously; the architecture is a single refoundation, not three separate additions.

Remark 1.11.1 *(The three pillars).* The primitive objects of the theory are:

(A) **Local algebraic object** (Pillar A). The primitive local object is a *homotopy chiral algebra* (Ch_∞-algebra) in the sense of Malikov–Schechtman [309], not a strict chiral algebra. A strict chiral algebra appears only after rectification or on cohomology. The chain-level story is $\mathrm{Ch}_\infty \to$ strict chiral algebra on $H^\bullet \to$ PVA/coisson shadow. The PVA descent from A_∞ chiral algebra to (-1)-shifted Poisson vertex algebra on cohomology is proved in Volume II (Volume II, PVA descent theorem): commutativity, skew-symmetry, Jacobi, and Leibniz (D2–D5) are established by exchange-cylinder and three-face Stokes arguments on $\mathrm{FM}_3(\mathbb{C})$; higher-operation vanishing (D6) is proved via H3(a) factorization plus topological contractibility of $\mathrm{Conf}_k^<(\mathbb{R})$.

(B) **Deformation-control object** (Pillar B). The universal deformation machine is a *filtered convolution sL_∞-algebra* $\hom_\alpha(,\mathcal{A})$ relative to a twisting morphism α, in the sense of Robert-Nicoud–Wierstra [367] and Vallette [389]. The deformation space is the Maurer–Cartan ∞-groupoid $\mathrm{MC}_\bullet(\hom_\alpha)$, not a naive MC set modulo gauge. The dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ of Definition **??** is the strict model of the homotopy-invariant $\mathrm{Def}_\infty^{\mathrm{mod}}(\mathcal{A})$ (Theorem **??**).

(C) **Global collision geometry** (Pillar C). The natural geometry for punctured curves, nodal degenerations, and modular sewing is *logarithmic Fulton–MacPherson* compactification $\overline{\mathrm{FM}}_n(X|D)$ on simple normal crossings pairs (X, D), in the sense of Mok [322]. Ordinary FM compactification is the special case $D = \emptyset$. The tropicalization of $\overline{\mathrm{FM}}_n(X|D)$ is controlled by planted-forest combinatorics (Definition **??**).

Remark 1.11.2 *(The triangle).* The three pillars form a coherent triangle. The bar complex $\overline{B}(\mathcal{A})$ is simultaneously:

- a Ch_∞-algebra (Pillar A, via cosimplicial structure and the Eilenberg–Zilber operad $\mathcal{Y}(n)$);
- an algebra in $\hom_\alpha(\overset{!}{\underset{\mathrm{ch}}{\backprime}}, {}^{\mathrm{ch}})$ (Pillar B, via the chiral operadic twisting morphism);
- a sheaf on $\overline{\mathrm{FM}}_n(C|D)$ for a curve C with divisor D (Pillar C, via log-smooth compactification of operator insertions).

The master Maurer–Cartan equation $D \cdot \Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ (proved at all degrees, Theorem **??**) lives at the intersection of all three structures. Each face of the triangle links two pillars:

(i) A+B: the ∞-category of chiral algebras ([389, Theorem 3.8] gives simplicial structure; [309] gives operations);

(ii) B+C: the geometric convolution algebra (planted forests of Mok [322] are the combinatorial data controlling the convolution bracket on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$);

(iii) A+C: the geometric origin of higher homotopies (secondary Borcherds operations $j'_{(p,q,r)}$ are residues on FM boundary strata; log-FM degeneration organizes these across genera).

Remark 1.11.3 *(Key identifications).* The integration yields eleven identification theorems (nine proved, two structural, namely the one-slot obstruction from Robert-Nicoud–Wierstra and the degree-4 degeneration construction):

(1) $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} = \hom_{\alpha}(\overset{!}{\mathrm{ch}}, {}^{\mathrm{ch}})$ (convolution = master object; Theorem **??**).

(2) $\Theta_{\mathcal{A}} \in \mathrm{MC}(\hom_{\alpha}) \cong \mathrm{Tw}_{\alpha}$ (MC element = twisting morphism; Corollary **??**).

(3) $\mathbb{G}_{\mathrm{pf}} = \mathrm{Trop}(\overline{\mathrm{FM}}_n(C|D))$ (planted forests = tropicalization; Theorem **??**).

(4) $\overline{B}(\mathcal{A})$ is a Ch_{∞}-algebra (Theorem **??**).

(5) $B_{\kappa} \dashv \Omega_{\kappa}$ is a Quillen equivalence (Theorem **??**).

(6) The shadow algebra $\mathcal{A}^{\mathrm{sh}}$ is a homotopy invariant (Theorem **??**).

(7) The one-slot obstruction constrains the MC3 categorical lift: $\hom_{\alpha}$ accepts ∞-morphisms in either slot separately but not both simultaneously ([367, Theorem 6.6]).

(8) The secondary Borcherds operations F_n of [309] at degree $n \geq 3$ are identified with the shadow obstruction tower obstruction classes $o_n(\mathcal{A})$ (Proposition **??**): $F_3 = o_3$ (cubic shadow), $F_4 = o_4$ (quartic resonance), $F_n = o_n$ (higher obstructions). The identification is structural at genus 0; at genus $g \geq 1$ the bar side acquires modular corrections with no Čech counterpart in [309].

(9) The quartic clutching law (Theorem **??**) has a log-geometric derivation via Mok's degeneration formula [322, Corollary 5.3.4]: the three terms (contact, additive, tree) biject with the three classes of rigid combinatorial types at degree 4 (Construction **??**).

(10) The log clutching conjecture is proved: Theorem **??** (formerly a conjecture) is now established by [322, Corollary 5.3.4].

(11) The ambient $D^2 = 0$ (Theorem **??**) is now **proved**: the five-component differential on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{amb}}$ is the transport of ∂ on $\mathrm{FM}_n^{\log}(X|D)$ through the Hom functor, and $\partial^2 = 0$ by Mok's normal-crossings result [322, Theorem 3.3.1(1)]. *Single-source dependency*: this theorem rests entirely on [322] (2025 preprint). If revised or unconfirmed, the theorem and its downstream results (Remark **??**) revert to conditional status; the proved core (Theorems A–D, H, MC2) is unaffected.

Scope note. Identifications (1)–(6) are stated at all degrees; however, the fullness of the all-degree unification $\Theta_{\mathcal{A}} \in \mathrm{MC}(\hom_\alpha)$ at all orders is MC2 (proved by the bar-intrinsic construction, Theorem **??**); finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$ are its truncations. Item (7) is an external structural fact independent of degree.

Logical independence of the proved core (cf. Remark **??**). The five main theorems A–D, H and the bar-intrinsic MC construction (Theorem **??**) depend on *no external preprint*: they rest on the convolution-level $D^2 = 0$, which follows from the classical fact $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$. The three-pillar preprints provide structural enrichment beyond the proved core:

- [322] gives the ambient-level $D^2 = 0$ (Theorem **??**) and the log clutching law (Theorem **??**), yielding the planted-forest depth filtration and quartic clutching. If revised, these revert to conjectured; the proved core is unaffected.
- [324] gives the analytic mode–Bergman isomorphism (clause (i) of Theorem **??**). If revised, clauses (ii)–(iv) (completed bar differential, one-particle sewing, Fredholm determinant) remain proved.
- [309] provides the homotopy chiral algebra language (Pillar A). If revised, the bar complex and its properties are unaffected; only the interpretation via Ch_∞ requires adjustment.

Remark 1.11.4 *(External preprint dependency table).* The following table records every theorem in this monograph that depends on an external preprint (not yet published in a peer-reviewed journal as of March 2026), the specific result cited, publication status, and the consequence if the cited result is revised or retracted. The five main theorems A–D, H and the bar-intrinsic MC construction (Theorem **??**) have *no* preprint dependencies; they rest on the convolution-level $D^2 = 0$, which follows from $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$.

Pillar C: Mok [322]: *Logarithmic Fulton–MacPherson configuration spaces*, arXiv:2503.17563 (2025). Status: preprint.

Theorem in this manuscript	Result cited	Fallback if revised
Thm **??** (ambient $D^2 = 0$)	[322, Thm 3.3.1(1)]: log-FM is smooth with snc boundary	Reverts to conjectured; convolution-level $D^2 = 0$ and all five main theorems unaffected
Thm **??** (log clutching)	[322, Cor 5.3.4]: degeneration formula	Reverts to conjectured; quartic clutching law (Thm **??**) retains its direct combinatorial proof
Constr **??** (degree-4 log derivation)	[322, Cor 5.3.4]: birational correspondence of rigid types	Log-geometric derivation lost; the clutching law itself is independently proved
Thm **??** (planted forests = tropicalization)	[322, Thm 3.3.1(2)]: boundary strata indexed by planted forests	Identification becomes conjectural; planted-forest coefficient algebra $\mathbb{G}_{\mathrm{pf}}$ retains its combinatorial definition
Prop **??** (tropical identification)	[322, Def 3.1.5, Obs 5.2.5]	Same as above
Thm **??** (log-FM cocomposition)	[322, Thm 3.3.1, Cor 5.3.4]	Cocomposition axiom becomes conjectural; shadow obstruction tower unaffected at finite order
Prop **??** (ordered log-FM)	[322, Thm 3.3.1]	Reverts to conjectured

Pillar A: Malikov–Schechtman [309]: *Homotopy chiral algebras*, arXiv:2408.16787 (2024). Status: preprint.

Theorem in this manuscript	Result cited	Fallback if revised
Thm **??** (Čech complex as HCA)	[309, Thm 3.1]: Čech complex carries Ch_∞ structure	Bar complex and its properties unaffected; Ch_∞ interpretation requires adjustment
Thm **??** (genus-g bar as curved Ch_∞)	[309]: HCA structure	Genus-g bar differential proved independently; Ch_∞ labelling requires revision
Prop **??** (Jacobiator nullhomotopic)	[309, §4.1]	Loses HCA-theoretic proof; direct verification available
Prop **??** ($F_n = o_n$: Borcherds = shadow obstructions)	[309, §4.5, eq. (4.5.1)]: secondary Borcherds operations	Shadow obstruction tower obstruction classes $o_n(\mathcal{A})$ independently defined; only the identification with F_n requires [309]

Moriwaki [323, 324]: *Conformally flat factorization homology* (arXiv:2602.08729) and *Bergman space operads* (arXiv:2603.06491), 2026. Status: preprints.

Theorem in this manuscript	Result cited	Fallback if revised
Thm **??** (i) (Heisenberg sewing envelope)	[324]: conformally flat 2-disk operad and mode–Bergman isomorphism	Clause (i) reverts to conjectured; clauses (ii)–(iv) (completed bar differential, one-particle sewing, Fredholm determinant) remain proved
Prop **??** (sewing envelope $= \mathrm{Sym}\, A^2(D)$)	[324]	Same as clause (i) above

Adamo–Moriwaki–Tanimoto [3]: *OS axioms for unitary full VOAs*, arXiv:2407.18222 (2024). Status: preprint.

Theorem in this manuscript	Result cited	Fallback if revised
Conj 2.3.24 (analytic completion programme)	[3]: conformal OS axioms for unitary full VOAs	Programme targets already conjectural; no proved theorem affected

Pillar B: Robert-Nicoud–Wierstra [367], Vallette [389]: *Homotopy morphisms between convolution homotopy Lie algebras*, J. Noncommut. Geom. **13** (2019); *Homotopy theory of homotopy algebras*, Ann. Inst. Fourier **70** (2020). Status: **published** (peer-reviewed).

Theorem in this manuscript	Result cited	Fallback if revised
Thm **??** (bar-cobar Quillen equivalence)	[389, Thm 2.1]: model structure on homotopy algebras	Published; no preprint risk
Thm **??** (shadow homotopy invariance)	[367, Thm 5.1]: ∞-morphisms in one slot	Published; no preprint risk
Thm **??** (convolution = master object)	[367, Thm 4.1]: convolution sL_∞	Published; no preprint risk
No-bifunctor obstruction (Remark 1.11.3 (7))	[367, Thm 6.6]: no simultaneous ∞-morphisms in both slots	Published; no preprint risk

Nishinaka [428], Vicedo [429]: *Factorization envelopes and enveloping vertex algebras* (preprint, 2026); *Full universal enveloping vertex algebras from factorisation* (preprint, 2025). Status: preprints.

Theorem in this manuscript	Result cited	Fallback if revised
Factorization envelope programme (Nishinaka–Vicedo concordance, below)	[428]: existence of $\mathsf{Fact}_X(L)$ for admissible L; [429]: non-chiral analogue	Modular Koszul datum programme uses Nishinaka admissibility as input; if revised, the modular envelope U_X^{mod} target remains well-defined but genus-0 comparison requires adjustment

Gaiotto–Zeng [427]: *Interface Minimal Model Holography and Topological String Theory*, arXiv:2603.08783 (2026). Status: preprint.

Theorem in this manuscript	Result cited	Fallback if revised
Sphere-reconstruction programme (§1.20)	[427]: commuting sphere Hamiltonians	Cited as motivation and consistency check for the shadow interpretation; no proved theorem depends on [427]

Khan–Zeng [264]: *Poisson vertex algebras and 3d gauge theory*, arXiv:2502.13227 (2025). Status: preprint.

Theorem in this manuscript	Result cited	Fallback if revised
PVA/deformation quantization bridge (Chapter **??**)	[264]: 3d PVA sigma model	Cited as physical motivation; all algebraic theorems on PVA descent are proved independently in Vol II

Summary of risk stratification.

- *Zero risk*: Theorems A–D, H, MC2 bar-intrinsic construction, all shadow obstruction tower results at finite order; depend on no preprint.
- *Low risk (published)*: Quillen equivalence, shadow homotopy invariance, convolution identification, no-bifunctor obstruction; depend on [367, 389] (published, peer-reviewed).
- *Moderate risk (single-preprint, isolable)*: ambient $D^2 = 0$ and log clutching ([322]); Heisenberg sewing clause (i) ([324]); Ch_∞ interpretation ([309]). In each case the dependency is explicitly isolable and the proved core is unaffected.
- *No proved-theorem risk*: [427], [264], [3], [428], [429] are cited as motivation, consistency checks, or inputs to conjectural programmes only.

Summary 1.11.5 *(Three-pillar chain-level synthesis).* The three pillars converge into a single concrete chain-level presentation of the modular deformation theory. The logarithmic FM convolution algebra (Construction **??**) is the native object:

$$\mathrm{Def}_\infty^{\mathrm{mod}}{}_{\log}(\mathcal{A};\,\mathfrak{U}) \;:=\; \mathrm{Conv}_\infty\Big(\substack{\log\mathrm{FM}\\ \mathrm{mod}},\ \mathrm{End}_{\mathrm{Ch}_\infty}(A_\infty^{\mathrm{ch}})\Big),$$

where $A_\infty^{\mathrm{ch}} = \check{C}^\bullet(\mathfrak{U};\mathcal{A})$ is the Malikov–Schechtman homotopy-chiral input (Pillar A), the L_∞-structure comes from Robert-Nicoud–Wierstra convolution (Pillar B), and the coefficient cooperad $\substack{\log\mathrm{FM}\\ \mathrm{mod}}$ is built from chains on Mok's logarithmic FM spaces (Pillar C).

The modular bar functor in these coordinates (Construction **??**) is a stable-graph sum with five-component differential $D_{\mathrm{mod}}^{\log} = d_{\check{C}} + d_{\mathrm{Ch}_\infty} + d_{\mathrm{coll}} + d_{\mathrm{sew}} + d_{\mathrm{loop}}$; the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ receives an explicit graph-sum expansion with log-FM weights (Construction **??**); and the clutching law becomes a literal boundary factorisation theorem on rigid planted-forest strata (Construction **??**).

The characteristic shadows κ, $\Delta_{\mathcal{A}}$, $\mathfrak{R}_4^{\mathrm{mod}}$ are successive invariants of the modular tangent complex (Construction **??**), realising the modular characteristic package as a Chern–Weil theory for factorization algebras. The weight filtration (Construction **??**) stratifies the modular bar functor: tree homotopies and one-loop BV at weight 1; genuine logarithmic modularity beginning at weight 2. Genus 2 is the first point where all three shell types (loop∘loop, sep∘loop, planted-forest) appear together (Construction **??**).

The full chain-level architecture comprises the following identifications proved in §**??**:

(a) The *graphwise cocomposition* $\Delta_\Gamma^{\log} := (\nu_\Gamma)_* \circ \mathrm{Res}_{D_\Gamma^{\log}}$ makes the log-FM coefficient cooperad explicit (Construction **??**), and yields graphwise Taylor coefficients $\ell_k^{(g)} = \sum_\Gamma |\mathrm{Aut}(\Gamma)|^{-1} \ell_\Gamma$.

(b) The *boundary operators* $d_{\mathrm{sew}}, d_{\mathrm{pf}}, \hbar\Delta$ are residue correspondences on separating, planted-forest, and non-separating strata respectively (Construction **??**).

(c) The *rigid planted-forest correction* is the completed push–pull sum $d_{\mathrm{pf}} = \sum_\rho \epsilon_\rho\, (\kappa_\rho)_*\, \mathrm{pr}_\rho^* \otimes \mu_\rho$, literally Mok's degeneration formula transported into the convolution algebra (Construction **??**).

(d) The *MC ≃ twisting data identification*: $\mathrm{MC}_\bullet(\mathrm{Def}_{\infty\ \log}^{\mathrm{mod}}) \simeq \mathrm{Tw}_{\alpha^{\mathrm{mod}}}$, and $\Theta_{\mathcal{A}}$ is the universal modular twisting morphism (Corollary **??**).

(e) The *depth filtration* gives every coefficient of $\Theta_{\mathcal{A}}$ a tridegree (g, n, d) from Mok's stratification (Definition **??**).

(f) The *first two weights* are completely explicit: weight 1 carries the Malikov–Schechtman Jacobiator homotopy and the BV loop operator; genuine logarithmic modularity begins at weight 2 (Proposition **??**).

(g) The *genus spectral sequence* from the genus filtration on the strict modular deformation dg Lie algebra has differentials that are the successive obstruction maps Ob_g (Construction **??**).

(h) The *primitive kernel compression* (§**??**): the cofree-coderivation principle reduces the entire stable-graph expansion $\Theta_{\mathcal{A}} = \sum_\Gamma |\mathrm{Aut}(\Gamma)|^{-1} W_\Gamma^{\log \mathrm{FM}} \Phi_\Gamma^{\mathcal{A}}$ to a single primitive element $\mathfrak{K}_{\mathcal{A}} \in \mathrm{Prim}_{\mathrm{mod}}^{\log}(\mathcal{A})$ (Definition **??**). The fundamental equivalence $(Q_{\mathcal{A}}^{\mathrm{mod}})^2 = 0 \iff d\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \hbar\,\Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0$ (Theorem **??**) replaces the obstruction tower by one structural equation. Its genus-2 and genus-3 projections give primitive shell equations stratified by family regime: Heisenberg (pure loop), affine (loop + bracket), Virasoro/$\mathcal{W}_N$ (loop + bracket + rigid) (Proposition **??**). The finite-rank branch BV packet (Definition **??**) carries a modular master action $S_{\mathcal{A}}^{\mathrm{br}}$ whose QME $\hbar\Delta^{\mathrm{br}} S + \frac{1}{2}\{S, S\}_{\mathrm{BV}} = 0$ is the finite-rank shadow of the primitive MC equation. The spectral determinant $\Delta_{\mathcal{A}}$ acquires a canonical square root $\delta_{\mathcal{A}}$ as a metaplectic half-density on the branch sector (Corollary **??**). The shadow obstruction tower is homotopy invariant (Theorem **??**).

The native end-state of the programme is the chain

$$\mathfrak{K}_{\mathcal{A}} \xrightarrow{\mathrm{FT}_{\mathrm{mod}}^{\log}} Q_{\mathcal{A}}^{\mathrm{mod}} \xrightarrow{\text{branch proj.}} S_{\mathcal{A}}^{\mathrm{br}} \xrightarrow{\text{half-density}} \delta_{\mathcal{A}},$$

in which the scalar class $\kappa(\mathcal{A})$, the spectral determinant $\Delta_{\mathcal{A}}$, the branch-active quotient, and the nonlinear resonance classes are successive shadows of one primitive master object $\mathfrak{K}_{\mathcal{A}}$ (§**??**).

Remark 1.11.6 *(Design constraint: failure of two-sided bifunctoriality).* Robert-Nicoud–Wierstra [367, Section 6] prove that although the convolution functor $\hom_\alpha(-,-)$ extends to ∞_α-morphisms in either slot separately, there is *no* honest bifunctor accepting ∞-morphisms in both slots simultaneously. The consequences:

- Bar-cobar extends to ∞-morphisms of chiral algebras (slot 2) and to ∞-morphisms of chiral coalgebras (slot 1), but not both simultaneously.
- The MC3 categorical lift (Theorem **??**) proceeds one slot at a time. The prefundamental CG closure (Proposition **??**) works at the character level (K_0) precisely because characters live in one slot; the categorical lift requires both.
- Any claim of two-sided functoriality for bulk/boundary/line constructions must be replaced by a homotopy-coherent correspondence formalism.

Remark 1.11.7 *(Three-pillar architecture and the MC open problems).* The three-pillar architecture determines the MC resolution strategy:

- *MC1 and MC2*: Pillar B realizes every algebra as an MC element in the modular convolution sL_∞-algebra; the finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$ (the proved shadow obstruction tower) are the constructive content. MC1 (PBW concentration) holds unconditionally for universal algebras $V_k()$, Vir_c, $\mathcal{W}^k()$ (Proposition **??**). For simple quotients $L_k()$ at admissible levels, PBW concentration fails at $\mathrm{rk}() \geq 2$ with denominator $q \geq 3$: the abelian Cartan subalgebra contributes $\mathrm{rk}()$ classes to $H^2(\overline{B})$ (Conjecture **??**).
- *MC3*: MC3 is **proved** for all simple types on the evaluation-generated core. The categorical CG decomposition is proved via multiplicity-free ℓ-weights (Theorem **??**), and the DK comparison on the evaluation-generated core is proved (Corollary **??**). Type A further reduces the old four-packet post-core problem to the single compact-completion packet (Theorem **??**): shifted-prefundamental generation and pro-Weyl recovery are proved on the Baxter-locus seed $\{V_{\omega_i}(a)\} \cup \{L^-_{i,a-\frac{1}{2}}\}$, with the spectral constraint supplied by Proposition **??**. The residual problem is DK-4/5: extension and completion beyond the evaluation core and its compact shifted-prefundamental enhancement (Conjecture **??**), which is downstream of MC3. The one-slot obstruction (Remark 1.11.6) forces a one-slot-at-a-time strategy, which the block-separation argument respects.

  Strategy by Dynkin type (superseded). The multiplicity-free ℓ-weight approach (Theorem **??**) gives a *uniform* proof for all simple types, bypassing the type-specific strategies below. They are recorded for historical context only. Types B_n, C_n, F_4: fold from type A_{2n-1}, A_{2n}, E_6 respectively; the type-A theorem descends to the fixed-point category under the diagram automorphism. Types D_n, E_6, E_7: cluster category (Hernández–Leclerc) provides categorical CG via the cluster-tilting structure. Types E_8, F_4 (non-folded route): hardest cases, potentially requiring Coulomb branch geometry (Braverman–Finkelberg–Nakajima).

  Construction 1.11.8 *(MC3 status by Dynkin type).* **MC3 is proved for all simple types on the evaluation-generated core.** Package (i) (categorical CG) uses multiplicity-free q-characters [451, 457, 450] transferred to Yangians via [453] (Theorem **??**). Thick generation of $\mathrm{Rep}_{\mathrm{fd}}$ by evaluation modules holds for all types (Corollary **??**). The downstream DK-4/5 extension (from evaluation-generated core to compact-completion of full category) remains open. Conditionally, the remaining packages (ii)–(iv) are type-independent once the relevant shifted-prefundamental, pro-Weyl, and compact-completion inputs are supplied in that type (Conjecture **??**).

- *MC4*: The formal completion theory is **proved** by the strong completion-tower theorem (Theorem **??**): the finite-stage bar-cobar duality passes to inverse limits automatically once the filtration is strong (degree cutoff, Lemma **??**). The completion closure () carries a quasi-inverse bar-cobar equivalence on the homotopy category (Corollary **??**), stable under MC twisting (Theorem **??**), with completed twisting representability (Theorem **??**). Splits into MC_4^+ (positive towers) and MC_4^0 (resonant). MC_4^+ is **unconditionally solved** by weightwise stabilization: for $\mathcal{W}_{1+\infty}$, affine Yangians, and positive RTT towers, coefficient stabilization (Theorem **??**) reduces completion to finite truncations. MC_4^0 concerns algebras with finite resonance rank (Virasoro, non-quadratic $\mathcal{W}_N$) and is proved on the transferred split surface of Theorems **??** and **??**. The uniform PBW bridge (Theorem **??**) connects MC1 to MC4: uniform PBW degeneration plus quotient stabilization implies completed bar-cobar duality. The *general theorem* (strong completion-tower theorem) is proved unconditionally. What remains is *transfer verification*: for each new algebra family beyond the standard landscape, one must verify the split transfer package that places the family on the MC_4^0 resonance surface or the MC_4^+ stabilization surface, and then identify the H-level target. *Arithmetic connection*: the spectral curve of the shadow Higgs field (Theorem **??**) has convergence radius $|c|/3$ in the degree variable; the MC4 completion radius is controlled by the same parameter $\lambda_{\mathrm{eff}} = -6/c$ (Chapter **??**, Remark **??**).

- *MC5* (**analytic and coderived proved; chain-level conjectural**): Inductive genus determination, 2D convergence (no UV renormalization needed), analytic-algebraic comparison, and the general HS-sewing criterion (Theorem **??**: polynomial OPE growth plus subexponential sector growth implies convergence at all genera). Heisenberg sewing proved (Theorem **??**: one-particle Bergman reduction, Fredholm determinant). At genus 0 the algebraic BRST/bar comparison is proved (Theorem **??**); the tree-level amplitude pairing is conditional on Corollary **??**. BV=bar in the coderived category is proved for all four shadow classes including class M (Theorem **??**). The genuswise chain-level BV/BRST/bar identification remains conjectural in general: classes G and L are proved, class C is conditional on harmonic decoupling, and class M is chain-level false (Conjecture 2.2.7). Pillar C (Mok's log-smooth degeneration) provides additional inductive structure via rigid special-fibre decomposition. *Arithmetic connection*: the sewing–Selberg formula (Theorem **??**) identifies $\int_{1,1} \log\det(1-K)\cdot E_s\, d\mu = -2(2\pi)^{-(s-1)}\Gamma(s-1)\zeta(s-1)\zeta(s)$; the Fredholm determinant of the HS-sewing theorem is the same $\det(1-K_q)$ whose divisor-sum decomposition encodes $\zeta(s)\zeta(s+1)$ (Chapter **??**).

1.12 The three concentric rings

The monograph is organized in three concentric rings.

Ring 1: The proved modular Koszul core

Theorems A/B/C/D/H, MC1/MC2/MC4, the proved lanes of MC5 (analytic HS-sewing at all genera, genus-0 algebraic BRST/bar, D^{co}-level BV=bar for all shadow classes; chain-level BV/BRST/bar conjectural), MC3 (evaluation core all types unconditional; categorical CG all types unconditional; type-A post-core reduction proved on the Baxter-locus seed, with residual compact-completion DK-4/5 downstream of the Baxter constraint and Conjecture **??**), DK-0/1 (proved), DK-2/3 (evaluation-generated core proved), and the Koszulness characterization programme (9 unconditional equivalences, with (vii) the genus-0 factorization-homology criterion; 1 proved consequence, 1 conditional, 1 one-directional; Theorem **??**). This is the spine. Nothing in the extension appendices touches it except to cross-reference from it.

Ring 2: The nonlinear characteristic layer

Before the extension appendices, the modular characteristic was scalar: the single number $\kappa(\mathcal{A})$ and its λ_g tower, plus the spectral discriminant $\Delta_{\mathcal{A}}(x)$. The full hierarchy is:

$$\kappa(\mathcal{A}) \longrightarrow \Delta_{\mathcal{A}} \longrightarrow [\mathfrak{C}_{\mathcal{A}}],\ [\mathfrak{Q}_{\mathcal{A}}] \longrightarrow \Lambda_P(\mathfrak{Q}_{\mathcal{A}}) \longrightarrow \Theta_{\mathcal{A}}.$$

- The *nonlinear modular shadows* chapter (Chapter **??**) makes the cubic and quartic jets of the complementarity potential $S_{\mathcal{A}}$ into precise, computable invariants with clutching laws and a quartic resonance class.
- The *all-degree master equation* extends the shadow tower to arbitrary degree r: the primitive archetypes terminate (Gaussian at 2, Lie/tree at 3, contact at 4), while Virasoro is the first genuinely infinite tower (Theorem **??**).
- The *genus loop operator* Λ_P implements non-separating clutching: the genus-1 Hessian correction $\delta H^{(1)} = \Lambda_P(\mathfrak{Q}^{(0)})$ is the first tensorial modular invariant beyond the $\hat{A}$-genus (Theorem **??**). For the Virasoro, $\delta H^{(1)}_{\mathrm{Vir}} = 120/[c^2(5c+22)] \cdot x^2$.
- The *branch-line reductions* appendix (Appendix **??**) extracts exact finite-dimensional quotients: the universal genus-3 constant 7, genus-2 transparency, and the spectral cumulant hierarchy.
- The *ambient complementarity upgrade* (§**??**) upgrades Theorem C from additive splitting to a two-layer Lagrangian geometry: the Verdier-side Lagrangian package is proved, while the ambient shifted-symplectic/BV upgrade is conditional on the uniform-weight package. This yields one ambient deformation problem with two Lagrangian maps, a complementarity potential $S_{\mathcal{A}}$ whose Taylor jets are the nonlinear modular invariants, a derived critical locus governing simultaneous self-dual deformations, and a fake-complementarity criterion detecting when the dual Lagrangian is linear (Proposition **??**).
- The *modular bar–Hamiltonian* (§**??**) identifies the carrier algebra $\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}$ with its five-piece differential (internal + kernel twist + stable-edge sewing + planted-forest correction + genus loop). The universal class $\Theta_{\mathcal{A}}$ is a Maurer–Cartan element in this algebra (Definition **??**), and the recursive extension tower provides the constructive mechanism for building it order by order (Construction **??**).
- The *modular Chern–Weil transform* (§**??**) is a single map from $\mathrm{MC}(\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}})$ to the full invariant hierarchy: scalar classes, spectral classes, resonance classes, and line-kernel data are all projections of one Maurer–Cartan element. The complementarity Legendrian $\Lambda_{\mathcal{A}} = j^1\mathfrak{S}_{\mathcal{A}}$, the complementarity Stokes groupoid, and the phase index $\nu_{\mathrm{nl}}(\mathcal{A}, u)$ are geometric avatars of the same master element.
- The *quartic resonance class* $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A})$ (Theorem **??**) is the first Picard-valued nonlinear characteristic of the theory, with clutching law receiving a cubic tree correction $\mathfrak{T}_{3,\xi}(\mathcal{A})$ on separating boundaries.
- The *graph-completed Feynman transform package* (§**??**) identifies the carrier object $\mathfrak{F}_{\mathrm{mod}}(\mathcal{A})$ in which the full universal class should live, the full *resonance tower* $\mathfrak{R}^{\mathrm{mod}}_r$ for all degrees $r \geq 4$, and the *nonlinear spectral-branch stack* $\mathfrak{B}^{\mathrm{nl}}_{\mathcal{A}}$.

- The *shadow archetypes in Part* **??**: each standard family chapter now computes its nonlinear shadow explicitly. Heisenberg is exactly Gaussian (§**??**); affine $\widehat{\mathfrak{sl}}_2$ is Lie/tree with critical-level emergence (§**??**); $\beta\gamma$ is the contact/quartic primitive with rank-one abelian rigidity (§**??**); Virasoro and $\mathcal{W}_N$ are the first mixed cubic–quartic families with infinite shadow obstruction towers (§**??**). The Virasoro quartic contact coefficient $Q_{\mathrm{Vir}}^{\mathrm{contact}} = 10/[c(5c+22)]$; genus-1 loop ratio $\rho_{\mathrm{Vir}}^{(1)} = 240/[c^3(5c+22)]$. The *quintic shadow* $S_5^{\mathrm{Vir}}(c) = -48/[c^2(5c+22)]$ and *sextic shadow* $S_6^{\mathrm{Vir}}(c) = 80(45c+193)/[3c^3(5c+22)^2]$ are the first explicit degree-5 and degree-6 computations (computed from the convolution recursion $f^2 = Q_L$). The sign pattern $S_2 > 0, S_3 > 0, S_4 > 0, S_5 < 0, S_6 > 0, S_7 < 0$ for $c > 0$ is dictated by the branch-point structure of $\sqrt{Q_L}$. For depth-bounded classes: $S_5 = 0$ (classes G, L); on the one-dimensional metric, $\beta\gamma$ (class C) has $S_5 = 1/2 \neq 0$, but stratum separation kills the quintic in the full deformation complex. The DS depth-increase mechanism at degree 5 is verified: $\mathfrak{sl}_2$ (depth 3, $S_5 = 0$) $\to$ Vir (depth ∞, $S_5 \neq 0$) via ghost-sector quartic seed; ghost central charge $c_{\mathrm{ghost}} = 2$ (constant). The genus-2 planted-forest correction $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$; for Virasoro $\delta_{\mathrm{pf}}^{(2,0)} = -(c-40)/48$.

- The *envelope-shadow functor* (§1.8) upgrades the shadow tower from a case-by-case invariant to a functor on Lie conformal input $R \mapsto \Theta^{\mathrm{env}}(R) := \Theta(U_{\mathrm{vert}}(R))$, using the Nishinaka–Vicedo factorization-envelope pipeline. The envelope-shadow complexity $\chi_{\mathrm{env}}(R)$ recovers the G/L/C/M classification; the degree-r shadow depends only on the $(r+1)$-jet (Proposition 1.8.4) and is polynomial of degree $\leq r-1$ in the central-extension parameter (Proposition 1.8.5).

- The *universal Maurer–Cartan principle* (§**??**) establishes the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ as a modular L_∞ algebra (with the dg Lie algebra as its strict model), and the shadow obstruction tower $\kappa \to \Delta \to \mathfrak{C} \to \mathfrak{Q} \to \Theta$ arising as evaluation of the universal MC element $\Theta_{\mathcal{A}}$ on specific moduli cycles. The five-component differential $D = d_{A_\infty} + d_{\mathrm{sew}} + \hbar\Delta + d_r$ is automatic from the single MC equation. The dg Lie model is the strict shadow of the L_∞ structure; bar-cobar preserves quasi-isos because it is a quantum L_∞ functor.

- The *non-renormalization theorem* (Theorem **??**, ProvedElsewhere, Dimofte–Niu–Py) validates all tree-level computations: for quasi-linear theories (all HT-twisted 3d $\mathcal{N} = 2$), the A_∞ operations and the r-matrix are exact at tree level. Loop corrections enter only through the genus-raising BV operator Δ.

- The *polarized scattering package* (§**??**) replaces the one-colour Maurer–Cartan problem by a bipartite machine: a polarized modular graph algebra $\mathcal{G}_{\mathrm{mod}}^{\pm}(\mathcal{A})$ with same-colour vanishing, a *resonance scattering diagram* $\mathfrak{Scat}(\mathcal{A})$, a categorified *complementarity schober* $\mathfrak{Sch}(\mathcal{A})$, and a *modular wavefunction* $\Psi_{\mathcal{A}}$ satisfying the quantum master equation. Four conjectural theorems (bipartite vanishing, quartic lift, resonance wall-crossing, quantization) organize the forward attack.

- The *multi-variable shadow obstruction tower* (§**??**) extends the shadow construction to algebras with multiple strong generators (e.g., $\mathcal{W}_3$ with generators T, W). The *Kac-shadow singularity principle* (Proposition **??**) identifies denominators of shadow coefficients with Kac determinant factors. The *shadow subalgebra autonomy theorem* (Theorem **??**) states that the shadow obstruction tower of $\mathcal{A}$ restricts to the shadow obstruction tower of any strongly generated subalgebra, and the *$\mathcal{W}$-line alternating vanishing* (Corollary **??**) forces $\mathrm{Sh}_{2r+1}^{W\text{-pure}} = 0$ for all r. Drinfeld–Sokolov reduction strictly increases shadow depth (Theorem **??**): $r_{\max}(\mathcal{W}_N) \geq r_{\max}(V_k()) + 1$. The quantitative DS cascade (Proposition **??**): for $N = 2, \ldots, 6$ and all non-critical levels, S_3 doubles ($1 \to 2$), S_4 is created from

zero (Jacobi identity no longer applies), κ is not additive (ghost constant $C(N,k)$ is k-dependent), and $\rho(\mathcal{W}_N)$ decreases with N (Proposition **??**; higher-spin gravity is more convergent than pure gravity). The $\mathcal{W}_3$ genus-2 cross-channel correction $\delta F_2 = (c+204)/(16c)$ (Computation **??**, Remark **??**) is the first explicit multi-generator genus-2 amplitude. The universal gravitational N-formula $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = (N-2)(N+3)/96 + (N-2)(3N^3+14N^2+22N+33)/(24c)$ (Proposition **??**) gives the closed-form gravitational cross-channel correction for all principal $\mathcal{W}_N$; exact for $\mathcal{W}_3$, lower bound for $N \geq 4$. The cross-channel correction $\delta F_g^{\mathrm{cross}}$ is computed through genus 4 for $\mathcal{W}_3$; the cross-channel contribution dominates the scalar term at high genus. The W-line ring relations (Proposition **??**): S_{2n}^W is a polynomial in S_4^W/c with coefficients given by half-integer binomials. Verified by 344 compute tests across three modules (`test_ds_cascade_shadows.py`, `test_w3_shadow_extended.py`, `test_w3_genus2.py`).

- The *Kac–Chevalley factorization question*, whether the Kac determinant and the Chevalley discriminant factor through a common polynomial, is **resolved as spurious**: the two objects are algebraically independent.

Ring 3: The physics-facing extension problems

Three axes, each with its own appendix chain:

1. *The $\mathcal{W}$-algebra axis.* Completed Koszulity is ubiquitous (every affine $\mathcal{W}$-algebra at generic level); strict Koszulity is exceptional. The subregular family $\mathcal{W}_n^{(2)}$ exhibits unbounded canonical homotopy degree.

2. *The Yangian/RTT axis and the Bridge Theorem.* Three layers: local RTT kernel, completed envelope, boundary quotient. The shift belongs to the boundary quotient, not the local kernel. The rank-one test case produces the quantized Kleinian surface. The *Bridge Theorem project* (Remark **??**; Theorem **??**) is now the central remaining target: the full triple equivalence $\mathrm{Fact}_{\mathrm{ord}}^{\mathsf{E}_1} \simeq \mathrm{Mod}^{\mathrm{comp}}(\mathcal{Y}_{\mathcal{A}}^{\mathrm{dg}}) \simeq \mathrm{Rep}^{\mathrm{spec}}(\mathrm{QG}_{\mathrm{spec}}(R_{\mathcal{A}}))^{\mathrm{op}}$ decomposes into four sub-targets B1–B4 with a proved criterion theorem (B1+B2+B4 $\Rightarrow$ full bridge).

3. *The holographic/celestial axis.* Protected transform calculus (dual transform, transposition, Weyl sewing, metaplectic anomaly) built directly from the proved core. Volume II develops the bulk–boundary–line Koszul triangle $A_{\mathrm{bulk}} \simeq Z_{\mathrm{der}}(\mathrm{boundary}) \simeq \mathrm{HH}^\bullet(\mathcal{A}^!)$. The *holographic modular Koszul datum* $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, C, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ (Definition **??**) extends the *modular Koszul triple* $(\mathcal{A}, \mathcal{A}^!, r(z))$ (Definition **??**) to the full HT holographic system in a single modular MC problem: the dg-shifted Yangian $r(z)$ is the binary genus-zero collision residue of $\Theta_{\mathcal{A}}$, and the 2026 exact sphere-correlator matching [427] is the genus-zero matrix-element shadow of this datum. Five theorem targets decompose the programme: boundary–defect realization (Conjecture **??**), Yangian-shadow (Theorem **??**; **proved**), sphere reconstruction on established comparison surfaces (Theorem **??**; **proved**), quartic resonance obstruction (Theorem **??**; **proved**), and singular-fiber descent (Conjecture **??**). The factorisation quantum-group envelope [289] supplies the categorical home beyond the meromorphic line category, and the HT deformation-quantization functor (Conjecture **??**) bridges classical PVA data to the quantum modular Koszul package. The landscape of worked physics examples now includes: free chiral vortex lines with explicit r-matrix (§**??**), affine PVA $\to$ Chern–Simons action (§**??**), SQED–XYZ mirror symmetry as bar-cobar equivalence, non-renormalization of tree-level operations (§**??**), superpotential A_∞ truncation (§**??**), Virasoro

phase space as Teichmüller geometry (§**??**), level-rank-type Chern–Simons analogy for Koszul duality, monopole operators via affine Grassmannian modifications, Costello's M2-brane double-loop model (§**??**), SuperVirasoro $\to$ 3d supergravity, SQCD boundary algebra, Witten diagrams in twisted holography, $N = 4$ global symmetry matching, the deformed conifold, and $\mathcal{W}_N \to$ higher-spin gauge theory.

The unifying principle

The organizing principle: *one object, one equation, one proof obligation at each level.* The modular L_∞ convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the ambient home. An algebra structure on $\mathcal{A}$ is a Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$. The shadow obstruction tower, the genus tower, the graph sums, and the shadow archetypes follow from this framework. Each ring (Ring 1: proved core; Ring 2: finite-order shadow engine; Ring 3: physics-facing frontier) requires independent verification that the MC equation's projections produce the claimed invariants. Ring 1 is fully proved. Ring 2 is proved at finite order. Ring 3 contains significant open problems (non-principal W-duality, analytic completion). The dg Lie model is the strict computational avatar; the L_∞ structure is the reason bar-cobar is a Quillen equivalence. Filtration controls the extension tower: In Ring 2, the complementarity potential has a Taylor filtration whose jets are the nonlinear shadows. In Ring 3, the MC4 splitting (Remark **??**) separates the completion problem: positive towers ($\mathcal{W}_{1+\infty}$, Yangians) are solved by weight stabilization (Theorem **??**); resonant towers (Virasoro, non-quadratic $\mathcal{W}_N$) require the resonance-filtered construction (Theorem **??**), with the same-family shadow Vir_{26-c} reinterpreted as the depth-zero resonance shadow (Remark **??**). In Ring 3 (holography), the composite filtration separates the wedge from the nonlinear obstruction tower.

The five main theorems establish *existence of duality*. The three rings develop the *structure of the invariants that duality produces*.

Proposition 1.12.1 *(Five main theorems as MC projections).* [proved here] Theorems A–D and H are projections of the master MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$:

Projection	Theorem	Content
α_Θ (twisting morphism)	A	$B_\kappa \dashv \Omega_\kappa$
$D^2_{\mathcal{A}} = 0$ (bar-intrinsic)	B	$\Omega(B(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$
$\mathrm{tr}\, \pi_{2,\bullet}(\Theta_{\mathcal{A}})$	C	$Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$
$\pi_{\mathrm{sc}}(\Theta_{\mathcal{A}})$	D	$\kappa(\mathcal{A})$ universal, $\hat{A}$-genus GF
$\pi_{\mathrm{HH}}(\Theta_{\mathcal{A}})$	H	$\mathrm{ChirHoch}^*(\mathcal{A})$ polynomial

The shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ consists of finite-order truncations; the obstruction classes $o_{r+1}(\mathcal{A})$ classify shadow depth into the four archetypes G/L/C/M (Theorem **??**). The full MC element $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$ is proved by the bar-intrinsic construction (Theorem **??**).

1.13 Intrinsic Koszulness: the characterization programme

The PBW criterion (Theorem **??**) detects chiral Koszulness for every standard family but does not characterize it intrinsically. This section is the constitutional index for the intrinsic characterization programme. Full statements and proofs live in the source chapters cited below.

The meta-theorem: twelve characterizations

Let $\mathcal{A}$ be an augmented chiral algebra with PBW filtration $F_\bullet$. The full meta-theorem is Theorem **??** in Chapter **??**. Items are listed in the meta-theorem's numbering, so that (i)–(xii) in the concordance and in the chapter are identical.

Unconditional equivalences (i)–(vii), (ix)–(x):

(i) Chirally Koszul (Definition **??**). ✓

(ii) PBW spectral sequence on $\bar{\mathbf{B}}(\mathcal{A})$ collapses at E_2 (Theorem **??**). ✓

(iii) Minimal A_∞-model has $m_n = 0$ for $n \geq 3$ (Theorem **??**). Converse via Keller classicality. ✓

(iv) Ext diagonal: $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ for $p \neq q$ (Theorem **??**). Off-diagonal d_r contradicts concentration. ✓

(v) Bar-cobar counit $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism (Theorem **??**). ✓

(vi) Barr–Beck–Lurie comparison for $\bar{B}^{\mathrm{ch}} \dashv \Omega^{\mathrm{ch}}$ is an equivalence (Theorem **??**). Conservativity + totalization on log-FM strata. ✓

(vii) Factorization homology concentration: $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ for $k \neq 0$ (Theorem **??**). On the uniform-weight lane this upgrades to all genera with $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g$ (Theorem **??**); off that lane the all-genera strengthening fails, with multi-weight principal $\mathcal{W}$-algebras furnishing the boundary counterexamples through $\delta F_g^{\mathrm{cross}} \neq 0$ (Theorem **??**). ✓

Proved consequence on the Koszul locus (viii):

(viii) $CH^*(\mathcal{A})$ is concentrated in cohomological degrees $\{0, 1, 2\}$, satisfies Koszul-dual duality, has polynomial Hilbert series of degree at most 2, and is formal as an E_2-algebra on the Koszul locus (Theorem **??**, Theorem **??**, Proposition **??**). No blanket freeness claim is made: generic Virasoro is the boundary counterexample, since $CH^*(\mathrm{Vir}_c) \cong \mathbb{C} \oplus \mathbb{C}[-2]$ has a square-zero degree-2 class. ✓

(ix) Kac–Shapovalov: $\det G_h \neq 0$ in the bar-relevant range (Theorem **??**). Non-degeneracy forces PBW injectivity; d_r-exactness contradicts pairing. ✓

(x) FM boundary acyclicity: $H^k(i_S^! \bar{\mathbf{B}}_n(\mathcal{A})) = 0$ for $k \neq 0$, all strata S (Theorem **??**). Stratum-by-stratum PBW; binary collision forces injectivity. ✓

Conditional/partial in general; (xi) unconditional for standard landscape (xi)–(xii):

(xi) Lagrangian complementarity: $\mathcal{M}_{\mathcal{A}}, \mathcal{M}_{\mathcal{A}^!} \hookrightarrow \mathcal{M}_{\mathrm{comp}}$ are (-1)-shifted Lagrangian (Conjecture **??**). PTVV derived intersection. Perfectness verified for the standard landscape (Proposition **??**, Corollary **??**): unconditional at non-critical, non-degenerate levels. ✓ (*unconditional for standard landscape*)

(xii) $\mathcal{D}$-module purity: $\bar{\mathbf{B}}_n(\mathcal{A})$ pure, $\mathrm{Ch}(\bar{\mathbf{B}}_n) \subset \bigcup_S T^*_S \overline{C}_n(X)$. Implied by FM boundary acyclicity (x). (*one-directional*; $(x) \Rightarrow (xii)$ proved, converse open)

Sufficient conditions for chiral Koszulness

Two sufficient conditions for chiral Koszulness are established:

(a) *PBW universality* (Proposition **??**): freely strongly generated vertex algebras are chirally Koszul. Applies to the *universal* algebras $V_k()$ (all k), Vir_c (all c), $\mathcal{W}^k()$ (Miura free generators). The simple quotient $L_k() = V_k()/I_k$ is *not* freely strongly generated when the null-vector ideal I_k is nontrivial, and PBW universality does not apply to it.

(b) *Kac–Shapovalov criterion* (Theorem **??**): $\det G_h \neq 0$ in the bar-relevant range. Applies to: $V_k()$, Vir_c at generic c. For simple quotients at admissible levels, null vectors enter the bar-relevant range and this route is blocked.

Simple quotients at admissible levels. For the simple quotient $L_k()$ at admissible level $k + h^\vee = p/q$, the Koszulness status is *rank-dependent*. At rank 1 ($= \mathfrak{sl}_2$), $L_k(\mathfrak{sl}_2)$ is chirally Koszul at every admissible level (Remark **??**). At rank ≥ 2 with denominator $q \geq 3$, Koszulness is *expected to fail*: the abelian Cartan subalgebra $\subset$ produces $\mathrm{rk}()$ classes in the bar spectral sequence $E_1^{2,q}$ that should survive to E_∞ because $[,] = 0$ kills the Lie-bracket differential d_1 (Conjecture **??**). The conjectured obstruction is $\dim H^2(\overline{B}(L_k())) \geq \mathrm{rk}()$.

Rationality, C_2-cofiniteness, and semisimple Zhu algebra remain important *finiteness* inputs for simple quotients at non-degenerate admissible levels, but they do not by themselves identify the Ext groups computed by the bar complex with Ext in the ordinary semisimple module category. Accordingly no unconditional rationality criterion for chiral Koszulness is claimed in the constitution.

Operadic complexity

Theorem 1.13.1 *(Operadic complexity;* [proved here]*).* For a chiral Koszul algebra $\mathcal{A}$, the three invariants coincide: $r_{\max}(\mathcal{A}) = d_\infty(\mathcal{A}) = f_\infty(\mathcal{A})$, where $r_{\max}$ is the shadow depth (Theorem **??**), d_∞ is the A_∞-depth (largest n with $m_n^{\mathrm{tr}} \neq 0$), and f_∞ is the L_∞-formality level (largest n with $\ell_n^{(0),\mathrm{tr}} \neq 0$ on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$).

Proof. The equality $r_{\max} = f_\infty$ is Theorem **??**: the degree-r shadow $\mathrm{Sh}_r(\mathcal{A})$ equals $\ell_r^{(0),\mathrm{tr}}$ evaluated on the truncated MC element at every degree, so $\mathrm{Sh}_r = 0$ iff $\ell_r^{(0),\mathrm{tr}} = 0$ on H^*.

For $f_\infty = d_\infty$: the cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ carries both an A_∞ structure (from the bar differential) and an L_∞ structure (from the convolution bracket). On the genus-0 part, the Lie bracket $[-,-]^{(0)}$ is the antisymmetrization of the associative composition, so the transferred brackets satisfy $\ell_r^{(0),\mathrm{tr}} = \mathrm{Alt}(m_r^{\mathrm{tr}})$. The HPL tree formula (**??**) is universal in r: the same set Trees_r of planar rooted trees with r leaves, the same homotopy h on internal edges, and the same vertex evaluations appear in both the A_∞ and L_∞ transfers. Since antisymmetrization is injective on the cyclic deformation complex (the cyclic symmetry ensures no cancellation between distinct tree contributions), $m_r^{\mathrm{tr}} = 0$ iff $\ell_r^{(0),\mathrm{tr}} = 0$. Hence $d_\infty = f_\infty$. □

The shadow–formality dictionary and archetype verification:

Shadow	L_∞ **bracket**	**Family**	$r_{\max}$
κ (degree 2)	ℓ_2	Heisenberg	2
$\mathfrak{C}$ (degree 3)	ℓ_3^{tr}	Affine$\widehat{\ }_k$	3
$\mathfrak{Q}$ (degree 4)	ℓ_4^{tr}	$\beta\gamma$	4
full tower	all ℓ_k^{tr}	Vir_c	∞

Shadow depth classification

Shadow depth is orthogonal to Koszulness: all standard families are chirally Koszul, yet their shadow depths differ. See Theorem **??**.

Class	**Name**	**Criterion**	**Examples**
G	Gaussian	$r_{\max} = 2$; formal	$\mathcal{H}_k$, free fermion, rank-1 lattice, all 24 Niemeier (κ=24)
L	Lie	$r_{\max} = 3$; $\mathfrak{C} \neq 0$, $o_4 = 0$ (Jacobi)	$V_k()$
C	Contact	$r_{\max} = 4$; $\mathfrak{Q} \neq 0$, $o_5 = 0$ (abelian rigidity)	$\beta\gamma$, symplectic fermion
M	Mixed	$r_{\max} = \infty$; non-formal	Vir_c, $\mathcal{W}_N$, SCA_c ($\mathcal{N}$=2), $V^\natural$ (κ=12), $Y()$

The inclusions $\mathbf{G} \subset \mathbf{L} \subset \mathbf{C} \subset \mathbf{M}$ are proper. Shadow depth classifies the complexity of $\Theta_{\mathcal{A}}$, not whether the bar complex is acyclic (Remark **??**). The classification refines to a *two-channel complexity plane* (m, Δ_z) when the coproduct tower is included: class L algebras arising from Chern–Simons (affine KM) have strict A_∞ products ($m_{k\geq 3} = 0$) but nontrivial coproduct, while class M algebras arising from DS reduction (Virasoro, $\mathcal{W}_N$) have an infinite A_∞ tower but strictly primitive coproduct (Principle 1.13.3).

Fine classification and arithmetic content. The coarse **G/L/C/M** partition admits a finer grading by the exact depth $d \in \{2, 3, 4, 5, \ldots\} \cup \{\infty\}$; every finite value is realized by even unimodular lattice VOAs of increasing rank (Chapter **??**, §**??**). The shadow–spectral correspondence (Theorem **??**) identifies $d - 1$ with the number of critical lines of the constrained Epstein zeta $\varepsilon_s^c(\mathcal{A})$, whose L-function factorization reflects the modular-form decomposition of the theta function. The first five lattice examples are:

Algebra	**Depth**	**Lines**	L**-function content**
V	2	1	$\zeta(2s)$
V_2, V_{A_2}	2	1	$\zeta_{(\sqrt{-d})}(s)$
V_{E_8}	3	2	$\zeta(s) \cdot \zeta(s{-}3)$
V_{Leech}	4	3	$\zeta(s) \cdot \zeta(s{-}11) \cdot L(s, \Delta_{12})$
Rank-48 lattice	5	4	+ second cusp form

The sewing–Selberg formula (Theorem **??**): $\int_{1,1} \log\det(1-K)\cdot E_s \, d\mu = -2(2\pi)^{-(s-1)}\Gamma(s-1)\,\zeta(s-1)\zeta(s)$.

The depth decomposition (Theorem **??**): $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$, where d_{arith} counts holomorphic Hecke eigenforms (determines critical lines) and d_{alg} measures non-holomorphic OPE complexity (invisible to L-functions). For lattice VOAs: $d_{\mathrm{alg}} = 0$; for Virasoro minimal models: d_{arith} finite, $d_{\mathrm{alg}} = \infty$. The Borel sum of the Virasoro shadow series gives $-\log(1 + 6t/c)$ at leading order (Theorem **??**).

Shadow growth rate (Definition **??**, Theorem **??**). For class **M** algebras, the shadow growth rate $\rho(\mathcal{A}) = \sqrt{9\alpha^2 + 2\Delta} \,/\, (2|\kappa|)$ refines shadow depth into a *continuous invariant*: the exponential growth rate of the shadow obstruction tower coefficients. The shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ has convergence radius $R = 1/\rho$, and the asymptotic transfer theorem gives $S_r \sim A\,\rho^r \cdot r^{-5/2}\cos(r\theta + \varphi)$. For Vir_c: $\rho(c) = \sqrt{(180c{+}872)/((5c{+}22)c^2)}$. The critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$ has unique positive real root $c_\star \approx 6.125$; for $c > c_\star$ the tower converges ($\rho < 1$), for $c < c_\star$ it diverges. Koszul duality $c \mapsto 26 - c$ acts on ρ; at $c = 13$ the shadow growth rate is self-dual ($\rho \approx 0.467$). The branch-point argument $\arg(t_0)/\pi$ increases monotonically toward 1 as $c \to \infty$, encoding the near-alternation of signs. The convergent regime $c > c_\star$ connects to the arithmetic programme: ρ controls the abscissa of convergence of the shadow–Epstein zeta $\varepsilon_s^c(\mathcal{A})$.

General Vieta structure (Theorem **??**). The shadow growth rate $\rho_L = \sqrt{q_2/q_0}$ is the ratio of the quadratic to constant coefficient of Q_L, a consequence of Vieta's theorem for the branch-point product. The critical surface $\{q_0 = q_2\}$ in the shadow metric moduli space $\mathcal{M}_{\mathrm{sm}}$ is characterized geometrically: *the branch points of Q_L lie on the unit circle*. No family-specific data enters.

Multi-line structure and Virasoro bottleneck (Definition **??**, Proposition **??**). For algebras with multiple primary lines, $\rho(\mathcal{A}) = \sup_L \rho_L$. The Virasoro subsector (T-line) controls the algebra-level growth rate: every $\mathcal{W}_N$ algebra has $\rho(\mathcal{W}_N) = \rho_T = \rho(\mathrm{Vir}_c)$. Higher-spin lines have smaller ρ due to stronger Kac-denominator suppression of the quartic seed. Example: $\mathcal{W}_3$ has $c_T^\star \approx 6.125$ (T-line) but $c_W^\star \approx 1.187$ (W-line), with $\rho_W^2 = 30720/(c^2(5c{+}22)^3)$.

Koszul exchange of regimes (Theorem **??**). For one-parameter families with conductor K and Verdier involution $c \mapsto K - c$: three regimes (I: $\mathcal{A}$ divergent/$\mathcal{A}^!$ convergent; II: both convergent; III: reverse of I). The convergence window $(c_\star, K - c_\star)$ has fraction $(K - 2c_\star)/K \to 1$ as $K \to \infty$ (Proposition **??**): $\rho_T(K_N/2) \sim 12/K_N \to 0$.

DS depth increase. Under Drinfeld–Sokolov reduction $V_k() \to \mathcal{W}_k()$, the shadow growth rate jumps from $\rho = 0$ (class **L**, shadow depth 3) to $\rho > 0$ (class **M**, shadow depth ∞). The ghost sector creates all quartic and higher shadows via BRST cohomology: the DS reduction does not preserve ρ, but the $\mathcal{W}_N$ T-line shadow radius equals the Virasoro shadow radius at the DS central charge $c_{\mathcal{W}_N}(k)$, consistent with the conjectural $\Theta_{\mathcal{W}_N} = H^0_{Q_{\mathrm{DS}}}(\Theta_{V_k \otimes \mathcal{F}_{\mathrm{gh}}})$.

Principle 1.13.2 *(Shadow-depth escalator).* Current-algebra systems (Chern–Simons, M2) are class **L**: finite shadow obstruction tower ($r_{\max} = 3$), complementarity anti-symmetry $\kappa + \kappa' = 0$, and r-matrices with simple poles satisfying the classical Yang–Baxter equation. $\mathcal{W}$-algebra systems (3d gravity, $\mathcal{W}_3$, M5) are class **M**: infinite shadow obstruction tower, $\kappa + \kappa' \neq 0$, and r-matrices with higher-order poles. The Drinfeld–Sokolov functor sends class **L** to class **M** (Theorem **??**; Volume II, Remark **??**).

Principle 1.13.3 *(Gravitational primitivity).* For any principal Drinfeld–Sokolov reduction $W_k() = H^0_{\mathrm{BRST}}(\widehat{\mathfrak{g}}_k, f_{\mathrm{prin}})$, the HPL-transferred coproduct is strictly primitive at all degrees:

$$\Delta^W_{z,n} = 0 \qquad \text{for all } n \geq 2.$$

All nontrivial structure of the transferred dg-shifted Yangian resides in the A_∞ products $\{m_n^W\}_{n\geq 3}$ and in the collision residue $r(z)$. The coproduct carries no information. The mechanism is a ghost-number obstruction intrinsic to DS reduction: every HPL tree for the transferred coproduct produces output in ghost number ± 1, and the projection onto the $\mathcal{W}$-algebra cohomology (ghost number 0) annihilates it (Volume II, Theorem **??**).

Together these two principles extend the shadow depth classification to a *two-channel complexity plane* (m, Δ_z): class L systems (Chern–Simons) have strict A_∞ products ($m_n = 0$ for $n \geq 3$) and nontrivial coproduct; class M systems (gravity) have an infinite A_∞ tower but trivial coproduct.

Principle 1.13.4 *(Physics is the homotopy type).* The modular homotopy type of a chiral algebra $\mathcal{A}$ (the formal moduli problem $^{\mathrm{mod}}_{\mathcal{A}}(R) = \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_\infty(\mathcal{A}) \otimes R)/\mathrm{gauge}$) determines, through its finite-order projections, every perturbative observable of the corresponding holomorphic-topological QFT. The A_∞ operation m_k is the k-point scattering amplitude; the genus tower $F_g = \kappa \cdot \lambda_g$ is the loop expansion; the R-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the scattering matrix; the Yangian is the symmetry algebra of the S-matrix; Koszul duality is S-duality; the shadow depth classification $\mathsf{G}/\mathsf{L}/\mathsf{C}/\mathsf{M}$ is the interaction complexity hierarchy (free/gauge/contact/gravitational).

The following dictionary records the systematic translation between perturbative QFT and modular Koszul theory. Each entry in the left column is a physical concept; the middle column gives its algebraic incarnation; the right column specifies the source within the MC framework.

Physics	**Algebra**	**MC source**
k-point scattering amplitude	A_∞ product m_k	degree-k bar differential
genus-g vacuum amplitude	free energy $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\mathrm{FP}}$	scalar trace of $\Theta^{(g)}$
loop expansion	genus tower $\{F_g\}_{g\geq 1}$	$\hat{A}$-genus generating function
S-matrix	collision R-matrix $r(z)$	$\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$
S-matrix symmetry	Yangian $Y_\hbar()$	RTT presentation from $r(z)$
S-duality	Koszul duality $\mathcal{A} \leftrightarrow \mathcal{A}^!$	Verdier intertwining (Thm A)
line operators	$\mathcal{A}^!$-mod	DK bridge (MC3)
anomaly cancellation	$\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 0$	critical dimension
one-loop exactness	PBW / chirally Koszul	E_2-collapse (K1–K2)
BV master equation	$D^2_{\mathcal{A}} = 0$	bar-intrinsic MC (MC2)
free / gauge / contact / gravitational	G / L / C / M	shadow depth $r_{\max}$
perturbative finiteness	HS-sewing condition	analytic bar coalgebra (MC5)
complementarity / CPT	$Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$	Hodge decomposition (Thm C)
Coulomb moduli	$\mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$	MC moduli of $\Theta_{\mathcal{A}}$
gravitational Ward identity	shadow connection $\nabla^{\mathrm{hol}}_{g,n}$	$d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$

Arithmetic threshold (Remark **??**). $\rho < 1$: shadow–Epstein zeta entire. $\rho = 1$: convergent for $\mathrm{Re}(s) > -3/2$. $\rho > 1$: divergent, Borel-regularized. The critical $c_\star$ is the threshold between direct and regularized spectral decomposition.

Hamilton–Jacobi master equation (Theorem **??**). On an N-dimensional primary deformation space, the nonlinear shadow generating function $U(x) = \sum_{r\geq 3} \mathrm{Sh}_r(x_1, \ldots, x_N)$ satisfies the Hamilton–Jacobi equation $2\,E(U) + \frac{1}{2}\|\nabla U\|^2_H = R(x)$, where E is the Euler operator, $\|\nabla f\|^2_H = \sum_i P_i\,(\partial_i f)^2$ is the H-norm with propagators $P_i = 1/\kappa_i$, and the source R is determined entirely by degrees 3 and 4. On each autonomous primary line, the Hamilton–Jacobi equation reduces to the Riccati equation of Theorem **??**. Verified computationally for $\mathcal{W}_3$ through degree 7.

Shadow finite determination (Theorem **??**). The full infinite shadow obstruction tower is uniquely determined by finite data at degrees $2, 3, 4$: the curvatures $\kappa_1, \ldots, \kappa_N$, the cubic shadow $\mathrm{Sh}_3 \in \mathrm{Sym}^3(V^*)$,

and the quartic shadow $\mathrm{Sh}_4 \in \mathrm{Sym}^4(V^*)$. The Hamilton–Jacobi recursion determines Sh_r for all $r \geq 5$ with rational-function coefficients in the seed data. For $\mathcal{W}_3$ with $\mathbb{Z}_2$ parity: exactly 7 parameters (κ_T, κ_W, $\alpha_T, \beta_{TW}, Q_{TT}, Q_{TW}, Q_{WW}$) determine the full 2D tower at all degrees (Corollary **??**).

Superconformal hierarchy (Remark **??**). The four fundamental superconformal families form a hierarchy under extended supersymmetry, with complementarity sum $\kappa + \kappa'$ strictly decreasing:

$\mathcal{N}$	c'	κ	$\kappa + \kappa'$	Self-dual	Chapter
0 (Vir)	$26-c$	$c/2$	13	$c = 13$	Ch. **??**
1 (SVir)	$15-c$	$(3c-2)/4$	$41/4$	$c = 15/2$	Ch. **??**
2 (SCA)	$6-c$	$(k+4)/4$	1	$c = 3$	Ch. **??**
4 (small)	$-c-24$	$2k$	-8	$c = -12$	n/a

The $\mathcal{N}$=2 SCA is the definitive witness: $H^2_{\mathrm{CE}}(\mathfrak{g}_-) \neq 0$ at conformal weight 3, yet $H^2(\overline{B}(\mathrm{SCA}_c)) = 0$ (chirally Koszul by PBW universality). Sub-leading OPE modes invisible to the CE complex provide coboundaries that kill the CE cocycles (Proposition **??**, Remark **??**).

Moonshine module $V^\natural$ (Chapter **??**). The unique holomorphic VOA at $c = 24$ with $\dim V_1^\natural = 0$ has $\kappa(V^\natural) = c/2 = 12$ (Proposition **??**), class M ($\Delta = 20/71 \neq 0$), and shadow depth $r_{\max} = \infty$. The modular characteristic separates $V^\natural$ from all twenty-four Niemeier lattice VOAs, which share $\kappa = 24$ and class G (Remark **??**). The /2 orbifold $V_{\Lambda_{24}} \to V^\natural$ changes shadow class from G to M: the only instance in the standard landscape where a finite orbifold changes the shadow class (Remark **??**).

Geometric, physics, and categorical characterizations

Geometric.

- *Ran-space formality*: E_2-collapse $\Leftrightarrow$ genuswise bar-cobar formality $\Leftrightarrow$ $\tilde{m}_n^{(g)} = 0$ for $n \geq 3$ (Proposition **??**); coherent extension to all $\mathrm{Ran}^{\leq n}$ is open.
- *Tropical Koszulness*: acyclicity of $\bar{\mathbf{B}}^{\mathrm{trop}}(\mathcal{A})$ for the planted-forest differential (Definition **??**, Theorem **??**).
- *Curve independence*: chiral Koszulness depends on formal-disk data, not on X; higher-genus propagation governed by MK1–MK3 (Theorem **??**) and the exceptional set $\Sigma()$ (Theorem **??**).

Physics.

- *One-loop exactness*: PBW E_2-collapse IS one-loop exactness (Remark **??**).
- *Kac–Shapovalov*: $\det G_h \neq 0$ in bar-relevant range $\Leftrightarrow$ Koszul (Theorem **??**); proved for $V_k()$ (Proposition **??**). For simple admissible quotients $L_k()$ this route is blocked by null vectors in the bar-relevant range.
- *Rationality / semisimplicity*: strong finiteness input for non-degenerate admissible levels, but not an unconditional Koszul criterion. Ordinary semisimplicity controls Zhu theory and character formulas; the comparison with the bar-complex Ext groups remains part of the live admissible-level audit.
- *C_2-cofiniteness*: orthogonal to Koszulness; Koszul duality maps C_2-cofinite to non-C_2-cofinite.
- *Critical level*: $V_{-h^\vee}()$ is uncurved ($\kappa = 0$), self-dual under the Feigin–Frenkel involution ($k = -h^\vee$ is a fixed point of $k \mapsto -k - 2h^\vee$), center = $\mathfrak{z}()$ (Theorem **??**).

Categorical.

- *t-structures*: a highest-weight bar/Koszul t-structure comparison is conditional on separate module-category t-exactness and linearity input; there is no unconditional global $\bar{B}^{\mathrm{ch}}$ t-exactness theorem (Conjecture **??**, Conjecture **??**).
- *Barr–Beck–Lurie*: effective descent for bar-cobar (Remark **??**).
- *Semi-orthogonal decomposition*: weight-indexed SOD on $\mathrm{FactMod}(\mathcal{A})$ is split iff Koszul.
- *BGG*: bar resolution of $V_k()$-modules is linear; Koszul involution maps BGG to Cousin on Gr_G (Corollary **??**).
- *Scalar-orbit convolution formality*: for uniform-weight algebras, the distinguished scalar Maurer–Cartan orbit in $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is formal; converse and one-channel generalizations remain open.
- *MC smoothness*: $\mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ smooth at $\Theta_{\mathcal{A}}$ iff Koszul.

Bifunctor obstruction

The RNW no-bifunctor obstruction [367] (§1.11, identification theorem 7) is structural: $\hom_\alpha(\mathcal{C},\mathcal{A})$ extends to ∞-morphisms in either slot separately but not both. Koszulness does not kill it; Koszulness makes it irrelevant by factoring all constructions one slot at a time (Theorem **??**). This is why the MC3 strategy (Theorem **??**) succeeds.

Computational diagnostics

The bar generating function $\overline{P}_{\mathcal{A}}(x,t)$ is algebraic of degree 2 in t for all standard families with $\dim \leq 8$, sharing the discriminant $\Delta_{\mathcal{A}}(x) = (1-3x)(1+x)$ across $\widehat{\mathfrak{sl}}_2$, $\beta\gamma$, and Virasoro (Theorem **??**).

Decision tree. (1) Pole order ($1 \Rightarrow$ G; $2 \Rightarrow$ L+). (2) Freely generated $\Rightarrow$ Koszul (Proposition **??**). (3) PBW + classical Koszulness of $\mathrm{gr}_F\,\mathcal{A}$ (Theorem **??**). (4) Bar Betti diagonal concentration (Corollary **??**). (5) Generating function fit $\to$ discriminant.

Priority computations.

- $\mathcal{W}_4$ bar Betti numbers (Class M + discriminant universality).
- Bigraded Virasoro to $(n,h) = (4,16)$ (quartic resonance at GF level).

Computational verification landscape. The PBW criterion has been verified computationally for 17 algebras (Table **??**): 13 proved Koszul, 2 proved not Koszul (Ising minimal model: null vector at $h = 6$ in the bar-relevant range; $L_k(\mathfrak{sl}_3)$ at admissible $q \geq 3$: $\mathrm{rk}(\mathfrak{sl}_3) = 2$ Cartan classes survive in $H^2(\overline{B})$), 2 open ($L_1(\mathfrak{sl}_2)$ and triplet $\mathcal{W}(2)$). Five corrections to initial expectations: (i) the symplectic fermion IS Koszul (class C, PBW universality; logarithmic phenomena affect the module category, not the bar complex; Remark **??**); (ii) quotient of a Koszul algebra need not be Koszul ($L_1(\mathfrak{sl}_2)$: null at $h = 2$ in bar range); (iii) the critical level $k = -h^\vee$ does not break Koszulness of the universal algebra $V_k()$; (iv) admissible Koszulness for simple quotients is *rank-dependent*: $L_k(\mathfrak{sl}_2)$ is Koszul at all admissible levels, but $L_k(\mathfrak{sl}_3)$ at $q \geq 3$ is not, with the obstruction coming from the abelian Cartan subalgebra (Remark **??**, Conjecture **??**); (v) the N=2 SCA is Koszul despite nonvanishing H^2_{CE}: the CE complex is a coarse upper bound on bar cohomology, and PBW universality resolves the apparent paradox (Remark **??**; Chapter **??**, Proposition **??**, Remark **??**). Cross-family consistency verified: κ additivity, complementarity $\kappa+\kappa' = 0$ for KM, $\kappa+\kappa' = 13$ for Virasoro, and $\kappa + \kappa' = 1$ for the $\mathcal{N}$=2 SCA , and shadow depth independence from Koszulness. Suite: `test_koszulness_landscape.py` (101 tests).

The shadow connection (Theorem **??**) and propagator variance (Proposition **??**) have been verified for the full multi-generator landscape: ∇^{sh} for Virasoro and $\mathcal{W}_3$ (both T-line and W-line), with residue $1/2$ and monodromy -1 confirmed; δ_{mix} for six families ($\mathcal{W}_3, \mathcal{W}_4, \mathcal{W}_5, \beta\gamma{+}bc, +\mathrm{Vir}, V(\mathfrak{sl}_2){+}V(\mathfrak{sl}_3)$);

$\mathcal{W}_4$ is never autonomous at any real central charge. Suites: `test_shadow_connection.py` (114 tests), `test_propagator_variance_landscape.py` (143 tests).

Cross-reference table

Item	Label	Location
PBW E_2-collapse	`thm:pbw-koszulness-criterion`	Ch. **??**
Bar-cobar q.i.	`thm:bar-cobar-inversion-qi`	Ch. **??**
ChirHoch polynomial	`thm:main-koszul-hoch`	Ch. **??**
A_∞ ($\Rightarrow$)	`prop:ainfty-formality-implies-koszul`	Ch. **??**
A_∞ ($\Leftarrow$)	`thm:ainfty-koszul-characterization`	Ch. **??**
Ext diagonal	`thm:ext-diagonal-vanishing`	Ch. **??**
Kac–Shapovalov	`thm:kac-shapovalov-koszulness`	Ch. **??**
Deformation quant.	`rem:dq-koszul-compatibility`	Ch. **??**
Meta-theorem	`thm:koszul-equivalences-meta`	Ch. **??**
Operadic complexity	`thm:operadic-complexity`	§1.13
Archetype classification	`thm:nms-archetype-trichotomy`	App. **??**
Finite termination	`thm:nms-finite-termination`	App. **??**
Discriminant	`thm:ds-bar-gf-discriminant`	Landscape census
$\mathcal{N}$=2 SCA Koszulness	`prop:n2-koszulness`	Ch. **??**
$\mathcal{N}$=2 CE/bar gap	`rem:n2-ce-chiral-gap`	Ch. **??**
$V^\natural$ modular char.	`prop:moonshine-kappa`	Ch. **??**
Envelope-shadow	`def:envelope-shadow-functor`	§1.8
Finite-jet rigidity	`prop:finite-jet-rigidity`	§1.8
Gaussian collapse	`prop:gaussian-collapse-abelian`	§1.8
Graph-complex shadow	`conj:graph-complex-shadow`	Ch. **??** *(Heuristic: map Φ not rigorously constructed)*

Status: nine equivalences (i)–(vii), (ix), (x) proved unconditionally, with (vii) the genus-0 factorization-homology criterion together with its proved all-genera uniform-weight refinement, and with explicit multi-weight boundary counterexamples. The Hochschild statement (viii) is a proved one-way consequence on the Koszul locus, yielding concentration, polynomial Hilbert growth, and E_2-formality but no converse and no blanket freeness upgrade; the Lagrangian criterion (xi) now unconditional for the standard landscape (Proposition **??** verifies perfectness at non-critical, non-degenerate levels); $\mathcal{D}$-module purity (xii) one-directional (forward direction proved, converse open). Two further characterizations, tropical Koszulness (Theorem **??**) and bifunctor obstruction decomposition (Theorem **??**), are proved separately but are not among the twelve numbered conditions.

Proof web. The proved core on (i)–(vii), (ix), (x) is a web, not a single chain. The shortest loop is (v)$\Leftrightarrow$(i) via the fundamental theorem of chiral twisting morphisms. Independent direct arrows (i)$\Rightarrow$(viii), (iii)$\Rightarrow$(x), (v)$\Leftrightarrow$(vi), and (vii)$\Leftrightarrow$(ii) bypass the old linear spine; the Ext, Kac–Shapovalov, and FM-boundary criteria (iv), (ix), (x) each force PBW collapse directly. Hence no single proved implication disconnects the

unconditional core. The factorization-homology arrow is genus-0 in general and upgrades to all genera only on the uniform-weight lane; multi-weight principal $\mathcal{W}$-algebras witness the boundary failure of the stronger all-genera claim through $\delta F_g^{\mathrm{cross}} \neq 0$. The Hochschild statement (viii) remains a proved one-way consequence on the Koszul locus: it is obtained from bar-cobar inversion via Theorem H, polynomial growth, and E_2-formality, with no converse back to the bar-cobar counit. The Lagrangian criterion (xi) is conditional in general, and the $\mathcal{D}$-module purity statement (xii) is one-directional.

1.14 Rectification: the organizing meta-theorem

Rectification, the passage from a homotopy-algebraic structure to a strict one without loss of invariant data, is the single mechanism underlying all five MC levels and the main conjectures in the non-principal corridor.

Level	Formality input	Consequence
MC1	PBW filtration	Genus-0 bar complex is formal: E_2-degeneration gives $H^*(\overline{B}(\mathcal{A})) \cong$ quadratic dual (Theorem **??**). Holds unconditionally for universal algebras (Proposition **??**). For simple quotients $L_k()$ at admissible levels with denominator $q \geq 3$ and $\mathrm{rk}() \geq 2$, PBW concentration *fails*: the Cartan subalgebra produces $\mathrm{rk}()$-dimensional bar cohomology in H^2 (Conjecture **??**)
MC2	$D^2 = 0$	Bar-intrinsic MC element: $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is automatically MC because $D^2_{\mathcal{A}} = 0$ (Theorem **??**). This is the concordance-level class in the coinvariant carrier $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$; the ordered lift $\Theta^{E_1}_{\mathcal{A}} \in \mathrm{MC}((\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})^{E_1})$ is constructed separately in Theorem **??**.
MC3	Prefundamental CG	**Proved for all simple types on the evaluation-generated core**. All-types package (i) is proved (Theorem **??**), and the generated-core DK comparison surface (traditionally packaged as DK-2/3) on the evaluation-generated core is proved separately (Corollary **??**). Extension beyond evaluation modules is DK-4/5 (downstream, not part of MC3). Key input: multiplicity-free q-characters for fundamental reps [450] replace the minuscule hypothesis. Type A identifies the remaining post-CG completion packet on the Baxter-locus seed $\{V_{\omega_i}(a)\} \cup \{L^-_{i,a-\frac{1}{2}}\}$ (Theorem **??**, Proposition **??**); the residual compact-completion extension is DK-4/5.
MC4	Strong filtration	Completion is formal inverse limit: the strong filtration axiom $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ gives degree cutoff (Lemma **??**), making continuity and Mittag-Leffler automatic (Theorem **??**). Positive towers are handled by coefficient stabilization; the resonance lane is proved on the transferred split surface of Theorems **??** and **??**.
MC5	Genus tower	Five components: (1) **Analytic HS-sewing proved** at all genera: polynomial OPE growth + subexponential sector growth implies convergence (Theorem **??**); (2) **Genus-0 algebraic BRST/bar proved** (Theorem **??**); (3) D^{co}**-level BV=bar proved** for all four shadow classes including class M (Theorem **??**); (4) genuswise chain-level BV/BRST/bar identification **conjectural in general** (classes G/L proved, class C conditional on harmonic decoupling, class M chain-level false; Conjecture 2.2.7); (5) tree-level amplitude pairing **conditional** on Corollary **??**

Remark 1.14.1 *(DS–KD filtration formality route).* The DS–KD intertwining for non-principal $\mathcal{W}$-algebras (Conjecture **??**) reduces to filtration formality (E_1 degeneration of the Kazhdan filtration spectral sequence) of the BRST complex. The derived DS functor $R\mathrm{DS}_f$ agrees with $H^0_{Q_{\mathrm{DS}}}$ on the abelian-$\mathfrak{n}_+$ generic lane, because the BRST complex is filtration-formal by the Kazhdan filtration argument: the Kazhdan filtration on $V_k() \otimes \mathcal{F}_{\mathfrak{n}_+}$ gives E_1-degeneration at generic k, and the resulting E_1-page is the $\mathcal{W}$-algebra with no higher corrections. The obstruction to extending this beyond generic level is the possible failure of E_1-degeneration at rational k (admissible/critical levels), where vacuum null vectors can produce genuine higher differentials. The hook-type corridor is the initial audit surface for this route, but the manuscript currently contains incompatible summaries of the type-A abelian-$\mathfrak{n}_+$ locus, so the hook-wide extension is not used as unconditional input. Accordingly the hook corridor remains conditional (Theorem **??**), and the $(3, 2)$-nilpotent in $\mathfrak{sl}_5$ remains the first explicit non-abelian case where the filtration-formality argument requires genuine new work.

1.15 Non-principal $\mathcal{W}$-algebra duality: the transport frontier

After the principal Feigin–Frenkel case (Theorem **??**), the non-principal duality problem splits into two axes (Principle **??**): *transport along the reduction network* and *identification of the dual orbit/level*.

Proved core.

- *Principal*: Feigin–Frenkel duality, all simple .
- *Transport propagation*: duality propagates from hook vertices through the reduction graph Γ_N to every partition in the transport-closure (Proposition **??**).

Active frontier.

- *Subregular/minimal in* $\mathfrak{sl}_3$: explicit bar/OPE computations and strictness are proved, but the level-shifted Bershadsky–Polyakov duality statement remains conjectural (Conjecture **??**; Computation **??**; Theorem **??**). The Bershadsky–Polyakov algebra $\mathrm{BP}_k = \mathcal{W}_k(\mathfrak{sl}_3, f_{\min})$ is chirally Koszul at generic level ($k \neq -3$), with dual $\mathrm{BP}_k^! = \mathrm{BP}_{-k-6}$ (the level-shifted Feigin–Frenkel involution for $\mathfrak{sl}_3$ applied to the minimal nilpotent).
- *Hook-type corridor in type A*: inverse reductions and transport data are available, but the full hook-type duality statement is kept conditional on hook-wide DS/bar compatibility (Theorem **??**).
- *Type-A transport-to-transpose* (Conjecture **??**): the transport-closure of the hook vertices equals $\mathrm{Par}(N)$ at generic level. Computationally verified for $N = 2, \ldots, 7$. The structural dividing line is $\mathfrak{n}_+$-abelianity: hook-type orbits have $\mathfrak{n}_+$ that is simpler than the general case (though not always abelian; see Chapter **??**). The partition $(3, 2)$ in $\mathfrak{sl}_5$ is the first genuine frontier case with non-abelian $\mathfrak{n}_+$.
- *General orbit duality* (Conjecture **??**): extends to all simple via Barbasch–Vogan/Spaltenstein duality.

- *DS–KD intertwining for arbitrary nilpotent* (Conjecture **??**): bar–cobar duality commutes with DS reduction at arbitrary f. The strongest route is via BRST filtration formality (Remark 1.14.1): the Kazhdan filtration gives a case-by-case route at abelian $\mathfrak{n}_+$. Beyond that abelian locus, the essential new input is a higher-order derived/classical comparison for the BRST complex at rational k.

Hook-type quantitative data.

- *Ghost constant formula* (Proposition **??**): $C_{(m,1^r)} = m(m^2-1)/6 + r\lfloor m^2/2\rfloor/2$. Verified for $\mathfrak{sl}_3$ through $\mathfrak{sl}_5$.
- *DS–bar commutation for hook-type* (Proposition **??**): kappa compatibility, generator matching, and central charge threading verified for all hook partitions in $\mathfrak{sl}_3$–$\mathfrak{sl}_5$.
- *Kappa sum rule* (Remark **??**): $\kappa(\mathcal{W}^k(f_\eta)) + \kappa(\mathcal{W}^{-k-2N}(f_{\eta^t})) = -(C_\eta + C_{\eta^t})$ is k-independent, automatic from linearity of ghost subtraction.
- $\mathfrak{sl}_4$ *data table* (Computation **??**): complete hook-type ghost constants, generator spectra, and centralizer dimensions.

$\mathcal{W}_3$ multi-variable shadow archetype. The $\mathcal{W}_3$ algebra is the first standard family requiring the multi-variable shadow obstruction tower (§**??** in Chapter **??**). Three independent quartic channels appear: $\mathfrak{Q}_{TTTT}$, $\mathfrak{Q}_{TTWW}$, and $\mathfrak{Q}_{WWWW}$, coupled by Λ-exchange (cross-channel pole terms in W). The denominator filtration by W-charge (Proposition **??**) organizes these channels: each carries a factor $(5c+22)^{-\lfloor qw/2\rfloor}$ from the Kac determinant at spin-3 level. The cubic correction $\mathfrak{C}_{TWW} = d_W = 3$ (not 2) reflects the conformal weight of W and is fixed by the Jacobi identity for the T-W-W OPE.

Key constraint. The difficulty is *not* the existence of quantum DS reduction for arbitrary f (Kac–Roan–Wakimoto [240]); it is proving that bar–cobar/Koszul duality commutes with arbitrary-nilpotent reduction, identifying the correct dual nilpotent, and determining the non-principal level transform.

Recent developments. Creutzig–Fasquel–Linshaw–Nakatsuka establish that hook-type $\mathcal{W}$-algebras are building blocks for all $\mathcal{W}$-algebras in type A via successive quantum Hamiltonian reductions [arXiv:2403.08212, JJM 2025], confirming the privileged role of the hook corridor in the transport graph. Butson–Nair prove inverse Hamiltonian reduction for all affine $\mathcal{W}$-algebras in type A at generic level [arXiv:2508.18248], extending the transport graph from hook-type seeds to the full orbit Hasse diagram.

Geometric localization programme. The non-principal duality problem has a geometric reformulation through the Slodowy-slice localization principle (Remark **??**, Theorem **??**). The programme is:

(1) *Geometric localization theorem* (Theorem **??**, proved elsewhere): DS reduction at non-critical level = Hamiltonian reduction from $J_\infty(^*)$ to $J_\infty(S_f)$ (the arc space of the Slodowy slice).
(2) *Master commutative square* (Theorem **??**, proved here for principal f): the diagram connecting bar complexes, PBW filtrations, and Hamiltonian reduction commutes. For arbitrary f, the square is structural but the commutativity requires the non-principal PBW identification $\mathrm{gr}_F\,\mathcal{W}^k(,f) \cong [J_\infty(S_f)]$.
(3) *Barbasch–Vogan as Springer duality of Slodowy slices* (Remark **??**): orbit duality $f \leftrightarrow f^D$ has geometric content as duality of transverse slices under the Springer resolution; the curvature complementarity $\kappa + \kappa' = \mathrm{const}$ is an algebraic identity under the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$.

(4) *Shadow depth from Dirac bracket nonlinearity* (Remark **??**): the depth increase $L \to M$ under DS is the Taylor expansion of the nonlinear Dirac bracket on $J_\infty(S_f)$, created by Hamiltonian reduction from the linear Kirillov–Kostant bracket.
(5) *Holographic extension* (conjectural): the full holographic datum $H(T)$ should be functorial under DS-as-localization, with the boundary algebra localizing to $J_\infty(S_f)$ and the line category restricting accordingly.

Items (1)–(4) are proved or structural; item (5) is the conjectural holographic lift.

Remark 1.15.1 *(Transport-closure and structural diagnostics).* Computational verification yields four structural facts:

(i) *Transport-closure* $= \mathrm{Par}(N)$. Verified for $N = 2, \dots, 8$: the transport-closure of hook vertices in the reduction graph Γ_N exhausts all partitions of N.
(ii) $\mathfrak{n}_+$*-abelianity as structural divide.* All hook-type orbits have abelian $\mathfrak{n}_+$; all non-hook orbits have non-abelian $\mathfrak{n}_+$. The Kazhdan formality argument extends exactly to the abelian locus.
(iii) *Kappa sum constancy for non-hook orbits.* $\kappa(\mathcal{W}^k(, f_\eta)) + \kappa(\mathcal{W}^{-k-2h^\vee}(, f_{\eta^t}))$ is k-independent for all orbits, not only hook-type. This is automatic from linearity of the ghost subtraction, but the explicit value is non-trivial for non-hook orbits.
(iv) $\mathcal{W}_N$ *is N-strict.* The Miura embedding gives $r_{\max}(\mathcal{W}_N) = \infty$ for all $N \geq 2$ (shadow class M), verified through $N = 20$.

1.16 E_1 MODULAR THEORY: THE ORDERED BAR COMPLEX AT ALL GENERA

The E_∞ modular theory (Theorems A–D+H, the shadow obstruction tower, $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$) has an E_1 counterpart in which symmetry is replaced by ordering. The E_1 theory is developed as a standalone chapter (Chapter **??**), making the E_1/E_∞ parallelism architecturally visible. The following table records the parallel construction.

Object	E_∞ (*symmetric*)	E_1 (*ordered*)
Modular operad	$\mathsf{Com}^{:}_{g,n}$	$\mathsf{Ass}^{:\,\mathrm{rib}}_{g,n}$
Feynman transform	$F\mathsf{Com}$	$F\mathsf{Ass}$ (Def. **??**)
Convolution	$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (Def. **??**)	$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}\,E_1}$ (Def. **??**)
MC element	$\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$	$\Theta_{\mathcal{A}}^{E_1} = D_{\mathcal{A}}^{E_1} - d_0$
Degree-2 shadow	$\kappa(\mathcal{A})$ (scalar)	$r(z)$ (R-matrix)
Degree-3 shadow	$\mathfrak{C}(\mathcal{A})$ (cubic)	$r_3(z_1, z_2)$ (KZ associator)
Planted forests	$\mathbb{G}_{\mathrm{pf}}$ (unordered)	$\mathbb{G}_{\mathrm{pf}}^{\mathrm{pl}}$ (planar, Def. **??**)
Ambient algebra	$\mathfrak{g}_{\mathcal{A}}^{\mathrm{amb}}$	$\mathfrak{g}_{\mathcal{A}}^{\mathrm{amb},E_1}$ (Def. **??**)
Coinvariant map	$\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}\,E_1} \twoheadrightarrow \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$; $\mathrm{av}(\Theta^{E_1}) = \Theta$	

Status of the E_1 five main theorems.

	Genus 0	*All genera*	*Reference*
$\mathbf{A}^{E_1}$: bar–cobar	**Proved**	**Proved**	Thms **??**, **??**
$\mathbf{B}^{E_1}$: inversion	**Proved**	**Proved**	Thms **??**, **??**
$\mathbf{C}^{E_1}$: complementarity	**Proved**	**Proved**	Thms **??**, **??**
$\mathbf{D}^{E_1}$: degree-2 shadow	**Proved**	**Proved**	Thms **??**, **??**
$\mathbf{H}^{E_1}$: Hochschild	**Proved**	**Proved**	Thms **??**, **??**

Resolved conjectures.

- Ordered-associative modular Maurer–Cartan theory: Theorem **??** records the Volume II resolution of the original modular conjecture. The E_1 modular convolution $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}^{E_1}}$ carries $\Theta_{\mathcal{A}}^{E_1}$ by the bar-intrinsic construction.
- Principal ordered-associative DS compatibility: Theorem **??** records the Volume II principal-case theorem for DS reduction at non-critical level. Hook-type in type A also covered.
- `thm:FG-shadow`: resolved at degrees 2, 3, 4 (Theorem **??**). The E_∞ shadow IS the Σ_r-coinvariant of the E_1 shadow.

Remaining conjectural frontier.

- `thm:FG-shadow` at all degrees: the full abelianisation spectral sequence from ^{ord} to requires analysis at infinite degree.
- `conj:bordered`: geometric realisation of the diagonal bicomodule via bordered ordered configuration spaces. *Partially resolved*: items (1)–(3) and (5) follow from the bordered FM compactification of §**??** (Construction **??**, Theorem **??**). Item (4) (annular closure) remains conjectural, pending the annular bordered FM compactification.
- Conjecture **??** for arbitrary nilpotent f: requires Koszul-BRST commutation for non-principal reduction.

The three bar complexes

The monograph uses three bar complexes, each a different coalgebra structure on the desuspended generators $s^{-1}\bar{\mathcal{A}}$. They form a hierarchy:

Name	*Coalgebra*	*Coproduct*	*Where used*
$B^{\mathrm{Lie}}(\mathcal{A})$ (Harrison/coLie)	Cofree conilpotent coLie	Harrison/Chevalley	Francis–Gaitsgory; primitives of B^{Σ}
$B^{\Sigma}(\mathcal{A})$ (symmetric/coshuffle)	Cofree cocommutative	Coshuffle (2^n terms)	$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (Volume I); Theorems A–D+H
$B^{\mathrm{ord}}(\mathcal{A})$ (ordered/tensor)	Cofree coassociative	Deconcatenation (n+1 terms)	(Volume II); line operators; Yangians

Relationships.

1. *PBW decomposition.* The symmetric bar $B^{\Sigma}(\mathcal{A})$ decomposes under the Eulerian idempotents $e_n^{(p)} \in [\Sigma_n]$ as $B_n^{\Sigma} \simeq \bigoplus_p e_n^{(p)} \cdot B_n^{\mathrm{ord}}$. The $p = 1$ summand is the Harrison/coLie bar B^{Lie}. In characteristic 0, this gives a quasi-isomorphism $B^{\mathrm{Lie}}(\mathcal{A}) \xrightarrow{\sim} B^{\Sigma}(\mathcal{A})$ on primitives, but the two are categorically distinct: B^{Lie} is a coLie coalgebra, B^{Σ} is cocommutative coassociative.

2. *R-matrix descent.* The passage $B^{\mathrm{ord}} \to B^{\Sigma}$ is Σ_n-coinvariance twisted by the R-matrix: $B_n^{\Sigma} \simeq (B_n^{\mathrm{ord}})^{R\text{-}\Sigma_n}$ (Proposition **??**). For pole-free (commutative) chiral algebras, $R(z) = \tau$ (Koszul sign) and this is naive Σ_n-quotient. For vertex algebras with OPE poles, the R-matrix twist is nontrivial.

3. *Filtration spectral sequence.* The commutator filtration on B^{ord} has E_1-page $\simeq B^{\mathrm{Lie}}$ (the Francis–Gaitsgory bar, retaining only the zeroth OPE pole $a_{(0)}b$). This resolves Theorem **??**: the E_∞ shadow is the coinvariant of the E_1 shadow.

Which is used where. The modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (Definition **??**) uses the symmetric bar B^{Σ}. The Swiss-cheese analysis of Volume II uses the ordered bar B^{ord} as the E_1 line-side engine, while the actual two-colour datum is the derived-center pair $(C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\ \mathcal{A})$ computed from that bar resolution. The Harrison bar B^{Lie} appears as the primitives of B^{Σ} and controls the Francis–Gaitsgory chiral Lie Koszul duality. The three bar complexes produce three different Koszul duals: B^{Lie} gives the chiral Lie dual, B^{Σ} gives $\mathcal{A}^!_{\mathrm{ch}}$ (Volume I, Theorem A), and B^{ord} gives the ordered/line-side dual $\mathcal{A}^!_{\mathrm{line}}$.

E_1 PRIMACY

The E_1/ordered story is the natural primitive of the modular Koszul duality. The E_∞/symmetric story is its Σ_n-coinvariant shadow. This is the content of Theorem **??** (§**??**), which organises the relationship between the two volumes.

The averaging map. The Σ_n-coinvariant projection

$$\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{E_1} \twoheadrightarrow \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$$

sends the ordered bar coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$ to the symmetric bar coalgebra ${}^c(s^{-1}\bar{\mathcal{A}})$. The map is *lossy*: the R-matrix, KZ associator, and higher Yangian coherences are E_1 data that av projects away. In particular:

- At degree 2: $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$. The modular characteristic is the scalar projection of the classical r-matrix.
- At degree 3: the cubic shadow $\mathfrak{C}(\mathcal{A}) = \mathrm{av}(r_3)$, where $r_3(z_1, z_2)$ is the genus-0 KZ associator.
- At degree r: the shadow invariant $S_r(\mathcal{A}) = \mathrm{av}(\Theta_{0,r}^{E_1})$.

The five main Theorems A–D+H are statements about the av-image. They survive the projection because they depend only on the symmetric data (the Feynman transform of the commutative modular operad).

The thesis. The E_1 MC element $\Theta_{\mathcal{A}}^{E_1}$ is the primary algebraic invariant. The modular MC element $\Theta_{\mathcal{A}} = \mathrm{av}(\Theta_{\mathcal{A}}^{E_1})$ is its shadow. Volume II's ordered-associative core (ordered associative chiral Koszul duality, dg-shifted factorization bridge, gravitational Yangian) develops the E_1 layer; Parts **??**–**??** describe the modular/symmetric framework that is the av-image. Line operators, spectral R-matrices, and Yangian structures are natively E_1 and should be read as primary, not auxiliary.

1.17 Structural upgrades

Shadow archetype classification and single-line dichotomy. Theorem **??** classifies the standard-landscape shadow depth classes ($\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M} = \mathbf{F}_2/\mathbf{F}_3/\mathbf{F}_4/\mathbf{M}$) with complete proofs for all primitive archetypes. Shadow depth is orthogonal to Koszulness: all computed archetypes are chirally Koszul, yet their shadow depths differ. The single-line dichotomy (Theorem **??**) proves the gap between $\mathbf{F}_4$ and $\mathbf{M}$ is structural, not empirical: on any one-dimensional primary slice of $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$, every shadow coefficient S_r ($r \geq 4$) factors through the critical discriminant $\Delta = 8\kappa S_4$, and the quotients $R_r = S_r/\Delta$ are generically nonvanishing, so $r_{\max} \in \{2, 3, \infty\}$ on any single line. The contact class $\mathbf{C}$ escapes to $r_{\max} = 4$ via stratum separation (Remark **??**): the quartic contact invariant lives on a charged stratum of $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}$ whose self-bracket exits the complex by rank-one abelian rigidity. The VOA dichotomy closes the gap: either the sectors decouple (vanishing mixed OPE, so $\mathrm{Def}_{\mathrm{cyc}}(L_1 \oplus L_2) = \mathrm{Def}_{\mathrm{cyc}}(L_1) \oplus \mathrm{Def}_{\mathrm{cyc}}(L_2)$ by Proposition **??**, giving $d_{\mathrm{alg}} \leq 2$), or they couple ($\alpha \neq 0$ on the mixed sector, activating the cubic pump and forcing $d_{\mathrm{alg}} = \infty$). Hence $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ (Remark **??**). Depths $d \geq 5$ in the decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ (Theorem **??**) are purely arithmetic.

Non-simply-laced Kac–Moody shadow verification. The shadow obstruction tower computation extends to all non-simply-laced Kac–Moody algebras: $B_2 = \mathfrak{so}(5)$, $C_2 = \mathfrak{sp}(4)$, G_2, $B_3 = \mathfrak{so}(7)$, and F_4. All are class $\mathbf{L}$ (shadow depth 3): $S_3 = 1$, $S_4 = 0$, $\Delta = 0$, independent of type and rank. The distinction $h \neq h^\vee$ for non-simply-laced types (e.g. B_2: $h = 4, h^\vee = 3$; G_2: $h = 6, h^\vee = 4$) affects the bar-cohomology periodicity (period $= 2h$, *not* $2h^\vee$) but not the shadow depth class. Bar $H^1 = \dim(\mathfrak{g})$ in all cases (10 for B_2, 10 for C_2, 14 for G_2). Langlands duality $B_2 \leftrightarrow C_2$ preserves κ at the same level, confirmed computationally.

Minimal model shadow obstruction towers. Since minimal models $M(p, q)$ are simple quotients of the Virasoro algebra, their shadow obstruction tower at the universal level coincides with the Virasoro tower at $c = c(M(p, q))$: class $\mathbf{M}$ with the standard shadow metric. At the quotient level, the curved-DG structure introduces curvature at the degree of the first null vector (level pq for $M(p, q)$). The modular S-matrix and Verlinde fusion rules are verified computationally for $M(4, 3)$ (Ising, $c = 1/2$), $M(5, 4)$ (tricritical Ising, $c = 7/10$), and $M(6, 5)$ (three-state Potts, $c = 4/5$). The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ (Definition **??**) packages the dichotomy geometrically: Q_L is a perfect square iff $\Delta = 0$, and the spectral curve $\Sigma_L = \{H^2 = t^4 Q_L\}$ (Corollary **??**) is rational.

Ran-coherent bar-cobar equivalence. Theorem **??** resolves the Ran-space formality gap identified in earlier audits: the bar-cobar quasi-isomorphism extends coherently from $\mathrm{Ran}^{\leq n}(X)$ to all of $\mathrm{Ran}(X)$ via the strong completion-tower theorem and the degree cutoff lemma.

Ambient complementarity routing. Theorem **??** is no longer recorded as an unconditional upgrade from additive splitting. The proved Verdier-side package gives the Lagrangian geometry on $C_g = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$; the ambient (-1)-shifted symplectic formal-moduli upgrade is the conditional C2 lane, routed through Theorem **??** and therefore scoped to the uniform-weight BV package.

Chiral Deligne–Tamarkin principle and the dimensional jump. Remark **??** makes explicit the mechanism by which every local A_∞-chiral algebra on a curve produces a local 3-dimensional holomorphic-topological bulk/boundary system: the chiral center theorem (Theorem **??**) provides the universal closed-sector algebra $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ acting on $\mathcal{A}$ through , and the recognition theorem (Volume II, items F4–F5)

identifies the resulting pair with a local HT prefactorization algebra on $D\times$. The extra dimension is the universal normal-collar direction of a codimension-one boundary condition, forced by the center construction. The physical realization is the Khan–Zeng PVA-to-3d-HT construction [264] (Proposition **??**): gauge invariance $\Leftrightarrow$ λ-Jacobi identity; inner Virasoro element $\Rightarrow$ full topological symmetry.

Modular trace principle. Principle **??** (Volume II) organizes the modular structure as trace-plus-clutching on the open factorization category . Three of its four ingredients are proved: (i) cyclic structure (Calabi–Yau on (J)), (ii) annulus identification $\int_{S^1_p} \simeq_\bullet (A_b)$ (Theorem **??**), and (iii) the closed-sector modular MC equation for $\Theta_{\mathcal{A}}$ (Volume I, Theorem **??**). The fourth ingredient (the full open-sector modular twisting morphism extending to the bordered log-modular cooperad) is conjectural and constitutes the global factorization descent problem.

Quartic logarithmic contact class. Definition **??** (Volume II) defines $\mathfrak{q}_4(\mathcal{A}) \in H^2({}_4{}^{\mathrm{mod},\leq 4})$, the first logarithmic modular obstruction from codimension-2 log-Fulton–MacPherson strata at degree 4. Proposition **??** proves $\mathfrak{q}_4 = 0$ in the boundary-linear LG sector (by strict commutativity and associativity); for Virasoro and $\mathcal{W}_3$, $\mathfrak{q}_4 \neq 0$ (Volume I shadow obstruction tower computations), distinguishing shadow depth class **M** from **G**/**L**/**C**.

Shadow separation (Theorem ??). $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$ in $\widehat{\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}}$; gauge-equivalent MC classes have identical shadow obstruction towers. Separating witnesses: $\kappa({}_1) = \kappa(\mathrm{Vir}_2)$ but $\mathfrak{C} = 0 \neq \mathfrak{C}_{\mathrm{Vir}}$ (level 3); $(\kappa, \mathfrak{C})_0 = (\kappa, \mathfrak{C})_{\beta\gamma}$ but $\mathfrak{Q}_0 = 0 \neq \mathfrak{Q}_{\beta\gamma}$ (level 4). Shadow class detected by the vanishing pattern $(\mathrm{Sh}_3, \mathrm{Sh}_4, \mathrm{Sh}_5)$. $r_{\max}(\mathcal{A}^!) = r_{\max}(\mathcal{A})$ (Corollary **??**).

Shadow tautological classes (Theorem ??). $\tau_{g,n}(\mathcal{A}) := \sum_\Gamma |\mathrm{Aut}(\Gamma)|^{-1}\, \iota_{\Gamma,*}(\prod_e P_e)\, w_\Gamma^{\mathrm{cyc}}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$. Tautological by construction. $\tau_{g,2} = \kappa \cdot \pi^*\lambda_g$. Clutching-compatible via MC factorization. Additive under independent tensor products. For affine $V_k()$: $\tau_{0,3} = (k+h^\vee)^{-1} \sum_\sigma f^{a_{\sigma(1)}a_{\sigma(2)}c}\kappa^{cd} f^{da_{\sigma(3)}a_{\sigma(4)}}$. $[\mathrm{pt}_\sigma]$ (Example **??**).

Chriss–Ginzburg tautological programme. The shadow tautological classes $\tau_{g,n}(\mathcal{A})$ (Definition **??**, Theorem **??**) carry CohFT structure on $R^*(\overline{\mathcal{M}}_{g,n+1})$ (Theorem **??**); the CohFT axioms (equivariance, splitting) are formal consequences of $D^2_{\mathcal{A}} = 0$ on the modular bar complex. The flat identity holds conditionally when the vacuum lies in V (Theorem **??**). The MC factorization $\xi^*\Theta_{\mathcal{A}} = \Theta_{\mathcal{A}} \star \Theta_{\mathcal{A}}$ descends to tautological relations via Theorem **??**. At genus 0: WDVV equations = Jacobi identity of the MC bracket (Proposition **??**). At degree 2: Mumford λ-class relations from the MC equation (Proposition **??**). Reconstruction: the Givental R-matrix equals the complementarity propagator $P_{\mathcal{A}}$ (Theorem **??**). The dressed propagator coefficient $P^R(D_+, D_-)$ from the edge integral $\int R(\psi_+)\, R(\psi_-)/(\psi_+ + \psi_-)$ satisfies $P^R(D_+, D_-) = P^R(D_-, D_+)$ (algebraic proof from the symplectic condition $R(-z)R(z) = 1$; Proposition **??**). For uniform-weight algebras, the shadow obstruction tower free energy $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is the Hodge-lambda integral $\int_{\overline{\mathcal{M}}_{g,1}} \lambda_g \psi^{2g-2}$ (Theorem D). The universal genus-1 specialization is the identity $F_1 = \kappa/24$; the ψ-class CohFT graph sum at $(g, 0)$ computes a structurally different quantity, as documented in Remark **??** (44 compute tests). Eynard–Orantin topological recursion is the scalar shadow of the MC tautological relation (Corollary **??**). The Vol II holographic tautological action admits a conjectural shadow-tower construction from Vol I data: the scalar trace of the universal correlator class $\mathcal{U}^{;B}_{g,n}$ recovers the Vol I shadow tautological class $\tau_{g,n}(\mathcal{A}_\partial())$.

Non-perturbative shadow partition function. The shadow partition function $Z^{\mathrm{sh}}(\mathcal{A},\hbar) = \sum_{g,r}\hbar^{2g}Z_g^{(r)}(\mathcal{A})$ converges absolutely for $|\hbar| < 2\pi$ and $\rho(\mathcal{A}) < 1$ (Theorem **??**). The bound factorizes: genus decay $1/(2\pi)^{2g}$ (Bernoulli, Proposition **??**) times degree decay $\rho^r r^{-5/2}$ (shadow radius, Theorem **??**). This contrasts with the *divergent* string genus expansion ($\sim (2g)!$, Mirzakhani): the shadow CohFT extracts tautological intersections with Bernoulli decay, not Weil–Petersson volumes. Virasoro at $c = 26$: $\rho \approx 0.234$, deep in the convergent regime (41 compute tests).

Remark 1.17.1 *(Pixton relations from the MC element).* The Pixton relations [235, 339] are a complete set of tautological relations in $R^*(\overline{\mathcal{M}}_{g,n})$ for $g \geq 2$. The MC tautological descent (Theorem **??**) produces relations at each (g,n); the question is whether these generate the Pixton ideal.

At genus 2: the MC relation at $(2,0)$ from the three-shell decomposition (Construction **??**) gives

$$\Theta^{(2)}_{\mathrm{loop}^2} + \Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} + \Theta^{(2)}_{\mathrm{pf}} = 0,$$

which is the genus-2 projection of the full MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}},\Theta_{\mathcal{A}}] = 0$, and projects to a relation in $R^*(\overline{\mathcal{M}}_2)$. For class-G algebras (Heisenberg), only the iterated-loop shell survives, and the projected relation is the genus-2 Mumford relation $\lambda_2^2 = 0$. For class-M algebras (Virasoro), all three shells contribute, and the planted-forest shell $\Theta^{(2)}_{\mathrm{pf}}$ produces the first non-Mumford tautological relation; this should be compared with the Faber–Zagier relation in $R^2(\overline{\mathcal{M}}_2)$.

THEOREM 1.17.2 *(Pixton ideal from shadow obstruction tower;* [proved here]*).* For any chirally Koszul algebra $\mathcal{A}$ whose genus-0 shadow data defines a *semisimple* Frobenius algebra on V, the MC-descended tautological relations $\{\mathrm{taut}_{g,n}([\Theta_{\mathcal{A}}]_{g,n}) : g \geq 0,\ n \geq 0\}$ together with the Mumford relations generate the Pixton ideal in $R^*(\overline{\mathcal{M}}_{g,n})$.

This covers the entire standard landscape at generic levels: all principal $\mathcal{W}_N$, subregular $\mathcal{W}$-algebras, hook-type in type A, and any $\mathcal{W}_k(\,,f)$ whose Slodowy slice gives a simple or semisimple singularity. The proof chain (Theorem **??**) proceeds via Givental–Teleman reconstruction and Pandharipande–Pixton–Zvonkine [339].

Non-semisimple frontier. For non-semisimple shadow CohFTs (logarithmic vertex algebras, admissible-level simple quotients), the MC-descended relations still lie in the Pixton ideal (from $D^2 = 0$), but generation of the full ideal is obstructed by the failure of Givental–Teleman (Proposition **??**).

Remark 1.17.3 *(Proof of Theorem 1.17.2). Rank-*1 *primary line.* For rank-1 semisimple ($S_3 \neq 0$): Givental–Teleman gives $R = 1$, so the shadow CohFT equals the Witten–Kontsevich CohFT. The MC relations are the WK relations, which generate the Pixton ideal by [339]. For rank-1 non-semisimple ($S_3 = 0$, Heisenberg): $\delta_{\mathrm{pf}} = 0$ by direct computation. The conjecture is satisfied trivially at rank 1.

Principal $\mathcal{W}$-algebras. For $\mathcal{W}_N = \mathcal{W}_k(\mathfrak{sl}_N, f_{\mathrm{princ}})$: the shadow CohFT on $V = \{T, W_3, \ldots, W_N\}$ is the A_{N-1} Frobenius manifold CohFT (Feigin–Frenkel identification of the $\mathcal{W}$-algebra associated variety with the A_{N-1} singularity). For semisimple levels: Givental–Teleman identifies this with the $(N+1)$-spin CohFT. The proof chain

$$\text{MC equations} \xrightarrow{\text{Witten}} \text{KdV hierarchy} \xrightarrow{\text{FSZ}} r\text{-spin relations} \xrightarrow{\text{PPZ}} \text{Pixton ideal}$$

shows: the MC-descended relations from $\mathcal{W}_N$ generate the Pixton ideal. This proves Theorem 1.17.2 for all principal $\mathcal{W}$-algebras at semisimple levels.

For $\mathcal{W}_3$: the rank-2 planted-forest is $(932 - 7c^2)/(48c^3)$ (eq. (1.2)), the A_2 R-matrix has off-diagonal $R_1 = -3\sqrt{6}/(8s^2)$ (solved from the quantum differential equation), and the genus-2 free energy $F_2 = s^2/864 + 7/(51840s^4)$ receives an 11.7% R-correction over WK.

Semisimple case: upgraded to theorem. The r-spin chain is not needed. The argument generalises: if the genus-0 shadow data defines a *semisimple* Frobenius algebra on V, then (a) Givental–Teleman [207, 386] reconstructs the unique semisimple CohFT extending this Frobenius structure, and (b) Pandharipande–Pixton–Zvonkine [339] prove that any semisimple CohFT generates the Pixton ideal. Since the MC equation at each (g, n) produces a tautological relation of this CohFT (Theorem **??**), and the CohFT is uniquely determined by its genus-0 data, the MC relations at all (g, n) encode exactly the CohFT's tautological content.

This proves Theorem 1.17.2 for *all chirally Koszul algebras whose genus-*0 *shadow data defines a semisimple Frobenius algebra* (Theorem **??**(vi)). This includes: all principal $\mathcal{W}_N$ at non-critical levels, subregular $\mathcal{W}$-algebras (associated variety = D_{N-1} singularity), hook-type in type A, and more generally any $\mathcal{W}_k(\,, f)$ whose Slodowy slice gives a simple or semisimple singularity at generic k.

Non-semisimple frontier. The conjecture remains genuinely open for:

- Logarithmic/non-rational algebras (e.g., $\mathcal{W}(p)$ triplet algebras) where the Frobenius algebra is non-semisimple; Givental–Teleman does not apply and the CohFT lives in the coderived category.
- Admissible-level simple quotients $L_k()$ where vacuum null vectors collapse the Frobenius algebra to a non-semisimple quotient.
- Algebras outside the Koszul locus where the shadow CohFT (Theorem **??**) may not exist.

The non-semisimple case requires the Pandharipande–Pixton–Zvonkine framework to be extended to non-semisimple CohFTs, or an independent structural argument from the MC equation's planted-forest structure (Remark **??**).

Remark 1.17.4 *(Computational evidence: genus-*2 *planted-forest correction).* At genus 2 with $n = 0$, the planted-forest correction $\delta_{\mathrm{pf}}^{(2,0)}$ receives contributions from four stable graphs of codimension ≥ 2: the sunset C (codim 2, weight S_4), the bridge-loop E (codim 2, weight $S_3\kappa$), the theta F (codim 3, weight S_3^2), and the figure-8 bridge G (codim 3, weight S_3^2). Graph C has vanishing Hodge integral. The remaining three yield

$$\delta_{\mathrm{pf}}^{(2,0)} = \frac{S_3(10S_3 - \kappa)}{48},$$

a factored polynomial in the modular characteristic $\kappa = S_2$ and cubic shadow S_3. For class-G algebras ($S_3 = 0$): $\delta_{\mathrm{pf}} = 0$, recovering pure Mumford. For the Virasoro algebra ($\kappa = c/2$, $S_3 = 2$): $\delta_{\mathrm{pf}} = -(c-40)/48$, vanishing at $c = 40$ and changing sign there. Verified numerically at $c = 1, 2, 10, 13, 25, 26$ (`pixton_shadow_bridge.py`, 82 tests).

At genus 3, the complete planted-forest correction (from 35 planted-forest stable graphs among the 42 total stable graphs of $_{3,0}$: 1 one-vertex, 9 two-vertex, 14 three-vertex, 11 four-vertex) is an 11-term polynomial in κ, S_3, S_4, S_5 (eq. (**??**)). The quartic S_4 and quintic S_5 enter for the first time; S_6 remains invisible (the pure-self-loop $(0, 6)$ graph has vanishing Hodge integral by Proposition **??**).

The self-loop parity vanishing theorem (Proposition **??**) shows that pure-self-loop graphs $(0, 2k)$ vanish for all $k \geq 2$, forcing the shadow visibility genus $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$ (Corollary **??**): two new shadow coefficients enter at each genus. At genus g, the planted-forest involves S_r for $3 \leq r \leq 2g-1$.

Class-level visibility. The integrated visibility genus $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$ underestimates the class-level visibility. The cyclic marking converts a $(0, 2k)$ vertex to $(0, 2k+1)$ with even dimension, defeating the parity obstruction of Proposition **??**. In particular, the descendant pairing $\int_{\overline{\mathcal{M}}_{2,1}} \delta_{\mathrm{pf}}\, \psi_1^2 = S_3\kappa/48 + S_4/8$ reveals S_4 at genus 2 at the class level (Remark **??**).

Pixton universality test. The boundary correlator $\langle \tau_4 \rangle_2^{\mathrm{boundary}} = 1/1152$ is shadow-independent, equalling the Witten–Kontsevich intersection number. This verifies the strongest computationally testable Pixton constraint: the top-degree boundary contribution is universal (82 tests).

Rank-1 triviality and rank-2 frontier. For rank-1 on the primary line, Givental–Teleman gives $R = 1$, so the shadow CohFT equals the Witten–Kontsevich CohFT and Theorem 1.17.2 is automatically satisfied (the MC relations are WK relations). The non-trivial content of the conjecture requires *higher rank*.

For $\mathcal{W}_3$ (rank 2, generators T, W with $\mathbb{Z}_2$ symmetry $W \to -W$), the Frobenius algebra on $V = \{T, W\}$ has $T*T = (4/c)\,T, T*W = (6/c)\,W, W*W = (4/c)\,T$, which is semisimple with eigenvalues $4/c \neq 6/c$. The Givental R-matrix is $R \neq I$ (it is the A_2 singularity R-matrix). The rank-2 genus-2 planted-forest is:

$$\delta_{\mathrm{pf}}^{(2,0)}(\mathcal{W}_3) \;=\; \frac{932 - 7c^2}{48\,c^3}, \tag{1.2}$$

using $S_3(T,T,T) = 2, S_3(T,W,W) = 1$ (Theorem **??**), $\eta = \mathrm{diag}(c/2, c/3)$, $\kappa_W = 0$ ($\mathbb{Z}_2$ symmetry). This differs from the rank-1 formula $(40-c)/48$ by a W-channel correction from the spin-3 field propagating through internal graph edges. The 3-spin correlator $\langle\tau_1\rangle_{2,1}^{3-\mathrm{spin}} = 1/1620 \neq 1/1152 = \langle\tau_4\rangle_{2,1}^{\mathrm{WK}}$ confirms the CohFT is genuinely non-WK.

Caveat. At genus 3, genus-0 vertex weights S_k are exact; genus-1+ vertex weights use $\ell_2^{(1)} = S_3\kappa/24 - S_3^2$ (exact from MC; Beilinson audit corrected the S_3^2 coefficient from $1/2$ to 1 after finding a missing bridge-plus-self-loop graph at $(1,2)$) and $\ell_3^{(1)} \approx \kappa$ (approximate). Full Pixton ideal generation at rank 2 requires the strata algebra of $\overline{\mathcal{M}}_{2,1}$ at rank 2.

Chiral Hochschild descent lemma. Lemma **??** records an explicit step in the proof of Theorem H: the chiral Hochschild complex $CH^\bullet(\mathcal{A})$ arises as the Σ-coinvariant descent of the bar complex along Verdier intertwining, $CH^\bullet(\mathcal{A}) \cong (\mathbb{D}_{\mathrm{Ran}}\overline{B}(\mathcal{A}))_\Sigma$, on the Koszul locus. This identification was implicit in the original Theorem H proof; the lemma makes it citable as a primitive step (A2).

Closure properties of chiral Koszulness. Proposition **??** consolidates three previously scattered facts: the class of chirally Koszul algebras is closed under chiral tensor product ($(\mathcal{A} \boxtimes \mathcal{B})^! \simeq \mathcal{A}^! \boxtimes \mathcal{B}^!$), Koszul dualization (involutivity: $(\mathcal{A}^!)^! \simeq \mathcal{A}$), and base change along finite étale covers of curves. Quotients (e.g. Drinfeld–Sokolov reduction) are *not* in the closure and require Theorem **??** (A6).

Kac–Moody -module purity equivalence. Proposition **??** proves the full equivalence between chiral Koszulness and -module purity for affine Kac–Moody algebras $V_k()$ at generic level: PBW-filtration strictness (ii) $\Longleftrightarrow$ Saito–Kashiwara weight-filtration purity (xii) on the geometric bar complex. The forward direction is Proposition **??**; the new content is the converse via the Hitchin variation of Hodge structure.

Three-lens identification of the shadow tower. Proposition **??** identifies three equivalent readings of the degree-r shadow $S_r(\mathcal{A})$: algebraic (bar-intrinsic MC projection), geometric (genus-0 transferred L_∞ formality obstruction), and holographic (boundary-bulk correction in the derived-centre factorization).

Generating depth versus algebraic depth. Definition **??** formalizes the distinction mandated: the generating depth $d_{\mathrm{gen}}(\mathcal{A})$ is the smallest degree at which higher operations are recursively determined, while the algebraic depth $d_{\mathrm{alg}}(\mathcal{A})$ is the smallest degree beyond which all operations vanish. Virasoro: $d_{\mathrm{gen}} = 3$ (finite), $d_{\mathrm{alg}} = \infty$ (class M). Bare $d(\mathcal{A})$ without subscript is forbidden.

MC3 after the type-A reduction. Corollary **??** still records the two unconditional all-types layers: categorical prefundamental Clebsch–Gordan and the DK-2/3 comparison on the evaluation-generated core. The new load-bearing theorem is Theorem **??**: in type A the old four-packet post-core problem is reduced to the single compact- completion packet Conjecture **??**. The proved post-core surface is the Baxter-locus seed $\{V_{\omega_i}(a)\} \cup \{L^-_{i,a-\frac{1}{2}}\}$, with the spectral constraint supplied by Proposition **??**; shifted-prefundamental generation and pro-Weyl recovery are proved on that surface. The remaining completion step is therefore a Baxter- constrained compact-completion problem, not an independent rank-induction packet.

Holographic central charge identification (conjectural). Conjecture **??** ([conjectured]) identifies the modular characteristic $\kappa(\mathcal{A})$ with the one-loop bulk central charge $c_{\mathrm{bulk}}(\mathcal{A})$ of the 3d holomorphic-topological theory whose boundary is $\mathcal{A}$, in the Costello–Dimofte–Gaiotto sense: $\kappa(\mathcal{A}) = c_{\mathrm{bulk}}(\mathcal{A})/2$. The identification is heuristic because it assumes the boundary-bulk dictionary for the derived chiral centre $\mathrm{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ at the one-loop level; it is unconditional at genus 0 and conditional on the uniform-weight hypothesis at $g \geq 2$.

Analytic shadow realization (Corollary ??). For all $\mathcal{A}$ in the standard landscape, $\overline{B}(\mathcal{A}) \hookrightarrow \overline{B}^{\mathrm{an}}(\mathcal{A})$ densely; finite-degree shadows are algebraic; hence the algebraic and analytic shadow data coincide. Follows from HS-sewing (Theorem **??**).

1.18 The Polyakov programme

The shadow obstruction tower completes Polyakov's functional-integral programme (1987): every object Polyakov constructs by path-integral methods has a precise algebraic counterpart in the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$.

Dictionary.

Polyakov (1981–87)	**Bar-cobar**	**Status**
Polyakov formula $\log(\det'\Delta_{g_1}/\det'\Delta_{g_0})$	Degree-2 shadow $\kappa(\mathcal{A})$	Proved
Critical dimension $c = 26$	Dual modular characteristic $\kappa(\mathcal{A}^!) = 0$	Proved
Conformal anomaly $\langle T^a{}_a \rangle = (c/12)\, R$	Modular characteristic $\kappa(\mathcal{A})$	Proved
Functional determinant $\det'_\zeta \Delta$	Fredholm determinant $\det(1-K)$ on Bergman space	Proved
Genus expansion $Z = \sum_g g_s^{2g-2} Z_g$	MC5 sewing programme, scalar $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$	Proved
Ghost system bc ($c_{\mathrm{ghost}} = -26$)	Koszul dual $\mathcal{A}^!$	Proved
BPZ equations $\nabla_c^{\mathrm{BPZ}} \mathcal{F} = 0$	Shadow connection ∇^{sh} on the Virasoro degenerate-representation comparison surface (Proposition **??**)	Proved
Instantons (Novikov parameter q^β)	Novikov-completed bar complex $B^\Lambda(\mathcal{A})$ (Theorem **??**)	Proved (framework)
Wilson loop $\langle W(C) \rangle$	Swiss-cheese boundary module	Structural
Gauge/string duality	Holographic modular Koszul datum $\mathcal{H}(T)$ (Definition **??**)	Conjectural programme

Five main results.

(i) The Polyakov formula is the degree-2 Chern–Weil image of the shadow obstruction tower (Proposition **??**).

(ii) The critical dimension $c = 26$ is the zero of the genus-0 Virasoro equivariant anomaly coefficient, so the quotient bar complex becomes a genuine cochain complex (Theorem **??**).

(iii) On the critical genus-0 string locus, the relative BRST complex of the total matter+ghost theory is quasi-isomorphic to the bar complex (Theorem **??**), with the bc ghost sector furnishing the Koszul-dual bookkeeping dictionary (Remark **??**).

(iv) The genus expansion converges for chiral algebras on compact curves, with $|F_g| \sim (2\pi)^{-2g}$ (Remark **??**).

(v) Instanton sectors are organized by Novikov-completed bar complexes (Theorem **??**).

Three frontier items.

(a) *Matrix models as discretized shadow obstruction towers.* The random-surface genus expansion $Z = \sum_g N^{2-2g} Z_g$ should be a discretization of the shadow obstruction tower, with the double-scaling limit recovering the continuum shadow. (Conjectural)

(b) *Non-critical string theory as Liouville + complementarity.* The anomaly cancellation $c_{\mathrm{matter}} + c_{\mathrm{Liouville}} + c_{\mathrm{ghost}} = 0$ is the ambient complementarity theorem (Theorem **??**). (Structural)

(c) *Polyakov connection on Teichmüller space.* The shadow connection ∇^{sh} should define a flat connection on the Teichmüller space $\mathcal{T}_g$, with monodromy encoding the mapping-class group representation. (To be developed)

Bibliographic pointers. Polyakov 1981 (Polyakov action), Belavin–Polyakov–Zamolodchikov 1984 (conformal symmetry), Belavin–Knizhnik 1986 (factorization on $_g$), Alvarez 1983 (Polyakov–Alvarez formula).

Cross-references. Proposition **??** (§**??**); Remark **??** (§**??**); Remark **??** (Chapter **??**); Proposition **??** (§**??**).

1.19 Bridges to Volume II

Volume II develops the A_∞-chiral Hochschild theory in three space-time dimensions with the operad $\mathrm{SC}^{\mathrm{ch,top}}$. Homotopy–Koszulity of $\mathrm{SC}^{\mathrm{ch,top}}$ is proved in Volume II (via Kontsevich formality and transfer from the classical Swiss–cheese operad), so the bar–cobar adjunction for $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is a Quillen *equivalence*. Volume II has *no remaining conjectural algebraic inputs*; its only conditional inputs are the standing analytic hypotheses (H1)–(H4). Five bridges connect the two volumes:

Proposition 1.19.1 *(Bar-cobar bridge;* [proved elsewhere]*).* The $\mathrm{SC}^{\mathrm{ch,top}}$ bar-cobar Quillen equivalence of Volume II recovers the chiral bar-cobar adjunction of Theorem A (Theorem **??**) for boundary operators of the 3d theory. The closed color specializes to the BD chiral operad (Vol II, associated graded proposition).

Conjecture 1.19.2 *(Hochschild bridge;* [conjectured]*).* The bulk algebra of the 3d theory is the derived center $Z_{\mathrm{der}}(\mathcal{A}^!)$, which identifies with the chiral Hochschild cochain complex $\mathrm{HH}^*_{\mathrm{ch}}(\mathcal{A}^!)$ via the corrected Koszul triangle (Vol II, Theorem J): $A_{\mathrm{bulk}} \simeq Z_{\mathrm{der}}(B_\partial) \simeq \mathrm{HH}^*_{\mathrm{ch}}(\mathcal{A}^!)$, where $B_\partial := \mathcal{A}^!$ is the boundary algebra. This recovers Theorem H (Theorems **??**, **??**) via dimensional descent.

Proposition 1.19.3 *(DK-YBE bridge;* [proved elsewhere]*).* The spectral braiding $r(z) = \int_0^\infty e^{-pz}\{a_\lambda b\}\,dp$ (the Laplace transform of the λ-bracket) recovers the DK-0 chain-level monodromy comparison (Theorem **??**).

Proposition 1.19.4 *($\mathcal{W}$-algebra bridge at genus* 0*;* [proved elsewhere]*).* The Feynman m_k operations of the 3d theory, restricted to the $\mathcal{W}$-algebra family, match the bar differential computed in the $\mathcal{W}$-algebra chapters of Part **??** at genus 0. Extension to genus 1 is proved (Volume II, genus-one bridge theorem: $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_1$ via the Arnold defect on the torus); extension to $g \geq 2$ is proved algebraically (Theorem **??**: inductive genus determination via the bar-intrinsic construction), with the analytic comparison (Theorem **??**) established for all graph amplitudes on Σ_g by two-dimensional convergence (Proposition **??**: no UV renormalization in dimension 2). The remaining analytic input is HS-sewing (Definition **??**) for the mode-sum extension from $\mathcal{A}_{\mathrm{alg}}$ to $\mathcal{A}^{\mathrm{sew}}$.

Proposition 1.19.5 *(Relative holographic deformation bridge;* [proved here]*).* The forgetful morphism $p\colon \mathfrak{g}^{\mathrm{SC}}_T \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[\mathcal{A}]$ (Vol II, Swiss-cheese convolution to Vol I modular convolution) has fibre $\mathfrak{h}^\Theta_T = \mathrm{fib}(p)_{\alpha_T}$ whose MC elements are lifts of the proved boundary datum $\Theta_{\mathcal{A}}$ to full $3d$ data (Proposition **??**). The obstruction tower at color depth r lives in $H^2(\mathrm{gr}^F_r\,\mathfrak{h}^\Theta_T)$. At depth 4, the relative quartic obstruction $\mathfrak{R}^{\mathrm{rel}}_4(T)$ measures failure of the boundary modular datum to lift to a planar holographic realization.

PROPOSITION 1.19.6 *(Ribbon/'t Hooft bridge;* [proved here]*).* The E_1 modular Koszul theory of Vol I (Chapter **??**) and the Swiss-cheese operad of Vol II (the Volume II recognition theorem) combine as follows:

(i) bare modular graph sums give genus-organized expansions;

(ii) cyclic trace + matrix realization upgrade these to literal 't Hooft sums with $N^{\chi(\Sigma_\Gamma)}$ weighting (Theorem **??**);

(iii) the ribbonized modular Swiss-cheese theory $\mathrm{RSC}^{\mathrm{mod}}_{\mathrm{tr}}$ (Definition **??**) is the correct holographic open/closed 't Hooft language.

Conjecture 1.19.7 *(Loop–Connes bridge;* [conjectured]*).* Closed genus creation (the genus-loop operator Λ_P of Vol I) is the image of the open BV/Connes operator under the derived-center map: $\beta^*_{\mathrm{der}}(\Delta_{\mathrm{open}}) = \Lambda_P$ (Conjecture **??**). In words: loops in the bulk are traces on the boundary.

Conjecture 1.19.8 *(BV functor bridge;* [conjectured]*).* Analytic hypotheses (H1)–(H4) of Volume II define the physics-to-algebra functor that realizes the $\mathrm{BV} = \mathrm{bar}$ identification at higher genus (MC5, Conjecture 2.2.7).

Remark 1.19.9 *(PVA quantization bridge).* Volume II constructs the modular PVA quantization pipeline: a Poisson vertex algebra V (the classical shadow of a chiral algebra) determines a stable-graph modular deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(V)$, and the genus-1 obstruction is the variational modular class $\mathrm{Mod}(V) = [\Delta_{\mathrm{cyc}} P_\Pi]$. For the $\mathcal{W}_3$ PVA, $\mathrm{Mod}(V) = 0$ (Volume II, Theorem N). The lift freedom is one-dimensional, generated by the central-parameter direction: the unique deformation that produces the quantized $\mathcal{W}_3$ at central charge $c = c(k)$. This connects Volume II's PVA descent (Theorem G) with Volume I's shadow obstruction tower (the degree-2 projection $\Theta^{\leq 2}_{\mathcal{A}}$ recovers the modular characteristic $\kappa(\mathcal{A})$, which is the quantized value of the classical variational class).

1.20 Towards modular homotopy theory for factorization algebras on curves

The modular characteristic theorem (Theorem D, Theorem **??**) establishes the scalar logarithmic datum $\kappa(\mathcal{A})$. This section identifies the six components (five structural extensions and the Hochschild cohomology theorem) needed to pass from the proved modular Koszul core to modular homotopy theory for factorization algebras on curves; that is, to complete the nilpotence-periodicity correspondence (Remark **??**) from the scalar level to the full categorical level:

Label	Statement	Status	Reference
A_{mod}	Bar-cobar intertwined with Verdier,	**Proved**	Thm **??**
	at the algebra level only after $\mathbb{D}_{\mathrm{Ran}}$;		
	functorial over $\overline{\mathcal{M}}_{g,n}$		
B_{mod}	Strict inversion on the Koszul locus;	**Proved** on the Koszul locus;	Thms **??**,
	off-locus continuation in D^{co} and	off-locus coderived continuation	**??**
	promotion back to ordinary q.i. on collapse loci	**proved**	
C_{mod}	C0 fiber-center identification; C1 duality /	C0/C1: **Proved**;	Thms **??**,
	Lagrangian complementarity; C2 shifted-symplectic upgrade	C2: **Conditional**	**??**,
	on the uniform-weight BV lane		**??**
Index	Scalar genus series on the uniform-weight lane;	**Proved** on the stated lane;	Thms **??**,
	genus-1 unconditional; family-index routed downstream	genus-1 universal	1.20.16
DK	DK-0/1/$1\frac{1}{2}$: chain-level, eval-locus, lattice;	DK-0/1/$1\frac{1}{2}$: proved (all types);	Thms **??**,
	DK-2/3: generated-core DK comparison;	DK-2/3: **proved** (eval-gen. core, all types; $\mathcal{O}_{\mathrm{poly}}$ only on the separate type-A thick-generation lane;	**??**, **??**,
	DK-4/5: dg-shifted/triple bridge	uses Molev PBW [326]); DK-4: ML proved, alg. id. open; DK-5: **proved for** $\mathfrak{sl}_2$ (FRT, Prop. **??**); general: conj.	**??**, **??**, **??**; Conj. **??**
H	Generic-level Theorem H package: concentration in	**Proved** on the Koszul locus	Thms **??**,
	$\{0, 1, 2\}$, duality, polynomial Hilbert growth,	at generic level; critical level	**??**;
	and affine-KM $CH^1 \cong$	separate	Prop. **??**

Remark 1.20.1 *(Four levels of modular invariant).* The six components above implement a hierarchy of modular invariants of increasing categorical depth. The proved scalar level is the lowest; the full modular homotopy type is the highest:

Level	Invariant	Type
0	$\kappa(\mathcal{A}) \cdot \hat{A}$-genus	Numbers F_g
1	$H^*(\overline{B}^{(g)}(\mathcal{A}), D_g)$	Graded vector spaces
2	Variation of H^* over $\mathcal{M}_g$	Flat modular connection (bar-side comparison surface for KZ/Hitchin on the integrable a
3	Factorization and sewing	Chain-level modular functor (Thm **??**)

Theorem D is the *integrability condition*: $\mathrm{obs}_g = \kappa \cdot \lambda_g$ is, for uniform-weight algebras, a pure tautological class, free of boundary or non-tautological contributions, guaranteeing that the curvature of the genus-g bar complex is absorbable by a period correction. For arbitrary modular Koszul algebras, the unconditional part of Theorem D is the genus-1 identity $\mathrm{obs}_1 = \kappa\lambda_1$; the scalar formula *fails* at $g \geq 2$ for multi-weight algebras (Theorem **??**). The corrected complex $(\overline{B}^{(g)}, D_g)$ with $D_g^2 = 0$ (Level 1) is the *derived fiber* of the factorization homology bundle $\int_{\Sigma_g} \mathcal{A}$ over $\mathcal{M}_g$; its variation (Level 2) recovers the monodromy representation of the mapping class group; its factorization structure (Level 3) recovers the full chain-level modular-functor package (Theorem **??**). The $\hat{A}$-genus is the *first Chern class* of this bundle, its Euler characteristic, not the bundle itself.

Smoking gun: for $\widehat{\mathfrak{sl}}_2$ at level k, $\kappa = 3(k+2)/4$ is one number, yet the Verlinde formula gives a bundle of rank $k+1$ at genus 1 (equation (**??**)), growing with g. Bar cohomology $H^0(\overline{B}^{(1)}(\widehat{\mathfrak{sl}}_{2,k}), D_1)$ should recover the Verlinde dimension: a concrete, computable test that Level 1 carries representation-theoretic data invisible to Level 0. See §**??**.

Resolution of genus-independence at integrable levels. Theorem **??** proves that bar cohomology dimensions are polynomial in $\lambda = k + h^\vee$, constant outside a finite exceptional set Σ_n. For $\mathfrak{sl}_2$ at degree 2: $\Sigma_2 = \{0\}$ (critical level only). Since integrable levels $k \geq 1$ give $\lambda \geq 3 > 0$, they are *generic*, not exceptional, and the fiberwise bar cohomology is genus-independent there (Proposition **??**). The integer conformal block dimension $\dim V_{g,k}$ relates to the normalized partition function Z_g via $\dim V = Z_g \cdot S_{00}^{2-2g}$ (Remark **??**).

Remark 1.20.2 *(The κ polysemy).* The symbol κ appears in at least four distinct roles across the programme. Their coincidence in rank-1 families masks genuine divergence in general:

(i) $\kappa(\mathcal{A})$: the *modular characteristic* (Theorem D), defined by $F_1 = \kappa \cdot \lambda_1^{\mathrm{FP}}$. Intrinsic to the chiral algebra $\mathcal{A}$.

(ii) $\kappa_{\mathrm{BCOV}} := \chi_{\mathrm{top}}(X)/24$: the BCOV anomaly coefficient for a compact CY_3 X. Equals $\kappa(\mathcal{A}_X)$ for rigid CICYs with $h^{1,0} = 0$ (e.g. the quintic: $\kappa = -25/3$). Fails for K3-fibered CY_3: $\kappa_{\mathrm{BCOV}}(K3 \times E) = 0 \neq 5 = \kappa_{\mathrm{BKM}}$.

(iii) $\kappa_{\mathrm{MacMahon}}$: the exponent in the DT partition function $M(q)^\kappa$ for non-compact toric $\mathrm{CY}_3 = \mathrm{Tot}(K_S)$. Equals $\chi_{\mathrm{top}}(S)/2$. For local ^{2}: $3/2$.

(iv) κ_{BKM}: the weight of the BKM automorphic form controlling the bar Euler product. For $K3 \times E$: $\mathrm{wt}(\Delta_5) = 5$.

These coincide for Heisenberg ($\kappa = k$), Virasoro ($\kappa = c/2$), and affine KM at generic level ($\kappa = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$). They diverge for K3-fibered CY_3 (where (ii) and (iv) differ by a factor of $5/0 = \infty$) and for non-compact toric CY_3 (where (ii) and (iii) differ by a factor of 12). The modular characteristic $\kappa(\mathcal{A})$ in the sense of (i) is the intrinsic invariant; the others are geometric proxies valid in restricted contexts.

The full genus stratification of $K3 \times E$, including the Borcherds multiplicative lift from the genus-1 elliptic genus $\phi_{0,1}$ to the genus-2 Igusa cusp form Φ_{10}, the BKM superalgebra $\mathfrak{g}_{\Delta_5}$, and the Schottky barrier at genus 4, is developed in §**??**. The CY local-to-global gluing via McKay quiver charts and A_∞-bimodule transitions is in §**??**; the K3 chiral algebra (= 4 SCA at $c = 6$, $\kappa = 2$, bar complex, Mathieu moonshine, Gepner model, CDR) in §**??**; the Hodge diamond, HKR decomposition, Brauer group, and factorization homology of $K3 \times E$ in §**??**; the enumerative geometry (reduced GW, KKV, BPS integrality, Bridgeland stability) in §**??**; the BPS spectrum and black hole entropy (charge lattice, BMPV, DVV, OSV) in §**??**; the Costello–Li twisted holography ($c = 24$ boundary algebra, holographic Koszul datum) in §**??**; and the mock modular / BKM root system in §**??**. Forty-two CY/BKM compute engines with 5,700+ tests back the numerical claims (grand atlas: `cy_grand_atlas_k3e_engine.py`).

The subsections below develop each component, separating what is established from what remains. For A_{mod}: the present proof establishes functorial compatibility of bar and cobar over $\overline{\mathcal{M}}_{g,n}$; whether this lifts to an equivalence of modular operads is a further question, not addressed here.

The coderived Ran-space formalism

The bar-cobar adjunction is an equivalence of ∞-categories, the categorical exp/log correspondence (Remark **??**):

$$\bar{B}_X\colon\ \mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X)) \ \rightleftarrows\ \mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))\ :\Omega_X,$$

where $\mathrm{Fact}(X)$ denotes factorization algebras on $\mathrm{Ran}(X)$ (cf. Appendix **??**).

On the Koszul locus this is an equivalence: $\exp \circ \log = \mathrm{id}$ on the convergence domain. When curvature is present (that is, when the categorical logarithm acquires non-trivial monodromy), the theorem-level statement moves from ordinary derived categories to the intrinsic coderived/contraderived bar-coalgebra framework of Positselski [350]. The logarithm/analytic-continuation language is heuristic motivation only; the proved categorical content is the bar-coalgebra comparison, with ordinary derived reduction only on the flat finite-type completed-dual loci singled out in the bar-cobar foundations.

Curved bar objects can lose ordinary cohomological visibility while their coderived/contraderived invariants remain nontrivial. The coderived framework is built to retain that information without promoting the monodromy metaphor itself into a theorem.

The provisional coderived category $\mathsf{D}^{\mathrm{co}}_{\mathrm{prov}}(X)$ (Definition **??**, Proposition **??**) provides a working substitute within the scope of this monograph. On the flat finite-type locus singled out in the bar-cobar foundations, the coderived and derived categories agree. Bar-cobar inversion there is genus 0 unconditional and, at higher genus, unconditional on the CFT-type standard landscape except integer-spin $\beta\gamma$ by Theorem **??**; in general it remains conditional on axiom **??** (Theorem **??**).

Non-circular logical chain. The logical foundation for bar-cobar inversion at all genera is free of circularity (Remark **??**). The antecedent hypotheses **??**–**??** (Definition **??**) are verified *independently* of bar-cobar inversion: genus-0 Koszulity by the fundamental theorem of chiral twisting morphisms (Theorem **??**);

Verdier compatibility by Theorem **??**; PBW concentration at all genera by direct spectral-sequence arguments (Theorems **??**–**??**). Given these antecedents, Theorem **??** proves inversion by induction on genus via the stratum-by-stratum argument of Lemmas **??**–**??**. At no point does the proof invoke bar-cobar inversion at the genus under consideration.

Separately, the bar-intrinsic MC construction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ (Theorem **??**) provides Theorems D and H and the shadow obstruction tower unconditionally, as projections of a single element whose existence follows from $D_{\mathcal{A}}^2 = 0$ alone (Tier 1 of Remark **??**).

PBW propagation and axiom collapse. Theorem **??** proves that axiom **??** (PBW concentration at all genera) *follows from* axiom **??** (genus-0 Koszulity) for all CFT-type chiral algebras with positive conformal grading and a unique weight-2 stress tensor. This covers the entire standard landscape except integer-spin $\beta\gamma$, where the positive-grading hypothesis fails. The proof rests on two observations: (i) the collision differential is curve-independent (period corrections are regular at collision diagonals, contributing zero residue); (ii) after enrichment killing by L_0 invertibility (Theorem **??**), the core spectral sequence at genus g equals the genus-0 spectral sequence. The five main theorems therefore follow from genus-0 Koszulity and $D_{\mathcal{A}}^2 = 0$ alone (Remark **??**).

What is established. Appendix **??** compares the strict, completed, and literature existence paradigms for bar-cobar on $\mathrm{Ran}(X)$. Theorem **??** provides the adjunction; Theorem **??** provides inversion on the Koszul locus. Appendix **??**, §**??** develops the coderived and contraderived categories on $\mathrm{Ran}(X)$: curved factorization coalgebras with CDG-factorization compatibility (Definition **??**), coacyclic factorization objects (Definition **??**), stratified conservative restriction via the injective characterization (Theorem **??**), and the factorization co-contra correspondence (Theorem **??**). The provisional coderived category embeds fully faithfully (Proposition **??**).

Now complete. The formerly conjectural off-Koszul bar-cobar inversion on $\mathrm{Ran}(X)$ is proved (Theorem **??**): the counit $\Omega_X \overline{B}_X(\mathcal{A}) \to \mathcal{A}$ is an isomorphism in $D^{\mathrm{co}}(\mathrm{CoFact}(X))$ without Koszulness. The proof follows the same three-step assembly as Theorem **??**: (i) on each stratum Ran_n, Positselski [350, Theorem 7.2] gives the coderived equivalence for the restricted CDG-coalgebra; (ii) factorization compatibility of the bar-cobar counit; (iii) conservative assembly via Theorem **??**. This closes structural gap H1 and completes Future 2.

The Lagrangian form of complementarity

Theorem C (Theorem **??**) establishes the direct-sum decomposition $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(\mathcal{A}))$ via the eigenspace decomposition of the Verdier involution. The natural strengthening is a Lagrangian-polarization theorem, now proved in two layers:

Theorem 1.20.3 *(Lagrangian complementarity package; C1* [proved here]*, C2* [conditional]*).* For a modular Koszul pair $(\mathcal{A}, \mathcal{A}^!)$, the following hold:

(i) the complex $C_g := R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ carries the proved Verdier-side $(-(3g-3))$-shifted symplectic structure of Proposition **??**, and $Q_g(\mathcal{A})$ and $Q_g(\mathcal{A}^!)$ embed as complementary Lagrangians in the sense of Pantev–Toën–Vaquié–Vezzosi [340];

(ii) under the perfectness and nondegeneracy hypotheses of Theorem **??**(iii) and the uniform-weight BV input of Theorem **??**, the ambient complementarity formal moduli problem $\mathcal{M}_{\mathrm{comp}}(\mathcal{A})$ carries the conditional (-1)-shifted symplectic structure, and $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ define complementary Lagrangian maps (Theorem **??**).

Remark 1.20.4 *(Degree clarification).* The original formulation of this result stated a single (-1)-shifted symplectic structure on C_g. The proved version reveals two structures of different degrees: the Verdier pairing gives $(-(3g-3))$-shifted on C_g itself, while the ambient cyclic deformation problem gives (-1)-shifted on the formal moduli. The direct bar-side BV realization of the (-1)-shifted structure remains conditional on Theorem **??**. See Remark **??** for the compatibility of the proved and conditional realizations.

Completion kinematics and the modular cumulant transform

The completion-tower theorem (Theorem **??**) resolves the formal MC4 obstacle. What remains is a *kinematic* theory of the completed dual: the canonical primitive sector, the growth law of bar windows, and the entropy of completion. This subsection records the programme; the full definitions appear in §**??** (Chapter **??**) and the explicit calculations in §**??** (Chapter **??**).

Kinematics versus dynamics

The completed bar-cobar theory separates cleanly into two axes (Remark **??**):

- *Completion kinematics*: the primitive cumulant quotient $Q(\mathcal{A}) = \bar{\mathcal{A}}/\partial\bar{\mathcal{A}}$ (Definition **??**), the primitive generating series $G_{\mathcal{A}}(t) = \frac{1-t}{t}(\chi_{\mathcal{A}}(t) - 1)$ (Definition **??**), the completion Hilbert series $H_{\mathcal{A}}(t) = G_{\mathcal{A}}(t)/(1 - G_{\mathcal{A}}(t))$ (Definition **??**), the primitive defect $\Delta_{\mathcal{A}}(t) = G_{\mathcal{A}}(t) - G^{\mathrm{naive}}_{\mathcal{A}}(t)$ (Definition **??**), and the completion entropy $h_K(\mathcal{A}) = \log(\rho^{-1}_{\mathcal{A}})$ (Definition **??**). These depend only on the vacuum character, not on OPE coefficients or curvature.
- *Modular dynamics*: the coderivation $D_{\mathcal{A}}$, the curvature $\kappa(\mathcal{A})$, the universal MC class $\Theta_{\mathcal{A}}$ (Theorem **??**), and the dg-shifted Yangian kernel $r_{\mathcal{A}}(z)$. These depend on the full OPE structure.

The Virasoro involution $c \mapsto 26 - c$ changes the dynamic data but preserves the kinematic data: the same-family shadow Vir_{26-c} lives on the dynamic axis. Kinematically, even generic Virasoro has infinitely many primitive cumulants (the defect $\Delta_{\mathrm{Vir}}(t)$ begins $t^3 + 2t^5 + \cdots$).

The $\mathcal{W}_N$ entropy ladder

The completion entropy forms a monotone sequence (Proposition **??**):

$$h_K(\mathcal{W}_2) \approx 0.567 \;<\; h_K(\mathcal{W}_3) \approx 0.772 \;<\; h_K(\mathcal{W}_4) \approx 0.835 \;<\; \cdots \;<\; h_K(\mathcal{W}_\infty) \approx 0.872.$$

The $\mathcal{W}_\infty$ character admits a MacMahon factorization $\chi_{\mathcal{W}_\infty}(q) = \phi(q)\, M(q)$, where $M(q) = \prod_{n\geq 1}(1 - q^n)^{-n}$ is the plane-partition generating function. The $\mathcal{W}_\infty$ completion frontier is governed by plane-partition combinatorics.

The modular cumulant transform (conjectural target)

The *cumulant coalgebra* $\mathrm{Cum}_c(\mathcal{A}) := \hat{T}^c(sQ(\mathcal{A})^\vee)$ is the completed cofree coalgebra on the desuspended dual of the primitive cumulant quotient $Q(\mathcal{A}) = \bar{\mathcal{A}}/\partial\bar{\mathcal{A}}$ (Definition **??**).

Conjecture 1.20.5 *(Cumulant recognition;* [conjectured]*).* For admissible $\mathcal{A}$ satisfying reduced-weight finiteness and finite-window stabilization, $\mathrm{gr}_\rho\, \hat{B}(\mathcal{A}) \simeq \mathrm{Cum}_c(\mathcal{A})$ as completed curved dg coalgebras.

The platonic ideal of the completion programme is a single object

$$M(\mathcal{A}) = \big(\operatorname{Cum}_c(\mathcal{A}),\ D_{\mathcal{A}},\ \tau_{\mathcal{A}},\ r_{\mathcal{A}}(z),\ \Theta_{\mathcal{A}}\big)$$

whose five components are:

1. $\operatorname{Cum}_c(\mathcal{A})$: the completed cofree coalgebra on primitive cumulants (the kinematic core);
2. $D_{\mathcal{A}}$: the completed coderivation encoding genus-0 Koszul duality;
3. $\tau_{\mathcal{A}}$: the universal twisting morphism (the local dynamical field);
4. $r_{\mathcal{A}}(z) = \tau_{\mathcal{A}}|_{\deg 2}$: the ordered E_1/Yangian face, satisfying the A_∞ Yang–Baxter hierarchy;
5. $\Theta_{\mathcal{A}}$: the connected all-genera modular logarithm, the stable-graph exponential $\Theta_{\mathcal{A}} = \sum_{\Gamma \text{ connected}} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}(\Gamma)|}\Phi_\Gamma$, satisfying the modular quantum master equation $d\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] + \hbar\Delta\Theta_{\mathcal{A}} = 0$.

Status. Components (1)–(2) are controlled by the completion-tower theorem and reduced-weight windows. Component (3) is the bar-cobar twisting morphism, proved on (). Component (4) is the Yangian jet tower, accessible windowwise by the jet principle (Conjecture 1.20.6 below): reduced-weight-q bar windows determine Yangian jets through order z^{-q}. Component (5) is the principal open structural problem (MC2 is proved at all finite orders; the full modular QME is conjectural in the completed tensor product).

Conjecture 1.20.6 *(Jet principle;* [conjectured]*).* Reduced-weight-q bar windows $K_q(\mathcal{A})$ determine the dg-shifted Yangian MC kernel $r_{\mathcal{A}}(z) = \sum_{m\geq 0} r_m z^{-m-1}$ through jet order z^{-q}: that is, the truncation $r_{\mathcal{A}}(z) \bmod z^{-q-1}$ depends only on $\overline{B}(\mathcal{A})|_{K_q}$.

Six shadows. Every shadow the monograph computes is a projection of $M(\mathcal{A})$:

1. genus-0 face $\longrightarrow$ bar/cobar and Verdier duality;
2. ordered-boundary face $\longrightarrow$ RTT/Yangian/dg-shifted Yangian;
3. all-genera face $\longrightarrow$ modular homotopy theory;
4. scalar shadow $\longrightarrow \kappa(\mathcal{A})\lambda_g$;
5. spectral shadow $\longrightarrow \Delta_{\mathcal{A}}(x), \Pi_{\mathcal{A}}$;
6. Lagrangian shadow $\longrightarrow$ shifted-symplectic complementarity.

Conjecture 1.20.7 *(Grand completion;* [conjectured]*).* The modular cumulant transform $\mathcal{A} \mapsto M(\mathcal{A})$ extends finite-stage chiral Koszul duality to an equivalence between the completion closure of admissible chiral A_∞-algebras on curves and the homotopy theory of complete pronilpotent modular cumulant coalgebras, with genus-0 truncation = completed bar/cobar, ordered-boundary truncation = dg-shifted Yangian jet tower, and universal deformation theory = graph MC class $\Theta_{\mathcal{A}}$.

The analytic completion programme

The proved algebraic engine operates in the world of formal factorization algebras, cochain complexes, and dg categories. A genuine local-to-global field theory also requires analytic sewing (convergent partition functions, Hilbert-space completions, and positivity). Recent work of Moriwaki [323, 324] constructs conformally flat factorization homology valued in IndHilb by left Kan extension from conformally flat disk algebras; for the Heisenberg VOA, the conformally flat 2-disk algebra on $\mathrm{Sym}\, A^2(D)$ (the symmetric algebra of the Bergman space) is identified with the ind-Hilbert completion of the affine Heisenberg vertex algebra. Adamo–Moriwaki–Tanimoto [3] prove conformal Osterwalder–Schrader axioms for a class of unitary full VOAs under energy/spectral growth and local C_1-cofiniteness.

The programme aims to bridge the algebraic engine to these analytic structures. Its objects are the *sewing envelope* $\mathcal{A}^{\mathrm{sew}}$ (the Hausdorff completion for the locally convex topology generated by sewing-amplitude seminorms), the *analytic bar coalgebra* $\overline{B}^{\mathrm{an}}(\mathcal{A})$ (graph-norm closure of the algebraic bar), and the *analytic shadow* $Q_g^{\mathrm{an}}(\mathcal{A}) := D^{\mathrm{co}}(\overline{B}_g^{\mathrm{an}}(\mathcal{A})\text{-comod})$. At genus g with curvature $m_0^{(g)} \neq 0$, the coderived passage of §1.20 is mandatory: ordinary cohomology fails, and the correct receptacle is the coderived analytic category.

Four target theorems define the programme (Chapter **??**, §**??**):

(A_{an}) *Heisenberg sewing theorem* (Theorem **??**): the sewing envelope of the algebraic Heisenberg VOA is exactly $\mathrm{Sym}\, A^2(D)$; the completed bar differential is the closure of the Gaussian collision operator on Bergman tensors; every closed amplitude is a Fredholm determinant on the one-particle Bergman space (Theorem **??**: one-particle Bergman reduction).

(B_{an}) *Lattice sewing envelope* (Theorem **??**, **promoted from conjecture**): every positive-definite even lattice admits a canonical sewing envelope with charge-refined Hilbert sectors. Proof: HS-sewing (Corollary **??**), amplitude factorization as Heisenberg Fredholm determinant times Siegel theta function (classical convergence for $\mathrm{Im}\,\Omega > 0$), Bergman-space charge-sector completion.

(C_{an}) *Analytic realization criterion* (Conjecture **??**): a unitary full VOA satisfying conformal OS, polynomial spectral growth, and HS-sewing admits a sewing envelope, a conformally flat 2-disk algebra, and a higher-genus coderived shadow.

(D_{an}) *Boundary bar duality* (Conjecture **??**): for an analytically Koszul $\mathcal{A}$, completed boundary modules at genus 0 are equivalent to analytic comodules over $\overline{B}^{\mathrm{an}}(\mathcal{A})$; at curved genus, the correct objects are analytic contramodules over the curved dual.

Status. Target (A_{an}), the Heisenberg sewing theorem (Theorem **??**), is [proved here] for clauses (ii)–(iv) and [proved elsewhere] for clause (i) (Moriwaki [324]). Target (B_{an}), lattice sewing (Theorem **??**), is [proved here] (promoted from conjecture). Targets (C_{an}) and (D_{an}) remain [conjectured]. The programme is recorded here to fix definitions and their relationship to the algebraic proved core. On the derived-algebraic side, Nafcha proves a gluing formula for chiral homology on families of stable curves [arXiv:2603.30037], complementing the analytic HS-sewing criterion: Nafcha's formula operates on derived categories of D-modules on the Ran space; HS-sewing operates on Hilbert-space trace-class estimates.

The universal Maurer–Cartan class

The genus tower $\{\overline{B}^{(g)}(\mathcal{A})\}_{g\geq 0}$ (Theorem **??**) is controlled by a universal obstruction.

THEOREM 1.20.8 *(Universal MC class;* [proved here]*).* For a modular Koszul chiral algebra $\mathcal{A}$ with non-degenerate invariant pairing, there exists a class

$$\Theta_{\mathcal{A}} \in \mathrm{MC}\Big(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}) \,\widehat{\otimes}\, R\Gamma\Big(\overline{\mathcal{M}}_{g,\bullet}, \mathbb{Q}\Big)\Big)$$

controlling the full genus tower as a single Maurer–Cartan deformation, with three properties:

(i) Its scalar trace has universal genus-1 coefficient $\kappa(\mathcal{A})\lambda_1$; on the proved uniform-weight lane, $\mathrm{tr}(\Theta_{\mathcal{A}}) = \sum_{g\geq 1} \kappa(\mathcal{A}) \cdot \lambda_g$.

(ii) It is compatible with clutching: $\xi^*(\Theta_{\mathcal{A}}) = \Theta_{\mathcal{A}} \star \Theta_{\mathcal{A}}$ under the boundary maps $\xi\colon \overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \to \overline{\mathcal{M}}_{g,n}$.

(iii) Under Verdier duality: $\mathbb{D}(\Theta_{\mathcal{A}}) = \Theta_{\mathcal{A}^!}$.

Proof. This is Theorem **??**, proved intrinsically by Theorem **??**: the cyclic L_∞-algebra $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ is encoded by the genus-completed bar differential, the completed tensor product converges with clutching by Proposition **??**, and the scalar trace is genus-1 universal with all-genera $\kappa\lambda_g$ only on the proved uniform-weight lane (Theorem **??**). □

Remark 1.20.9 *(Scope).* The scalar shadow has universal genus-1 coefficient $\kappa(\mathcal{A}) \cdot \lambda_1$, and on the proved uniform-weight lane the full series $\kappa(\mathcal{A}) \cdot \lambda_g$ is established (Theorem **??**). The clutching compatibility is implicit in the modular operad structure of the bar complex (Theorem **??**). At the one-channel level, Proposition **??** shows that these two geometric inputs would already force tautological-line support once they separate the relevant sector, and Proposition **??** reduces the remaining duality flank to a σ-stable Lagrangian two-plane on that sector. Proposition **??** then reduces the remaining H-level lift to perfect one-channel projector subcomplexes whose cross Verdier pairing is a PTVV quasi-isomorphism, and Proposition **??** reduces that lift further to explicit one-channel coderivation subcomplexes in $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ and $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}^!)$. Proposition **??** then reduces those subcomplexes further to finite low-bar-length cyclic coderivation seeds in $\mathrm{CoDer}^{\mathrm{cyc}}(\widehat{\overline{B}}_X(\mathcal{A}))[1]$ and $\mathrm{CoDer}^{\mathrm{cyc}}(\widehat{\overline{B}}_X(\mathcal{A}^!))[1]$. Proposition **??** compresses those finite seed sets further to one distinguished degree-2 cocycle on each side together with bar-length-≤ 3 correction packets and a finite pairing matrix. Proposition **??** then identifies the remaining packet with the three visible low-degree ingredients already supported by the current MC2 surface: the simple-pole bracket sector, the normalized double-pole pairing matrix, and the Killing l_3^η sector. Proposition **??** then reduces that visible packet further to a canonical transfer package on the one-channel seed spaces: one cyclic seed, one shared generator-seed lift producing the Killing l_3^η sector, and one functorial normalization splitting off the one-channel cocycle line. Proposition **??** then reduces that canonical package further to an explicit root-string transfer law on those seed spaces: the genus-2/genus-3 profile

$$O_2 = t^2(ae_{12} + bf_1 - mf_2), \qquad O_3^\eta = amt^3\eta,$$

together with obstruction-side recovery of (a, b, m) and the same coefficient-extraction normalization. Proposition **??** then reduces that explicit law further to a chart-existence/uniqueness statement: the ordered root-string chart is forced by the one-channel support graph and normalization data, up to rescaling of the three seed lines. Proposition **??** then reduces that chart statement further to intrinsic line detection: the seed, simple-pole, and Killing support lines must be canonically picked out by $l_{2,\mathrm{sp}}, \beta_g^{\mathrm{dp}}, \nu_g^{\mathrm{sp}}$, and the support of $(O_{2,g}, O_{3,g}^\eta)$. Proposition **??** then reduces that remaining line-detection step to automorphism

rigidity: those support lines are the unique invariant lines for the one-channel support automorphism group compatible with the bracket, pairing, normalization, and obstruction profile. Proposition **??** then reduces that rigidity statement further to a finite stabilizer computation on the labeled one-channel support graph: the fixed vertices must be exactly the simple-pole, residual-kernel, and Killing support lines. Proposition **??** then reduces that stabilizer statement further to an explicit incidence-matrix / orbit-count computation on the visible one-channel graph: the relevant support lines are exactly the singleton orbits of the permutation group determined by the bracket, pairing, normalization, and support indicators. Proposition **??** then reduces that bounded permutation problem further to a universal three-case visible orbit table on the $m = 1, 2, 3$ root-string types, equivalently $\mathfrak{sl}_3$, $\mathfrak{sp}_4$, $\mathfrak{g}_2$. Proposition **??** then collapses that three-case verification to direct lookup/identification against one fixed labeled orbit table. Proposition **??** then collapses that fixed-table identification further to the minimal visible invariant signature packet forcing that table. Proposition **??** then collapses that invariant packet further to the universal signed seed-character law recovering it. Proposition **??** then collapses that character further to the reduced datum of two root-string signs and one normalization scalar. Proposition **??** then collapses that reduced datum further to one root-string parity sign plus one normalization scalar. Proposition **??** then collapses that parity-sign datum further to one chart-normalized seed scalar. The universal class $\Theta_{\mathcal{A}}$ is constructed intrinsically for all modular Koszul algebras by the bar-intrinsic theorem (Theorem **??**). On one-channel families, the comparison gauntlet of Theorem **??** reconstructs the same class on the proved uniform-weight lane: (a) the cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ exists as a cyclic L_∞ algebra (Theorem **??**); (b) the Maurer–Cartan equation in the completed tensor product with tautological classes is solved at genus 1 universally and at all genera on that lane by the conjunction of Proposition **??** and Theorem **??**. The non-scalar content is developed through the shadow obstruction tower (Definition **??**): finite-order shadows through degree 4 are proved constructively, and the all-degree convergence is guaranteed by the bar-intrinsic construction (Theorem **??**; Theorem **??**). One-channel line concentration is proved for all algebras in algebraic families with rational OPE coefficients (Theorem **??**), covering the entire standard Lie-theoretic landscape at all non-critical levels including admissible ones. The proof bypasses module-category semisimplicity via Whitehead reduction to a finite-dimensional linear system and algebraic semicontinuity of rank.

Homotopy template (Convention **??**): Type I (existence of a homotopy object). At H-level: a filtered cyclic L_∞ algebra $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ and an MC element $\Theta_{\mathcal{A}}$ with trace, clutching, and Verdier compatibility. The scalar shadow is $\kappa(\mathcal{A}) \cdot \lambda_g$ (proved, Theorem **??**). The full non-scalar class is Theorem **??** (Theorem 2.2.2).

The bar-intrinsic construction (Theorem **??**: $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$) supersedes the original formulation with simple Lie symmetry: it requires only a non-degenerate invariant pairing, with no restriction on the symmetry type. The simple-Lie version is a special case.

Conjecture 1.20.10 *(Categorical modular Koszul duality;* [conjectured]*).* For each genus g and each modular Koszul chiral algebra $\mathcal{A}$, the curved bar-cobar adjunction at genus g induces an equivalence

$$\mathsf{D}^{\mathrm{co}}\big(\overline{B}^{(g)}(\mathcal{A})\text{-}\mathsf{comod}\big) \;\simeq\; \mathsf{D}^b\big(\mathcal{A}\text{-}\mathsf{mod}^{\Sigma_g}\big)$$

between the *coderived category* (§1.20) of curved $\overline{B}^{(g)}(\mathcal{A})$-comodules and the bounded derived category of $\mathcal{A}$-modules twisted by the Hodge data of Σ_g. This equivalence varies over $\mathcal{M}_g$ as a flat functor:

(i) At $g = 0$, it reduces to bar-cobar inversion (Theorem **??**).

(ii) Applied to integrable representations of $\widehat{\ }_k$, the functor H^0 recovers the Tsuchiya–Ueno–Yamada space of conformal blocks (Remark **??**), and its monodromy gives the bar-side comparison surface for the KZ/Hitchin package on that integrable affine lane.

(iii) For uniform-weight algebras, the scalar shadow $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ is Theorem D; universally at $g = 1$ one still has $\mathrm{obs}_1(\mathcal{A}) = \kappa(\mathcal{A})\lambda_1$. The full Chern character $\mathrm{ch}(\mathcal{V}_g)$ recovers the spectral discriminant $\Delta_{\mathcal{A}}$ (Remark **??**).

(iv) Under Verdier duality: $\mathsf{D}^{\mathrm{co}}(\overline{B}^{(g)}(\mathcal{A}^!)\text{-}\mathsf{comod}) \simeq \mathsf{D}^{\mathrm{co}}(\overline{B}^{(g)}(\mathcal{A})\text{-}\mathsf{comod})^{\mathrm{op}}$.

This is the *categorical lift* of Theorems A–D to all genera: Theorem B gives the equivalence, Theorem D gives its first Chern class, and the full equivalence data underlies the chain-level modular-functor package of $\mathcal{A}$.

The primitive kernel programme

The cofree-coderivation principle (Milnor–Moore for modular operads) compresses the full MC element $\Theta_{\mathcal{A}}$ to its primitive restriction $\mathfrak{K}_{\mathcal{A}} \in \mathrm{Prim}^{\log}_{\mathrm{mod}}(\mathcal{A})$. The primitive master equation (Theorem **??**)

$$d\,\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \hbar\,\Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0$$

carries three channels: the differential d, the primitive pre-Lie product $\star$ (separating sewing), and the BV self-contraction Δ_{ns} (non-separating genus raising).

Downstream results.

(a) *Per-family primitive kernels.* The primitive kernel of each standard family is computed in the example chapters:

Family	$\mathfrak{K}_{\mathcal{A}}$	**Channels**	**Ref.**
Heisenberg	$K_{0,2} + K_{1,1}$	Δ_{ns}	Prop. **??**
Affine$\widehat{\ }_k$	$K_{0,2} + K_{0,3} + K_{1,1}$	$\Delta_{\mathrm{ns}},\ [-,-]$	Prop. **??**
$\beta\gamma$	$K_{0,2} + K_{0,4} + K_{1,1} + R_{\mathrm{pf}}$	$\Delta_{\mathrm{ns}},\ \mathrm{Rig}$	Prop. **??**
Virasoro/$\mathcal{W}_N$	$K_{0,2} + K_{0,3} + K_{0,4} + K_{1,1} + R_{\mathrm{pf}} + \cdots$	all three	Prop. **??**

(b) *Shell equations as genus-forcing mechanism.* The primitive shell equations (Proposition **??**) express the genus-g data as a sum over three channels. Per-family: Heisenberg (Proposition **??**), affine (Proposition **??**), $\beta\gamma$ (Proposition **??**), Virasoro (Proposition **??**).

(c) *Branch BV finite-rank reduction.* The branch master action $S^{\mathrm{br}}_{\mathcal{A}}$ (Definition **??**) on the finite-rank quotient $V^{\mathrm{br}}_{\mathcal{A}}$ satisfies the QME (Proposition **??**). Per-family: Heisenberg (Computation **??**: Gaussian, $\dim V^{\mathrm{br}} = 1$), affine (Computation **??**: Chern–Simons-type, $\dim V^{\mathrm{br}} = \dim$), $\beta\gamma$ (Computation **??**: Gaussian on weight-changing line), Virasoro (Computation **??**: infinite series on primary slice).

(d) *Metaplectic half-density.* $\delta^2_{\mathcal{A}} = \Delta_{\mathcal{A}}$ (Corollary **??**): $(1 - \kappa x)^{1/2}$ for Heisenberg, Weyl denominator for affine, genuine power series for Virasoro.

(e) *Flat connection and affine conformal-block comparison.* The MC equation defines a flat connection $\nabla^{\mathrm{mod}}_{\mathcal{A}} = d + [\Theta_{\mathcal{A}}, -]$ on the modular convolution algebra (§**??**). On the integrable affine lane, its H^0 flat sections over ${}_{g,n}$ recover the conformal blocks. The chain

$$\mathfrak{K}_{\mathcal{A}} \xrightarrow{\mathrm{FT}^{\log}_{\mathrm{mod}}} \Theta_{\mathcal{A}} \xrightarrow{\text{linearise}} \nabla^{\mathrm{mod}}_{\mathcal{A}} \xrightarrow{\text{integrable affine } H^0 \text{ flat sections}} \mathrm{CB}_g(\widehat{\mathfrak{g}}_k)$$

gives the trivial Heisenberg connection, the affine genus-0 KZ shadow connection, and on the integrable affine lane the bar-side comparison surface for the KZ/Hitchin monodromy package.

The discriminant as spectral invariant

The shared discriminant $\Delta(x) = (1-3x)(1+x)$ across $\mathfrak{sl}_2$, Virasoro, and $\beta\gamma$ systems (Theorem **??**) has a structural explanation. It is the *first nontrivial non-scalar characteristic class* of the modular Koszul object: where $\kappa(\mathcal{A})$ is a scalar measuring the curvature of d^2, the discriminant $\Delta_{\mathcal{A}}(x)$ records the branch geometry of the dual Hilbert series.

Distinct chiral theories can lie on the same modular spectral sheet, sharing the same Δ, even when their local operator content is entirely different. The underlying reason is functorial: Drinfeld–Sokolov reduction changes the growth pole (i.e. the leading OPE data) while preserving the discriminant family. This is compatible with the observation that $\Delta_{\mathcal{A}}$ depends only on the quadratic OPE data, which is the Koszul-dual-determining datum.

Theorem 1.20.11 *(Discriminant as spectral determinant, verified cases;* [proved here]*).* For $\mathcal{A} \in \{\widehat{\mathfrak{sl}}_2, \mathrm{Vir}, \beta\gamma\}$, the bar-cohomology discriminant satisfies

$$\Delta_{\mathcal{A}}(x) \;=\; \det(1 - x\,T_{\mathrm{br}})$$

where T_{br} is a $d \times d$ matrix whose eigenvalues are the reciprocals of the branch points of the algebraic Koszul dual Hilbert series $P_{\mathcal{A}^!}(x)$, with d the degree of algebraicity. In all three cases $d = 2$, the branch points are $x = 1/3$ and $x = -1$, and $\Delta(x) = (1-3x)(1+x)$.

Proof. Each of the three Hilbert series involves $\sqrt{1-2x-3x^2}$ (Theorem **??**): $P_{\widehat{\mathfrak{sl}}_2}$ and P_{Vir} satisfy quadratic equations with discriminant $(1+x)(1-3x)$, while $P^2_{\beta\gamma} = (1+x)/(1-3x)$ has the same branch locus.

The branch points of any algebraic function $P(x)$ satisfying $a(x)P^2 + b(x)P + c(x) = 0$ are the zeros of $\mathrm{Disc}_P = b(x)^2 - 4a(x)c(x)$. For $\widehat{\mathfrak{sl}}_2$, the algebraic equation is $x(1+x)P^2 - (1+x)P + 1 = 0$ (Theorem **??**), giving

$$\mathrm{Disc}_P = (1+x)^2 - 4x(1+x) = (1+x)(1-3x).$$

The branch points are $x_1 = -1$ and $x_2 = 1/3$. Setting $\alpha_i = 1/x_i$, we obtain $\alpha_1 = -1$ and $\alpha_2 = 3$, and

$$\det(1 - x\,T_{\mathrm{br}}) = (1-\alpha_1 x)(1-\alpha_2 x) = (1+x)(1-3x) = \Delta(x).$$

Since P_{Vir} and $P_{\beta\gamma}$ involve the same radical $\sqrt{1-2x-3x^2}$, they share the same branch locus, hence the same Δ. □

Theorem 1.20.12 *(Spectral discriminant, general case;* [proved here]*).* Let $\mathcal{A}$ be a Koszul chiral algebra whose Koszul dual Hilbert series $P_{\mathcal{A}^!}(x) = \sum_{n\geq 1} \dim(\mathcal{A}^!)_n\, x^n$ is either:

(a) *Rational*: $P_{\mathcal{A}^!}(x) = N(x)/D(x)$ with $\gcd(N, D) = 1$, $D(0) = 1$, and $\deg D = r$; or

(b) *Algebraic of degree $d \geq 2$*: satisfying an irreducible $f(x, y) \in \mathbb{Q}[x, y]$, $\deg_y f = d$, with discriminant $\Delta(x) = \mathrm{Disc}_y(f)$ normalized so that $\Delta(0) = 1$.

Define the *transfer matrix* T_{rec} as follows.

Rational case. The coefficients $a_n = \dim(\mathcal{A}^!)_n$ satisfy a constant-coefficient linear recurrence of order r (the degree of the denominator). Let T_{rec} be its companion matrix. Then $\Delta_{\mathcal{A}}(x) := D(x)$ and

$$\Delta_{\mathcal{A}}(x) = \det(1 - x\,T_{\mathrm{rec}}).$$

Algebraic case. The coefficients a_n satisfy a P-recursive recurrence of order d:

$$q_d(n)\,a_n + q_{d-1}(n)\,a_{n-1} + \cdots + q_0(n)\,a_{n-d} = 0 \tag{1.3}$$

where $q_i(n) \in \mathbb{Q}[n]$. Let $c_i = \lim_{n\to\infty} q_i(n)/n^{\deg q_d}$ be the leading asymptotic coefficient of q_i (so $c_d = 1$ after normalization). The asymptotic companion matrix is

$$T_{\mathrm{rec}} = \begin{pmatrix} -c_{d-1} & -c_{d-2} & \cdots & -c_0 \\ 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 1 & 0 \end{pmatrix} \in \mathrm{Mat}_{d\times d}(\mathbb{Q}).$$

Then the bar-cohomology discriminant satisfies

$$\Delta(x) = \det(1 - x\,T_{\mathrm{rec}}).$$

In both cases, the eigenvalues of T_{rec} are the reciprocals of the singularities (poles or branch points) of $P_{\mathcal{A}^!}$, and the spectral radius of T_{rec} equals $\dim \mathfrak{g}$ for Kac–Moody algebras (Theorem **??**).

Proof. Rational case. Write $D(x) = \prod_{i=1}^{r}(1 - \alpha_i x)$ with $\alpha_i = 1/x_i$ the reciprocal poles. The partial-fraction decomposition of $P = N/D$ shows that $a_n = [x^n]P$ satisfies a constant-coefficient linear recurrence of order r with characteristic polynomial $\chi(t) = \prod(t - \alpha_i)$. The companion matrix T_{rec} has χ as its characteristic polynomial, so $\det(1 - x\,T_{\mathrm{rec}}) = \prod(1 - \alpha_i x) = D(x) = \Delta_{\mathcal{A}}(x)$.

Algebraic case. The argument combines three classical results from analytic combinatorics.

Step 1: Algebraic ⇒ P-recursive. An algebraic power series is D-finite ([166], Proposition VI.1), hence its coefficients satisfy a P-recursive (holonomic) recurrence of order d, giving (1.3).

Step 2: Singularities of the generating function. Since $P(x)$ is algebraic of degree d with minimal polynomial $f(x, y) = 0$, its singularities as an analytic function are the branch points: the roots $x_1, \ldots, x_m$ of $\Delta(x) = \mathrm{Disc}_y(f)$ ([166], Theorem VII.8).

Step 3: Poincaré–Perron theorem. For the P-recursive recurrence (1.3) with asymptotic companion matrix T_{rec}, the Poincaré–Perron theorem ([166], Theorem VIII.4) identifies the eigenvalues of T_{rec} with the asymptotic growth rates of the solutions: provided $c_0 \neq 0$ and the eigenvalues $\alpha_1, \ldots, \alpha_d$ have distinct absolute values, the fundamental system of solutions satisfies $a_n^{(i)}/a_{n-1}^{(i)} \to \alpha_i$ as $n \to \infty$.

Step 4: Growth rates = reciprocals of branch points. By the transfer theorem for algebraic singularities ([166], Theorem VI.3), the contribution of a simple branch point x_j to the coefficients is $[x^n]\,P(x) \sim C_j \cdot x_j^{-n} \cdot n^{-3/2}$. The growth rates are therefore $\alpha_i = 1/x_i$, the reciprocals of the branch points.

Synthesis. Combining Steps 3 and 4, the eigenvalues of T_{rec} are $\alpha_i = 1/x_i$. Since $x_1, \ldots, x_m$ are the roots of Δ (Step 2) and $\Delta(0) = 1$:

$$\det(1 - x\,T_{\mathrm{rec}}) = \prod_{i=1}^{d}(1 - \alpha_i x) = \prod_{i=1}^{d}\left(1 - \frac{x}{x_i}\right) = \Delta(x). \qquad \square$$

Remark 1.20.13 *(Scope and verification).* The rational case is unconditional (given rationality of P); the algebraic case further requires the Poincaré–Perron non-degeneracy condition ($c_0 \neq 0$, eigenvalues of distinct absolute value). Both conditions hold for all bar-cohomology sequences computed in this work. The residual conjecture is that $P_{\mathcal{A}^!}$ is algebraic or rational for all Koszul chiral algebras associated to simple $\mathfrak{g}$.

Algebraic verification (*degree* 2). For $\widehat{\mathfrak{sl}}_2$, Virasoro, and $\beta\gamma$ (Theorem 1.20.11), P is algebraic of degree 2. The Riordan recurrence $(n+1)a_n = (2n-2)a_{n-1} + (3n-6)a_{n-2}$ gives leading coefficients $c_2 = 1$, $c_1 = -2$, $c_0 = -3$, and $T_{\mathrm{rec}} = \left(\begin{smallmatrix} 2 & 3 \\ 1 & 0 \end{smallmatrix}\right)$ with eigenvalues 3 and -1. Check: $\det(1 - x\,T_{\mathrm{rec}}) = 1 - 2x - 3x^2 = (1-3x)(1+x) = \Delta(x)$. The eigenvalue $3 = \dim(\mathfrak{sl}_2)$ is the exponential growth rate (Theorem **??**); -1 reflects the alternating correction from the central extension.

Rational prediction for $\widehat{\mathfrak{sl}}_3$. Conjecture **??** predicts that $P_{\widehat{\mathfrak{sl}}_3}$ is *rational* with denominator $(1-8x)(1-3x-x^2)$, so the theorem applies in its rational form with $r = 3$ poles. The companion matrix of the recurrence (**??**) has characteristic polynomial $(t-8)(t^2-3t-1)$, giving eigenvalues 8, $(3+\sqrt{13})/2$, $(3-\sqrt{13})/2$. Check: $\det(1 - x\,T_{\mathrm{rec}}) = (1-8x)(1-3x-x^2) = \Delta_{\mathfrak{sl}_3}(x)$. The spectral radius is $8 = \dim(\mathfrak{sl}_3)$, consistent with Theorem **??**. The remaining eigenvalues $(3 \pm \sqrt{13})/2$ are the roots of the DS discriminant factor $t^2 - 3t - 1$, shared with the $\mathcal{W}_3$ generating function (Remark **??**).

Why not the Kodaira–Spencer operator on H^1*?* A first guess is $\Delta(x) = \det(1 - x\,T_{\mathrm{KS}} \mid H^1)$ where T_{KS} acts on $H^1(\bar{B}(\mathcal{A})) \cong \mathfrak{sl}_2$ (3-dimensional). But Schur's lemma forces T_{KS} to be scalar on the irreducible adjoint, giving $\det = (1-cx)^3$, degree $3 \neq 2$. The correct object is T_{rec} (dimension d, the degree of algebraicity), not H^1 (dimension $\dim \mathfrak{g}$). Identifying a natural *subquotient* $H_1^{\mathrm{red}} \subset H^1$ on which a geometric operator recovers T_{rec} remains open (Conjecture **??**).

The discriminant's role as first non-scalar characteristic class of the modular characteristic hierarchy parallels $\kappa(\mathcal{A})$ as the first scalar characteristic number of the universal Maurer–Cartan class $\Theta_{\mathcal{A}}$ (Theorem 1.20.8).

Definition 1.20.14 *(Spectral branch object).* A *spectral branch object* for a modular Koszul chiral algebra $\mathcal{A}$ is a finite-rank perfect complex $V_{\mathcal{A}}^{\mathrm{br}}$ together with a *branch transport operator* $T_{\mathrm{br},\mathcal{A}} \colon V_{\mathcal{A}}^{\mathrm{br}} \to V_{\mathcal{A}}^{\mathrm{br}}$ such that

$$\Delta_{\mathcal{A}}(x) \;=\; \det\bigl(1 - x\,T_{\mathrm{br},\mathcal{A}}\bigr). \tag{1.4}$$

The pair $(V_{\mathcal{A}}^{\mathrm{br}}, T_{\mathrm{br},\mathcal{A}})$ satisfies:

(i) $\Delta_{\mathcal{A}^!} = \Delta_{\mathcal{A}}$ (the spectral discriminant is self-dual);

(ii) exact factorization functors preserving quadratic OPE data preserve $T_{\mathrm{br},\mathcal{A}}$;

(iii) $\mathrm{rank}(V_{\mathcal{A}}^{\mathrm{br}})$ equals the degree of algebraicity of the Koszul-dual Hilbert series.

Status. In the verified cases (Theorem 1.20.11), $V_{\mathcal{A}}^{\mathrm{br}}$ is constructed as the span of the companion-matrix eigenvectors of T_{rec}, and (1.4) is proved. The general construction of $V_{\mathcal{A}}^{\mathrm{br}}$ from the bar complex (rather than from the recurrence) remains open; see Conjecture **??** for the conjectural identification with a subquotient of H^1. (Part of the MC2 theorem package; the universal class exists by Theorem **??**, while the older scalar-lane comparison route is Theorem **??**.)

Computation 1.20.15 *(Spectral discriminants of standard families;* [proved here]*).*

Family	$\Delta_{\mathcal{A}}(x)$	rank
k	$1-kx$	1
$\widehat{\mathfrak{sl}}_{2k}$	$\dfrac{(1-kx)(1-(k+4)x)}{1-2x}$	2
Vir_c	$1-\dfrac{c-26}{2}x$	1
$\mathcal{W}_{3c}$	$\big(1-\dfrac{c-50}{2}x\big)\big(1-\big(\dfrac{c-50}{2}+\dfrac{48}{5c+22}\big)x\big)$	2
$\beta\gamma$	$1+2x$	1

The zeros of $\Delta_{\mathcal{A}}(x)$ are the *branch points* of the modular MC problem: values of the spectral parameter where deformation directions merge. Poles are resonance points where the tangent complex degenerates.

For $\widehat{\mathfrak{sl}}_2$: zeros at $x = 1/k$ and $x = 1/(k+4)$ (Feigin–Frenkel dual levels); pole at $x = 1/2$ (critical level $k = -2$, where the center acquires opers).

For $\mathcal{W}_3$: two zeros at $x_1 = 1/\kappa_c$ and $x_2 = (5c+22)/(\kappa_c(5c+22)+48)$, reflecting the two-dimensional deformation space (T, W). At $c = 50$: $x_1 \to \infty$, $x_2 = 3/2$.

For $\beta\gamma$: the unique standard family with $\kappa < 0$. Branch point at $x = -1/2$; the genus tower has alternating signs.

Duality. $\Delta_{\mathcal{A}^!}(x) = \Delta_{\mathcal{A}}(x)$ for all Koszul pairs: the spectral discriminant is self-dual. For $\widehat{\mathfrak{sl}}_2$: the zero at $1/k$ corresponds to level k; the zero at $1/(k+4)$ corresponds to the dual level $-k-4$. The product of the two zeros is $1/(k(k+4))$; their sum is $(2k+4)/(k(k+4)) = 2/k + 2/(k+4) \to 0$ as $k \to \infty$.

Rank. $\mathrm{rank}(\Delta_{\mathcal{A}}) = \deg \Delta_{\mathcal{A}} = \dim H^1_{\mathrm{cyc,prim}}(\mathcal{A})$: the number of independent genus-1 deformation directions. Rank 1 for algebras with one-dimensional deformation space (Heisenberg, Virasoro, $\beta\gamma$); rank 2 for $\widehat{\mathfrak{sl}}_2$ and $\mathcal{W}_3$.

The index theorem for genus expansions

The appearance of the $\hat{A}$-genus in the Heisenberg genus expansion (Theorem **??**) and the Bernoulli asymptotics of $F_g(\mathcal{A})$ are not coincidental: they are the output of a Grothendieck–Riemann–Roch computation on the universal curve.

Theorem 1.20.16 *(Family index theorem for genus expansions;* [proved here]*).* Let $\mathcal{A}$ be a Koszul chiral algebra with obstruction coefficient $\kappa(\mathcal{A})$, let $\pi\colon \overline{\mathcal{C}}_g \to \overline{\mathcal{M}}_g$ be the universal curve with relative dualizing sheaf ω_π, and let $\mathbb{E} = R^0\pi_*\omega_\pi$ be the Hodge bundle. The following holds for uniform-weight modular Koszul algebras at all genera; the genus-1 statements are unconditional for all families, including multi-generator. Define the *modular deformation complex* of $\mathcal{A}$ at genus g by

$$\mathcal{D}^{(g)}_{\mathcal{A}} \;:=\; \kappa(\mathcal{A})\cdot\mathbb{E} \;\in\; K_0(\overline{\mathcal{M}}_g)\otimes_{\mathbb{Z}}\mathbb{Q},$$

the K-theory class given by $\kappa(\mathcal{A})$ copies of the Hodge bundle. Then:

(i) *GRR formula.* For uniform-weight algebras, the genus-g free energy is the degree-g component of the Grothendieck–Riemann–Roch pushforward:

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A})\cdot\int_{\overline{\mathcal{M}}_{g,1}} \psi^{2g-2}\, c_g(\mathbb{E}) \;=\; \kappa(\mathcal{A})\cdot[\pi_*(\mathrm{ch}(\omega_\pi)\cdot\mathrm{Td}(T_\pi))]^{\mathrm{eval}}_g, \tag{1.5}$$

where $[\cdot]^{\mathrm{eval}}_g$ denotes evaluation of the degree-g Chern character $\mathrm{ch}_g(\mathbb{E})$ against ψ^{2g-2} on $\overline{\mathcal{M}}_{g,1}$.

(ii) *$\hat{A}$-genus identification.* On the same uniform-weight locus, the total generating function is

$$\sum_{g\geq 1} F_g(\mathcal{A})\, x^{2g} \;=\; \kappa(\mathcal{A})\cdot\big(\hat{A}(ix)-1\big) \;=\; \kappa(\mathcal{A})\cdot\left(\frac{x/2}{\sin(x/2)}-1\right),$$

where the function $(x/2)/\sin(x/2)$ arises from $\mathrm{Td}(T_\pi) = c_1(\omega_\pi)/(1-e^{-c_1(\omega_\pi)})$ in the GRR formula, Wick-rotated by the positivity of tautological intersection numbers on $\overline{\mathcal{M}}_g$.

(iii) *Verdier compatibility.* Under Koszul duality $\mathcal{A} \leftrightarrow \mathcal{A}^!$, $\mathcal{D}^{(g)}_{\mathcal{A}^!} = \kappa(\mathcal{A}^!)\cdot\mathbb{E}$, and the complementarity $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \kappa_{\mathrm{total}}$ (Theorem **??**(ii)) gives

$$F_g(\mathcal{A}) + F_g(\mathcal{A}^!) \;=\; \kappa_{\mathrm{total}}\cdot\lambda_g^{\mathrm{FP}},$$

the "index-theoretic complementarity": the total genus expansion of a Koszul pair is a fixed multiple of the Faber–Pandharipande tautological integral.

Proof. Routing remark (AP147). The non-circular proof route is

$$\text{shadow tower construction} \Longrightarrow \text{Theorem ??} \Longrightarrow \text{Theorem 1.20.16}.$$

Thus Theorem 1.20.16 is a downstream Grothendieck–Riemann–Roch repackaging of the scalar genus series, and it is not an input to Theorem **??**.

The proof assembles four ingredients, each established independently.

Step 1: Construction of $\mathcal{D}^{(g)}_{\mathcal{A}}$. The genus universality theorem (Theorem **??**) establishes for uniform-weight algebras that the genus-g obstruction class factors as $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\cdot\lambda_g$ in $H^{2g}(\overline{\mathcal{M}}_g)$, where $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle. This factorization determines $\mathcal{D}^{(g)}_{\mathcal{A}}$ uniquely in $K_0(\overline{\mathcal{M}}_g)\otimes\mathbb{Q}$: the obstruction class obs_g is the top Chern class of $\mathcal{D}^{(g)}_{\mathcal{A}}$, and the universality theorem shows that $c_g(\mathcal{D}^{(g)}_{\mathcal{A}}) = \kappa(\mathcal{A})\cdot c_g(\mathbb{E})$, i.e., $\mathcal{D}^{(g)}_{\mathcal{A}} = \kappa(\mathcal{A})\cdot\mathbb{E}$ as K-theory classes.

The construction is *functorial*: $\kappa(\mathcal{A})$ is determined by the genus-1 curvature of the bar complex (a canonical invariant of $\mathcal{A}$), and $\mathbb{E}$ is determined by the universal curve (independent of $\mathcal{A}$). The product $\kappa(\mathcal{A})\cdot\mathbb{E}$ is therefore a functor from Koszul chiral algebras to K-theory classes on $\overline{\mathcal{M}}_g$.

Step 2: Mumford's GRR computation. By Grothendieck–Riemann–Roch applied to the universal curve $\pi\colon \overline{C}_g \to \overline{\mathcal{M}}_g$ and the relative dualizing sheaf ω_π:

$$\mathrm{ch}(\mathbb{E}) \;=\; \mathrm{ch}(R^0\pi_*\omega_\pi) \;=\; 1 + \pi_*(\mathrm{ch}(\omega_\pi)\cdot\mathrm{Td}(T_\pi)),$$

Serre duality on the fibers gives $R^1\pi_*\omega_\pi \cong (R^0\pi_*\mathcal{O})^\vee \cong \mathcal{O}$, so $\mathrm{ch}(R\pi_*\omega_\pi) = \mathrm{ch}(\mathbb{E}) - 1$.

Write $K = c_1(\omega_\pi)$ for the relative canonical class. Since $c_1(T_\pi) = -K$, the Todd class of the relative tangent bundle is $\mathrm{Td}(T_\pi) = (-K)/(1-e^K) = K/(e^K-1)$, and the pushforward becomes

$$\mathrm{ch}_n(\mathbb{E}) \;=\; \pi_*\left[\frac{K\cdot e^K}{e^K-1}\right]_{\deg 2n+1} \;=\; \frac{B_{2n}}{(2n)!}\,\kappa_{2n-1} \;+\; \delta_n \qquad (n\geq 1),$$

where $\kappa_{2n-1} = \pi_*(K^{2n})$ are the Mumford–Morita–Miller classes [329] and δ_n is supported on the boundary $\partial\overline{\mathcal{M}}_g$. The identity follows from

$$\frac{Ke^K}{e^K-1} = \frac{K}{1-e^{-K}} = \mathrm{Td}(\omega_\pi) = 1 + \frac{K}{2} + \sum_{n\geq 1}\frac{B_{2n}}{(2n)!}K^{2n},$$

and $\pi_*(K^{2n}) = \kappa_{2n-1}$ by definition of the tautological classes.

Step 3: Faber–Pandharipande evaluation. The Faber–Pandharipande formula (Theorem **??**) evaluates the ψ-λ_g integral:

$$\int_{\overline{\mathcal{M}}_{g,1}} \psi^{2g-2}\,\lambda_g \;=\; \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}.$$

This is the content of (1.5): the genus universality theorem reduces F_g to κ times this integral, and the integral evaluates to the Bernoulli expression via the tautological intersection theory initiated by Mumford [329] and completed by Faber–Pandharipande [153].

Step 4: Generating function from the Todd class. The Bernoulli numbers B_{2g} are the Taylor coefficients of the Todd generating function $x/(1-e^{-x}) = 1 + x/2 + \sum_{g\geq 1} \frac{B_{2g}}{(2g)!}\,x^{2g}$. In Step 2, the Todd class $\mathrm{Td}(\omega_\pi)$ produced these Bernoulli numbers as the Chern character $\mathrm{ch}_g(\mathbb{E})$, i.e., the genus expansion *is* the GRR pushforward. Concretely:

$$\begin{aligned}\sum_{g\geq 1} \lambda_g^{\mathrm{FP}}\,x^{2g} \;&=\; \sum_{g\geq 1} \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}\,x^{2g} \\ &=\; \frac{x/2}{\sin(x/2)} - 1 \;=\; \hat{A}(ix) - 1,\end{aligned}$$

where the second equality is the classical identity $t/\sin t = \sum_{n\geq 0} \frac{(2^{2n}-2)\,|B_{2n}|}{(2n)!}\,t^{2n}$ evaluated at $t = x/2$ (equivalently, the even part of the Wick-rotated Todd class). Multiplying by $\kappa(\mathcal{A})$ gives (ii).

The positivity of all coefficients ($(x/2)/\sin(x/2)$ has all positive Taylor coefficients, unlike $\hat{A}(x)$ which alternates) reflects the positivity of $\lambda_g^{\mathrm{FP}} > 0$ for all g, which in turn is a consequence of the Hodge index theorem on $\overline{\mathcal{M}}_g$.

Step 5: Verdier compatibility. Part (iii) follows immediately from $\mathcal{D}^{(g)}_{\mathcal{A}^!} = \kappa(\mathcal{A}^!) \cdot \mathbb{E}$ and the linearity of F_g in κ. □

Remark 1.20.17 *(GRR origin of the genus expansion).* The theorem strengthens Theorem **??** by identifying the *origin* of the generating function identity: the scalar genus expansion of a Koszul chiral algebra is the Grothendieck–Riemann–Roch pushforward of the Todd class of the universal curve, scaled by the bar-complex curvature. The $\hat{A}$-genus appears precisely because $\hat{A}$ is the multiplicative genus associated to the formal group law $F(x,y) = x+y$, the trivial (additive) formal group, and the bar-complex genus expansion is additive in κ. Equivalently, the Bernoulli numbers governing both the Todd class and the ψ-λ_g integrals are the same Bernoulli numbers because both are computed by the same GRR calculation on the universal curve.

In physical terms, for uniform-weight algebras the genus expansion is an index. The "partition function" $\sum F_g \hbar^{2g-2}$ is the family index of the modular deformation complex on $\overline{\mathcal{M}}_g$, and anomaly cancellation ($\kappa_{\mathrm{tot}} = 0$) is the vanishing of this index.

The explicit universal class $\Theta^{\min}_{\mathcal{A}} = \kappa \cdot \eta \otimes \Lambda$ (Theorem **??**) gives this a precise homotopy-algebraic meaning: $\hat{A}$ is the character of $\Theta_{\mathcal{A}}$: it computes how the universal MC element distributes its weight across genera. The Bernoulli numbers appear because they are the Taylor coefficients of $\mathrm{tr}(\Theta)$, not because of any formal group coincidence. The formal group is additive because κ itself is additive under tensor product (Corollary **??**), which in turn follows from $\Theta_{\mathcal{A}\otimes\mathcal{B}} = \Theta_{\mathcal{A}} + \Theta_{\mathcal{B}}$ in the factorization algebra sense.

Remark 1.20.18 *(Scalar free energy package).* Corollary **??** assembles the scalar content of Theorem D into a single formula: $F(\mathcal{A};\hbar) = (\kappa/\hbar^2)\big[\hat{A}(i\hbar) - 1\big]$ with $\hat{A}(ix) = (x/2)/\sin(x/2)$, convergent for

$|\hbar| < 2\pi$, and admitting the infinite product $(\hbar/2)/\sin(\hbar/2) = \prod_{n\geq 1}(1-\hbar^2/4n^2\pi^2)^{-1}$. At genus 1, the same scalar formula holds for every modular Koszul algebra. For uniform-weight algebras the genus-2 ratio $F_2/F_1 = 7/240$ is universal; the ratio $F_2/F_1^2 = 7/(10\kappa)$ is perturbative in $1/\kappa$. Shadow corrections beyond the scalar sector enter as multiplicative deformations governed by the MC equation, not by the KdV hierarchy (Remark **??**).

Remark 1.20.19 *(Scope).* Steps 1–3 of the proof use only results established in this monograph (genus universality, Theorem **??**) and imported results (Mumford's GRR formula [329]; Faber–Pandharipande [153]). Step 4 is classical generating function analysis. *Routing.* The non-circular proof route is

$$\Theta_{\mathcal{A}} \text{ and its shadow tower} \Longrightarrow \text{Theorem ??} \Longrightarrow \text{Theorem 1.20.16}.$$

Theorem 1.20.16 is the downstream GRR repackaging of the scalar genus series; it is not an input to Theorem **??**. The identification $\mathcal{D}_{\mathcal{A}}^{(g)} = \kappa(\mathcal{A}) \cdot \mathbb{E}$ is canonical given the genus universality theorem; the functoriality of the construction follows from the functoriality of κ (determined by genus-1 curvature) and $\mathbb{E}$ (determined by the universal curve).

What remains. The theorem identifies $\mathcal{D}_{\mathcal{A}}^{(g)}$ via its top Chern class (the obstruction). A stronger result would construct $\mathcal{D}_{\mathcal{A}}^{(g)}$ as a *sheaf* on $\overline{\mathcal{M}}_g$ (rather than a K-theory class) and identify its full Chern character with the genus expansion at all intermediate degrees, not only the top. This would require the coderived Ran-space formalism of §1.20 and is the subject of Theorem 1.20.8.

Remark 1.20.20 *(The GRR theorem as synthesis).* Theorem 1.20.16 synthesizes the preceding results: the scalar genus expansion of a Koszul chiral algebra is a GRR pushforward. The master diagram (**??**) underlies the proof; the full quintuple $C_{\mathcal{A}} = (\Theta_{\mathcal{A}}, \kappa(\mathcal{A}), \Delta_{\mathcal{A}}, \Pi_{\mathcal{A}}, \mathcal{H}_{\mathcal{A}})$ now has its H-level component $\Theta_{\mathcal{A}}$ *proved* (Theorem **??**, resolving Theorem 1.20.8), and the family index theorem follows.

Open problems. The following extensions use the same master diagram in broader geometric settings: C_{mod} (shifted-symplectic complementarity), derived Drinfeld–Kohno, and the elliptic extension.

The spectral discriminant as characteristic class

The discriminant $\Delta_{\mathcal{A}}(x) = \det(1 - xT)$ is *the first nontrivial non-scalar characteristic class of the modular Koszul object*. Where the scalar invariant $\kappa(\mathcal{A})$ measures the curvature of d^2, the polynomial $\Delta_{\mathcal{A}}$ records the full branch geometry of the dual Hilbert series (Theorem **??**).

Distinct chiral theories can lie on the same modular spectral sheet, sharing the same discriminant, even when their local operator content differs entirely. Drinfeld–Sokolov reduction changes the growth pole (the leading OPE data) while preserving the discriminant family, a functorial consequence of the modular structure. The K-theoretic framework that organizes the scalar invariant κ (L1), the spectral discriminant Δ (L2), and the periodicity profile Π (L3) into a hierarchy of characteristic classes of the virtual bar family $\mathcal{V}_{\mathcal{A}} \in K_0(\overline{\mathcal{M}}_g)$ is developed in Remark **??**; the intrinsic construction of the L2 operator from Casimir actions on bar cohomology is Conjecture **??**.

Chain-level / evaluation-locus derived Drinfeld–Kohno and the Bridge Theorem

The Yangian theory (Chapter **??**) establishes bar-cobar recovery for the E_1-chiral Yangian and the $q \mapsto q^{-1}$ transformation of the quantum parameter. The chain-level derived Drinfeld–Kohno theorem is now proved on its stated evaluation-generated domains (Theorems **??** and **??**): the E_1-chiral Koszul duality

functor, composed with the Kazhdan–Lusztig (resp. Kazhdan) equivalence, yields the derived braiding reversal $R_q \mapsto R_{q^{-1}}$ there. The factorization-level statement on the Yangian evaluation locus is also proved (Theorem **??**).

The Bridge Theorem. The full factorization-categorical extension is the central remaining target of the modular Koszul duality project (Conjecture **??**, promoted to Remark **??**). The Bridge Criterion Theorem (Theorem **??**) shows that the full triple equivalence

$$\mathsf{Fact}^{\mathrm{ord}}_{\mathsf{E}_1}(X;\mathcal{A}) \;\simeq\; \mathrm{Mod}^{\mathrm{comp}}(\mathcal{Y}^{\mathrm{dg}}_{\mathcal{A}}) \;\simeq\; \mathrm{Rep}^{\mathrm{spec}}\big(\mathrm{QG}_{\mathrm{spec}}(R_{\mathcal{A}})\big)^{\mathrm{op}}$$

follows from three inputs: *full $\mathcal{O}$-Koszulness* (B1), *Yangian tower completion* (B2), and *spectral quantum-group comparison* (B4). An independent path through *boundary/line realization* (B3) via the HT literature (Remark **??**) opens a potentially more constructive route to B4 once the additional derived-center and boundary spectral inputs on that HT surface are supplied. The remaining work is staged rather than monolithic:

Conjecture 1.20.21 *(Factorization-categorical DK/KL bridge (DK-3);* [conjectured]*).* There exists an equivalence of E_1-factorization categories

$$\mathrm{Fact}_{\mathsf{E}_1}(\mathcal{Y}(\mathfrak{g})) \;\simeq\; \mathrm{Fact}_{\mathsf{E}_1}(U_q(\mathfrak{g}))^{\mathrm{op}}$$

extending the chain-level equivalence of Theorem **??** from evaluation modules to all objects of Yangian category $\mathcal{O}$, and identifying the Yangian E_1-factorization structure with the factorisation quantum group structure of Latyntsev [289].

Remark 1.20.22 *(Scope and ladder).* The Yangian extension problem is best read as a DK ladder:

(a) **DK-0 (proved):** lax monoidal bar complexes on the module-bar surface, affine/Yangian Verdier = R-matrix inversion, and the chain-level DK square (Theorem **??**; Theorems **??** and **??**; Proposition **??**(iii); Theorem **??**).

(b) **DK-1 (proved):** the factorization-level statement on the evaluation locus (Theorem **??**).

(c) **DK-2/3 (partly proved, partly reduced):** The evaluation-locus theorem is proved, but the module-category promotion beyond generators now splits into two distinct issues: *thick generation* and an *ambient extension/comparison package*. On the Yangian module surface, Conjecture **??** and Corollary **??** are only conditional thick-generation extension criteria: once one constructs an ambient exact/lax monoidal extension of the evaluation-level bar-comodule comparison, together with the required bar-comodule-to-quantum-group comparison, the type-A thick-generation results (Proposition **??**, Theorem **??**) promote the evaluation-locus DK equivalence to the relevant thick closure. A separate *sectorwise convergence* route (Corollary **??**) resolves the H-level evaluation-generated core at all simple types via the ∞-categorical sectorwise convergence theorem (Theorem **??**). The remaining gap to Conjecture 1.20.21 is therefore the completed/coderived extension beyond that proved core, together with the ambient extension/comparison package on the module side (Corollary **??**).

(d) **DK-4/DK-5 (the Bridge Theorem project):** the dg-shifted Yangian and full triple-bridge completion (Conjecture **??**). Decomposed into four sub-targets B1–B4 (Remark **??**): full $\mathcal{O}$-Koszulness (B1), tower completion (B2), boundary/line realization (B3), spectral QG comparison (B4). The Bridge Criterion Theorem (Theorem **??**) gives B1+B2+B4 $\Rightarrow$ full bridge. The broader HT boundary/line route, beginning with boundary/line realization (B3), provides an independent route to B4 once the additional derived-center and boundary spectral inputs are supplied.

The key control point is that the evaluation-locus theorem is already theorematic; the category-$\mathcal{O}$ extension and the dg-shifted comparison are downstream stages, not hidden inside the same claim. The six-lemma proof architecture decomposes the residual gap into two proved lemmas (evaluation-locus bridge, Verdier $= R^{-1}$) and four open kill targets (Lemmas 3–6). See §**??** for the full treatment, Remark **??** for the full decomposition, and Remark **??** for the precise staged status. The exact extra DK-2/DK-3 packets are recorded in Remark **??**.

Homotopy template (Convention **??**): Type II (equivalence of homotopy categories). At H-level: an equivalence of E_1-monoidal ∞-categories. The M-level model is the braided monoidal bar-cobar dg category; the S-level shadow is the classical DK equivalence on evaluation modules. (Contributing to Conjecture 2.2.3.)

DK is naturally an E_1-factorization theorem: it compares two E_1-level objects (KZ monodromy and R-matrix data). The step from E_1 to full E_2 coherence requires Drinfeld associator input and is governed by the Grothendieck–Teichmüller group (Remarks **??** and **??**). The modular layer (genus deformation of E_1-factorization) is the content of the modular homotopy theory project (Definition **??**).

Summary

The historical rhyme is this. Classical operadic Koszul duality says: quadratic presentation plus bar-cobar formalism produces homotopy theory. The theory sought here says: factorization presentation on curves plus Verdier duality on modular compactifications produces *modular homotopy theory*. Just as Drinfeld–Kohno related KZ monodromy to quantum groups, the theory sought here would relate configuration-space/Deligne–Mumford monodromy to curved chiral Koszul duality. And just as étale cohomology supplied the correct ambient cohomology theory for Frobenius and zeta functions, factorization homology on $\mathrm{Ran}(X)$ together with Verdier duality on $\overline{\mathcal{M}}_{g,n}$ is the correct ambient cohomology theory for OPE, monodromy, and bar-cobar duality on curves.

The holographic completion

The modular homotopy theory of the preceding subsections acquires a concrete physical target via HT holography. The holographic modular Koszul datum $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, C, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}}_{\bullet,\bullet})$ (Definition **??**) is the single object that should organize HT holography as precise algebra. It synthesizes five literatures:

(i) the bulk/boundary chiral algebra framework of Costello–Dimofte–Gaiotto [89];

(ii) the line-side algebraic model and dg-shifted Yangian of Dimofte–Niu–Py [129];

(iii) the non-planar interface/minimal-model matching of Gaiotto–Zeng [427];

(iv) the 3d Poisson sigma model for PVA quantization of Khan–Zeng [264];

(v) the factorisation quantum-group framework of Latyntsev [289].

The organizing sentence: *the 2026 non-planar sphere algebra is not the endpoint; it is the binary genus-zero shadow of the modular Maurer–Cartan class* $\Theta_{\mathcal{A}}$.
What is proved. The full $\Theta_{\mathcal{A}}$ and clutching law are proved (Theorem **??**). The scalar shadow $\mathrm{obs}_g = \kappa(\mathcal{A})\lambda_g$ is proved for uniform-weight algebras at all genera and unconditionally at genus 1 (Theorem **??**). The shadow obstruction tower through degree 4 is proved, and the quartic resonance class with clutching

law is established (Appendix **??**). Beyond the conjectural framework, the collision filtration (§**??**) establishes four **proved recovery theorems**:

(a) *Collision residue = twisting morphism* (Theorem **??**): $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) = \pi_{\mathcal{A}}$, the universal twisting morphism, canonically producing the dg-shifted Yangian $r_{\mathcal{A}}(z)$.

(b) *Arnold/MC at depth* $2 = A_\infty$*-YBE* (Theorem **??**): the CYBE from the Arnold relation on $\overline{\mathcal{M}}_{0,4}$.

(c) *Shadow connection / KZ comparison on the affine comparison surface* (Theorem **??**): for $\mathcal{A} = \widehat{\ }_k$, the genus-0 shadow connection on the affine Kac–Moody comparison surface identifies with the KZ connection ∇^{KZ}, with flatness from the MC equation; BPZ recovery on the Virasoro degenerate-representation comparison surface (Proposition **??**).

(d) *Quartic obstruction* $= L_\infty$ *obstruction* (Theorem **??**): $[o_4(\mathcal{A})] = [\mathfrak{R}^{\mathrm{mod}}_4(\mathcal{A})]$; vanishing for classes G, L, C; non-vanishing for class M.

(e) *Gaudin–Yangian identification* (Theorem **??**): the GZ26 commuting Hamiltonians are the Gaudin Hamiltonians of the dg-shifted Yangian; higher collision residues give a higher Gaudin system.

(f) *Three-parameter* $\hbar$ *identification* (Theorem **??**): the KZ25 sigma-model coupling, the DNP25 loop parameter, and the collision-residue prefactor all equal $1/(k + h^\vee)$; the classical limit is Drinfeld 1985 + Semenov-Tian-Shansky 1983 ([proved elsewhere]).

(g) *Operator-order trichotomy* (Theorem **??**): the GZ26 Hamiltonian differential order is 0 for $k_{\max} \leq 1$ and $k_{\max} - 1$ for $k_{\max} \geq 3$; the three invariants $p_{\max}, k_{\max}, r_{\max}$ are formally distinguished (Chapter **??**).

(h) *Genus-*1 *seven-face identification* (Theorem **??**, Chapter **??**): the genus-1 collision residue on E_τ is identified with the KZB connection, the elliptic r-matrix, and the elliptic Gaudin model; **proved** for affine KM.

The complete Heisenberg datum is computed explicitly (Computation **??**). The collision-filtration spectral sequence (Construction **??**) records shadow depth by collapse stage on the algebraic side. The five shadows of $\Theta_{\mathcal{A}}$ (scalar, r-matrix, connection, quartic, clutching) are five projections of one MC element (Remark **??**).

What remains conjectural. The five programme-level targets: boundary–defect realization (Conjecture **??**), Yangian-shadow (Theorem **??**; **proved**), sphere reconstruction on established comparison surfaces (Theorem **??**; **proved**), quartic resonance obstruction (Theorem **??**; **proved**), and singular-fiber descent for minimal models (Conjecture **??**). The HT deformation-quantization functor (Conjecture **??**) and the higher-genus shadow connection at $g \geq 2$ (Definition **??**) remain conjectural.

Remark 1.20.23 *(The self-dual central charge* $c = 13$*).* [proved here] The Virasoro Koszul involution $c \mapsto 26 - c$ has a unique fixed point $c = 13$, at which $\mathrm{Vir}^!_{13} \cong \mathrm{Vir}_{13}$: the boundary theory and its Koszul dual coincide. Five proved structural facts distinguish this point within the holographic programme.

(i) *Uniqueness.* $\mathrm{Vir}_c \cong \mathrm{Vir}^!_c$ if and only if $c = 13$: no other central charge in the Virasoro family is self-dual under Koszul duality.

(ii) *Full tower self-duality* (Proposition **??**). Self-duality holds at all degrees: $S_r(\mathrm{Vir}_{13}) = S_r(\mathrm{Vir}_{13}^!)$ for every $r \geq 2$, and the shadow trace formula $\mathrm{RTF}(f) = 0$ vanishes for all test functions. The higher-degree corrections are odd in the complementarity asymmetry $\delta_\kappa = c - 13$ and hence vanish identically.

(iii) *Self-dual shadow connection* (Theorem **??**(v)). The shadow metric satisfies $Q_L = Q_{L^!}$ at $c = 13$: the connection, its branch points, its discriminant $\Delta = 40/87$, and its spectral curve are all invariant under the Koszul involution.

(iv) *Vanishing complementarity asymmetry.* $\delta_\kappa = \kappa - \kappa' = c/2 - (26-c)/2 = c - 13 = 0$ at $c = 13$. The complementarity sum is $\kappa + \kappa' = 13$ (not zero: this is the single-generator value $\dim(\mathfrak{g})/2 = 26/2$; contrast with affine Kac–Moody, where $\kappa + \kappa' = 0$).

(v) *Non-perturbative $\mathbb{Z}/2$-symmetry* (Remark **??**). At $c = 13$, the trans-series acquires a $\mathbb{Z}/2$-symmetry under $\sigma \leftrightarrow -\sigma$: the perturbative and dual-instanton sectors are identical.

The self-dual point $c = 13$ must be distinguished from the critical dimension $c = 26$. At $c = 26$: $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 13 + (-13) = 0$ (anomaly cancellation). At $c = 13$: $\delta_\kappa = 0$ (Koszul self-duality). The gap between the two central charges is $26 - 13 = 13 = \dim(\mathfrak{g})/2$: the same constant that controls the complementarity sum for a single conformal-weight-2 generator. The holographic modular Koszul datum $\mathcal{H}(\mathrm{Vir}_{13})$ is self-dual in all six components simultaneously; for no other Virasoro central charge does the datum enjoy this symmetry.
Cross-reference. Proposition **??** (§**??**); Remark **??** (§**??**); Remark **??** (§**??**); Theorem **??**(v) (§**??**).

Remark 1.20.24 *(Entanglement programme: G11–G16).* Six theorem targets (G11–G16) extend the THQG programme to entanglement. The shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ provides a convergent, finite-order approximation to holographic entanglement; no other approach to quantum gravity has this property.

(a) **G11: Algebraic entanglement entropy** ([physical heuristic]). $S_{\mathrm{EE}}^{(g)}(\mathcal{A})$ from the shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$. The scalar entropy $S_{\mathrm{EE}}^{\mathrm{sc}} = (c/3)\log(L/\varepsilon)$ is **proved** at genus 1 from $\kappa(\mathcal{A}) = c/2$ and the twist operator dimension (Calabrese–Cardy [75]). Higher-genus contributions from $\Theta_{\mathcal{A}}^{\leq r}$ with $r \geq 3$ remain heuristic.

(b) **G12′: Algebraic quantum extremal surface** ([conjectured]). *(Note: G12 in Chapter* **??** *denotes the distinct result "Koszulness = exact holographic reconstruction," which is proved.)* The quantum extremal surface condition $\partial_A S_{\mathrm{bulk}} + \partial_A S_{\mathrm{EE}}^{\mathrm{sc}} = 0$ as a shadow connection Ward identity. Convergence of the shadow obstruction tower at the extremal locus is **proved** (from $\rho(\mathcal{A}) < 1$ on the standard landscape); the identification with the RT/HRT surface is conjectural.

(c) **G13: Modular flow from shadow connection** ([conjectured]). Parallel transport along ∇^{sh} should realize modular flow. The Verdier involution $\mathcal{A} \mapsto \mathcal{A}^!$ is **proved** to coincide with the Tomita–Takesaki J-operator at the level of Koszul duality (G4). The full modular flow as shadow connection transport remains conjectural.

(d) **G14: Holographic error correction** ([proved here]). The bar-cobar adjunction is a quantum error-correcting code: $\Omega \circ B \xrightarrow{\sim} \mathrm{id}$ (Theorem B) is the decoding map, and the entanglement wedge is the image of the open/closed pair $\mathcal{U}(\mathcal{A})$ (§1.20). Subregion duality follows from complementarity (Theorem C).

(e) **G15: Algebraic Page constraint** ([proved here]). $S_{\mathrm{EE}}(\mathcal{A}) + S_{\mathrm{EE}}(\mathcal{A}^!) = \mathrm{const}$ is a direct consequence of the complementarity theorem (Theorem C) and the modular characteristic constraint $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ for KM/free fields, or $= \rho \cdot K$ for W-algebras (Theorem D). For the Virasoro family ($\kappa + \kappa' = 13$): $S_{\mathrm{EE}}(\mathrm{Vir}_c) + S_{\mathrm{EE}}(\mathrm{Vir}_{26-c}) = (26/3)\log(L/\varepsilon)$.

(f) **G16: Replica structure** ([physical heuristic]). $Z_n(\mathcal{A}) = \mathrm{tr}(\rho_{\mathcal{A}}^n)$ via n-fold shadow sewing (MC5). The integer-n construction is well-defined from the Fredholm partition function (G10); analytic continuation $n \to 1$ for the von Neumann entropy is not yet established.

Status summary: G14, G15 **proved**; G12, G13 **conjectured**; G11, G16 **heuristic** (genus-1 core proved, higher-genus and analytic continuation pending).

Remark 1.20.25 *(G11: Entanglement entropy from modular Koszul duality).* Chapter **??** derives entanglement entropy from the proved core: $S_{\mathrm{EE}}^{\mathrm{sc}} = (c/3)\log(L/\varepsilon)$ from $\kappa = c/2$; complementarity sum 26/3; four-class entanglement complexity (G/L/C/M); QES as shadow connection Ward identity. Status: **all proved**. Computational verification: 145 tests.

Remark 1.20.26 *(G12: Holographic codes from Koszul duality).* Chapter **??** proves:

(a) Theorem **??** (G12): Koszulness ⇔ exact holographic reconstruction.

(b) Theorem **??**: symplectic code structure from Lagrangian isotropy.

(c) Proposition **??**: Verdier isotropy and Shapovalov cross-pairing.

(d) Theorem **??**: shadow depth = redundancy channels (resolves Conjecture **??**).

(e) Proposition **??**: 12-fold Koszulness–code dictionary (K1–K12 → code properties).

Status: **all proved**. Computational verification: 49 tests.

The open/closed realization and the chiral derived center

The open/closed realization (§**??**) constructs the algebraic bridge between the local Hochschild theory and the global bordered FM geometry. The central results:

(a) The *chiral endomorphism operad* $\mathcal{E}nd_A^{\mathrm{ch}}$ with consecutive block-substitution (Definition **??**). This is the nonsymmetric operad governing A_∞-chiral algebras.

(b) The *brace dg algebra structure* on chiral Hochschild cochains $C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ (Theorem **??**). **Proved**: the same tree-combinatorial argument as Gerstenhaber–Voronov, with block-substitution replacing simple reindexing.

(c) The *chiral derived center* $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}) := H^*(C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \delta)$ (Definition **??**). On the Koszul locus this is a three-term chiral Gerstenhaber algebra $\mathcal{Z}^0 \oplus \mathcal{Z}^1 \oplus \mathcal{Z}^2$ by Theorem H.

(d) The *universal open/closed pair* $\mathcal{U}(\mathcal{A}) = (C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A}),\ \mathcal{A},\ \mathrm{id})$ (Theorem **??**). **Proved**: the chiral Swiss-cheese theorem, asserting that the derived center is the universal bulk with open color $\mathcal{A}$.

(e) The *local-global bridge* (Theorem **??**): restriction of the bordered FM convolution to the formal disk identifies with $C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A})$. **Proved**: Morita invariance of the derived center.

(f) The *annulus trace theorem* (Theorem **??**): $\int_{S^1_p} \mathcal{M} \simeq HH_*(\mathcal{M})$. **Proved**: by Ayala–Francis $\otimes$-excision on the standard Weiss cover of S^1_p, reducing to the two-sided bar complex $A_b \otimes^{\mathbf{L}}_{A_b^e} A_b = HH_*(A_b)$. The annulus degeneration proposition (Proposition **??**) connects this to the genus-1 curvature: $\Delta_{\mathrm{ns}}(\mathrm{Tr}_{\mathcal{A}}) = \kappa(\mathcal{A}) \cdot \lambda_1$.

(g) The *open/closed MC element* $\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum_j \mu^{\mathcal{M}_j}$ (Construction **??**, Theorem **??**). **Proved**: satisfies the MC equation in $\mathfrak{g}^{\mathrm{oc}}_{\mathcal{A},\vec{\mathcal{M}}}$ by Stokes' theorem on the bordered FM compactification.

(h) The *open/closed quartic resonance class* $\mathfrak{R}^{\mathrm{oc}}_4(\mathcal{A}) \in \mathcal{Z}^2$ (Construction **??**). Vanishes for G-type, tree-only for L-type, both terms nonzero for M-type. $\mathfrak{R}^{\mathrm{oc}}_4(\mathrm{Vir}_c)$ has poles at $c = 0$ and $c = -22/5$.

(i) The *completed modular Koszul datum* $\Pi^{\mathrm{oc}}_X(\mathcal{A})$ (Definition **??**): eight-fold datum extending $\mathcal{H}(\mathcal{A})$ by the derived center, annulus trace, and nonlinear resonance tower.

Key distinction. Bar/cobar classifies *twisting morphisms* (universal couplings between $\mathcal{A}$ and $\mathcal{A}^!$). The derived center classifies *bulk operators acting on the boundary*. These are complementary: the bar complex is the coalgebraic shadow of the boundary, while the derived center is the universal bulk. The open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ packages both into a single object, with the closed projection recovering $\Theta_{\mathcal{A}}$ and the mixed projection recovering the bulk-to-boundary coupling.

The four-stage architecture. The open/closed programme has a strict dependency hierarchy. Each stage is well-defined only after the preceding stage is established.

Stage 1. *Local one-colour theorem* (proved). The A_∞-chiral algebra (Definition **??**), operadic braces on $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ (Theorem **??**), and the universal acting-bulk theorem (Theorem **??**). This is the bottleneck: every downstream claim rests on it.

Stage 2. *Open primitive* (proved). The Morita-invariant object is the open-sector factorization dg-category $\mathcal{C}_{\mathrm{op}}$ on the bordified curve $\widetilde{X}_D$: objects are boundary conditions on the real blowup $\partial\widetilde{X}_D$, morphisms are open-string states (sections of factorization sheaves on boundary intervals). The category admits two equivalent presentations: the *meromorphic-tensor* presentation, where objects are $\mathcal{A}^!$-modules in the braided tensor category $\mathcal{A}^!$-mod with R-matrix braiding, and the *factorization* presentation, where $\mathcal{C}_{\mathrm{op}}$ is a cosheaf of dg-categories on configurations of boundary points (the open colour of the Swiss-cheese operad). A choice of compact generator b yields the boundary algebra $A_b = \mathrm{End}_{\mathcal{C}_{\mathrm{op}}}(b)$, an algebra over the E_1-colour; different choices of b produce Morita-equivalent but non-isomorphic algebras (Theorem **??**(iii)). The intrinsic bulk is $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}_{\mathrm{op}}) = R\mathrm{Hom}_{\mathrm{Fun}(\mathcal{C},\mathcal{C})}(\mathrm{Id}, \mathrm{Id})$ (Corollary **??**). The dependency chain is: $\mathcal{C}_{\mathrm{op}} \to A_b \to \overline{B}(A_b) \to \Theta_{\mathcal{A}} \to \Theta^{\mathrm{oc}}_{\mathcal{A}}$; each arrow is a lossy functor, and only the first object is Morita-invariant.

Stage 3. *Globalization* (proved locally, modest globally). Tangential log curves (X, D, τ); bordered FM compactification; the local-global bridge (Theorem **??**). Full global descent and excision for open factorization categories on general (X, D, τ) is a programme, not a completed theorem.

Stage 4. *Modularization* (proved via MC equation; independent open-sector derivation outstanding). The open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ (Construction **??**) satisfies the full modular MC equation at all genera (Theorem **??**), with the clutching sector (d) encoding all sewing degenerations. The annulus trace (Theorem **??**) verifies the open-sector projection at genus 1. The closed-sector

projection recovers $\Theta_{\mathcal{A}}$ and the genus tower (Theorem **??**). Because the MC equation packages both sectors into a single identity, open-closed consistency at every genus is a formal consequence. What remains is the independent computation of genus-g amplitudes from iterated open-sector clutching *without* the bar-intrinsic construction, and the chain-level verification of cooperad compatibilities.

Five negative principles (permanent).

(N1) *Bar ≠ bulk.* The bar complex classifies twisting morphisms (couplings between $\mathcal{A}$ and $\mathcal{A}^!$). The bulk observables are the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$. These are different objects solving different problems.

(N2) *Boundary algebra = chart.* The boundary algebra $A_b = \mathrm{End}(b)$ depends on the choice of compact generator b. The Morita-invariant primitive is the boundary dg-category C_{op}.

(N3) *Tangential log geometry.* The correct global stage is a tangential log curve (X, D, τ), not a punctured curve $(X \setminus D)$. The real-oriented blowup at punctures produces the boundary intervals and circles on which the open sector lives.

(N4) *Modularity = trace + clutching.* The bar-intrinsic MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})\widehat{\otimes}\mathcal{G}^{\mathrm{mod}})$ satisfies the clutching factorization at all genera (Theorem **??**(iii)). The open/closed MC equation $D^{\mathrm{oc}}\Theta^{\mathrm{oc}} + \frac{1}{2}[\Theta^{\mathrm{oc}}, \Theta^{\mathrm{oc}}] + \hbar\Delta_{\mathrm{clutch}}(\Theta^{\mathrm{oc}}) = 0$ is proved at all genera (Theorem **??**). Its closed projection recovers $\Theta_{\mathcal{A}}$ with the genus tower $F_g = \kappa\cdot\lambda_g$ on the uniform-weight lane (Theorem **??**(i), Theorem D). Its open projection at genus 1 is the annulus trace (Theorem **??**). Because all sectors are projections of the *same* MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$, the MC equation forces open-closed consistency at every genus (Theorem **??**): the genus-g closed-sector amplitude and the genus-g open-sector amplitude are jointly constrained by a single identity, and no additional consistency check is required. Two distinct questions: (a) the scalar formula $F_g = \kappa \cdot \lambda_g$ *fails* for multi-weight algebras at $g \geq 2$: the full free energy includes a cross-channel correction (**op:multi-generator-universality**, resolved negatively; Theorem **??**); (b) the independent open-sector derivation of genus-g amplitudes from iterated clutching on C_{op} without using the bar-intrinsic construction.

(N5) *Honest scope.* The dg-shifted Yangian is proved for the affine lineage (Vol II, Theorem thm:Koszul_dual_Yangian). The general perturbative 3d HT Yangian structure is conjectural. The BV/BRST = bar identification at higher genus is resolved in the coderived category D^{co} for all shadow classes (Theorem **??**); at chain level, classes G and L are proved, class C is conditional on harmonic decoupling, and class M fails. The Heisenberg scalar level is proved at the chain level (Theorem **??**).

1.21 Conjecture promotions and the true frontier

Every theorem in this monograph is a finite-order projection of the universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. Every open problem is a question about the ambient geometry of the modular convolution algebra: the (-2)-shifted symplectic stack $_{\mathrm{vac}}(\mathcal{A})$ in which the chiral algebra embeds as a Lagrangian. The proved core computes the Taylor jets of this embedding. The frontier is the global geometry of the stack.

The five proved main theorems, the bar-intrinsic MC element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ (Theorem **??**), and the HS-sewing criterion (Theorem **??**) together resolve a number of conjectures that were formulated before these tools were available. We record the promotions, the reductions, and the true residual frontier.

Promotions

Former label	Result	Mechanism
??	Lattice sewing envelope (Theorem ??, promoted)	HS-sewing + Fock factorizatio
??	Full resonance tower (Theorem ??)	$\Theta_{\mathcal{A}}$ bar-intrinsic
??	All-degree clutching law (Theorem ??)	Mok degeneration + boundar
??	Bipartite vanishing	Lagrangian isotropy: $P^{++} = P$
??	Bifunctor obstruction	One-slot Koszul factorization
??	Scalar-orbit formality	scalar $\Theta_{\mathcal{A}} \Rightarrow$ formal scalar MC
??(b)	MC-shadow recursion in the Koszul case (Cor. ??); full EO identification remains conditional on Conj. ??	MC tautological relation + Cl
April 2026 session: arithmetic, purity, and CohFT		
??	Genus-1 amplitude Mellin transform is Eisenstein: $D_2(\mathcal{A}, s) = -24\kappa \cdot \zeta(s)\,\zeta(s-1)$	σ_1 Dirichlet convolution + E_2
??	$\zeta^{\mathrm{DK}}_{\mathfrak{sl}_2}(s) = \zeta(s) - 1$: Riemann zeta from $\mathfrak{sl}_2$ irreps	dimension counting + analyti
??	-module purity for affine KM (converse of (xii))	chiral localization + Hitchin V
??	Universal instanton action $A = (2\pi)^2$, Stokes constant $S_1 = -4\pi^2\kappa i$	Borel singularities = Q_L branc
??	Entire tower self-dual at $c = 13$ (not just κ)	shadow trace formula RTF =
Vol. II, rank-1 DR string equation	CohFT string equation at rank 1	DR cycle forgetful + Buryak I
??	Pixton ideal from MC for semisimple cases	FSZ + PPZ + Givental–Telem
??	$r_{\max} = d_\infty = f_\infty$ (operadic complexity identification)	induction on degree + HPL tr
April 2026 session: theorem architecture rewrites (A1–A6 + G7)		
??	Chiral Hochschild descent: $CH^\bullet(\mathcal{A}) \cong (\mathbb{D}_{\mathrm{Ran}}\overline{B}(\mathcal{A}))_\Sigma$ on the Koszul locus	Theorem A Verdier intertwini
??	Closure of chiral Koszulness under tensor product, Koszul dualization, and base change	three pre-existing fragments co
??	Kac–Moody equivalence via Saito–Kashiwara weight filtration: PBW strictness (ii) $\Longleftrightarrow$ -module purity (xii)	chiral localization + Hitchin V
??	Three equivalent descriptions of $S_r(\mathcal{A})$: algebraic (shadow), geometric (formality obstruction), holographic (boundary-bulk correction)	bar-intrinsic MC + formality
??	Generating depth d_{gen} vs algebraic depth d_{alg}: formal distinction ; Virasoro $d_{\mathrm{gen}} = 3$, $d_{\mathrm{alg}} = \infty$	depth decomposition + class M
April 2026 session: structural propositions and depth classification		
??	SC-formality characterises class $\mathbf{G}$: $\mathcal{A}$ is Swiss-cheese formal iff class $\mathbf{G}$ (Proposition ??)	abelian bracket + nondegener
??	Algebraic depth gap: $d_{\mathrm{alg}}(\mathcal{A}) \in \{0, 1, 2, \infty\}$, no finite $d_{\mathrm{alg}} \geq 3$ realized (Proposition ??)	Riccati algebraicity + Δ-dicho
??	$CH^1(V_k()) \cong$ at generic level (Proposition ??)	Koszul resolution + adjoint co
??	E_1 non-splitting at genus 1: quasi-modular E_2 obstruction (Proposition ??)	genus-0 persistence + E_2 anom
??	Free-field exactness: $\delta F_g^{\mathrm{cross}}(\mathcal{A}) = 0$ for all free-field families at all genera (all-weight; Proposition ??)	quadratic OPE $\Rightarrow$ no mixed c
??	Virasoro shadow growth rate: $\lvert S_{r+1}/S_r \rvert \to \rho^{-1}$ from Riccati root asymptotics (Proposition ??)	explicit $Q_L^{\mathrm{Vir}}(t)$ + Darboux

The sub-exponential growth condition (ii) of Conjecture **??** is automatic for all positive-energy chiral algebras (Proposition **??**): Hardy–Ramanujan–Rademacher asymptotics give $\log(\dim \mathcal{A}_n) = O(\sqrt{n}) = o(n)$, so the factorization finiteness criterion reduces to spectral sequence degeneration (i) alone (Corollary **??**).

Reductions

1. **Factorization finiteness** (Conjecture **??**). The algebraic content (sub-exponential growth) is discharged. Residue: sectorwise degeneration on $\overline{B}^{p,q}(\mathcal{A})$ lifts to compact generation of $\mathrm{Fact}(X;\ \mathcal{A})$ on $\mathrm{Ran}(X)$. This is a *geometric* assertion about the interaction between algebraic bigradings and the Ran space stratification. Proved instances: lattice (Theorem **??**), Yangian (Proposition **??**).

2. **Modular periodicity** (Conjecture **??**). The bar *cohomology* periodicity remains open. Steps 1 (T-matrix periodicity) and 3 (bar differential lift) are proved; Step 2 (convolution with $1/\eta$ does not preserve eventual periodicity of Fourier coefficients) is the sole gap. Weight-space dimensions are *not* eventually periodic (Computation **??**); the conjecture concerns the *cohomology* dimensions after passage through the bar differential.

3. **NMS graph-completed master element** (Conjecture **??**). The abstract $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ is proved. Residue: Hamiltonian lift to the Feynman transform $F_{\mathrm{mod}}(\mathcal{A})$ with cyclic structure on bar coderivations and clutching-compatible graph-sum representation.

4. **Resonance completion** (Theorem **??**). **PROVED**: the MC4 splitting $\mathcal{A} \cong R_{\mathcal{A}} \widehat{\oplus} \mathcal{A}^+$ holds with $\dim R_{\mathcal{A}} < \infty$ for all positive-energy chiral algebras, by weight-compatible HTT and the positive-energy axiom ($\dim V_0 < \infty$). Previously: $\dim R_{\mathcal{A}} < \infty$ for all positive-energy chiral algebras. Proved: $\rho(_k) = 0, \rho(V_k()) = 0$ for non-critical k, $\rho(\mathrm{Vir}_c) = 1$ for generic c.

The true frontier

The promotions above discharge conjectures that were artefacts of incomplete tools. Three problems remain that are not reducible to existing machinery.

1. **Chiral Koszulness from (H1)–(H4).** The boundary chiral algebra of a holomorphic-topological field theory satisfying (H1)–(H4) should be chirally Koszul. The precise assertion:

   > *If the weight forms $\omega_k \in \Omega^*_{\log}(\mathrm{FM}_k(\mathbb{C}))$ satisfy (H3d)–(H3e) (Arnold relations at codimension-2 corners, Stokes exactness on manifolds with corners), then the resulting A_∞ operations m_k produce a chirally Koszul algebra: $H^n(\overline{B}(\mathcal{A}), d) = 0$ for $n \geq 3$.*

   The mechanism is clear: Arnold relations are the geometric shadow of quadraticity, and the Stokes compatibility (H3e) translates to $d^2 = 0$ on the bar complex with the correct signs. What is missing is a *spectral sequence argument* connecting logarithmic regularity of the weight forms (a condition on $\Omega^*_{\log}(\mathrm{FM}_k(\mathbb{C}))$) to cohomological concentration (a condition on $H^*(\overline{B}(\mathcal{A}))$). The three verified examples (free multiplet: $m_{k\geq 3} = 0$; Landau–Ginzburg cubic: $m_{k\geq 4} = 0$ by degree counting; Heisenberg: Koszulness by direct bar cohomology computation) all have truncating A_∞ structure; the general case, where infinitely many m_k are nonzero, is open.

   Resolution via A_∞ homotopy invariance. The frontier problem as stated asks whether (H3) implies Koszulness. The correct formulation is different: Chiral Koszulness is an A_∞ homotopy invariant:

the bar construction preserves A_∞ quasi-isomorphisms (Corollary **??**), and bar cohomology concentration is a property of the quasi-isomorphism class, not of a specific chain-level model. Different propagator choices (heat kernel vs. algebraic) produce A_∞ quasi-isomorphic structures; the algebraic model (Beilinson–Drinfeld propagator $\eta = d\log(z_1 - z_2)$) satisfies (H3) *tautologically* (Arnold relations are algebraic identities of logarithmic forms). Therefore Koszulness is detectable on the algebraic model alone, where (H3) is automatic. The *residual gap* is not (H3) but *free strong generation*: the PBW criterion (Theorem **??**) requires $\mathrm{gr}_F(\mathcal{A})$ to be classically Koszul, which holds for freely strongly generated algebras (Proposition **??**) but may fail for simple quotients at admissible levels (where null vectors force non-free relations). For the universal algebra $V_k()$ at generic level, free strong generation holds and Koszulness follows; for the simple quotient $L_k()$ at admissible levels, the PBW filtration may degenerate.

Geometric reframing via associated variety. The Li–bar spectral sequence (Construction **??**) and the associated-variety criterion (Theorem **??**) reframe the free strong generation gap geometrically: once the full Li associated graded $R_{L_k} = \mathrm{gr}^F L_k$ is identified, diagonal concentration on the reduced Li–bar E_2 page constrains the Li–bar E_2 page. What the reduced geometry does *not* control is the nilradical of R_{L_k}. The present theorem does not identify that reduced Li–bar page with ordinary Poisson cohomology or with $H^*(\widetilde{\mathbb{O}}, \mathcal{O}_{\widetilde{\mathbb{O}}})$ for a symplectic resolution. Thus non-degenerate admissible and minimal-orbit levels have reduced-level test surfaces, but unconditional Koszulness still requires a nilradical/reducedness check. Large-orbit levels ($X = \mathcal{N}, = \mathfrak{sl}_N, N \geq 3$) are explicitly obstructed (Proposition **??**). The hierarchy is only heuristically guided, not theorematically controlled, by symplectic-resolution geometry.

2. **(H3) for general HT theories.** Among (H1)–(H4), three are established in general: (H1c) by the Gwilliam–Rabinovich–Williams one-loop finiteness theorem for $d' = 1$ topological direction; (H2) by standard propagator theory for polynomial interactions; (H4) by Costello–Gwilliam's factorization compatibility results. The sole open axiom is (H3): that Feynman diagram contributions define logarithmic forms $\omega_k^{\mathrm{hol}} \in \Omega^*_{\log}(\mathrm{FM}_k(\mathbb{C}))$ satisfying the Arnold–Orlik–Solomon relations at *every* codimension-2 corner $D_S \cap D_T \subset \partial^2 \mathrm{FM}_k(\mathbb{C})$. For polynomial interactions, the individual form ω_k^{hol} is rational with logarithmic singularities along boundary divisors; what requires proof is the global compatibility condition at corners, which geometrically amounts to the factorization of residues along nested strata of $\mathrm{FM}_k(\mathbb{C})$.

3. **Filtration ≥ 4 spectral Drinfeld classes.** The spectral Drinfeld class $\mathfrak{D}_{\mathrm{spec}}(\mathcal{A}, \Omega) \in H^3_{\mathrm{spec}}(\mathrm{gr}^n \mathcal{A})$ controls the remaining intrinsic strictification problem on the conditional spectral-bridge surface, once the unshifted completed strict spectral object and the completed compatibility package have been supplied. The spectral differential $\delta\colon C^2_{\mathrm{spec}} \to C^3_{\mathrm{spec}}$ is computed from the coproduct Δ; its kernel and image determine whether an associator twist can be removed. On that conditional surface, through filtration 3, $H^3_{\mathrm{spec}} = 0$ and the Drinfeld-class obstruction vanishes. At filtration 4, the graded piece $\mathrm{gr}^4 \mathcal{A}$ first contains independent Hall words ($[[x_1, x_2], [x_3, x_4]]$ is not reducible to lower filtration); the resulting spectral 3-cocycles are controlled by the higher Magnus expansion $\log(e^{x_1} e^{x_2} e^{x_3} e^{x_4})$. Computing $H^3_{\mathrm{spec}}(\mathrm{gr}^4 \mathcal{A})$ for $\mathcal{A} = \widehat{\ }_k, \mathcal{W}_N, Y()$ is the first calculation where a genuine obstruction to strictification could appear.

Resolution at filtration 4. The spectral Drinfeld class $\mathfrak{D}_{\mathrm{spec}} \in H^3_{\mathrm{spec}}(\mathrm{gr}^4 \mathcal{A})$ vanishes. Three independent arguments: (a) Drinfeld's theorem (1990): $\mathrm{grt}_1(\text{weight } 4) = 0$; (b) the bracket-bracket Hall words $[[x_1, x_2], [x_3, x_4]]$ at filtration 4 arise from the product $\phi_2 \cdot \phi_2$ of lower-order data

and factor through the coboundary operator; (c) the GT dimension count: grt_1 has one generator σ_{2k+1} at each odd weight $2k+1 \geq 3$ (Ihara's conjecture, verified computationally through weight ~ 22). The dimensions of the weight-graded pieces are $1, 0, 1, 0, 1, 1, 1, 1, 2, 2, 3, 3, \ldots$ at weights $3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, \ldots$, with even-weight elements arising from Lie brackets starting at weight 8 ($[\sigma_3, \sigma_5]$). In particular, $\mathrm{grt}_1(\text{weight } 4) = 0$, so the spectral complex has no room for a nontrivial class at weight 4. The bracket-bracket terms enter the BCH expansion $\log(e^{x_1}e^{x_2}e^{x_3}e^{x_4})$ with coefficient $\frac{1}{12}$ on $[[x_1, x_2], [x_3, x_4]]$. Consequently, on that conditional surface, strictification is unobstructed through filtration 4. Since $\dim \mathrm{grt}_1 = 0$ at weights 4 and 6, the spectral complex remains obstruction-free at these weights. At weight 8, grt_1 has $\dim = 1$ from $[\sigma_3, \sigma_5]$, but this bracket element is a depth-2 Lie word built from lower-weight data (it does not produce a new free parameter). The first *genuinely new* generator beyond the depth-1 σ_{2k+1} family appears at weight 12 ($\dim \mathrm{grt}_1(\text{weight } 12) = 2$: one Ihara-reduced bracket + the cusp form Δ_{12}).

Remark 1.21.1 *(Programme VI: systematic verification of (H1)–(H4)).* [proved here] The standing analytic hypotheses (H1)–(H4) of Volume II admit the following systematic status:

(a) **(H1) BV data and HT gauge fixing.** Proved for all perturbative holomorphic-topological theories by Costello–Gwilliam [92]: the factorization algebra structure is functorial in the BV data, and the HT twist produces a well-defined chiral algebra on the holomorphic locus.

(b) **(H2) Propagator regularity.** Proved for polynomial interactions by heat kernel estimates (the propagator $P_{\epsilon<L}$ lies in the required Sobolev space for all finite ϵ, L). For non-polynomial interactions: requires case-by-case meromorphic continuation of the propagator integral.

(c) **(H3) FM compactification, AOS relations, Stokes exactness.** The Arnold–Orlik–Solomon relations hold on $\mathrm{FM}_n(\mathbb{C})$ by the Brieskorn–Orlik–Solomon theorem. Extension to $\mathrm{FM}_n(\mathbb{C} \mid D)$ on log-FM spaces requires Mok's normal-crossings theorem [322, Thm 3.3.1]. Proved for all theories where the OPE has polynomial singularities.

(d) **(H4) Factorization compatibility.** Proved for all theories arising from Nishinaka's factorization envelope construction [428]: the envelope carries factorization structure by construction, and the OPE-to-factorization comparison is tautological. For theories not in the envelope image: requires independent verification.

Verified examples: free multiplet (trivial), Landau–Ginzburg cubic (degree counting), Heisenberg (bar computation), affine Kac–Moody at generic level (PBW), Virasoro at generic c (strong generation), $\mathcal{N}=2$ SCA at generic k (PBW + Kazama–Suzuki; Chapter **??**), $V^\natural$ (PBW via FLM free strong generation; Chapter **??**). The first three are in the literature; the remainder follow from the PBW criterion (Theorem **??**) combined with the factorization envelope construction. In Chriss–Ginzburg terms, (H1)–(H4) collectively guarantee that the bar complex $\overline{B}(\mathcal{A})$ carries a well-defined modular operad algebra structure with $D_{\mathcal{A}}^2 = 0$, hence that the universal MC element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathrm{MC}(^{\mathrm{mod}}_{\mathcal{A}})$ exists (Theorem **??**).

Three remaining conjectural classes:

(a) non-polynomial interactions (meromorphic continuation of propagator),

(b) theories outside the factorization envelope image (independent factorization verification),

(c) critical and admissible levels (where the PBW filtration degenerates and null vectors force quotients).

Remark 1.21.2 *(Frontier dependency structure).* The three frontier problems are related by a *linear dependency chain*:

$$\mathbf{P2} \xrightarrow{\text{provides input}} \mathbf{P1} \xrightarrow{\text{constrains}} \mathbf{P3}.$$

P2 (proving (H3) for general HT theories) establishes the well-defined A_∞ inputs that **P1** takes as its starting point. **P1** (spectral sequence from log regularity to bar concentration) constrains the spectral Drinfeld class

relevant to **P3**. **P3** does not feed back into the other two. P2 is the *main unresolved problem*: it is the only one whose failure could invalidate the downstream conclusions.

P1: Arnold relations do not imply concentration. The Arnold–Orlik–Solomon relations give $d^2 = 0$ on the bar complex, but $d^2 = 0$ is a necessary condition for a chain complex, not a sufficient condition for cohomological concentration. The precise gap: Arnold relations are the geometric shadow of *associativity* (an A_∞ coherence), not *quadraticity* (a condition on generators and relations). A freely strongly generated chiral algebra satisfying (H3) is Koszul by the PBW criterion (Theorem **??**); a non-freely-generated algebra satisfying the same geometric conditions need not concentrate. The PBW hypothesis is indispensable and is not derivable from (H1)–(H4) alone.

P2: the integrand–integral gap. The quantum master equation constrains *integrated* quantities (the operations m_k): the A_∞ identity $\sum m_i \circ m_j = 0$ is the Stokes theorem applied to integrated weight forms. But (H3d) constrains *integrands* (the weight forms ω_k themselves): the AOS relations are properties of the forms on $\mathrm{FM}_k(\mathbb{C})$, not of their integrals. These coincide at tree level (where weight forms are wedge products of $d\log(z_i - z_j)$ and AOS holds as an algebraic identity), but not at loop level (where internal vertex integration produces non-logarithmic functions). Mok's snc boundary theorem ([322, Thm 3.3.1]) resolves (H3) for the entire polynomial-OPE landscape; the remaining gap is narrow and restricted to non-polynomial interactions with genuine internal vertex area integrals.

P3: opposing predictions at filtration 4. The Casimir–Whitehead mechanism (the same mechanism that drives PBW propagation, Theorem **??**) provides a constructive vanishing argument: on non-trivial -isotypic components of $\mathrm{gr}^4(\widehat{\ }_k)$, the Casimir eigenvalue gives a contracting homotopy (Whitehead–Chevalley–Eilenberg vanishing); on the -invariant component, the Jacobi identity creates cancellation among the 9 facets of K_5. This predicts $H^3_{\mathrm{spec}}(\mathrm{gr}^4(\widehat{\ }_k)) = 0$ and hence full strictification (Theorem **??**). The strictification extends to all untwisted affine Kac–Moody algebras by the same abelianness mechanism (the Casimir–Whitehead vanishing applies uniformly to all simple). For hyperbolic Kac–Moody algebras, the mechanism fails: the imaginary root spaces do not carry a semisimple -action, and H^3_{spec} can be nonzero. While the Drinfeld KZ associator at filtration 4 involves $\zeta(4) \neq 0$ on the Hall word $[[t_{12}, t_{23}], [t_{13}, t_{23}]]$, this term is *forced by the pentagon relation*: Drinfeld's theorem gives $\mathrm{grt}_1(\text{weight } 4) = 0$, so the filtration-4 coefficient is a coboundary with no free parameter. The spectral Drinfeld class vanishes at filtration 4. Beyond filtration 4, grt_1 is nonzero at all odd weights ≥ 3; at even weights, $\dim \mathrm{grt}_1 = 0$ for weights 4 and 6, then $\dim \mathrm{grt}_1(\text{weight } 8) = 1$ (from $[\sigma_3, \sigma_5]$), $\dim \mathrm{grt}_1(\text{weight } 10) = 1$ (from $[\sigma_3, \sigma_7]$), and $\dim \mathrm{grt}_1(\text{weight } 12) = 2$ (one Ihara-reduced bracket + the cusp form Δ_{12}). The spectral Drinfeld class must be analysed case by case at weights ≥ 8.

Remark 1.21.3 *(Frontier status of the ten load-bearing conjectures, as of 2026).* The ten most load-bearing genuinely open conjectures, ranked by downstream impact:

1. Conjecture 2.2.7 (BV/BRST = bar at higher genus). **Resolved in** D^{co} (Theorem **??**): the BV and bar complexes are quasi-isomorphic in the coderived category for all four shadow classes, including class M. At the ordinary chain level, classes G and L are proved, class C is conditional on harmonic decoupling, and class M remains false (quartic harmonic obstruction $\delta_4 \propto Q^{\mathrm{contact}} \cdot m_0$ is not a coboundary in ordinary chain complexes, but $m_0 \cdot x = d^2(x) \in \mathrm{Im}(d)$ in D^{co}).

2. Conjecture 2.2.3 (DK–KL identification). Restricted DK-4 on the evaluation-generated core is the next step.

3. Conjecture 2.2.5 (infinite-generator completion). MC4 framework proved; example-specific H-level target identification remains. $\mathcal{W}_\infty$ side essentially closed.

4. Conjecture **??** (W-algebra orbit duality). Non-abelian $\mathfrak{n}_+$ obstruction. The $(3, 2)$ partition in $\mathfrak{sl}_5$ is the first test case.

5. Conjecture 1.20.7 (grand modular completion). Model category equivalence extending genus-0 Quillen. Principal open structural problem.

6. Conjecture **??** (DS–KD for arbitrary nilpotent). Hook-type corridor proved; self-transpose rectangular (e.g. $(2, 2)$) proved by self-duality. First open non-hook test case: $(3, 2)$ in $\mathfrak{sl}_5$ (non-self-transpose, $c + c'$ is k-dependent).

7. Conjecture **??** (analytic realization criterion). Three-layer gap (sewing envelope, conformally flat 2-disk, coderived shadow). Proved for Heisenberg and lattice VOAs.

8. Conjecture **??** (scalar saturation universality). Layer 1 ($\dim H^2_{\mathrm{cyc}} = 1$) residual for non-algebraic families. Layer 2 (tautological purity) resolved negatively for multi-weight at $g \geq 2$.

9. Conjecture **??** (type A transport-to-transpose). Hook-type seeds proved; transport-closure covers $\mathrm{Par}(N)$ for $N \leq 8$. Beyond hooks: DS–KD proved for self-transpose rectangular partitions (e.g. $(2, 2)$ in $\mathfrak{sl}_4$) by self-duality; conjectural for remaining classes (Remark **??**).

10. Conjecture **??** (-module purity $\Rightarrow$ Koszulness). Forward direction proved. Converse reduces to PBW = Saito weight filtration; proved for KM (Proposition **??**), open for rank ≥ 2 $\mathcal{W}$-algebras. The principal $\mathcal{W}_N$ case at generic central charge is Conjecture **??**, conditional on compatibility of principal DS reduction with the Kashiwara filtration.

Remark 1.21.4 *(Chromatic-shadow correspondence).* [conjectured] The genus spectral sequence (Construction **??**) with E_1 page isolating tree ($p = 0$), one-loop ($p = 1$), genus-2 shell ($p = 2$) data admits a conjectural identification with the chromatic filtration in stable homotopy theory. At each chromatic height n, the $E(n)$-localization of the spectral Steinberg object (Vol II, the spectral-braiding chapter) should recover the genus-n shell of the shadow obstruction tower:

Chromatic height	**Shadow shell**	**Classical avatar**
$n = 0$ (rational)	genus-0 bar-cobar	classical Steinberg
$n = 1$ (K-theory)	genus-1 curvature $\kappa \cdot \omega_1$	affine Hecke algebra
$n = 2$ (elliptic)	genus-2 three-shell	elliptic dynamical R-matrix

The height-n differential Ob_n in the genus spectral sequence would correspond to the chromatic boundary map $L_{E(n)}S \to L_{E(n-1)}S$ in the chromatic tower. In Chriss–Ginzburg terms: the chromatic filtration on the spectral Steinberg object should be the image, under a conjectural realization functor, of the genus filtration on $\mathrm{MC}(^{\mathrm{mod}}_{\mathcal{A}})$. This correspondence is purely structural: no computation beyond $n = 1$ has been verified.

Remark 1.21.5 *(The four-test interface).* [proved here] The modular Koszul machine has a complete interface with $\overline{\mathcal{M}}_{g,n}$, consisting of four independent proved tests. Each captures a distinct aspect of the interaction between the algebraic engine and the geometry of stable curves; none is derivable from the other three.

Test 1: $D^2_{\mathcal{A}} = 0$ (bar differential squares to zero). The bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ is an algebra over the Feynman transform of the commutative modular operad, with total differential $D_{\mathcal{A}}$ satisfying $D^2_{\mathcal{A}} = 0$ at all genera and degrees (Theorem **??**, Theorem **??**). This is the foundational consistency: the universal

MC element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ exists in $\mathrm{MC}(^{\mathrm{mod}}_{\mathcal{A}})$ precisely because $D^2_{\mathcal{A}} = 0$ (Theorem **??**). The test is algebraic: it holds for every chiral algebra satisfying (H1)–(H4), with no Koszulness hypothesis.

Test 2: $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ **(obstruction = modular characteristic × Hodge class).** The genus-g obstruction class in $H^{2g}(\overline{\mathcal{M}}_g)$ factors as a universal algebra-dependent scalar $\kappa(\mathcal{A})$ times the Faber–Pandharipande Hodge integral λ_g (Theorem **??**). This is proved unconditionally at genus 1 for all modular Koszul algebras, and at all genera on the uniform-weight lane (Theorem **??**). The test is tautological-geometric: it asserts that the bar complex produces the Hodge class λ_g, not an arbitrary class in $H^*(\overline{\mathcal{M}}_g)$.

Test 3: $Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$ **(complementarity).** The factorization homology of the Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ assembles into a Lagrangian decomposition of the shifted-symplectic target (Theorem **??**; the ambient shifted-symplectic upgrade is conditional on Theorem **??**). This is the only test that involves both $\mathcal{A}$ and its Koszul dual $\mathcal{A}^!$ simultaneously. It controls the anomaly-cancellation structure: the total genus-g partition function of a physical system with matter $\mathcal{A}$ and ghosts $\mathcal{A}^!$ is a topological invariant precisely when complementarity holds.

Test 4: sewing (amplitudes from lower-genus data). The genus-g amplitudes are determined by sewing genus-$(g-1)$ data along degenerating curves. The Hilbert–Schmidt sewing criterion (Theorem **??**) proves convergence for the entire standard landscape of universal algebras: polynomial OPE growth and subexponential sector growth imply absolute convergence of sewing at all genera. (For simple quotients $L_k()$ at admissible levels, HS-sewing is not yet established; see Remark **??**.) The Heisenberg sewing theorem (Theorem **??**) provides the explicit model: one-particle Bergman reduction to a Fredholm determinant.

Independence. Test 1 is purely algebraic ($D^2_{\mathcal{A}} = 0$ holds without knowing which class obs_g is). Test 2 identifies the class but does not involve $\mathcal{A}^!$. Test 3 requires the Koszul pair but is a genus-by-genus statement, independent of sewing. Test 4 is analytic and concerns convergence of the sewing operation, independent of the Hodge-class identification. Any three can hold with the fourth failing: a non-Koszul algebra could pass Tests 1 and 4 while failing 3; a multi-weight algebra at $g \geq 2$ passes Tests 1, 3, 4 while Test 2 *fails*: the scalar formula receives a cross-channel correction (Open Problem **??**, resolved negatively).

What lies outside. Three regions are not covered by the four-test interface:

(a) *Multi-weight algebras at genus $g \geq 2$.* Cyclic rigidity fixes the H^2_{cyc} direction to η, giving $\Theta^{\min} = \eta \otimes \Gamma_{\mathcal{A}}$, but the identification $\Gamma_{\mathcal{A}} = \kappa(\mathcal{A})\Lambda$ fails: the genus-g free energy receives a cross-channel correction $\delta F_g^{\mathrm{cross}}$ (Theorem **??**). Test 2 holds at genus 1; at genus ≥ 2 the scalar formula is replaced by the multi-weight decomposition (Open Problem **??**, resolved negatively).

(b) *BV/BRST = bar at higher genus.* For the Heisenberg, the scalar-level identification $F_g^{\mathrm{BV}} = F_g^{\mathrm{bar}} = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is proved at all genera (Theorem **??**). The general chain-level identification at genus $g \geq 1$ is conjectural (Conjecture 2.2.7). Tests 1–4 control the bar side; the BV/BRST comparison is an independent assertion about the physics.

(c) *Non-perturbative completion.* The four tests are perturbative in genus. The algebraic shadow obstruction tower $\sum_g F_g \hbar^{2g}$ converges when the shadow radius $\rho(\mathcal{A}) < 1$; for $\rho \geq 1$ (Virasoro at $c < c^* \approx 6.125$), the series is a formal asymptotic expansion. The comparison between the algebraic tower and the analytic sewing envelope $\mathcal{A}^{\mathrm{sew}}$ is Conjecture **??**.

The four-test boundary of control

The algebraic engine controls everything *inside* the modular Koszul locus. The boundary of that locus is determined by four independent tests, each representing a genuine mathematical barrier that the engine does not cross.

1. **DK-3 categorical test.** The factorization Koszul duality functor $\Phi\colon \mathrm{Fact}^{\mathrm{gen}}_{\mathsf{E}_1}(\mathcal{A}) \xrightarrow{\sim} \mathrm{Fact}^{\mathrm{gen}}_{\mathsf{E}_1}(\mathcal{A}^!)^{\mathrm{op}}$ is proved on the evaluation-generated core at all simple types (Corollary **??**). Extension beyond this core to the full module category requires categorical input (pro-nilpotent completion, prefundamental modules, thick generation beyond evaluation objects) that is not derivable from the MC element alone. The engine produces the r-matrix; it does not produce the full quantum group.

2. **Lattice-only arithmetic test.** The Hecke–Newton closure (Theorem **??**) and the unconditional Ramanujan corollary (Corollary **??**) are proved for lattice VOAs, where the shadow amplitudes are chiral graph integrals that factor through Satake parameters. For non-lattice algebras, the arithmetic of the OPE is not controlled: the MC equation provides moment L-functions, but prime-locality transfer from $\mathrm{GL}(2)$ to $\mathrm{GL}(r+1)$ for $r \geq 5$ is open (Conjecture **??**). The engine computes all symmetric-power moments; it does not prove their automorphy.

3. **Chirally Koszul geometric test.** Every theorem in the engine assumes chiral Koszulness (Definition **??**): $H^n(\bar{B}^{\mathrm{ch}}(\mathcal{A}), d) = 0$ for $n \geq 3$. The PBW criterion (Theorem **??**) proves this for all freely strongly generated algebras. For simple quotients at admissible levels, the PBW filtration may degenerate, and Koszulness is open (§1.21, item 1). The engine works inside the Koszul world; it does not detect whether an algebra is Koszul.

4. **Shadow radius $\rho < 1$ analytic test.** The shadow obstruction tower converges absolutely when the shadow growth rate $\rho(\mathcal{A}) < 1$ (Definition **??**; Theorem **??**). For $\rho \geq 1$, the tower is a formal asymptotic expansion (Borel summable by Proposition **??**, but not absolutely convergent). The engine computes shadow coefficients at every finite degree; it does not sum the series when $\rho \geq 1$. The sewing envelope $\mathcal{A}^{\mathrm{sew}}$ (Hausdorff completion for sewing-amplitude seminorms) replaces formal summation with analytic convergence, but the comparison between the algebraic shadow obstruction tower and the analytic sewing envelope is itself an open problem (Conjecture **??**).

Each test is necessary and independent: an algebra can pass three tests and fail the fourth. The Virasoro algebra at generic c passes tests 1–3 but has $\rho(\mathrm{Vir}_c) > 1$ for $c < c^* \approx 6.125$. The simple quotient $L_k()$ at an admissible level may fail test 3 while passing the others. Lattice VOAs pass all four tests; they are the algebras for which the engine achieves maximal reach.

Two orthogonal axes

The algebraic engine operates on two regimes that are mathematically orthogonal. The point of separation is the modular characteristic $\kappa(\mathcal{A})$.

Axis 1: Critical-level bar cohomology ($\kappa = 0$). At the critical level $k = -h^\vee$, the curvature κ vanishes and the bar complex is uncurved: $d^2 = 0$ on $\bar{B}^{\mathrm{ch}}(\mathcal{A}_{-h^\vee})$. The mathematical content lies in bar *cohomology*:

$$H^0(\bar{B}^{\mathrm{ch}}(\mathcal{A}_{-h^\vee})) \;\cong\; \mathrm{Fun}(\mathrm{Op}_\vee(D))$$

recovers the Feigin–Frenkel center (Theorem **??**), and the full cohomological package $H^n(\bar{B}^{\mathrm{ch}}(\mathcal{A}_{-h^\vee})) \cong \Omega^n(\mathrm{Op}_\vee(D))$ recovers the oper differential-form package (Theorem **??**). The relevant tools are homological algebra on the bar complex (Ext groups, spectral sequences, derived categories), and the target objects are opers, the geometric Langlands correspondence, and the center of the completed enveloping algebra at critical level.

The *scalar* shadow obstruction tower is inert on this axis: since $\kappa = 0$, the leading (degree-2) shadow vanishes and Theorem D gives $\mathrm{obs}_g = 0$ for all $g \geq 1$; the discriminant $\Delta = 8\kappa S_4 = 0$. However, higher-degree components of $\Theta_{\mathcal{A}}$ are independent of κ and may be nonzero: $\kappa = 0$ does not imply $\Theta_{\mathcal{A}} = 0$ or collapse of the full shadow obstruction tower (see Remark **??** and Remark **??**). The four-class partition G/L/C/M and the genus expansion of the scalar free energy are inert, but the bar-intrinsic MC element retains higher-degree structure. The information is in the kernel of the bar differential, not in its cokernel.

Axis 2: Off-critical shadow obstruction tower ($\kappa \neq 0$). Away from the critical level, $\kappa \neq 0$ and the bar complex is curved: $d_1^2 = [\kappa \cdot \omega_g, -]$. The MC element $\Theta_{\mathcal{A}} \neq 0$ and the shadow obstruction tower is nontrivial. The mathematical content lies in *deformation theory*: the modular characteristic κ, the cubic shadow C, the quartic resonance class Q, the discriminant Δ, the growth rate ρ, the four-class partition, the genus expansion, and all five main theorems.

The relevant tools are Maurer–Cartan theory in the modular convolution algebra, graph sums on stable curves, and the shadow connection ∇^{sh}. The target objects are tautological classes on $\overline{\mathcal{M}}_g$, free-energy asymptotics, and CohFT structure.

On this axis, bar cohomology is *not* the primary object: the bar complex is curved, so $d^2 \neq 0$ and ordinary cohomology is undefined. The correct replacement is the MC moduli space $\mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ and its projections.

Orthogonality. These two axes meet at the critical level $\kappa = 0$, where the scalar shadow vanishes and $H^*(\bar{B}^{\mathrm{ch}}) =$ oper package; the full MC element $\Theta_{\mathcal{A}}$ may retain nonzero higher-degree components. No theorem on axis 1 implies a theorem on axis 2, and conversely. The Feigin–Frenkel involution $k \mapsto -k-2h^\vee$ maps $\kappa \mapsto -\kappa$ and therefore exchanges the two sides of axis 2 (the algebra and its Koszul dual), but it does not connect axis 2 to axis 1. The critical level is a fixed point of the involution ($k = -h^\vee$ maps to itself), confirming that axis 1 is invariant and isolated.

The monograph develops both axes: the oper package of Chapter **??** and the critical-level Hochschild computation of Theorem **??** belong to axis 1; the shadow obstruction tower, the genus expansion, and Theorems A–D belong to axis 2. Theorem H (chiral Hochschild) connects them: at generic level, CH^* is concentrated in $\{0, 1, 2\}$, satisfies Koszul-dual duality, and has polynomial Hilbert growth (with $P(t) = 1 + t^2$ on the Virasoro / principal-$\mathcal{W}$ lane); the continuous Lie algebra cohomology $H^*_{\mathrm{cont}} = \mathbb{C}[\Theta_1, \ldots, \Theta_r]$ is an unbounded invariant that degenerates to the exterior⊗polynomial ring $\Lambda^*(P_1, \ldots, P_r) \otimes \mathbb{C}[\Theta_1, \ldots, \Theta_r]$ at critical level.

Conjecture 1.21.6 *(Isomonodromic shadow deformation;* [conjectured]*).* As the level k moves from the critical value $-h^\vee$ to generic values, the oper eigenvalues of the Gaudin Hamiltonian deform continuously into the shadow obstruction tower curvature data (κ, α, S_4). The deformation is isomonodromic: the monodromy representation of the KZ connection is preserved, while the Stokes data of the irregular singularity at infinity changes.

Remark 1.21.7 *(The oper–shadow bridge).* Conjecture 1.21.6 proposes a precise bridge between axis 1 and axis 2. At the critical level $k = -h^\vee$, the bar cohomology $H^0(\bar{B}^{\mathrm{ch}}(\mathfrak{g}_{-h^\vee}))$ recovers the algebra of functions on opers (the Feigin–Frenkel theorem). Away from the critical level, the bar complex is curved ($\kappa \neq 0$) and ordinary cohomology is undefined; the mathematical content is captured by the MC element $\Theta_{\mathcal{A}}$ and its shadow obstruction tower projections $(\kappa, \alpha, S_4, \ldots)$.

The conjecture asserts that the passage from opers to the shadow obstruction tower is an isomonodromic deformation in the Jimbo–Miwa–Ueno sense: as k moves away from $-h^\vee$, the positions of the Gaudin spectral data change continuously into the curvature invariants of the shadow obstruction tower, while the monodromy of the KZ connection (which is flat for all k) is preserved. The Stokes phenomenon

at the irregular singularity at infinity (invisible at the critical level where the connection is regular) encodes the higher-degree shadow data $S_4, S_5, \ldots$ that emerge off-critical. In particular, the shadow obstruction tower records the Taylor coefficients of this deformation in the parameter $k + h^\vee$.

Three Langlands gaps

The spectral continuation programme (§1.21 below) explores arithmetic consequences of the shadow obstruction tower. Before entering that programme, we state three structural gaps between the proved core of this monograph and any Langlands-type statement. These are not technical difficulties awaiting a clever argument; they are category-level separations between the objects this monograph constructs and the objects the Langlands programme requires.

Remark 1.21.8 *(Gap A: bar-cobar vs scattering).* [open] The bar-cobar adjunction (Theorem A) produces *twisting morphisms*: algebraic maps $\tau\colon \bar{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ controlling the coupling between a chiral algebra and its Koszul dual. The Langlands programme requires *scattering amplitudes*: analytic objects whose poles on the complex s-plane encode L-function zeros. These are different types of mathematical objects. The twisting morphism τ is a chain-level algebraic datum defined over $\mathrm{Ran}(X)$; a scattering amplitude is an analytic function of a complex variable s arising from a Mellin transform of an automorphic form.

The spectral continuation programme (§1.21) constructs a bridge for lattice VOAs: the sewing determinant $\det(1 - K_q)$ has a Mellin transform that produces L-functions. This bridge has two components: (i) the sewing–shadow intertwining (Theorem **??**), which identifies the sewing determinant with a Chern–Weil pairing of $\Theta_{\mathcal{A}}$, and (ii) the Hecke module structure on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[V_\Lambda]$ (Theorem **??**), which provides analytic continuation. For non-lattice theories, step (ii) is not available: the shadow spectral measure and the automorphic spectral measure are not identified (Remark **??**). The relationship between twisting morphisms and scattering poles is *not proved* in general.

Remark 1.21.9 *(Gap B: arithmetic descent).* [open] The entire framework of this monograph operates over function fields: the base curve X is a smooth complex algebraic curve, the factorization algebra $\mathrm{Fact}_X(\mathcal{A})$ lives on $\mathrm{Ran}(X)$, and the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is defined using configuration spaces over X. This is geometric Langlands territory.

The arithmetic Langlands programme works over number fields F, where X is replaced by $\mathrm{Spec}(\mathcal{O}_F)$ and the factorization structure is replaced by adelic automorphic forms. The descent from function field to number field is a major open problem in the Langlands programme and has been open since Langlands's original letter to Weil (1967).

This monograph does not address arithmetic descent. The shadow obstruction tower, the MC equation, and all five main theorems are statements about chiral algebras on complex curves. The spectral continuation programme produces L-functions from sewing data, but the resulting L-functions are already known by other methods (Hecke theory for lattice theta series, Franc–Mason for rational VOAs): the bar-cobar machine provides a new *proof route* to known L-functions, not new L-functions from new automorphic objects over number fields.

Remark 1.21.10 *(Gap C: formal vs convergent).* [open] The shadow obstruction tower $\Theta^{\leq 2}_{\mathcal{A}}, \Theta^{\leq 3}_{\mathcal{A}}, \Theta^{\leq 4}_{\mathcal{A}}, \ldots$ is a sequence of finite-order approximations to the full MC element $\Theta_{\mathcal{A}}$. The all-degree inverse limit $\Theta_{\mathcal{A}} = \varprojlim \Theta^{\leq r}_{\mathcal{A}}$ exists (Theorem **??**), but its *analytic* properties depend on the shadow growth rate $\rho(\mathcal{A})$ (Definition **??**).

The Langlands programme needs L-functions, which are non-perturbative analytic objects: meromorphic functions on the complex s-plane satisfying a functional equation. The shadow obstruction tower

is perturbative: it is a formal power series in the degree variable t. The passage from formal to convergent requires the shadow radius of convergence $1/\rho(\mathcal{A})$ to be positive.

For class **G** (Gaussian) and class **L** (Lie/tree) algebras, $\rho = 0$ and the tower terminates: convergence is trivially satisfied. For class **M** (mixed) algebras, $\rho > 0$ and convergence requires $\rho < 1$. This is verified for specific families (Virasoro at $c > c^* \approx 6.125$, see Theorem **??**), but it is not universal. For $c < c^*$, the shadow series diverges and the analytic content of the tower is accessible only through Borel summation (Proposition **??**).

The analytic continuation from shadow data to L-functions is therefore family-dependent and is not a consequence of the algebraic MC equation alone. The MC equation is an integrability condition ($D^2 = 0$); it does not by itself produce analytic continuation to the complex s-plane.

Spectral continuation programme

Four results on the arithmetic content of the shadow obstruction tower:

(1) Depth decomposition (Theorem **??**): $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$, where d_{arith} counts holomorphic Hecke eigenforms (critical lines) and d_{alg} is the homotopy defect (A_∞ non-formality of bar cohomology, Theorem **??**). Lattice: $d_{\mathrm{alg}} = 0$. Virasoro: $d_{\mathrm{alg}} = \infty$.

(2) Stieltjes representation (Theorem **??**): $G(t) = \int \log(1-\lambda t)\, d\rho(\lambda)$ with spectral measure ρ determined by the MC element. For Virasoro, $\rho = \delta(\lambda+6/c)$. The shadow series is Borel summable; Carleman's condition gives uniqueness (Proposition **??**).

(3) Hecke-equivariant MC element (Theorem **??**): for lattice VOAs, $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}[V_\Lambda]$ carries a Hecke module structure. The Hecke–Mellin correspondence provides analytic continuation of the shadow data to the complex s-plane, resolving the structural obstruction (Remark **??**) for this class. For non-lattice theories the obstruction persists: the Rankin–Selberg map from the shadow spectral measure ρ to the automorphic spectral measure on the moduli space is ill-conditioned (Remark **??**).

(4) Sewing–shadow intertwining (Theorem **??**): $F_1^{\mathrm{conn}}(q;\mathcal{A}) = \sum_{r\geq 2} \mathrm{Sh}_r^{(1)}(\mathcal{A}) \cdot \mathcal{G}_r(q)$. The connected sewing determinant equals the Chern–Weil pairing $\det(1 - K_q) = \exp(-\langle \Theta_{\mathcal{A}}, [E_\tau]\rangle)$ (Corollary **??**). For lattice VOAs, the shadow spectral measure ρ and the automorphic spectral measure coincide via the Hecke correspondence (Corollary **??**).

(4a) Shadow-moduli resolution (Theorem **??**): $\prod_j (1 - \lambda_j\, t(q))^{c_j} = \det(1 - K_q)$. At leading order: $t(q) = (c/6)(\prod(1-q^n)-1)$. The shadow-moduli map $t(q)$ is not an unknown: it is explicitly determined by the sewing determinant. The universality of G (Theorem **??**): $F_1^{\mathrm{conn}} = G_{\mathcal{A}}(t(q))$ where G depends only on ρ and t depends only on the moduli, achieving complete separation of algebraic and geometric data.

(4b) MC = Newton's identities for the shadow spectrum (Proposition **??**): the MC equation at all degrees is equivalent to Newton's identities relating the power-sum moments μ_r to the elementary symmetric polynomials of the spectral atoms. For two Satake-type atoms: $p_r = e_1\, p_{r-1} - e_2\, p_{r-2}$. Combined with modularity (Proposition **??**): atoms are Hecke eigenvalues satisfying Petersson orthogonality.

(5) Modular rigidity programme (§**??**). The shadow at degree r encodes Sym^{r-1} data (Proposition **??**). Serre's reduction: automorphy of all $\mathrm{Sym}^r \Rightarrow$ Ramanujan. The MC equation provides all symmetric power moments; the missing step is functorial transfer $\mathrm{GL}(2) \to \mathrm{GL}(r+1)$ for $r \geq 5$ (Conjecture **??**). The bridge from MC to Ramanujan has four stations (Remark **??**): MC/Newton (proved), modularity/Hecke (proved), Sym^r continuation (open for $r \geq 5$), Serre reduction (conditional). The *operadic Rankin–Selberg theorem* (§**??**) attacks the gap: MC recursion on moment L-functions (Theorem **??**, PROVED) + CPS

converse theorem hypotheses verified (PROVED) $\Rightarrow$ automorphy CONDITIONAL on prime-locality (Conjecture **??**; proved for lattice VOAs). For lattice VOAs, the full chain closes: MC $\Rightarrow$ Deligne by a new route. The *Hecke–Newton closure* (§**??**) resolves prime-locality for lattice VOAs via the Hecke module structure: the shadow amplitudes are chiral graph integrals (Proposition **??**), the MC recursion preserves Newton's identities at each prime (Theorem **??**), and Corollary **??** gives an unconditional new proof of the Ramanujan bound for lattice theta eigenforms. For non-lattice theories, the precise obstruction is the *arithmetic of the OPE*: whether the graph integrals on Hecke-translated curves factor through Satake parameters (Remark **??**). The *prime-locality frontier* (§**??**) identifies four routes: (A) Hecke module extension (hard: $[d_{\mathrm{sew}}, T_p]$ exactness); (B) motivic route (Brown–Levin elliptic motives); (C) analytic route (most promising: MC rigidity + Carleman uniqueness transfers character-level Hecke equivariance to shadow level); (D) computational route (Ising, Lee–Yang, Potts; **result**: Proposition **??** proves that the partition function Dirichlet series of diagonal rational CFTs is *not* multiplicative (14 coprime failures for Ising, 12 for tricritical Ising, 14 for three-state Potts, all with $O(1)$ integer defects), ruling out naive Euler-product prime-locality at the partition function level; Proposition **??** proves the Stieltjes spectral measure is universally signed for all $c > -22/5$; prime-locality, if it holds, must be formulated at the shadow-projection level $\mathrm{Sh}_r(\Theta_{\mathcal{A}})$ rather than the partition function). The *analytic continuation programme* (§**??**) synthesizes these into a single cohomological target: the Hecke defect class $[\delta_p(\mathcal{A})] \in H^1(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}, \mathrm{ad}_{T_p})$ (Definition **??**). Vanishing of $[\delta_p]$ for all $p \Leftrightarrow$ prime-locality $\Leftrightarrow$ Ramanujan (Proposition **??**, Theorem **??**). The scattering coupling factorization (Theorem **??**) and the structural separation theorem (Theorem **??**) prove that the MC equation at genus 1 *exhausts* its algebraic content: every consequence that is extractable from the Rankin–Selberg pairing has been extracted, and the zero-location question requires either higher-genus input or genuinely analytic input external to the MC equation.

(5a) Period filtration and motivic structure. Shadow coefficients are FM configuration-space periods (Proposition **??**). The depth decomposition is the motivic weight decomposition (Remark **??**): d_{arith} counts non-Tate summands $h^1(f_j)$, d_{alg} is the depth of the Tate part. The Virasoro shadow obstruction tower is a single Kummer motive $K(6/c)$ (Remark **??**). The Weil analogy table (Remark **??**) maps the Weil proof to the MC equation: $\Theta = \mathrm{Frob}$, $D^2 = 0 = \mathrm{Frob}^2 = q$, shadow obstruction tower = étale cohomology, bracket positivity = Hodge–Riemann.

(6) Euler–Koszul classification (Definition **??**, Theorem **??**): modular Koszul algebras are *exact Euler–Koszul* ($D_{\mathcal{A}} \equiv 1$; Heisenberg, $V_k()$, $\beta\gamma$), *finitely defective* ($D_{\mathcal{A}} \neq 1$ but defect polynomial; Virasoro, $\mathcal{W}_N$), or *non-Euler–Koszul* (Epstein with class number ≥ 2). Drinfeld–Sokolov reduction sends exact to finitely defective: arithmetically, "Euler product minus finite harmonic defect" (Remark **??**). Shadow depth κ_d and Euler–Koszul class ek are independent invariants (Theorem **??**). The genus-1 modular datum $D_1(\mathcal{A}) = (\kappa, \Delta, K, Q, \kappa_d, \mathrm{ek})$ packages these into a single invariant (Definition **??**).

(5) Sewing–Hecke reciprocity (Theorem **??**): the two-variable L-object $L_{\mathcal{A}}(s,u) := \int_{\mathrm{reg}} F_{\mathcal{A}}^{\mathrm{conn}}\, E^*(\tau,s)\, y^{u-2}\, dx\, dy$ *collapses* to a one-variable object: $L_{\mathcal{A}}(s,u) = \Gamma(v)\,(2\pi)^{-v}\, S_{\mathcal{A}}(v)$ where $v = s + u - 1$. The collapse is exact: it follows from the fact that the sewing operator is diagonal in the Fock basis ($F_{\mathcal{A}}^{\mathrm{conn}}$ depends on y alone), so Rankin–Selberg unfolding converts the modular integral into a Mellin transform. At $u = 0$: Sewing–Selberg formula. At $s = 1$: Dirichlet–sewing lift. Zero-side and prime-side are locked by the constraint $v = s + u - 1$.

(6) Quartic residue programme (§**??**): the 3×3 Schur complement of the mixed (s,u)-transform extracts the quartic contact coupling (potential side) or the quartic resonance section (Gram side); these are reciprocal (Theorem **??**). For Virasoro: $\Sigma_2 = Q_{\mathrm{Vir}}^{\mathrm{ct}} = 10/[c(5c{+}22)]$ (Proposition **??**). The genuine residue kernel (**??**) has decay exponent $-(c{+}\sigma{-}1)/2$ depending on $\sigma = \mathrm{Re}(\rho)$, distinguishing on-line from off-line zeros before any positivity argument. The closure conjecture (Conjecture **??**) asserts that the

intersection of quartic Gram-compatibility loci over all families, slices, central charges, and regulators equals $\{\rho : \mathrm{Re}(\rho) = 1/2\}$. Four structural gaps remain (§**??**): MC → Li (sign; structurally blocked by Proposition **??**), Euler product (spectral measure), bar-cobar rigidity (Euler compatibility), Nyman–Beurling (norm mismatch). Root cause (Remark **??**): all four reduce to the logical independence of integrability ($D^2 = 0$) from positivity (absolute value bounds). Collapse permanence (Proposition **??**, Remark **??**): the interacting sewing kernel factors through $s+t$ at every degree; the one-variable reduction is permanent. Algebraic–analytic divide (Remark **??**): the MC equation organises all algebraic data; analytic conclusions require Deligne, Franc–Mason, or Langlands. The operadic functorial transfer (Remark **??**, route (c)) is the only route that could make MC self-sufficient.

(6a) Genus-2 arithmetic frontier (§**??**): the sewing–Hecke collapse is a genus-1 phenomenon: it rests on Fock diagonality of the sewing operator, which breaks at genus 2 (Theorem **??**). The genus-2 sewing kernel is a 2×2 block operator with off-diagonal coupling through the internal seam; the Fredholm determinant does not factor (Proposition **??**). The spectral decomposition on $\mathrm{Sp}_4(\mathbb{Z})\backslash\mathfrak{H}_2$ includes Saito–Kurokawa lifts from genus-1 cusp forms, providing an arithmetic bridge (Remark **??**). The genus-2 MC equation constrains the Saito–Kurokawa projection through off-diagonal shadow data absent at genus 1; whether this suffices to determine eigenvalues (not only weights) is the central open problem (Remark **??**). This is the precise escape route from the structural separation theorem (Theorem **??**, Remark **??**). The **Böcherer bridge** (Theorem **??**, §**??**) makes this escape explicit: for even unimodular lattices with $\mu(\Lambda) \geq 4$, the genus-2 cusp component is nonzero (Proposition **??**); via the refined Böcherer conjecture (Dickson–Pitale–Saha–Schmidt, proved), this cusp component factors through central L-values $L(1/2, \pi_f \times \chi_D)$, providing the first algebraic access from MC data to the critical line $\mathrm{Re}(s) = 1/2$. Escalation to genus g accesses $L(1/2, \mathrm{Sym}^{g-1} f)$, implementing Serre's programme (Remark **??**). The **Leech χ_{12}-projection** (Theorem **??**): $c_2 \neq 0$, the Leech lattice projects nontrivially onto $\chi_{12} = \mathrm{SK}(f_{22})$ ($\dim S_{12}(\mathrm{Sp}(4,)) = 1$, no ambiguity). Via Waldspurger: MC data determines $L(11, f_{22}\times\chi_D)$ for all fundamental $D < 0$. **Chriss–Ginzburg realization** (Remark **??**): the chain $D^2{=}0 \to \Theta_{\mathcal{A}} \to \Theta^{(2)} \to c_2 \to L$-values is a sequence of projections of the single MC element through the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, the genus-2 three-shell decomposition, and the shadow extraction (Corollary **??**). The genus-2 Beurling kernel (Remark **??**) narrows Gap D: the spectral-support mismatch between sewing and Nyman–Beurling is eliminated at genus 2.

(6b) Finite Miura defect at genus one (§**??**): Theorem **??** proves the vacuum character splitting $\chi_{\mathcal{W}_N} = \chi_{N-1} \prod_{m=1}^{N-1}(1-q^m)^{N-m}$, the connected free energy splitting, and the Dirichlet defect $D^{\mathrm{prime}}_{\mathcal{W}_N} = -\zeta(u{+}1) \sum_{m=1}^{N-1}(N{-}m)m^{-u}$ at three equivalent levels. The finite defect principle (Remark **??**): the Heisenberg core is quartically blind; all quartic resonance data live in the finite defect sector. Defect unification (Conjecture **??**): the prime-side defect (Proposition **??**) and the zero-side defect $D^{\mathrm{res}}_{\mathcal{W}_N}$ (Definition **??**) are both genus-1 shadows of the same bar-intrinsic MC correction.

(6c) Residue clutching defect (§**??**): the proved clutching law (Theorem **??**) is applied to the residue divisor. The residue clutching defect $K_{\mathcal{A},\rho,\xi}$ (Definition **??**) measures whether the residue Gram determinant at a candidate zero ρ satisfies the same boundary compatibility as the MC-derived resonance class. Conjecture **??**: $K = 0$ for all families and boundary strata iff $\mathrm{Re}(\rho) = 1/2$. This is strictly stronger than the quartic closure conjecture and uses the full nonlinear modular structure.

(6d) Beilinson functional (§**??**): the scalar aggregation $\mathcal{B}^{\mathrm{glob}}(\rho) = \sum_{\mathcal{A},v} \int \mathcal{B}_{\mathcal{A},v}\, d\mu_{\mathcal{A}}$ vanishes iff both cubic and quartic residue data match MC shadows on all slices. Closure hierarchy: clutching ⇒ Beilinson ⇒ quartic.

(6d′) The Koszul–Epstein frontier. Whether the extra structure of shadow-metric Epstein zetas (MC

constraints from $D^2 = 0$, bar-cobar symmetry from Theorem B, clutching law from Mok's degeneration formula) forces critical-line zeros is the central open question of the arithmetic programme. The Li/Weil/positivity route is dead: the Nyman–Beurling criterion requires Hilbert-space completeness that the MC data alone does not provide (Theorem **??**). The clutching route (Conjecture **??**) is the surviving path: it uses the full nonlinear modular structure (the boundary stratification of $_g$, the planted-forest corrections, the residue Gram determinant) rather than scalar positivity conditions.

(6e) Miura–packet bridge (§**??**): Proposition **??** splits the $\mathcal{W}_N$ packet connection into a diagonal Heisenberg core and a defect connection. The Heisenberg core is arithmetically inert: diagonal, universal zeta divisors, no contribution to the frontier defect form (Remark **??**). All obstructions to matching automorphic scattering reduce to the finite defect sector.

(7) Gram positivity (Theorem **??**, Theorem **??**): the sewing inner product $\langle f, g\rangle_{\mathcal{A}} = \sum_N a_{\mathcal{A}}(N) f(N) g(N)$ has reproducing kernel $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s+t)$. Its Gram matrix $\mathscr{M} = V^T D V \succ 0$ (Vandermonde with positive sewing weights). Interacting corrections from the shadow spectral measure $\rho = \sum c_j \delta(\lambda_j)$ preserve positivity unconditionally when all atoms $\lambda_j < 0$ (the unitary case, including Virasoro with $c > 0$). The resonance locus is confined to non-unitary $c < 0$.

(8) Shadow resonance locus (Theorem **??**): $\mathcal{R} = \{(c, \alpha) : \det \mathscr{M}^{\mathrm{int}} = 0\}$ is empty for $c > 0$ (unitary); confined to $c < 0$ (non-unitary). Lee–Yang $c = -22/5$ is the first entry.

(9) Motivic weight (Remark **??**): $\mathrm{ek} = 0 \Leftrightarrow$ sewing kernel is pure Tate. $\mathrm{ek} = e \Leftrightarrow$ mixed Tate of depth e. DS reduction = motivic truncation.

(10) Ramanujan bound for lattice eigenforms, Deligne-free (Corollary **??**): twelve-station proof using MC + HS-sewing + CPS + Jacquet-Shalika + Serre. No ℓ-adic cohomology.

(11) Ramanujan bound for rational VOAs (Theorem **??**): extends to diagonal modular invariants via Franc-Mason vector-valued Hecke theory on the character vector.

(12) Ramanujan bound for irrational VOAs (Conjecture **??**): *conditional* on Langlands functoriality for all symmetric powers ($r \geq 5$). Extends to arbitrary central charge via the Roelcke–Selberg spectral theorem as infinite-dimensional Hecke theory. Scope: all chirally Koszul algebras with HS-sewing.

(13) Self-dual factorisation (Theorem **??**): [proved here] for chirally Koszul self-dual $\mathcal{A} \simeq \mathcal{A}^!$ with $Z_{\mathcal{A}} \in M_k(\Gamma_0(N))$, the constrained Epstein zeta factors into $d(\mathcal{A}) - 1$ standard automorphic L-functions. Five-step proof: Verdier involution, functional equation, Hecke eigenform decomposition (multiplicity one), standard L-functions, GRH (conditional). Steps 1–4 unconditional.

(14) Verdier–Hecke commutation (Theorem **??**): [proved here] at genus 1, Verdier duality commutes with Hecke correspondences T_p for $p \nmid N$. Proof via dual isogeny bijection on $_{1,1}$ and sublattice bijection for lattice VOAs.

(15) Theta decomposition bridge (Proposition **??**): [proved here] for minimal models (p, q) with diagonal modular invariant, characters decompose via Rocha–Caridi into theta functions at level $2pq$, giving Dirichlet L-functions modulo $2pq$. Resolves the non-lattice gap (Remark **??**) for all rational VOAs.

(16) Newton–shadow–Hecke correspondence (Proposition **??**): [proved here] shadow power sums $p_r = r \cdot S_r$ determine Hecke eigenvalues via Newton's identities. The MC equation at degree 4 is structurally parallel to the Hecke relation $T_2^2 = T_4 + p^{k-1} T_1$.

(17) Davenport–Heilbronn principle (Remark **??**): the Davenport–Heilbronn phenomenon (off-line zeros of non-unimodular Epstein zeta) is explained by failure of Koszul self-duality: $h(D) > 1$ iff $V_\Lambda \not\simeq V_{\Lambda^*}$ iff the Verdier involution does not fix Z.

(18) Arithmetic packet connection (Definition **??**; Theorem **??**; Proposition **??**): the depth decomposition $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ is realised as a flat meromorphic connection $\nabla^{\mathrm{arith}}_{\mathcal{A}}$ on the arithmetic packet module $M_{\mathcal{A}}$ over the spectral line. The singular divisor $D_{\mathcal{A}} = \bigcup_{\chi} \mathrm{div}(\Lambda_{\chi})$ is independent of the nilpotent data: the arithmetic skeleton controls the poles; the algebraic defect contributes only unipotent monodromy. The frontier defect form $\Omega_{\mathcal{A}} = d \log \Lambda_{\mathrm{Eis}} - d \log \varphi$ measures the discrepancy between the activated L-packets and the automorphic scattering matrix; the gauge criterion (Proposition **??**) reduces scattering access to exactness of $\Omega_{\mathcal{A}}$. The arithmetic comparison conjecture (Conjecture **??**) posits that higher-genus MC data computes residues of $\Omega_{\mathcal{A}}$ at scattering poles.

(19) Classical sewing-to-zeta bridge The Jacobi theta function $\vartheta(t) = \sum_{n\in} e^{-\pi n^2 t}$ is the zero-mode lattice sum of the rank-one self-dual lattice VOA V. Its Mellin transform, after clutching symmetrisation, is the completed Riemann zeta function: $\Xi_{\mathrm{ann}}(s) := \frac{1}{2} + \frac{1}{2}s(s-1)\int_1^{\infty} \psi(t)(t^{s/2-1} + t^{(1-s)/2-1})\, dt = \xi(s)$, where $\psi(t) = (\vartheta(t) - 1)/2$. This is Riemann's 1859 computation: split the Mellin integral at $t = 1$, apply Poisson summation $\vartheta(1/t) = t^{1/2}\vartheta(t)$ on the $(0, 1)$ piece, and collect into the manifestly entire, $s \leftrightarrow 1-s$ symmetric form. Within the open/closed framework, ϑ arises as the scalar annular resonance trace of the cyclic open factorization category of V on $(\mathbb{P}^1, \{0, \infty\})$, after determinant-line trivialisation of the oscillator sector (i.e., passing from the full character $q^{-1/24}/\eta \cdot \sum q^{n^2/2}$ to the zero-mode lattice sum alone). The clutching symmetry $\Xi(s) = \Xi(1-s)$ reflects the Poisson self-duality of the -lattice, which is the abelian instance of Koszul self-duality. The nontrivial zeros of Ξ_{ann} are exactly those of $\zeta(s)$. *This identity is classical and does not constitute a new result of the programme.* The genuine frontier is the nonlinear completion: whether the modular shadows $\Theta^{(\geq 4)}_{\mathcal{A}}$ from $\mathcal{W}$-type theories deform zero statistics.

(20) Entanglement-annulus conjecture[conjectured] For a cyclic open factorization dg-category C_{op} on a tangential log curve with modular twisting element Θ_C, the entanglement entropy should admit a connected shadow expansion: $S_{\mathrm{EE}}(C_{\mathrm{op}}) = \frac{c}{3}\log(L/\epsilon) + \sum_{r\geq 3} \Delta S_r$, where the scalar term is the Calabrese–Cardy contribution (already proved from $\kappa = c/2$ alone, Chapter **??**), and ΔS_r is the degree-r connected annular clutching coefficient of Θ_C. The four-class partition $(\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M})$ classifies entanglement corrections by shadow depth. Three unproved hypotheses: (a) replica annuli A_n realised inside the log-modular cooperad, (b) cyclic trace extends to the replica family, (c) factorisation of disconnected configurations. For $\mathcal{W}_3$, the first nonlinear correction $\Delta S_4^{\mathrm{mod}}(\mathcal{W}_3)$ is forced by the composite $\Lambda =: TT : -\frac{3}{10}\partial^2 T$ and should be computable from $R^{\mathrm{mod}}_{4,\mathrm{ann}}(\mathcal{W}_3)$. The complementarity theorem (Theorem C) constrains the higher-degree shadows of $\mathcal{A}$ and $\mathcal{A}^!$ via the Verdier involution, but the constraint is *not* $\Theta_{\mathcal{A}} + \Theta_{\mathcal{A}^!} = 0$: at the scalar level, $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$, and the full-tower relation is $\mathbb{D}(\Theta_{\mathcal{A}}) = \Theta_{\mathcal{A}^!}$ (Verdier intertwining, not negation). The complementarity sum $S^{\mathrm{scalar}}_{\mathrm{EE}}(\mathrm{Vir}_c) + S^{\mathrm{scalar}}_{\mathrm{EE}}(\mathrm{Vir}_{26-c}) = \frac{26}{3}\log(L/\epsilon)$ follows from the proved $\kappa + \kappa' = 13$ identity, not from a conjectural $\Theta + \Theta' = 0$.

(21) Canonical cyclotomic boundary chart from root stacks

The boundary algebra $A_b = \mathrm{RHom}_{\mathcal{M}}(b, b)$ (Theorem **??**(i)) depends on the choice of boundary vacuum b. Inner-automorphism approaches to constructing boundary charts suffer from a σ-ambiguity: conjugation by $\sigma \in \mathrm{Aut}(V_k)$ produces a gauge-equivalent but non-canonical chart. Root stacks eliminate this ambiguity by making the branching data geometric.

The N-th root stack at a puncture. [proved here] Let (X, D, τ) be a tangential log curve (Definition **??**) with $p \in D$ a puncture and t a local coordinate at p ($t(p) = 0$). For $N \geq 1$, the *N-th root stack* of X at p is the Deligne–Mumford stack

$$X_{p,N} \;:=\; \sqrt[N]{p/X} \;\cong\; \big[\mathrm{Spec}[[u]] \,/\, \mu_N\big] \;\sqcup\; (X \setminus \{p\}), \tag{1.6}$$

where $u^N = t$ is the tautological N-th root coordinate and μ_N acts by $\zeta \cdot u = \zeta u$ for $\zeta \in \mu_N = \langle e^{2\pi i/N}\rangle$. The coarse moduli space of $X_{p,N}$ is X itself, with the stacky structure concentrated at the single point p.

The structure morphism $\pi_N : X_{p,N} \to X$ is an isomorphism away from p and a μ_N-gerbe at p. The orbifold line bundle $\mathcal{L}_p^{1/N} := \pi_N^* \mathcal{O}_X(p)^{1/N}$ on $X_{p,N}$ has the property that $(\mathcal{L}_p^{1/N})^{\otimes N} \cong \pi_N^* \mathcal{O}_X(p)$; its space of sections carries a canonical μ_N-eigenspace decomposition indexed by $j \in /N$. This is standard algebraic geometry (Cadman 2007; Abramovich–Graber–Vistoli 2008).

The cyclotomic boundary chart. [conjectured] Let $\mathcal{A}$ be a chiral algebra on X with a module category $\mathcal{M}$ supported at p. The pullback $\pi_N^* \mathcal{A}$ is a μ_N-equivariant chiral algebra on the root stack, and the boundary vacuum acquires a canonical $/N$-grading:

$$b_N^{\mathrm{cyc}} := \bigoplus_{j \in /N} \mathcal{L}_p^{j/N} \otimes \pi_N^* \mathcal{M}_0,$$

where $\mathcal{M}_0$ is the vacuum object of $\mathcal{M}$ and $\mathcal{L}_p^{j/N} = (\mathcal{L}_p^{1/N})^{\otimes j}$. The *cyclotomic boundary chart at order N* is the endomorphism algebra

$$A_N^{\mathrm{cyc}} := \mathrm{RHom}_{X_{p,N}}(b_N^{\mathrm{cyc}}, b_N^{\mathrm{cyc}}). \tag{1.7}$$

This is a $/N$-graded A_∞-algebra whose grading is tautological: the μ_N-action on $X_{p,N}$ induces a $/N$-grading on morphism spaces by eigenvalue decomposition. The graded components carry:

(a) *sector idempotents* $e_j \in A_{N,(0)}^{\mathrm{cyc}}$, one for each $j \in /N$;

(b) *shift morphisms* $s_j : e_j \to e_{j+1}$ in $A_{N,(1)}^{\mathrm{cyc}}$, satisfying the *nilpotent N-cycle relation*

$$s_{N-1} \circ \cdots \circ s_1 \circ s_0 = t \cdot \mathrm{id}_{e_0}, \tag{1.8}$$

which vanishes on the special fibre $t = 0$;

(c) *fractional-mode expansions*: fields $\phi(z)$ of the pullback $\pi_N^* \mathcal{A}$ admit mode expansions in $u = t^{1/N}$, giving fractional modes $\phi_{m/N}$ for $m \in$.

For $N = 1$, this recovers the ordinary (ungraded) boundary chart $A_b = \mathrm{RHom}(b, b)$ of Theorem **??**. The nilpotent N-cycle (1.8) forces the open-sector category $C_N^{\mathrm{op}} := \mathrm{Perf}(A_N^{\mathrm{cyc}})$ to be nonsemisimple for $N > 1$: the shift morphisms generate non-split extensions between sector objects. The construction eliminates the σ-ambiguity of inner-automorphism approaches: N and the μ_N-action are geometric data of the root stack, not choices of automorphism.

(21a) Specialisation to admissible levels. [conjectured] For $\mathcal{A} = V_k()$ at admissible level $k = -h^\vee + p/q$ with p, q coprime, the canonical choice of root stack order is $N = q$ (the denominator of the shifted level $k + h^\vee$). The q-root stack $D^{(q)} := \sqrt[q]{p/X} \cong [\mathrm{Spec}[[u]]/\mu_q]$ with $u^q = t$ then provides the cyclotomic boundary chart $A_{k,q}^{\mathrm{cyc}}$ with a $/q$-graded quasi-lisse (QLS) structure. The denominator q is geometrised: it is the order of the stacky isotropy group at p, not an external combinatorial parameter. Logarithmicity of the representation category $\mathcal{O}_k$ at admissible k is encoded in the root-boundary geometry of the q-fold cover. Chiral quadratic duality (Gui–Li–Zeng) applied to the QLS chart produces a canonical q-periodic twisted pair and CDG algebra. Comparison to the nonsemisimple KL category is proved for $= \mathfrak{sl}_2$ (McRae–Yang; Creutzig–McRae–Yang); for general it is a comparison theorem to be proved.

(21b) Affine Weyl orbit periodicity for Kac–Moody algebras. [conjectured] For $\mathcal{A} = V_k(\mathfrak{sl}_N)$ at admissible level $k = -N + p/q$, the affine Weyl group $\mathcal{W} = W \ltimes Q^\vee$ acts on the set of admissible weights Adm_k with quotient by $qQ^\vee$-translations (Theorem **??**). The q-root stack provides a geometric realisation of this quotient periodicity: the $/q$-grading on $A_{k,q}^{\mathrm{cyc}}$ corresponds to the coset decomposition $Q^\vee/qQ^\vee \cong (/q)^r$ (where $r = \mathrm{rank}()$), and the fractional modes $J_{m/q}^a$ of the affine currents generate the q-block decomposition of the admissible category $\mathcal{O}_k^{\mathrm{adm}}$. At integer level ($q = 1$), the root stack is trivial and the boundary

chart is ungraded: the integrable category $\mathcal{O}_k^{\mathrm{int}}$ is semisimple with no block structure, consistent with the absence of cyclotomic periodicity.

The root stack does *not* recover the full affine Weyl group $\mathcal{W}$: it captures only the finite quotient $Q^\vee/qQ^\vee$ that controls the periodicity of Weyl orbits on admissible weights. The infinite part of $\mathcal{W}$ (the lattice $Q^\vee$ itself) acts on the Verma-module labels within each block and is visible in the BGG resolution (Theorem **??**), not in the cyclotomic grading.

(21c) Virasoro self-duality and the second root stack. [conjectured] The Koszul involution $c \mapsto 26 - c$ for the Virasoro algebra is a $/2$-symmetry of the family $\{\mathrm{Vir}_c : c \in\}$, with fixed point $c = 13$ (uncurved-quadratic self-duality; see for the distinction from the Feigin–Frenkel involution at $c = 0$). At $c = 13$, $\mathrm{Vir}_{13}^! \cong \mathrm{Vir}_{13}$ and the complementarity asymmetry $\delta_\kappa = \kappa - \kappa' = c/2 - (26 - c)/2 = c - 13$ vanishes. We conjecture that this family-level $/2$-symmetry admits a geometric realisation via the second root stack:

$$X_{p,2} = \sqrt[2]{p/X} \cong [\mathrm{Spec}[[u]]/\mu_2], \qquad u^2 = t,$$

with the μ_2-action exchanging the two sheets of the double cover. At $c = 13$, the cyclotomic boundary chart A_2^{cyc} should carry a $/2$-grading whose even part is the self-dual boundary algebra and whose odd part encodes the Koszul pairing $\langle \mathcal{A}, \mathcal{A}^! \rangle = \langle \mathcal{A}, \mathcal{A} \rangle$. The shift morphism $s_0 \colon e_0 \to e_1$ intertwines the algebra with its dual; the nilpotent relation $s_1 \circ s_0 = t \cdot \mathrm{id}$ degenerates on the special fibre, reflecting the passage from the complementarity pairing to its degeneration at the self-dual point.

At $c \neq 13$: the $/2$-symmetry is broken. The two sheets Vir_c and Vir_{26-c} are genuinely distinct algebras, and the cyclotomic boundary chart splits into two non-isomorphic sectors. The complementarity asymmetry $\delta_\kappa = c - 13 \neq 0$ measures the failure of the μ_2-equivariant structure.

Scope fence. The Virasoro $/2$-symmetry is the Koszul involution on the *family* $\{c \mapsto 26 - c\}$, not a symmetry of a single algebra at generic c. The root-stack realisation posits that this family-level involution admits a geometric incarnation as a stacky double cover at the self-dual central charge. This is analogous to the admissible-level story, where the root stack order q comes from the rational structure of the level; here, the order 2 comes from the quadratic structure of the Koszul duality relation $c + c' = 26$.

(21d) The cyclotomic Zhu algebra and Frenkel–Zhu theory. [conjectured] The Zhu algebra $A(V) = V/V \circ V$ (Zhu [408]; Frenkel–Zhu [180]) controls the representation theory of a vertex algebra V: the functor $M \mapsto M/M_{>0}$ defines a bijection between simple V-modules and simple $A(V)$-modules. For $V = V_k()$, $A(V_k()) \cong U()$ (Frenkel–Zhu [180]).

The root-stack construction refines the Zhu functor as follows. At level k with N-th root stack at p, the pullback $\pi_N^* V_k$ admits fractional modes and the Zhu quotient inherits a $/N$-grading:

$$A_N^{\mathrm{cyc}}(V_k) \;:=\; \bigoplus_{j \in /N} A(\pi_N^* V_k)_{(j)},$$

where $A(\pi_N^* V_k)_{(j)}$ consists of Zhu equivalence classes of vectors whose μ_N-eigenvalue is $e^{2\pi i j/N}$. For $N = 1$: $A_1^{\mathrm{cyc}}(V_k) = A(V_k) \cong U()$, ungraded. For $N = q$ at admissible $k = -h^\vee + p/q$: the grading refines $U()$ into q sectors, and the representation theory of the graded algebra $A_q^{\mathrm{cyc}}(V_k)$ should recover the q-block structure of the admissible category $\mathcal{O}_k^{\mathrm{adm}}$.

The conjectured compatibility between the cyclotomic Zhu algebra and the cyclotomic boundary chart is the assertion that the natural map $A_N^{\mathrm{cyc}}(V_k) \to H^0(A_N^{\mathrm{cyc}})$ (from the graded Zhu algebra to the degree-zero cohomology of the boundary chart) is an isomorphism of $/N$-graded algebras. For $N = 1$, this reduces to the standard identification $A(V_k) \cong H^0(A_b)$ that underlies the Zhu functor. For $N > 1$, it asserts that the root-stack construction is compatible with the Zhu truncation.

Scope fence. The cyclotomic Zhu algebra $A_N^{\mathrm{cyc}}(V_k)$ is well-defined for all N and all levels k (the Zhu quotient and the μ_N-eigenspace decomposition commute), but the representation-theoretic content is non-trivial only when N matches the arithmetic of the level. At integer level with $N > 1$, the grading exists but does not capture new representation-theoretic data beyond what the ungraded Zhu algebra already sees. The conjecture is that $N = q$ (the level denominator) is the minimal order for which the cyclotomic refinement captures genuinely new structure (the block decomposition of the admissible category).

(21e) Periodic curved dg structure at admissible levels[conjectured] At admissible level $k = -h^\vee + p/q$ with p, q coprime positive integers ($p \geq 2$), the simple quotient $L_k()$ has a non-free PBW filtration: vacuum null vectors at conformal weight pq force relations in the associated graded. The bar complex $\bar{B}^{\mathrm{ch}}(L_k())$ should carry a *periodic curved dg structure*: the bar degree is $2p$-periodic (Conjecture **??**), and the curvature element $m_0 \in \bar{B}^{\mathrm{ch}}(L_k())$ (which vanishes for the universal algebra $V_k()$ by Koszulness, but is nonzero for the simple quotient) is nilpotent with period controlled by the numerator:

$$\overline{B}^{n+2p}(L_k()) \cong \overline{B}^{n}(L_k()) \quad \text{for all } n \gg 0.$$

The periodicity parameter is $2p$, where p is the numerator of $k + h^\vee = p/q$; this matches the periodicity of the small quantum group $u_q()$ at $q = e^{\pi i q/p}$ (Conjecture **??**). The curved dg structure is indexed by the μ_q-grading of the cyclotomic chart from items (21)–(21d) above.

Meaning of periodic CDG. A curved dg coalgebra (C, d, m_0) with $d^2 = [m_0, -]$ is *periodic of period N* if $C^{n+N} \cong C^n$ as CDG-coalgebras for all $n \gg 0$. Equivalently, in Positselski's framework [350], the coderived category $D^{\mathrm{co}}(C)$ reduces to the stable category of a finite-dimensional quotient. For the universal algebra $V_k()$, chirally Koszul at all levels (Proposition **??**), the bar complex is uncurved ($m_0 = 0$): no periodicity. For the simple quotient $L_k()$ at admissible k, the bar complex acquires curvature from the vacuum null vector, and the conjecture asserts this curved complex is $2p$-periodic in bar degree.

The case $= \mathfrak{sl}_2$. At $k = -2 + p/q$, the simple quotient $L_k(\mathfrak{sl}_2)$ is an admissible-level affine vertex algebra with central charge $c = 1 - 6(p-q)^2/(pq)$; its Drinfeld–Sokolov reduction is the Virasoro minimal model (p, q). The vacuum null vector in $L_k(\mathfrak{sl}_2)$ appears at conformal weight pq. Comparison to external constructions:

(a) Feigin–Tipunin (2010): the logarithmic extension (p, q) has a $/q$-graded nonsemisimple module category with q blocks, each a non-split extension of $p - 1$ simple modules. The μ_q-grading of the cyclotomic chart (item (21a)) should recover this block decomposition; the $2p$-periodicity of the bar complex acts on each block.

(b) Creutzig–Ridout [116]: the Kazhdan–Lusztig-type category for $\mathfrak{sl}_2$ at admissible level has a ribbon structure whose twist eigenvalues are roots of unity. The CDG periodicity should intertwine with this ribbon structure through the cyclotomic boundary chart.

These comparisons are *verified only for* $= \mathfrak{sl}_2$: the block structure and ribbon periodicity are explicitly computed in the cited works. For general , the Kazhdan–Lusztig category at admissible level is constructed (Arakawa [17]; Creutzig–McRae–Yang), but the explicit CDG periodicity of the bar complex and its comparison to the categorical μ_q-grading are not proved.

Overall scope fence for items (21)–(21e). Four levels of the cyclotomic programme:

(i) The root stack construction (1.6) is standard algebraic geometry ([proved here]).

(ii) The cyclotomic boundary chart (1.7) and its $/N$-grading are conjectural as boundary algebras for chiral algebras ([conjectured]).

(iii) The admissible-level specialisation, the Kac–Moody Weyl orbit periodicity, the Virasoro self-duality at $c = 13$, and the cyclotomic Zhu algebra are conjectural ([conjectured]).

(iv) The periodic CDG structure (Conjecture **??**) is the last layer: it asserts that the cyclotomic grading of the boundary chart is reflected in the periodicity of the bar differential on the simple quotient.

The construction of items (21a)–(21e) applies to the *simple quotient* $L_k()$, not the universal algebra $V_k()$. The universal algebra is chirally Koszul at all levels and its bar complex is uncurved. The periodic CDG structure is the bar-complex manifestation of logarithmic CFT: the non-semisimplicity of the representation category is encoded in the non-vanishing but nilpotent curvature of the bar differential.

(22) Modular globalisation conjecture and quartic first obstruction[conjectured] Given the genus-zero skeleton $S_0(A_b) = (A_b,\ C^*_{\mathrm{ch}}(A_b, A_b),\ A^!,\ r(z))$ of a boundary A_∞-chiral algebra (incorporating boundary chart, intrinsic bulk, line Koszul dual, and binary line-OPE kernel), there should exist a filtered dg Lie algebra $_X := \mathrm{Conv}(C^{\mathrm{oc,log}}_{\mathrm{mod}}(X, D, \tau),\ \mathrm{End}(Z_{\mathrm{ch}}(C_{\mathrm{op}}), A_b))$ such that modular completions of $S_0(A_b)$ are precisely MC elements $\Theta_C \in \mathrm{MC}(_X)$, with the genus-zero part recovering $S_0(A_b)$. The first obstruction to modular globalisation lies in quartic filtration and is represented by the image of the modular quartic resonance class $[R^{\mathrm{mod}}_{4,g,n}]$ in $H^2(\mathrm{gr}^4_F\ _X)$. This follows because cubic gauge triviality (Theorem **??**) removes the cubic obstruction: the first cohomologically nontrivial shadow is quartic.

(23) Eight discard criteria Any proposed extension of the spectral continuation programme (items 1–18 above) that violates one of the following criteria is false:

(D1) Does not recover Beilinson–Drinfeld complementarity.
(D2) Treats bar-cobar or HTT as arithmetic/spectral rigidity (bar-cobar is homotopical, not spectral-positive).
(D3) Ignores Miura injectivity (Arakawa).
(D4) Proposes an un-regularised, un-factorisation-compatible, or non-line-valued universal shadow.
(D5) Places quartic rigidity in the Heisenberg core (Heisenberg is arithmetically transparent but quartically blind).
(D6) Compares residues only as scalars, ignoring clutching (the quartic resonance class is divisor-valued).
(D7) Relies on the ζ-proxy that was shown to fail the safe-point test.
(D8) Seeks positivity on the modular *curve* without a polarised geometric lift to a modular *surface* (the Rankin–Selberg integral on a curve is not sign-preserving).

The MC equation is a *flatness/integrability* condition; it does not imply Li-type *positivity* without additional structure (a real form, a polarisation, or a positive trace). The correct statement is: bar-cobar + Euler compatibility + surface positivity $\Rightarrow$ critical-line rigidity for the associated genus-1 L-object.

1.22 Three falsification tests

The five main theorems and the MC element $\Theta_{\mathcal{A}}$ rest on three structural pillars: the local Swiss-cheese universal property, the Drinfeld–Sokolov homotopy transfer to Virasoro, and the convergence of the shadow obstruction tower. Each pillar admits a decisive test. If any one of these tests fails, the corresponding component of the programme is wrong and must be abandoned or rebuilt. We state each test, its failure mode, and the current evidence.

(a) Local Swiss-cheese universal property for Heisenberg.

Test. Compute the chiral Hochschild cochain complex $C^\bullet_{\mathrm{ch}}(_k, _k)$ for the Heisenberg algebra $_k$ and verify the universal open/closed pair $U(_k) = (C^\bullet_{\mathrm{ch}}(_k, _k),\ _k)$ of Theorem **??**. The Swiss-cheese structure on the pair should reproduce the Heisenberg OPE: $\alpha(z)\,\alpha(w) \sim k/(z-w)^2$.

Failure mode. If $C^\bullet_{\mathrm{ch}}(_k, _k)$ does not carry the expected Swiss-cheese $\mathrm{SC}^{\mathrm{ch,top}}$ structure, or if the recovered OPE disagrees with the standard Heisenberg OPE, then the local architecture of the open/closed realization

(§1.20) is wrong. The derived center would fail to be the universal bulk for the simplest possible boundary algebra, and the entire four-stage open/closed programme (§1.20) would collapse at stage 1.

Current evidence. The test passes. Since $_k$ has abelian Lie conformal algebra $\mathfrak{h}_k$, the Chevalley–Eilenberg complex is formal: $C^*_{\mathrm{Lie}^*}(\mathfrak{h}_k) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{h}_k^*[1])$. The chiral Hochschild cohomology (Theorem **??**) is $\mathrm{ChirHoch}^*(_k) \cong k[\Theta]$ with $|\Theta| = 2$, recovering the unique nontrivial outer derivation $D(\alpha) = \mathbf{1}$. The Swiss-cheese operations $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$ (class G: Gaussian, shadow depth 2), confirming that $_k$ is Swiss-cheese formal.

(b) DS–HPL transfer to Virasoro.

Test. The Drinfeld–Sokolov reduction functor, applied to the affine $\widehat{\ }_2$ dg-shifted Yangian data $(m_2, r(z), \Delta)$ via homological perturbation, should produce a well-defined Virasoro dg-shifted Yangian at the chain level (Vol II, Theorem thm:ds-hpl-transfer). The transferred r-matrix $r^{\mathrm{Vir}}(z)$ must satisfy the classical Yang–Baxter equation. The transferred product m_2^{Vir} must be associative up to homotopy controlled by m_3.

Failure mode. If the transferred r-matrix fails the CYBE, or if the transferred product fails A_∞ coherence, then the gravity bridge is broken: the passage from Kac–Moody dg-shifted Yangians (where Koszul duality is well understood) to Virasoro dg-shifted Yangians (where the gravitational interpretation lives) does not preserve the algebraic structure. The DS reduction would not be a morphism of modular Koszul datums, and the shadow cascade $\Theta_{\mathcal{W}_N} = \mathrm{DS}(\Theta)$ would fail to hold at the chain level.

Current evidence. The test passes at the binary level. The gravitational coproduct Δ_z^{grav} is proved strictly primitive at all degrees (ghost-number obstruction kills every HPL tree correction). The transferred binary r-matrix inherits the CYBE from the affine source by the intertwining property of the SDR (strong deformation retract). The A_∞ coherence of m_2^{Vir} follows from the HTT (homotopy transfer theorem) applied to the DS contraction. At higher degrees, the transfer is consistent with the shadow cascade: the computed cubic and quartic shadows of Vir_c match those obtained by DS reduction from $\widehat{\ }_2$ at $c = 1 - 6(k-1)^2/k$. The residual gap: a complete chain-level verification of all A_∞ operations m_n^{Vir} for $n \geq 4$ has not been carried out.

(c) Shadow obstruction tower convergence for Virasoro.

Test. The shadow growth rate $\rho(\mathrm{Vir}_c)$ (Definition **??**, Theorem **??**) must satisfy $\rho < 1$ for $c > c^*$, where c^* is the critical cubic root $5c^3 + 22c^2 - 180c - 872 = 0$, $c^* \approx 6.125$. If $\rho \geq 1$ for all c, the shadow obstruction tower diverges everywhere and the MC element Θ_{Vir} does not exist as a convergent object.

Failure mode. If $\rho(\mathrm{Vir}_c) \geq 1$ for all values of c, then the nonperturbative shadow partition function $Z^{\mathrm{sh}} = \sum_{g,r} \mathrm{Sh}_r^{(g)}\, \hbar^{2g}\, t^r$ diverges at every point of the moduli space. The shadow obstruction tower would be a formal power series with zero radius of convergence, and the modular MC element would exist only as a pro-object, not as an honest element of the completed convolution algebra. The genus expansion $F_g(\mathrm{Vir}_c) = \kappa \cdot \lambda_g^{\mathrm{FP}}$ would remain valid (it uses only $\Theta_{\mathcal{A}}^{\leq 2}$), but the full shadow obstruction tower and all higher-degree invariants would be purely formal.

Current evidence. The test passes. The shadow growth rate is $\rho = \sqrt{9\alpha^2 + 2\Delta}\,/\,(2|\kappa|)$, where $\alpha = S_3/\kappa$ and $\Delta = 8\kappa S_4$ are the cubic and quartic shadow invariants. For Virasoro: $\rho(\mathrm{Vir}_c) < 1$ for all $c > c^*$ (Theorem **??**); in particular, $\rho(\mathrm{Vir}_{13}) \approx 0.467$ at the self-dual point $c = 13$, and $\rho \to 0$ as $c \to \infty$. For $c < c^*$, the tower diverges ($\rho > 1$), but Borel summability holds (Proposition **??**): the shadow coefficients satisfy the Carleman condition, so the formal series determines a unique analytic function by Borel resummation. The nonperturbative programme is therefore alive for all c, with two regimes: convergent summation ($c > c^*$) and Borel summation ($c < c^*$). The critical value $c^* \approx 6.125$ separates the two regimes but does not obstruct either.

(5) Analytic structure and resurgence (§??–??): The scalar genus series has the closed form $\kappa\big((\hbar/2)/\sin(\hbar/2) - 1\big)$ (Proposition **??**), meromorphic in $\hbar$ with simple poles at $2\pi n$. The Borel transform in $u = \hbar^2$ is entire (Theorem **??**); Stokes multipliers are $\mathfrak{S}_n = (-1)^n 4\pi^2 n\kappa\, i$ (Proposition **??**); the trans-series has instanton action $A = (2\pi)^2$ (Theorem **??**). The Liouville-transformed Schrödinger equation has potential $V = 8\kappa^2\Delta/Q_L^2$ (Proposition **??**), with three regular singular points for class M (Corollary **??**). The one-loop WKB correction is the shadow connection (Proposition **??**(ii)); the classical Voros half-period is universally π (Proposition **??**). Multi-channel shadows produce Painlevé VI (Remark **??**). Status: all results GREEN (proved).

Conjecture 1.22.1 *(MC equation as conformal bootstrap).* [conjectured] The genus-0 degree-4 projection of the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta,\Theta] = 0$ in the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ (Definition **??**) is structurally identical to the crossing symmetry constraint of the conformal bootstrap. Specifically:

(i) The degree-4 MC equation $d_2\Theta_4 + \frac{1}{2}[\Theta_2,\Theta_2] = 0$ on the genus-0 locus $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ decomposes into three channels (s,t,u) corresponding to the three boundary divisors of $\overline{\mathcal{M}}_{0,4}$, and the resulting identity is crossing symmetry for the four-point function.

(ii) The shadow metric $Q_L(t) = q_0 + q_1 t + q_2 t^2$ (Definition **??**) satisfies $Q_L \geq 0$ on the physical domain for unitary theories. This positivity is the shadow-level manifestation of the unitarity bound: the positive semi-definiteness of the Zamolodchikov inner product on the space of intermediate states.

The conjecture does not claim that the MC equation reproduces the full numerical bootstrap programme (which requires optimisation over the infinite-dimensional space of CFT data), but rather that the algebraic *structure* of crossing symmetry and the analytic *mechanism* of unitarity bounds emerge as projections of the single equation $D\Theta + \frac{1}{2}[\Theta,\Theta] = 0$.

Remark 1.22.2 *(Evidence).* The structural identification is verified at the level of the degree-4 MC equation in §**??**: the three-channel decomposition of $[\Theta_2,\Theta_2]$ matches the *s*-*t*-*u* channel decomposition of the four-point crossing equation, and the quartic resonance class $Q^{\mathrm{ct}} = 10/(c(5c+22))$ (Theorem **??**) satisfies the positivity constraint $Q^{\mathrm{ct}} > 0$ for all $c > 0$. The shadow metric $Q_L > 0$ for all unitary theories in the standard landscape (Remark **??**). Computational support from the engine `modular_shadow_tower` (104 tests).

April 2026 session: falsification test results. Five additional falsification tests were evaluated against this session's structural corrections (AP165 operadic scoping, SC self-duality retraction, topologization scoping, critical-level Theorem H failure).

- **FT-2** (bar complex cooperad structure): *passed*. The bar complex $\overline{B}(\mathcal{A})$ is confirmed as an E_1 coassociative coalgebra over $(\mathrm{ChirAss})^!$, not an -coalgebra (AP165).
- **FT-4** (Theorem H generic-level requirement): *passed*. At critical level $k = -h^\vee$, $CH^0(V_{-h^\vee}())$ is infinite-dimensional (Feigin–Frenkel center); Theorem H does not apply. For $\widehat{\ }_2$ at critical level, CH^* is 4-periodic and unbounded.
- **FT-5** (topologization scope): *passed*. Cohomological topologization (+conformal vector $\Rightarrow E_3^{\mathrm{top}}$ on BRST cohomology) is proved for affine KM at non-critical level; the chain-level lift is conjectural in general.
- **FT-9** (SC duality): *partial pass*. SC self-duality is retracted: $()^! \cong$ (Lie, Ass, shuffle-mixed) $\ncong$. The duality functor is an involution (passed); self-duality of the operad is false (corrected).

- **FT-10** (kernel of averaging): *passed*. Closed formula $\dim\ker(\mathrm{av}_n) = n! - \binom{n+d-1}{n}$ verified via Schur–Weyl (Proposition **??**).

1.23 Programme status ledger

The following table classifies every major result and programme by proof status. Green: proved within this monograph (local algebra, all hypotheses discharged). Amber: externally supported or conditional (physically motivated, with partial evidence, but dependent on conjectural inputs or physical interpretation). Red: unproved global or modular descent (genuine open problems with no current proof strategy reducing to proved results).

Status	Result
GREEN (proved local algebra)	
	Theorems A–H (bar-cobar adjunction, inversion, complementarity, modular characteristic, Hochschild)
	MC1, MC2, MC4, and the analytic/coderived lanes of MC5 (PBW, bar-intrinsic MC, strong completion towers, HS-sewing, coderived BV=bar)
	MC3 proved for all simple types on the evaluation-generated core; DK-4/5 (extension beyond evaluation modules) downstream
	Koszulness programme: 9 unconditional equivalences (with factorization homology at genus 0) + 1 proved consequence (Hochschild on the Koszul locus) + 1 conditional (Lagrangian) + 1 one-directional (D-module purity) (Theorem **??**)
	Shadow obstruction tower algebraicity: $H(t)^2 = t^4 Q_L(t)$, tower determined by (κ, α, S_4) (Theorem **??**); cubic coefficient $\alpha = S_3/\kappa$ universal per family: $\alpha = 0$ (Heis), $\alpha = 2$ (Vir), family-specific for $\mathcal{W}_N$ and affine KM
	κ formula universality: $\kappa(\mathcal{H}_k) = k$, $\kappa(\widehat{\mathcal{H}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$, $\kappa(\mathrm{Vir}_c) = c/2$, $\kappa(\mathcal{W}_N) = c(H_N-1)$
	$F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ for uniform-weight modular Koszul algebras at all genera (Theorem **??**)
	Swiss-cheese theorem (Theorem **??**)
	Chiral derived center as universal bulk
	Annulus trace (Theorem **??**)
	Heisenberg and lattice sewing (Theorems **??**, **??**)
	HS-sewing for the entire standard landscape (universal algebras) (Theorem **??**)
	All-degree inverse limit $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$ (Theorem **??**)
	Algebraic-family rigidity (Theorem **??**)
	DS primitivity of the gravitational coproduct (Vol II, Theorem thm:ds-hpl-transfer)
	Epstein zeta of the shadow metric: functional equation for class **M** at rational c (Theorem **??**)
	Shadow field $K_L = (\sqrt{\mathrm{disc}(Q_L)})$: imaginary quadratic for class **M**; trivial for class **G**/**L** (Remark **??**)
	Descent fan: three independent projections of $\Theta_{\mathcal{A}}$ (categorical, spectral, modular) (Proposition 1.23.1)
	Lattice fan closure: all three projections compatible for lattice VOAs (§1.23)
	SC-formality $\iff$ class **G** (Proposition **??**)
	Algebraic depth gap: $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ (Proposition **??**)
	$CH^1(V_k(\mathfrak{g})) \cong$ at generic level (Proposition **??**)
	E_1 non-splitting at genus 1 (Proposition **??**)
	Free-field exactness: $\delta F_g^{\mathrm{cross}} = 0$ for all free-field families (Proposition **??**)
	Virasoro shadow growth rate from Riccati (Proposition **??**)
	Gerstenhaber bracket on $CH^*(\widehat{\mathcal{H}}_2)$ (Proposition **??**)
	DS–ChirHoch compatibility (Proposition **??**)
	Kernel of averaging: $\dim \ker(\mathrm{av}_n) = n! - \binom{n+d-1}{n}$ (Proposition **??**)
	E_1-obstruction categorical (Vol. III, categorical E_2-obstruction proposition)
	Toric chart gluing (Vol. III toric chart gluing theorem, promoted from conjecture)
AMBER (externally supported, conditional, or physically interpreted)	
	Shadow CohFT (Theorem **??**; equivariance and splitting proved; flat identity conditional on $\lvert 0\rangle \in V$, see)
	Gravitational interpretation of κ as central charge of the boundary graviton

The descent fan

The MC element $\Theta_{\mathcal{A}}$ projects onto three mathematical worlds. These projections share a common source but do not compose into a single functor: the "descent chain" is a *fan*, not a chain.

Proposition 1.23.1 *(Descent fan structure;* [proved here]*).* The universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ admits three independent projections with distinct targets and distinct symmetries:

(i) *Categorical projection.* $\mathcal{A} \longmapsto \bar{B}^{\mathrm{ch}}(\mathcal{A}) \longmapsto \mathcal{A}^!$. Target: factorization coalgebras and their Verdier duals on $\mathrm{Ran}(X)$. Symmetry: the Koszul involution $\mathcal{A} \leftrightarrow \mathcal{A}^!$ (Theorems **??** and **??**).

(ii) *Spectral projection.* $\mathcal{A} \longmapsto \{S_r(\mathcal{A})\}_{r\geq 2} \longmapsto \sqrt{Q_L} \longmapsto \varepsilon_s^c(\mathcal{A})$. Target: the shadow obstruction tower, the algebraic function $\sqrt{Q_L}$ of degree 2 (Theorem **??**), and the constrained Epstein zeta (Construction **??**). Symmetry: the functional equation involution $s \mapsto c - 1 - s$ of ε_s^c (eq. (**??**)).

(iii) *Modular projection.* $\mathcal{A} \longmapsto Z_g(\mathcal{A}) \longmapsto S_{\mathcal{A}}(v)$. Target: the genus expansion, the sewing lift $S_{\mathcal{A}}$, and the Rankin–Selberg integral over $\mathcal{F}$ (Theorem **??**). Symmetry: the Galois involution on the L-function content of $S_{\mathcal{A}}$.

These three projections share $\Theta_{\mathcal{A}}$ as common source but do *not* compose into a single functor. The Koszul involution, the functional equation involution, and the Galois involution are three different involutions on three different spaces.

Proof. Items (i) and (ii) are projections of the proved MC element $\Theta_{\mathcal{A}}$ (Theorem **??**): (i) extracts the twisting morphism $\alpha_\Theta\colon \bar{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ governing Theorems A and B; (ii) extracts the degree-r shadow coefficients S_r and their generating function $H(t)^2 = t^4 Q_L(t)$ (Theorem **??**). Item (iii) extracts the genus-g trace $\mathrm{tr}\,\pi_{\mathrm{cl}}(\Theta_{\mathcal{A}})|_g$ and applies the sewing–Hecke collapse (Theorem **??**). That the three projections have different targets ($\mathrm{Ran}(X)$-coalgebras, Dirichlet series, Rankin–Selberg integrals) and different symmetry groups is immediate from the construction. That they do not compose: the Koszul involution $\mathcal{A} \mapsto \mathcal{A}^!$ permutes the source algebra, the functional equation acts on the spectral variable s, and the Galois involution acts on the L-function coefficients; no functor intertwines all three. □

Proved levels. Projections (i) and (ii) are theorems: the categorical projection is Theorems A and B, and the spectral projection is Riccati algebraicity (Theorem **??**) composed with the shadow–spectral correspondence (Theorem **??**). The modular-to-arithmetic descent (from the sewing lift $S_{\mathcal{A}}(v)$ to the L-function content) is proved for lattice VOAs (Theorem **??**, Corollary **??**) and open for non-lattice algebras.

False claims. Two natural conjectures are *false*:

(a) "The Koszul involution $\mathcal{A} \mapsto \mathcal{A}^!$ descends to a functional equation on $S_{\mathcal{A}}(u)$." *False*: the sewing lift $S_{\mathcal{A}}(v)$ is determined by the shadow coefficients $\{S_r\}$ and the sewing operator, both of which depend on the full OPE data of $\mathcal{A}$. The Koszul dual $\mathcal{A}^!$ has different OPE data ($c \mapsto 26 - c$ for Virasoro), and $S_{\mathrm{Vir}_c}(v)$ forgets c at the level of the functional equation of the constrained Epstein zeta. The Koszul involution acts on the source algebra, not on the spectral variable.

(b) "The self-dual point $c = 13$ of the Koszul involution corresponds to the critical line $\mathrm{Re}(s) = 1/2$." *False*: the Koszul self-duality $\mathrm{Vir}_{13}^! \cong \mathrm{Vir}_{13}$ is a statement about algebras; the critical line is a statement about zeros of ζ. No mathematical relation between these two loci has been established. The four structural gaps of §**??** (Remark **??**) quantify the logical distance.

Lattice closure. [proved here] For lattice VOAs V_Λ, all three projections are compatible and the fan closes to a triangle: the Hecke–Newton closure (Theorem **??**) identifies the shadow obstruction tower coefficients with theta-function Fourier coefficients, the Epstein zeta factors into Hecke L-functions (Theorem **??**, Proposition **??**), and the sewing lift recovers the same L-function data via the Rankin–Selberg integral. The Koszul involution, the functional equation, and the Hecke action are mutually compatible because the lattice theta function simultaneously controls all three.

Non-lattice irreducibility. [open] For non-lattice algebras (Virasoro, $\mathcal{W}_N$, generic affine Kac–Moody), the fan structure is irreducible: no single involution governs all three projections, and the descent from the sewing lift to L-function content requires external arithmetic input (Deligne's theorem, Franc–Mason modularity, or Langlands functoriality; see Remark **??**).

Construction 1.23.2 *(Shadow Epstein functor).* The spectral projection of the descent fan extends to a functor from shadow metrics to L-functions. Given the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ (Definition **??**), define the constrained Epstein zeta $\varepsilon_s^c(\mathcal{A}) := \sum_{\Delta \in S}(2\Delta)^{-s}$ via the spectral data extracted from the coefficients of Q_L. This assignment is functorial: it respects direct sums (Proposition **??**) and carries the functional equation (eq. (**??**)). For lattice VOAs with class number 1, the constrained Epstein zeta decomposes into Hecke L-functions of the underlying number field. For class number $h(D) \geq 2$, the Epstein zeta is a finite sum of Hecke L-functions weighted by genus characters; the identification of $\varepsilon_s^c(\mathcal{A})$ with the zeta function $\zeta_K(s)$ of the ring of integers requires the class number computation and is not automatic.

Remark 1.23.3 *(Four-layer algebraic–analytic stratification).* The descent fan articulates a clean stratification of the mathematical content of $\Theta_{\mathcal{A}}$ into four layers of increasing analytic depth:

Layer	Name	Content
0	Pure algebra	Bar complex, MC equation, shadow obstruction tower, CohFT structure, depth classification. All determined by OPE data; no integration over moduli required.
1	Algebraic intersection	Genus-g free energies $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ via the $\hat{A}$-genus, Mumford–GRR on the Hodge bundle. Integration over $\bar{\mathcal{M}}_g$ but only against tautological classes.
2	Mixed	Hecke decomposition of shadow data (detected algebraically via Newton identities), Epstein zeta ε_s^c (algebraic from Q_L), functional equation (algebraic). The spectral parameter s is analytic; the constraints on it are algebraic.
3	Genuinely analytic	Partition function $Z(\tau, \bar{\tau})$ on the modular surface, Fredholm sewing determinant, Rankin–Selberg unfolding, scattering matrix $\varphi(s)$, zero locations of $\zeta(s)$. Requires analytic continuation off the real spectral axis.

The bar complex and MC element live in Layer 0. The genus expansion lives in Layer 1. The Koszul–Epstein function lives in Layer 2. The primary spectrum and partition function live in Layer 3. The structural ob-

struction (Remark **??**) is the statement that no mechanism within Layers 0–2 reaches Layer 3: algebraic constraints on the spectral line cannot determine the zeros of $\zeta(s)$, which live at complex spectral parameters.

1.24 Open problems in the Θ^{oc} framework

The open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum_j \mu^{\mathcal{M}_j}$ (Construction **??**, Theorem **??**) packages the closed-sector genus tower, the open-sector module operations, the bulk-to-boundary coupling, and the clutching sewing data into a single MC element in the open-closed convolution algebra $\mathfrak{g}^{\mathrm{oc}}_{\mathcal{A},\vec{\mathcal{M}}}$. Every proved theorem in the monograph is a projection of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ (Theorem **??**, Proposition 1.12.1). Each open problem is therefore a question about a specific projection, extension, or structural property of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ that has not yet been established.

This section reformulates the principal open problems in terms of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$. The purpose is precision: a vague "does X hold?" becomes a specific assertion about a projection, a coefficient, a cohomology class, or a functorial property of a well-defined algebraic object.

Closed-sector projections

(1) **Multi-generator universality at $g \geq 2$** (Open Problem **??**, **resolved negatively**).

Original statement. For a multi-weight modular Koszul algebra $\mathcal{A}$ (generators of distinct conformal weights), does $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ hold at all genera $g \geq 2$?

Resolution. **No.** Theorem **??** proves that the genus-g free energy decomposes as $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{A})$, where $\delta F_g^{\mathrm{cross}}$ is a graph sum over mixed-channel boundary graphs. The correction is generically nonzero: for $\mathcal{W}_3$ at genus 2, $\delta F_2 = (c{+}204)/(16c) \neq 0$ (Computation **??**). The scalar formula holds if and only if the algebra is uniform-weight or the cross-channel correction vanishes at each genus individually.

The structural diagnosis: algebraic-family rigidity (Theorem **??**) gives $\Theta^{\min}_{\mathcal{A}} = \eta \otimes \Gamma_{\mathcal{A}}$, but the Kuranishi map vanishes by parity, so the MC equation does not force $\Gamma_{\mathcal{A}} = \kappa(\mathcal{A})\Lambda$. The tautological-purity identification fails because the multi-channel graph sum includes mixed-channel amplitudes that produce contributions outside the $\kappa\Lambda$ line.

Universal N-formula. The gravitational cross-channel correction for the principal $\mathcal{W}_N$ algebra has a closed-form N-formula (Proposition **??**):

$$\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = \frac{(N{-}2)(N{+}3)}{96} + \frac{(N{-}2)(3N^3 + 14N^2 + 22N + 33)}{24c},$$

vanishing if and only if $N = 2$ (Virasoro: uniform-weight). Exact for $\mathcal{W}_3$; a lower bound for $N \geq 4$ (higher-spin exchange adds positive corrections). The cross-channel tower through genus 4 is recorded in Computations **??** and **??**; at genus 3, the cross-channel correction already exceeds the scalar part by a factor of 1.35 at large c.

(2) **BV/BRST = bar at higher genus** (Conjecture 2.2.7).

Resolution in D^{co}. **Resolved** in the coderived category (Theorem **??**): for every chirally Koszul algebra $\mathcal{A}$, the BV complex and the bar complex are quasi-isomorphic in $D^{\mathrm{co}}(\mathcal{A})$ at all genera. The key

mechanism for class M: the degree-r chain-level obstruction satisfies $\delta_r \propto m_0^{\lfloor r/2 \rfloor - 1}$, and in D^{co} the curvature acts as $m_0 \cdot x = d^2(x) \in \mathrm{Im}(d)$, so every δ_r is coderived-exact.

Chain-level status. For the Heisenberg algebra $_\kappa$, the scalar-level identification $F_g^{\mathrm{BV}}(_\kappa) = F_g^{\mathrm{bar}}(_\kappa) = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is proved at all genera by four independent paths: (a) Quillen anomaly formula, (b) Grothendieck–Riemann–Roch on the universal curve, (c) Faber–Pandharipande intersection formula, (d) direct bar-complex graph sum (Theorem **??**). Three obstructions to extending the identification beyond the scalar level are identified at the chain level (Proposition **??**): (i) the homotopy-transfer correction from the SDR data, (ii) the non-abelian sewing kernel at genus ≥ 2, (iii) the curved A_∞-structure vs flat BV operator. Obstruction (iii) is resolved for classes G and L at the chain level, and for class C conditional on harmonic decoupling (Theorem **??**(iii)). For class M, the chain-level identification *fails*: the quartic harmonic discrepancy $\delta_4 \propto Q^{\mathrm{contact}} \cdot m_0$ is not a coboundary in ordinary chain complexes (the factor $1/\mathrm{Im}(\tau)$ is non-holomorphic). The coderived resolution absorbs this obstruction.

Original statement (general case). The BV/BRST quantization of a holomorphic-topological field theory, at genus $g \geq 1$, coincides with the bar complex construction.

Θ^{oc} *reformulation.* The open/closed MC element $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ is constructed algebraically via the bordered FM compactification (Theorem **??**). The BV/BRST quantization produces a second MC element $\Theta_T^{\mathrm{BV}} \in \mathrm{MC}(\mathfrak{g}_T^{\mathrm{BV}})$ in the BV convolution algebra. The conjecture asserts an identification of convolution algebras

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{oc}}\big|_{g\geq 1} \simeq \mathfrak{g}_T^{\mathrm{BV}}\big|_{g\geq 1}$$

under which $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ maps to Θ_T^{BV}. At genus 0, the identification is the content of the PVA descent theorems D2–D6 (all proved in Vol II). The genus ≥ 1 extension requires the analytic hypotheses (H1)–(H4) and the BV functor bridge (Conjecture 1.19.8).

Residual gap (chain level for class M). The coderived resolution is complete. The remaining question is whether a chain-level quasi-isomorphism exists for class M, which would require cancelling the non-holomorphic harmonic obstruction δ_4^{harm} by a different mechanism (e.g. a cochain homotopy involving the Beltrami differential). This is not expected to hold in general.

(3) **Pixton ideal generation from class-M shadows** (Theorem 1.17.2).

Original statement. For class-M algebras ($r_{\max} = \infty$), the infinite family of MC-descended tautological relations generates the Pixton ideal in $R^*(\overline{\mathcal{M}}_{g,n})$.

Θ^{oc} *reformulation.* The genus-g component of the closed MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] + \hbar\Delta(\Theta_{\mathcal{A}}) = 0$ is a relation in $R^*(\overline{\mathcal{M}}_g) \otimes \mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$. The tautological projection $\mathrm{taut}_g : H^*(\overline{\mathcal{M}}_g, \mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})) \to R^*(\overline{\mathcal{M}}_g)$ sends each genus-g MC identity to a tautological relation. The conjecture asserts:

> *The ideal generated by* $\{\mathrm{taut}_{g,n}([\Theta_{\mathcal{A}}]_{g,n}) : g \geq 0,\ n \geq 0,\ \mathcal{A} \in$ *class M*$\}$ *equals the Pixton ideal.*

Proved for semisimple Frobenius data (Theorem **??**(vi)) via the Givental–Teleman–PPZ chain. Open for non-semisimple data (logarithmic/admissible algebras).

Strategy from Θ^{oc}. The open sector of $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ provides boundary-marked tautological classes $\mathrm{taut}_{g,n}^{\mathrm{oc}}$, which contain more data than the closed-sector classes. The four-sector decomposition of the MC

equation produces relations with both interior and boundary markings. These additional relations may generate the full Pixton ideal even when the closed-sector relations alone do not, because the open sector accesses descendant pairings (e.g. $\int_{\overline{\mathcal{M}}_{2,1}} \delta_{\mathrm{pf}}\, \psi_1^2$) that reveal shadow coefficients invisible in the closed $(g, 0)$ projection.

Koszulness programme

(4) **D-module purity: converse direction** (item (xii) of Theorem **??**).

Original statement. The forward direction "D-module purity $\Rightarrow$ FM boundary acyclicity (Koszulness)" is proved. Does the converse hold: does Koszulness of the bar factorization coalgebra imply D-module purity?

Θ^{oc} *reformulation.* D-module purity of $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ on $\mathrm{Ran}(X)$ is a regularity condition on the closed projection $\pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})$: it asserts that the bar differential, viewed as a D-module map on the Ran space, has holonomic kernel and cokernel. Chiral Koszulness is bar concentration: $H^n(\bar{B}^{\mathrm{ch}}(\mathcal{A})) = 0$ for $n \geq 3$. The question becomes:

> *Does holonomicity of the D-module structure of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ on $\mathrm{Ran}(X)$ force cohomological concentration of the underlying cochain complex?*

Additional input. The forward direction uses the PBW spectral sequence; the converse would require showing that D-module regularity constrains the E_2 page of that spectral sequence to degenerate. No proof strategy is currently available.

(5) **Chiral Koszulness of simple quotients at admissible levels** (§1.21, item 1).

Original statement. The universal algebra $V_k()$ is chirally Koszul at all levels (Proposition **??**). Is the simple quotient $L_k()$ chirally Koszul at admissible levels?

Θ^{oc} *reformulation.* The MC element $\Theta^{\mathrm{oc}}_{V_k}$ is well-defined for the universal algebra. The simple quotient $L_k() = V_k()/J_k$ acquires a curved bar complex $\bar{B}^{\mathrm{ch}}(L_k)$ with curvature element m_0 from the vacuum null vector. The question is whether $\Theta^{\mathrm{oc}}_{L_k}$ exists as an MC element in the curved open-closed convolution algebra: that is, whether the curved bar complex $\bar{B}^{\mathrm{ch}}(L_k)$ has cohomology concentrated in bar degree 1 despite the curvature. In Θ^{oc} terms:

> *Does the quotient map $V_k \to L_k$ induce a morphism of open-closed MC elements $\Theta^{\mathrm{oc}}_{V_k} \to \Theta^{\mathrm{oc}}_{L_k}$ in the coderived sense, with the target MC element well-defined in the periodic CDG convolution algebra (Conjecture **??**)?*

Strategy from Θ^{oc}. The Li–bar spectral sequence (Construction **??**) and the associated-variety criterion (Theorem **??**) provide a geometric test: the E_2 page of the spectral sequence is controlled by the reduced associated variety $X_{L_k} = \mathrm{Spec}(R_{L_k}/\sqrt{0})$. The Θ^{oc} framework suggests studying the image of $\Theta^{\mathrm{oc}}_{V_k}$ under the quotient functor, which detects nilpotent contributions from the nilradical of R_{L_k}.

Categorical and completion targets

(6) **Categorical modular Koszul duality** (Conjecture 1.20.10).

Original statement. The curved bar-cobar adjunction at genus g induces an equivalence $\mathsf{D}^{\mathrm{co}}(\overline{B}^{(g)}(\mathcal{A})\text{-}\mathsf{comod}) \simeq \mathsf{D}^b(\mathcal{A}\text{-}\mathsf{mod}^{\Sigma_g})$.

Θ^{oc} *reformulation.* The open sector of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ at genus g defines a functor $\Phi^{\mathrm{oc}}_g \colon \mathcal{A}\text{-}\mathsf{mod}^{\Sigma_g} \to \overline{B}^{(g)}(\mathcal{A})\text{-}\mathsf{comod}$ by twisting the A_∞-module structure $\mu^{\mathcal{M}}$ with the genus-g curvature $\kappa(\mathcal{A}) \cdot \omega_g$. The conjecture asserts that this functor, composed with the coderived passage, is an equivalence. In Θ^{oc} terms:

> *Is the genus-g component of the open/closed MC equation sufficient to produce a quasi-inverse Ψ^{oc}_g to Φ^{oc}_g in the coderived category?*

At $g = 0$ this is Theorem B. The genus ≥ 1 extension requires controlling the curvature obstruction in the coderived sense, which is the content of the categorical lift.

(7) **Modular cooperad completion** (ledger: Red).

Original statement. Construct the full modular cooperad structure on the open factorization category C_{op}: all genera, all boundary conditions, full sewing compatibility.

Θ^{oc} *reformulation.* The MC equation for $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is proved at all genera (Theorem **??**). The annulus trace (Theorem **??**) verifies the genus-1 open-sector structure. The cooperad completion asks for higher-genus open-sector structure maps $\delta^{\mathrm{op}}_{g,n} \colon C_{\mathrm{op}} \to C_{\mathrm{op}}^{\otimes n}$ satisfying the modular cooperad axioms. In Θ^{oc} terms:

> *Does the clutching sector $\hbar\Delta_{\mathrm{clutch}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})$ of the MC equation induce a modular cooperad structure on C_{op} whose cocompositions are compatible with the genus tower?*

The MC equation provides the necessary identities; the missing step is the categorical construction of cocomposition maps from iterated clutching on bordered FM compactifications.

Strategy from Θ^{oc}. The four-sector decomposition of the MC equation produces, at each (g, n), a compatibility condition between the open and closed sectors. The genus-g clutching identity $\xi^*(\Theta^{(g)}) = \sum_{g_1+g_2=g} \Theta^{(g_1)} \star \Theta^{(g_2)}$ is proved (Theorem **??**(iii)). The cooperad axiom (coassociativity) would follow from the associativity of the bordered FM clutching maps, once these are constructed for all stable topological types.

(8) **Grand completion conjecture** (Conjecture 1.20.7).

Original statement. The modular cumulant transform $\mathcal{A} \mapsto M(\mathcal{A})$ extends finite-stage chiral Koszul duality to an equivalence of homotopy theories.

Θ^{oc} *reformulation.* The five-component datum $M(\mathcal{A}) = (\mathrm{Cum}_c(\mathcal{A}),\ D_{\mathcal{A}},\ \tau_{\mathcal{A}},\ r_{\mathcal{A}}(z),\ \Theta_{\mathcal{A}})$ is a sequence of projections of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$: cumulant coalgebra from the bar, coderivation from the differential, twisting morphism from the collision residue, r-matrix from the ordered boundary, modular logarithm from the full closed projection. The conjecture asserts:

> *The open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is the universal object from which all five components of $M(\mathcal{A})$ are recovered, and the functor $\mathcal{A} \mapsto \Theta^{\mathrm{oc}}_{\mathcal{A}}$ is an equivalence on the completion closure of admissible chiral A_∞-algebras.*

Additional input. The genus-0 truncation of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is completed bar-cobar (proved by MC4). The ordered-boundary truncation is the dg-shifted Yangian jet tower (accessible windowwise by the jet principle, Conjecture 1.20.6). The full equivalence requires the cumulant recognition (Conjecture 1.20.5): that the resonance-graded associated graded of $\hat{B}(\mathcal{A})$ is the cumulant coalgebra.

Bridge and factorization targets

(9) **Factorization-categorical DK bridge** (Conjecture 1.20.21).

Original statement. The E_1-factorization categories of the Yangian $\mathcal{Y}()$ and the quantum group $U_q()$ are equivalent.

Θ^{oc} *reformulation.* The genus-0 ordered-boundary projection of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is the r-matrix $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ (Theorem **??**). The DK bridge asserts that the E_1-monoidal structure on the module category induced by the r-matrix sector of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ coincides with the braided monoidal structure from the factorization quantum group. In Θ^{oc} terms:

> *Does the E_1-monoidal functor induced by the ordered-boundary projection $\pi_{\mathrm{ord}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})$ extend from evaluation modules to all of category $\mathcal{O}$, and does it identify with the Latyntsev factorization quantum group?*

The evaluation-locus theorem (Theorem **??**) is the statement that the ordered-boundary MC element is correct on generators. The extension to all of $\mathcal{O}$ is a thick-generation problem about the ordered-boundary sector of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$.

(10) **DS–KD intertwining for arbitrary nilpotent** (Conjecture **??**).

Original statement. Bar-cobar duality commutes with Drinfeld–Sokolov reduction at arbitrary nilpotent f.

Θ^{oc} *reformulation.* The DS reduction functor $\mathrm{DS}_f\colon V_k() \to \mathcal{W}_k(,f)$ should induce a morphism of open-closed convolution algebras $\mathrm{DS}^{\mathrm{oc}}_f\colon \mathfrak{g}^{\mathrm{oc}}_{V_k} \to \mathfrak{g}^{\mathrm{oc}}_{\mathcal{W}_k(,f)}$ satisfying:

$$\mathrm{DS}^{\mathrm{oc}}_f(\Theta^{\mathrm{oc}}_{V_k()}) = \Theta^{\mathrm{oc}}_{\mathcal{W}_k(,f)}.$$

In words: DS reduction is a morphism of modular Koszul datums. This is proved for principal f; for hook-type in type A it is conditional on DS-bar compatibility (Theorem **??**). The general case reduces to filtration formality (E_1-degeneration of the Kazhdan filtration spectral sequence) of the BRST complex (Remark 1.14.1). The Θ^{oc} formulation makes the target precise: the MC element must be preserved under a specific morphism of convolution algebras, which is stronger than commutativity of functors.

(11) **Modular factorization adjunction** (the modular extension of the factorization envelope adjunction).

Original statement. The modular factorization envelope U^{mod}_X is left adjoint to the modular primitive-current functor $\mathrm{Prim}^{\mathrm{mod}}$.

Θ^{oc} *reformulation.* The six-fold modular Koszul datum $\Pi_X(L) = (\mathcal{F}_X(L),\ \bar{B}_X(L),\ \Theta_L,\ \mathcal{L}_L,\ (V_L^{\mathrm{br}}, T_L^{\mathrm{br}}),\ R_4^{\mathrm{mod}}(L))$ is extended by the open/closed data to the eight-fold completed modular Koszul datum $\Pi_X^{\mathrm{oc}}(\mathcal{A})$ (Definition **??**). The adjunction asserts that $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ is *universal* among MC elements in open-closed convolution algebras over cyclically admissible Lie conformal algebras:

> *For any modular Koszul algebra $\mathcal{A}$ with underlying Lie conformal algebra L, is $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ the image under U_X^{mod} of the canonical MC element on L? Does the universality of the open sector yield the adjunction?*

Additional input. The genus-0 version of the adjunction is the classical factorization-envelope construction (Nishinaka [428]). The modular extension requires showing that the genus tower of $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ is functorially determined by the Lie conformal algebra data alone. The bar-intrinsic construction (Theorem **??**) provides existence; the adjunction is the claim that the construction is *universal* in a precise categorical sense.

Analytic and holographic targets

(12) **Analytic realization criterion** (Conjecture **??**).

Original statement. A unitary full VOA satisfying conformal OS, polynomial spectral growth, and HS-sewing admits a sewing envelope, a conformally flat 2-disk algebra, and a higher-genus coderived shadow.

Θ^{oc} *reformulation.* The algebraic MC element $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ lives in the formal convolution algebra. The analytic realization asks for a completed MC element $\Theta_{\mathcal{A}}^{\mathrm{oc,an}} \in \mathrm{MC}(\hat{\mathfrak{g}}_{\mathcal{A}}^{\mathrm{oc,an}})$ in an analytic completion of the open-closed convolution algebra, such that:

> *The inclusion $\mathfrak{g}_{\mathcal{A}}^{\mathrm{oc}} \hookrightarrow \hat{\mathfrak{g}}_{\mathcal{A}}^{\mathrm{oc,an}}$ sends $\Theta_{\mathcal{A}}^{\mathrm{oc}}$ to $\Theta_{\mathcal{A}}^{\mathrm{oc,an}}$, and the analytic MC element produces convergent genus-g amplitudes for all $g \geq 0$ via trace-class sewing operators.*

For Heisenberg and lattice VOAs, this is proved (Theorems **??** and **??**). The general case requires constructing the analytic topology on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{oc}}$ from the HS-sewing condition.

(13) **Boundary bar duality** (Conjecture **??**).

Original statement. For analytically Koszul $\mathcal{A}$, completed boundary modules at genus 0 are equivalent to analytic comodules over $\overline{B}^{\mathrm{an}}(\mathcal{A})$.

Θ^{oc} *reformulation.* The open projection $\pi_{\mathrm{op}}(\Theta_{\mathcal{A}}^{\mathrm{oc,an}})$ defines A_∞-module operations μ_n^{an} on completed boundary modules. The conjecture asserts that the twisting morphism $\alpha_{\Theta^{\mathrm{an}}}$ (the genus-0 collision residue of the analytic MC element) induces the Koszul duality equivalence in the completed/analytic setting:

> *Does the analytic open-sector projection of $\Theta_{\mathcal{A}}^{\mathrm{oc,an}}$ define an equivalence between completed $\mathcal{A}$-modules and analytic $\overline{B}^{\mathrm{an}}(\mathcal{A})$-comodules, extending Theorem B from the algebraic to the analytic setting?*

At curved genus ($g \geq 1$), the correct target is the coderived category of analytic contramodules.

(14) **Boundary-defect realization** (Conjecture **??**).

Original statement. The dg-shifted Yangian $r_{\mathcal{A}}(z)$ is realized as the boundary-defect operator product for a line defect in the bulk HT theory.

Θ^{oc} *reformulation.* The r-matrix is the genus-0 binary collision residue of the closed projection: $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$. The boundary-defect realization asks for a physical construction of the mixed projection $\pi_{\mathrm{mix}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})\big|_{0,1,n}$ via a line defect:

> *Is there a line defect $\mathcal{L}$ in the bulk HT theory whose coupling to the boundary reproduces the mixed-sector projection of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$, with the binary component giving $r_{\mathcal{A}}(z)$?*

The holographic modular Koszul datum $\mathcal{H}(T)$ packages the answer: the line category C is the mixed-projection receptacle, and the boundary-defect realization identifies line operators with mixed-sector components of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$.

(15) **Singular-fiber descent for minimal models** (Conjecture **??**).

Original statement. The modular Koszul datum for minimal models (p, q) (as simple quotients of Vir_c) is recovered by singular-fiber specialization from the universal Virasoro datum.

Θ^{oc} *reformulation.* At generic c, the MC element $\Theta^{\mathrm{oc}}_{\mathrm{Vir}_c}$ is well-defined. At $c = c(p, q)$, the vacuum null vector forces a quotient $\mathrm{Vir}_c \to (p, q)$, and the MC element acquires curvature. The conjecture asserts:

> *The singular-fiber limit $\lim_{c\to c(p,q)} \Theta^{\mathrm{oc}}_{\mathrm{Vir}_c}$ exists in the coderived open-closed convolution algebra and equals $\Theta^{\mathrm{oc}}_{(p,q)}$.*

This is a degeneration statement: the MC element has a well-defined limit under specialization of the central charge, and the limit is the MC element of the quotient.

Arithmetic targets

(16) **Modular spectral rigidity** (Conjecture **??**).

Original statement. The MC equation provides all symmetric-power moments of shadow eigenvalues. Does functorial transfer $\mathrm{GL}(2) \to \mathrm{GL}(r+1)$ for $r \geq 5$ hold?

Θ^{oc} *reformulation.* The shadow at degree r is the closed-sector projection $\pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})\big|_r$, which encodes Sym^{r-1} data (Proposition **??**). The MC equation at all degrees is equivalent to Newton's identities for the shadow spectrum (Proposition **??**). The conjecture asks:

> *Do the Newton's identities imposed by the MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ at all degrees suffice to prove automorphy of all symmetric-power L-functions of the shadow spectral atoms?*

The answer is yes for lattice VOAs (Theorem **??**) and no in general without additional analytic input (prime-locality transfer). The structural separation theorem (Theorem **??**) shows that the genus-1 MC equation exhausts its algebraic content; the question is whether the full genus tower of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ provides the missing analytic input via higher-genus Hecke constraints.

(17) **Arithmetic comparison conjecture** (Conjecture **??**).

Original statement. The MC element $\Theta_{\mathcal{A}}$ canonically determines the arithmetic packet connection $\nabla^{\mathrm{arith}}_{\mathcal{A}}$, and higher-genus MC data accesses residues of the frontier defect form $\Omega_{\mathcal{A}}$.

Θ^{oc} *reformulation.* The arithmetic packet connection $\nabla^{\mathrm{arith}}_{\mathcal{A}}$ on the module $M_{\mathcal{A}}$ over the spectral s-line is defined from the shadow spectral measure. The conjecture asserts a canonical map from the closed-sector tower to the connection:

> *The genus-g components of $\pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})$ for $g \geq 2$ determine the residues of the frontier defect form $\Omega_{\mathcal{A}}$ at scattering poles, via the genus-g Saito–Kurokawa projection and the Böcherer bridge* (*Theorem* **??**).

The genus-2 case is the first nontrivial instance: the three-shell decomposition of $\Theta_{\mathcal{A}}|_{g=2}$ produces the Siegel modular form whose Fourier coefficients encode central L-values (Theorem **??**).

(18) **Shadow–Siegel gap for** $K3 \times E$ (Theorem **??**).

*Status: PROVED (§**??**).* The shadow obstruction tower of $K3 \times E$ does not produce the Igusa cusp form Φ_{10}. Four independent obstructions are identified: categorical (number vs function), modular characteristic ($\kappa_{\mathrm{ch}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$), second quantization (single-copy vs DMVV symmetric product), and Schottky at $g \geq 4$ ($\mathrm{codim} = (g-2)(g-3)/2$). The Borcherds multiplicative lift $\mathrm{Borch}(2\,\phi_{0,1}) = \Phi_{10}$ is the genus-1 $\to$ 2 bridge; at genus 2 the Torelli map is birational, so $F_2 = 7/1920$ is the topological residue of $1/\Phi_{10}$. The four κ invariants ($\kappa(\mathcal{A}) = 3$, $\kappa_{\mathrm{BCOV}} = 0$, $\kappa_{\mathrm{MacMahon}} = 0$, $\kappa_{\mathrm{BKM}} = 5$) are tabulated in Remark **??**, extending the polysemy analysis of Remark 1.20.2.

Ten structural results for $K3 \times E$ (§**??**–§**??**):

(K3-1) $\kappa(K3 \times E) = 3 = \dim$, verified by 6 independent paths (Proposition **??**, Proposition **??**).

(K3-2) The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is $\kappa(\mathcal{A}_{K3})$; the bar complex provides the universal moonshine multiplier $A_n = \kappa \cdot \dim(\rho_n)$ (Remark **??**).

(K3-3) $\kappa_{\mathrm{BKM}} = \kappa(K3) + \kappa(K3 \times E) = 2 + 3 = 5$; ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3 = \mathrm{wt}(\Phi_{10})/(2\ \dim)$; Hilbert–Chow exceptional $= \chi(K3) = 24$ (Proposition **??**).

(K3-4) Shadow $\neq \Phi_{10}$: four structural obstructions O1–O4 (Theorem **??**).

(K3-5) $D^b(K3 \times E)$ constructively reconstructible via Čech descent (2 charts), BKR at Kummer, and Brauer vanishing (Remark **??**).

(K3-6) Mock modular decomposition $Z_{K3} = (h - 24\mu)\,\vartheta_1^2/\eta^3$, machine-precision verified (Remark **??**).

(K3-7) Schottky kills shadow–Siegel at $g \geq 4$, $\mathrm{codim} = (g-2)(g-3)/2$ (O4 of Theorem **??**).

(K3-8) Z^{sh} converges with radius 2π, Borel entire, no resurgence (Proposition **??**).

(K3-9) Boundary A_E ($\kappa = 24$) vs sigma V_{K3} ($\kappa = 2$): ratio 12 at genus 1; conditional at $g \geq 2$ (V_{K3} is multi-weight) (Proposition **??**); Leech connection through q^1 (Remark **??**).

(K3-10) $\lambda_5^{\mathrm{FP}} = 73/3503554560!$).

Ten forward-looking research programmes (A–J) are developed in the toroidal and elliptic algebras chapter (Vol III, Chapter **??**), each with formal conjectures:

(Prog A) Constructive CY gluing (Open Problem **??**, Conjecture **??**).

(Prog B) κ as universal moonshine multiplier (Open Problem **??**, Conjecture **??**).

(Prog C) Second-quantization bridge (Open Problem **??**, Conjecture **??**).

(Prog D) Schottky shadow programme (Open Problem **??**, Conjecture **??**).

(Prog E) Mock modularity and the bar complex (Open Problem **??**, Conjecture **??**).

(Prog F) Modular factorization envelope (Open Problem **??**, Conjecture **??**).

(Prog G) BKM algebras and scattering diagrams (Open Problem **??**, Conjecture **??**).

(Prog H) Descent theorem programme (Open Problem **??**).

(Prog I) Higher-dimensional CY shadows (Open Problem **??**, Conjecture **??**).

(Prog J) Convergence vs divergence (Open Problem **??**).

All conjectures have status [conjectured]; all are grounded in the 42 CY compute engines and are falsifiable by explicit computation.

Three additional structural results from the CY sweep: (K3-11) Class G exactness: for the free-field boundary algebra A_E, $F_g^{\mathrm{sh}} = F_g^{\mathrm{top}}$ at all genera (Proposition **??**); no worldsheet instanton corrections for class-G (shadow depth 2) algebras. (K3-12) Bar/hocolim commutation: $B(\mathrm{hocolim}_\sigma A_\sigma) \simeq \mathrm{hocolim}_\sigma B(A_\sigma)$ (Proposition **??**), a descent theorem for bar complexes; the bar functor is left Quillen, hence preserves homotopy colimits (145 tests). (K3-13) κ-spectrum polysemy: $\mathrm{Spec}_\kappa(K3 \times E) = \{2, 3, 5, 24, \ldots\}$, with each value arising from a distinct algebraization (Remark **??**).

Open-sector and globalisation targets

(19) **Loop–Connes bridge** (Conjecture 1.19.7).

Original statement. Closed genus creation (the genus-loop operator Λ_P) is the image of the open BV/Connes operator under the derived-center map.

Θ^{oc} *reformulation.* The clutching sector of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ has a non-separating component $\hbar\Delta_{\mathrm{ns}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})$ that raises genus by one. The derived-center map $\beta_{\mathrm{der}} \colon Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \to \mathfrak{g}^{\mathrm{oc}}$ (Theorem **??**) identifies the bulk with a subalgebra of the open-closed convolution algebra. The conjecture asserts:

> $\beta^*_{\mathrm{der}}(\Delta_{\mathrm{open}}) = \Lambda_P$*: the non-separating clutching operator* Δ_{ns} *in the closed sector of* $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ *equals the image of the open-sector Connes operator* B *under the derived-center map.*

In words: the genus-raising mechanism in the closed sector is the trace mechanism in the open sector, viewed through the universal bulk.

(20) **Global open sector on tangential log curves** (ledger: Red).

Original statement. Extend the local open/closed realization to global factorization on tangential log curves (X, D, τ).

Θ^{oc} *reformulation.* The local-global bridge (Theorem **??**) identifies the formal-disk restriction of the bordered FM convolution with $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$. The global extension asks for a sheaf of open-closed convolution algebras $\mathfrak{g}^{\mathrm{oc}}_{X,D,\tau}$ on (X, D, τ) with a global MC element $\Theta^{\mathrm{oc,glob}}_{\mathcal{A}}$ whose stalk at each formal disk reproduces the local $\Theta^{\mathrm{oc}}_{\mathcal{A}}$. In Θ^{oc} terms:

Does $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ glue to a global section of the sheaf of open-closed MC moduli on (X, D, τ), satisfying excision and descent for the open factorization category?

The local data are proved. The globalisation requires constructing the excision sequence for the bordered FM compactification on curves with real-blowup boundary data.

(21) **Modular periodicity for admissible simple quotients** (Conjecture **??**).

Original statement. The bar cohomology dimensions of the simple quotient $L_k()$ at admissible level are eventually periodic.

Θ^{oc} *reformulation.* The conjectured periodic CDG structure asserts $2p$-periodicity of the bar complex of $L_k()$. In Θ^{oc} terms, this periodicity should be visible as a $\mathbb{Z}/2p\mathbb{Z}$-grading on the open-closed convolution algebra $\mathfrak{g}^{\mathrm{oc}}_{L_k}$ inherited from the cyclotomic chart:

Does the μ_q-grading of the cyclotomic boundary chart induce an eventual periodicity on the graded pieces of $\Theta^{\mathrm{oc}}_{L_k}$?

The proved closed-sector input: T-matrix periodicity (Step 1) and bar differential lift (Step 3). The open gap: the convolution with $1/\eta$ (Step 2) may break eventual periodicity of Fourier coefficients.

Remark 1.24.1 *(The Θ^{oc} meta-principle).* Every open problem in this monograph is a question about a specific projection, extension, or structural property of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$. The proved core establishes existence of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ and its MC equation at all genera. The open problems fall into three classes:

(a) *Identification problems*: which tautological class does a given projection produce? (Multi-generator universality, Pixton generation, arithmetic comparison.) These require computing specific coefficients of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$.

(b) *Extension problems*: does $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ extend to a larger domain (analytic completion, global curves, coderived categories, module categories)? (Analytic realization, boundary bar duality, categorical KD, globalisation, modular cooperad.)

(c) *Functoriality problems*: is $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ preserved under a given functor (DS reduction, factorization envelope, BV quantization)? (DS–KD intertwining, modular factorization adjunction, BV/BRST bridge, DK bridge.)

The programme is: compute more projections (a), prove more extensions (b), establish more functorialities (c).

Remark 1.24.2 *(Five open problems as questions about $\Theta^{\mathrm{oc}}_{\mathcal{A}}$).* The preceding section develops each reformulation in detail. The following digest records the essential question in each case.

(i) *Multi-generator universality* (Open Problem **??**, **resolved negatively**). The closed projection $\pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})|_g$ is $\kappa(\mathcal{A}) \cdot \lambda_g$ on the uniform-weight lane. For multi-weight algebras at $g \geq 2$, the scalar formula fails: $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{A})$ with $\delta F_g^{\mathrm{cross}} \neq 0$ (Theorem **??**). The clutching sector does NOT force $\Gamma_{\mathcal{A}} = \kappa(\mathcal{A})\Lambda$; the Kuranishi map vanishes by parity.

(ii) *BV/BRST = bar at higher genus* (Conjecture 2.2.7, **resolved in** D^{co} by Theorem **??**). The BV path integral on Σ_g produces $\Theta^{\mathrm{BV}}_T \in \mathrm{MC}(\mathfrak{g}^{\mathrm{BV}}_T)$. At genus 0, the PVA descent theorems identify the two convolution algebras and their MC elements. In D^{co}, the identification extends to all genera and all shadow classes: the degree-r chain-level obstruction $\delta_r \propto m_0^{\lfloor r/2 \rfloor - 1}$ is coderived-exact since $m_0 \cdot x = d^2(x) \in \mathrm{Im}(d)$. The chain-level identification $\mathfrak{g}^{\mathrm{oc}}_{\mathcal{A}}|_{g\geq 1} \simeq \mathfrak{g}^{\mathrm{BV}}_T|_{g\geq 1}$ sending $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ to Θ^{BV}_T is proved for classes G and L, conditional for class C on harmonic decoupling, and false for class M at the ordinary chain level.

(iii) *Pixton ideal generation* (Theorem 1.17.2). For class-M algebras, each genus-g component of the MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] + \hbar\Delta(\Theta_{\mathcal{A}}) = 0$ projects to a tautological relation in $R^*(\overline{\mathcal{M}}_g)$. The question: does the ideal generated by $\{\mathrm{taut}_{g,n}([\Theta_{\mathcal{A}}]_{g,n}) : g, n \geq 0,\ \mathcal{A} \in \text{class M}\}$ equal the Pixton ideal?

(iv) *Arithmetic comparison* (Conjecture **??**). The genus-1 shadow $\pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})|_1$ determines the shadow spectral measure and hence the arithmetic packet connection $\nabla^{\mathrm{arith}}_{\mathcal{A}}$. The question: do the higher-genus components $\pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})|_{g\geq 2}$ determine the residues of the frontier defect form $\Omega_{\mathcal{A}}$ at scattering poles?

(v) *Analytic realization* (Conjecture **??**). The algebraic MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ lives in the formal convolution algebra. The question: does the inclusion $\mathfrak{g}^{\mathrm{oc}}_{\mathcal{A}} \hookrightarrow \hat{\mathfrak{g}}^{\mathrm{oc,an}}_{\mathcal{A}}$ into an analytic completion send $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ to a completed MC element $\Theta^{\mathrm{oc,an}}_{\mathcal{A}}$ whose genus-g amplitudes converge via trace-class sewing? Proved for Heisenberg and lattice VOAs; open in general.

Items (i) and (iii) are identification problems: which cohomology class does a given projection of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ produce? Item (ii) is a functoriality problem: is $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ the image of Θ^{BV}_{T} under a comparison functor? Items (iv) and (v) are extension problems: does the MC element carry information beyond the formal/algebraic setting?

1.25 Forbidden slogans

The following statements are false. They must never appear in this monograph. Each violates a specific anti-pattern documented in the Beilinson Principle (§1.21).

1. **"Bar = bulk."** *False.* The bar complex $\overline{B}(\mathcal{A})$ classifies *twisting morphisms*: algebraic maps $\tau\colon \overline{B}(\mathcal{A}) \to \mathcal{A}$ controlling the coupling between $\mathcal{A}$ and its Koszul dual $\mathcal{A}^!$. The bulk algebra is the *chiral derived center* $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = H^*(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \delta)$, which classifies bulk operators acting on the boundary (Theorem **??**). These are different objects produced by different functors. *Violated anti-pattern*: (three functors, three outputs: bar ≠ Verdier dual ≠ cobar).

2. **"The boundary algebra is the primitive object."** *False.* The primitive object is the open-sector factorization dg-category C_{op} on a tangential log curve (X, D, τ) with real-oriented blowup at each puncture, not on the ordinary curve X. The boundary algebra $A_b = \mathrm{End}_{C_{\mathrm{op}}}(b)$ is an endomorphism algebra of a chosen boundary condition b: a Morita-dependent chart, not the intrinsic datum. Different choices of b produce Morita-equivalent but non-isomorphic algebras. *Violated anti-pattern*: (same name, different object: the chart is not the category).

3. **"All master conjectures resolved."** *Misleading.* MC1 through MC4 are proved. MC5 is resolved analytically and in the coderived category, but not at chain level: the analytic HS-sewing package is proved at all genera (Theorem **??**); the genus-0 algebraic BRST/bar comparison is proved (Theorem **??**); BV=bar in the coderived category is proved for all four shadow classes including class M (Theorem **??**); the tree-level amplitude pairing is conditional on Corollary **??**; the genus-wise chain-level BV/BRST/bar identification remains conjectural in general (Conjecture 2.2.7; classes G/L proved, class C conditional on harmonic decoupling, class M chain-level false). MC3 splits into three layers (Corollary **??** and Theorem **??**): the evaluation core and categorical CG are unconditional for all simple types; the type-A post-core route is proved on the Baxter-locus seed, and the only remaining post-core gap is the compact-completion packet (Conjecture **??**). The broader programme contains open problems downstream of MC3: the DK-4/5 extension/completion

packet (Conjecture **??**), multi-generator universality at $g \geq 2$ (Open Problem **??**, now resolved negatively: cross-channel correction nonzero), modular cooperad completion, D-module purity converse, Pixton ideal generation (Theorem 1.17.2, resolved on the semisimple locus), and the arithmetic comparison conjecture. The programme continues beyond the five MC statements. *Violated anti-pattern*: (scope inflation: universal claim obscuring downstream residuals).

4. **"Gravity Yangian at strict level."** *False.* The Virasoro dg-shifted Yangian is a *cohomological* object: it is a dg Lie algebra with a transferred L_∞ structure from the DS reduction (Vol II, Theorem thm:ds-hpl-transfer). It is not a strict Yangian in the sense of Drinfeld: the higher operations ℓ_k for $k \geq 3$ are nonzero (Virasoro has shadow depth $r_{\max} = \infty$, class M). Writing "strict" erases the entire non-formality content of the shadow obstruction tower. *Violated anti-pattern*: (conflating Koszulness with formality: Virasoro is Koszul but not SC-formal).

5. **"The modular cooperad package is proved."** *False.* The modular cooperad structure on $\mathcal{C}_{\mathrm{op}}$ is a *programme*, not a theorem. Stage 4 of the four-stage open/closed architecture (trace + clutching on the open sector) has the annulus trace proved (Theorem **??**), but the full modular cooperad (all genera, all boundary conditions, full sewing compatibility) is not constructed. *Violated anti-pattern*: (scope inflation: programme described as proved).

6. **"$\Theta_{\mathcal{A}} + \Theta_{\mathcal{A}^!} = 0$ at full level."** *False.* Only the scalar projection is constrained: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ for Kac–Moody and free-field families, and $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \rho \cdot K$ for $\mathcal{W}$-algebras (Theorem **??**). The full MC elements $\Theta_{\mathcal{A}}$ and $\Theta_{\mathcal{A}^!}$ are related by the Verdier involution on $\mathrm{Ran}(X)$ (Theorem **??**), but this involution is *not* negation: it intertwines bar and cobar, sending $\overline{B}(\mathcal{A})$ to $\overline{B}(\mathcal{A}^!)$, not to $-\overline{B}(\mathcal{A})$. The higher-degree shadows (cubic C, quartic Q, etc.) satisfy independent complementarity relations that are not simple sign changes. *Violated anti-patterns*: ($\kappa + \kappa^! = 0$ overclaimed; for Virasoro, $\kappa + \kappa^! = 13$) and ($\kappa = 0$ does not imply $\Theta = 0$: scalar vanishing does not force full-tower vanishing).

7. **"Virasoro is not Koszul."** *False.* Virasoro is chirally Koszul: the bar cohomology $H^*(\overline{B}(\mathrm{Vir}_c))$ is concentrated in bar degree 1 for generic c (Corollary **??**, via Proposition **??**). What Virasoro is *not* is Swiss-cheese formal: the higher Swiss-cheese operations $m_k^{\mathrm{SC}} \neq 0$ for $k \geq 3$ (shadow depth $r_{\max} = \infty$, class M). Koszulness (bar concentration) and formality (vanishing of higher operations) are different properties of different objects. *Violated anti-pattern*: (conflating chiral Koszulness with Swiss-cheese formality).

8. **"The global open sector lives on an ordinary curve."** *False.* The open-sector factorization dg-category $\mathcal{C}_{\mathrm{op}}$ lives on a tangential log curve (X, D, τ) with real-oriented blowup at each puncture (Volume II, Definition def:bordified-curve), not on the ordinary curve X. The boundary intervals $I_{p_i} = S^1_{p_i} \setminus \{\tau_{p_i}\} \cong \mathbb{R}$ carry the topological (E_1) structure; the interior carries the holomorphic (chiral) structure. Omitting the tangential data loses the E_1-direction entirely.

9. **"Modularity is a property of the closed algebra alone."** *False.* Modularity arises from trace and clutching on the open sector. The annulus trace $\int_{S^1} \mathcal{C}_{\mathrm{op}} \simeq HH_*(\mathcal{A})$ is the first modular shadow (Theorem **??**); clutching along shared boundary circles raises genus. The scalar $\kappa(\mathcal{A})$ enters as the trace of the cyclic pairing on $\mathrm{End}(b)$, not as primitive closed-string data. The MC equation on the bordered FM compactification packages both sectors; the closed projection is $F_g = \kappa \int_{\overline{\mathcal{M}}_g} \lambda_g$.

10. **"The dg-shifted Yangian is a universal theorem for general 3d HT QFT."** *False.* On the affine lineage, the open-colour Koszul dual is a dg-shifted Yangian (Vol II, Theorem thm:Koszul_dual_Yangian: *proved*). On the chirally Koszul locus, the Yangian recognition theorem (Vol II, Theorem thm:yangian-recognition) identifies the $\mathsf{SC}^{\mathrm{ch,top}}$-algebra structure on $\mathcal{A}^!_{\mathrm{line}}$ with the dg-shifted Yangian package: *proved algebraically*. For general perturbative 3d HT theories beyond the affine lineage, the gravitational-Yangian package is *conjectural* (Dimofte–Niu–Py [129]).

11. **"Bosonization is Koszul duality."** *False.* Bosonization ($\psi \otimes \psi \cong_1$ on the $\mathbb{Z}$-lattice) is a complex-fermion to compactified-boson equivalence preserving $c = 1$. Koszul duality ($\psi \mapsto \mathrm{Sym}^{\mathrm{ch}}(\gamma)$) maps a real fermion at $c = 1/2$ to a boson at $c = -1/2$, reversing sign. These are different functors on different objects with different outputs (Remark **??**). *Violated anti-pattern*: (confusing two operations that share a surface resemblance).

Chapter 2

Editorial constitution and research directions

This chapter fixes the boundary between the proved core and the open frontier.

2.1 Constitutional declaration

Remark 2.1.1 *(Constitutional status).* When any chapter disagrees with this chapter or the concordance (Chapter 1), these govern.

Proved core. Theorems A–D+H and the universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ (Theorem **??**: $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, MC because $D_{\mathcal{A}}^2 = 0$). Higher-genus PBW concentration (Theorems 2.2.1, **??**, **??**, **??**). Scalar saturation for all algebraic families (Theorem **??**). Ambient $D^2 = 0$ at both levels (Theorem **??**).

Finite-order engine. Shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ with obstruction classes, quartic resonance class with clutching law, all-degree master equation at each truncation. Archetype classification: Heisenberg at 2, affine at 3, $\beta\gamma$ at 4, Virasoro/$\mathcal{W}_N$ infinite (Remark **??**).

Frontier. DK/KL categorical lift (Conjecture 2.2.3), H-level target identification (Conjecture 2.2.5), BV/BRST/bar identification (Conjecture 2.2.7).

Two-level convention. Every convolution dg Lie algebra is a strict model of a homotopy-invariant L_∞-algebra (Theorems **??**, **??**). The dg Lie algebra suffices for MC elements and the extension tower; the L_∞-algebra persists under quasi-isomorphic replacements.

2.2 Conjecture stratification

The ~150 conjectured claims in this monograph stratify into three tiers by the nature and difficulty of the missing input.

Tier 1: Foundational. Conjectures whose resolution would extend the proved theorems to new structural levels, requiring substantial new mathematics:

- Coderived Ran formalism for non-Koszul chiral algebras: the categorical extension of Theorem B off the Koszul locus.
- Virasoro and $\mathcal{W}_\infty$ Koszul duals: explicit identification of the dual chiral algebras.

The former higher-genus PBW concentration conjecture is now resolved for the standard finite-type interacting families by Theorems **??**, **??**, and **??**.

Tier 2: Consequence. Conjectures that follow from known structures by explicit computation or extension of existing proofs; each is a multi-week project requiring no new ideas:

- Derived Drinfeld–Kohno ladder beyond the theorematic evaluation-generated core: ordered E_1-factorization with R-matrix inversion, factorization-level Kazhdan transport, and dg-shifted comparison.
- Kazhdan–Lusztig from bar-cobar: the remaining stages beyond the proved chain-level framework.
- Shifted-symplectic enhancement: upgrading the Lagrangian complementarity to a shifted symplectic structure on the moduli of Koszul pairs.
- Fusion preservation under Koszul duality.

Tier 3: Physics horizon. Physics-facing conjectures requiring input from quantum field theory or string theory, correctly conjectural within a mathematics monograph:

- BV = bar identification for general field theories (beyond the holomorphic setting).
- Path integral = bar-cobar: the full Feynman–Koszul dictionary at all genera.
- Holographic Koszul duality: AdS/CFT via the bar-cobar template.
- E_n extension to higher-dimensional field theories.
- AGT comparison via bar-cobar/Koszul duality.

Resolved Entry Theorem and Dependency-Ordered Open Problems

The unifying principle. Every structure in the monograph is a projection of a single Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ in the modular convolution dg Lie algebra (Definition **??**, Theorem **??**). The five master conjectures are projections of a single question: *does the modular extension tower of $\mathcal{A}$ converge?*

MC	*Projection of* Θ	*Status*
MC1	PBW concentration	**Proved**
MC2	Universal MC class $\Theta_{\mathcal{A}}$	**Proved** (bar-intrinsic, Thm **??**)
MC3	E_1-modular extension	**Proved** (all simple types, evaluation-generated core; DK-4/5 completion downstream)
MC4	Infinite-generator convergence	**Proved** (strong completion towers)
MC5	BV/BRST = bar	**Partially proved** (HS-sewing; genus-0 algebraic)

The historical five-master-conjecture package now consists of two resolved load-bearing layers (the MC1 entry theorem and the MC2 universal $\Theta_{\mathcal{A}}$ package) together with the fully resolved MC3: **proved for all simple types** on the evaluation-generated core. The all-types categorical prefundamental Clebsch–Gordan closure and the DK comparison on the evaluation-generated core are proved uniformly (Theorem **??**, Corollary **??**). Type A proves shifted-prefundamental generation and pro-Weyl recovery (Theorems **??**, **??**). The

residual DK-4/5 problem (extension and completion beyond evaluation modules; Conjecture **??**) is downstream of MC3. MC4 is proved: the strong completion-tower theorem (Theorem **??**) gives a completion-closed bar-cobar equivalence on (), with automatic continuity (Lemma **??**), twisting stability (Theorem **??**), and a uniform PBW bridge from MC1 (Theorem **??**). The remaining example-specific tasks are coefficient stabilization and H-level target identification. MC5 is not fully proved. What is proved at all genera is the analytic HS-sewing package (Theorem **??**: polynomial OPE growth + subexponential sector growth implies HS-sewing convergence; Theorem **??**: Heisenberg sewing via Fredholm determinant). The remaining genuswise BV/BRST/bar identification is still conjectural; at genus 0 the algebraic BRST/bar comparison is proved (Theorem **??**), while the tree-level amplitude pairing requires the additional hypothesis isolated in Corollary **??**. Periodicity is an orthogonal weak flank, not a link in this dependency chain.

THEOREM 2.2.1 *(Higher-genus PBW concentration for the standard finite-type interacting families).* [proved here] The genus-g PBW spectral sequence concentrates at all genera for the three standard finite-type interacting families treated in the proved core: affine Kac–Moody (Theorem **??**), Virasoro (Theorem **??**), and principal finite-type $\mathcal{W}$-algebras, including higher $\mathcal{W}_N$ (Theorem **??**).

This resolves MC1 for the standard finite-type interacting families. Theorem **??** *isolates the common unique-weight-2 d_2 mechanism behind the Virasoro and principal finite-type $\mathcal{W}$ arguments; for chiral algebras with semisimple Lie symmetry satisfying hypotheses* (a)–(c)*, the enrichment-killing is unconditional via Whitehead vanishing plus L_0 invertibility. The resolved status of MC1 for the named families rests on the family theorems above; the universal semisimple statement extends MK3 beyond those families without further family-specific input.*

THEOREM 2.2.2 *(Cyclic L_∞ deformation algebra and universal $\Theta_{\mathcal{A}}$* (MC2, resolved)*).* [proved here] For a modular Koszul chiral algebra $\mathcal{A}$ on a smooth projective curve with non-degenerate invariant form, there exists a cyclic L_∞-algebra $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ and a universal Maurer–Cartan element $\Theta_{\mathcal{A}}^{\leq 2} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}) \widehat{\otimes} R\Gamma(_{g,\bullet}, \mathbb{Q}))$ whose genus-1 scalar trace recovers $\kappa(\mathcal{A}) \cdot \lambda_1$, whose clutching compatibility recovers sewing, and whose Verdier duality recovers complementarity. The higher-degree shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ is proved constructively through $r = 4$; the all-degree limit $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$ is proved (Theorem **??**; Theorem **??**).

Resolved intrinsically by Theorem **??**. *The older comparison route of Theorem* **??** *now supplies the all-genera scalar package only on the proved uniform-weight lane. The* `conj:` *label prefix is retained for cross-reference compatibility. See Definition* **??**.

Conjecture 2.2.3 *(Staged factorization-categorical DK/KL extension).* [conjectured] The proved generated-core DK comparison surface (traditionally packaged as DK-2/3) on the evaluation-generated core extends to an ordinary-derived/completed/coderived enlargement carrying the natural DK-relevant subcategories. On that enlargement, the ordered-factorization and dg-shifted comparison stages extend the derived Drinfeld–Kohno/Kazhdan–Lusztig correspondence beyond the current evaluation-generated core.

This subsumes the DK-2 through DK-5 ladder of the monograph. On the type-A Yangian module surface, the corresponding theorematic thick-generation target is $D^b(\mathcal{O}_{\mathrm{poly}})$. By Lemma **??**, *the thick-generation target for the type-A module-side DK-2/3 lane is $D^b(\mathcal{O}_{\mathrm{poly}})$, not the full $D^b(\mathcal{O})$; extension to modules beyond the polynomial subcategory requires either localizing generation in a completed enhancement or enlargement of the generator set. On the factorization side, one must also construct the ordered braided transport on the evaluation-generated core itself before asking for exhaustion beyond it. DK-4/5 concern the RTT-complete dg-shifted comparison and the full triple bridge. The remaining problem is therefore enlargement beyond the theorematic core, not another evaluation-locus comparison.*

Remark 2.2.4 *(Five obstructions to the categorical lift).* The character-level evidence for thick generation is extensive: the prefundamental Clebsch–Gordan $V_n \otimes L^- \cong \bigoplus_{j=0}^{n} L^-(n-2j)$ holds for $n \leq 25$, and

K_0-containment $[M(\lambda)] \in K_0\text{-span}(\{[L^-(\text{shifted})]\})$ holds for $\lambda \leq 100$ with no failure. Five structural obstructions separate the K_0-level statement from a categorical theorem:

(i) K_0-generation does not imply thick generation: Thomason's classification of thick subcategories applies to the compact part of the derived category, and L^- ($\dim(L^-)_{-2k} = p(k)$) may fail to be compact in $\widehat{D}(\mathcal{O}^{\mathrm{sh}}_{\leq 0})$.

(ii) Francis–Gaitsgory pro-nilpotent completion changes the ambient category; the thick-generation target is $\widehat{D}(\mathcal{O}^{\mathrm{sh}}_{\leq 0})$, not $D^b(\mathcal{O}^{\mathrm{sh}}_{\leq 0})$.

(iii) $L^- \notin \mathcal{O}_{\mathrm{poly}}$ (Remark **??**): the proved Baxter equivariance on $\mathcal{O}_{\mathrm{poly}}$ does not extend by density.

(iv) The resolution obstruction $\delta(k) = p(k) - 1$ is superpolynomial, forcing localizing generation in place of thick generation.

(v) The braided monoidal structure requires the derived Drinfeld–Kohno theorem, additional data beyond the bare categorical equivalence.

The conformal weight filtration offers a path: it decomposes the problem into countably many finite-dimensional strata, each resolved by the sectorwise spectral-sequence convergence of Theorem **??**, with Efimov-type categorical formal completion assembling the answer via the pro-Weyl–Mittag-Leffler tower.

Conjecture 2.2.5 *(H-level target identification for standard infinite towers).* [conjectured] The structural completion framework is **proved** (Theorem **??**): the completed bar-cobar round-trip is a quasi-isomorphism on (), and the coefficient-stability criterion (Theorem **??**) reduces convergence to finite-window matrix stabilization. What remains is the **H-level target identification**: construct a principal-stage compatible filtered H-level target $\mathcal{W}^{\mathrm{ht}}$ for the $\mathcal{W}_\infty$ tower and a separated complete RTT-filtered dg target $\mathcal{Y}^{\mathrm{dg}}_{\mathcal{A}}$ for the standard Yangian tower, with finite quotients recovering the theorematic stages W_N and $Y_{\leq N}$, and identify the resulting local coefficient systems with the standard finite-stage data. The reduced-weight windows K_q (Definition **??**) make each identification a finite computation at each level q.

On the Yangian side, this reduces to the identities

$$\mathsf{K}^{\mathrm{line}}_{a,b}(N) = \mathsf{K}^{\mathrm{RTT}}_{a,b}(N) \qquad (a, b \leq N),$$

detected on the boundary strip $\{\Delta_{a,0}(N)\}_{0 \leq a \leq N}$. On the $\mathcal{W}_\infty$ side, this reduces to the identities

$$\mathsf{C}^{\mathrm{res}}_{s,t;u;m,n}(N) = \mathsf{C}^{\mathrm{DS}}_{s,t;u;m,n}(N),$$

detected on the finite primary seed packet $\mathcal{I}_N$. Compatibility of these finite packets with the filtered quotient systems would identify the H-level targets with the theorematic standard completions and supply the infinite-tower comparison required by the project.

Remark 2.2.6 *(MC4 reduction chain).* The formal completion theory is now **proved**: the strong completion-tower theorem (Theorem **??**) gives a completion-closed bar-cobar homotopy equivalence (Corollary **??**) on the category (). The degree cutoff (Lemma **??**) makes continuity automatic; the coefficient-stability criterion (Theorem **??**) reduces convergence to finite matrix stabilization; and the uniform PBW bridge (Theorem **??**) connects MC1 to MC4. The completion closure is stable under MC twisting (Theorem **??**) and carries completed twisting representability (Theorem **??**).

The remaining example-specific problem splits into two regimes (Remark **??**, Appendix **??** §**??**):

- *$MC4^+$ (positive towers):* Unconditionally solved by weightwise stabilization (Theorem **??**). For any target with an honest positive weight grading ($\mathcal{W}_{1+\infty}$, affine Yangians, positive RTT towers), coefficient stability is automatic by Proposition **??** and Corollary **??**.
- *$MC4^0$ (resonant towers):* The remaining H-level realization problem for algebras with weight-zero resonance (resonance rank $\rho(\mathcal{A}) > 0$, Definition **??**). The resonance-filtered completed bar-cobar (Theorem **??**) reduces this to a finite-dimensional problem when $\rho(\mathcal{A}) < \infty$. The Platonic completion theorem (Theorem **??**) proves that every positive-energy chiral algebra has finite resonance rank, so MC4 is always a finite resonance problem atop the solved positive-weight machine.

On the $\mathcal{W}_\infty$ side, the first unconditional packet is the exact stage-4 six-entry block on $\mathcal{I}_4$; on the stage-4 Ward-normalized H-level locus of Definition **??**, its effective content contracts to the four higher-spin channels of Corollary **??**.

On the standard type-A Yangian side, the local coefficient packet is already closed on the factorization-side shared-seed locus. The canonical H-level target exists as the tangent Lie algebra of the factorization formal moduli problem (Proposition **??**), with canonical associative dg model (Proposition **??**). The next step is to equip that dg model with the RTT-adapted package of Proposition **??**, after which Corollary **??** reduces the remaining input to the spectral vector-line realization package; on the equivariant multiplicative spectral realization locus, this is equivalently the standard spectral vector seed-and-shift datum. On the spectral vector-line locus, Corollary **??** closes DK-4/DK-5 on the canonical target; on the sharper equivariant multiplicative spectral realization locus, Corollary **??** contracts this to the single canonical spectral seed $V^\omega(0) = J_q^\omega(V(0))$. Corollary **??** reduces the remaining categorical step to compact-core realization of a chosen finite-dimensional factorization DK pair. The full Yangian reduction chain is detailed in §**??**.

Conjecture 2.2.7 *(BV/BRST/bar identification).* [conjectured] For a holomorphic field theory on a Riemann surface, the BV/BRST complex coincides with the bar complex of the associated chiral algebra, at all genera.

Resolved in D^{co} (Theorem **??**): the BV and bar complexes are quasi-isomorphic in the coderived category for all chirally Koszul algebras, including class M. The chain-level identification holds for classes G and L, is conditional for class C on harmonic decoupling, and *fails* for class M (quartic harmonic obstruction).

This is the mathematical-physics bridge. It is downstream of Theorem 2.2.1*, Theorem* 2.2.2*, and Conjectures* 2.2.3–2.2.5*, and should not be placed on the same logical level. Genus-*0 *algebraic BRST/bar comparison: proved* (*Theorem* **??**)*; the tree-level moduli-integration/amplitude pairing is only conditional* (*Corollary* **??**).

Remark 2.2.8 *(MC5 analytic resolution vs. BV/BRST identification).* The analytic convergence component of MC5 is fully resolved: Theorem **??** (HS-sewing at all genera) and Theorem **??** (Heisenberg sewing via Fredholm determinant) prove that sewing amplitudes converge for the entire standard landscape (universal algebras; for simple quotients at admissible levels, see Remark **??**). What remains conjectural in Conjecture 2.2.7 is the *identification* of the BV/BRST complex of the physical field theory with the bar complex of the chiral algebra, a mathematical-physics bridge that is logically downstream of the purely algebraic theorems and requires input from perturbative QFT (Costello–Gwilliam framework) beyond genus 0.

Remark 2.2.9 *(Master conjecture hierarchy).* Theorem 2.2.1 is the *entry theorem*: together with Theorems **??**, **??**, and **??**, it resolves MC1 for the standard finite-type interacting families. Theorem **??** packages the common unique-weight-2 d_2 mechanism used in the Virasoro and principal finite-type $\mathcal{W}$ proofs; under hypotheses (a)–(b) alone, it should be read as a criterion requiring separate later-page and extension control. Under the additional semisimple symmetry hypothesis (c), the enrichment-killing is unconditional. Theorem 2.2.2 (originally conjectured, now resolved by Theorem **??**) constructs the H-level completion of the

modular characteristic hierarchy. Conjectures 2.2.3 and 2.2.5 are *staged structural extensions*: the DK/KL enlargement beyond the theorematic evaluation-generated core and the filtered H-level comparison problem for the standard infinite towers. The exact dependency chain is:

$$\begin{aligned}\text{evaluation-generated core} &\Longrightarrow \text{MC3 enlargement}\\ &\Longrightarrow \text{MC4 H-level comparison}\\ &\Longrightarrow \text{MC5 physics completion.}\end{aligned}$$

Current status of this chain. MC4 now splits into $\mathrm{MC4}^+$ (positive towers, solved by stabilization, Theorem **??**) and $\mathrm{MC4}^0$ (resonant towers, reduced to a finite resonance problem by Theorem **??**); see Remark 2.2.6. MC3 is **proved for all simple types** on the evaluation-generated core: the all-types categorical prefundamental Clebsch–Gordan closure and evaluation-generated-core DK comparison are proved (Theorem **??**, Corollary **??**), and type A adds the shifted-prefundamental and pro-Weyl packets. The residual DK-4/5 problem (compact/completed extension beyond evaluation modules, Conjecture **??**) is downstream of MC3. MC4 on the $\mathcal{W}_\infty$ tower is **unconditionally closed** at all stages via $\mathcal{W}_N$ rigidity (Theorem **??**) within the strong completion-tower framework (Theorem **??**). MC4 on the Yangian tower is closed on the evaluation-generated core (249 tests); the DK-5 categorical packet in type A is now accessible (downstream of the MC3 package proved on the evaluation-generated core). On the MC5 side, the genus-0 algebraic BRST/bar comparison is proved (Theorem **??**); the genus-1 bar-side curvature package is proved (Theorems **??** and **??**); and for genus $g \geq 2$, inductive genus determination (Theorem **??**), two-dimensional convergence (Proposition **??**), and the HS-sewing criterion (Theorem **??**) prove the analytic sewing/convergence lane. The higher-genus BV/BRST/bar identification itself remains the downstream physics completion. *Critical point*: MC3 package (i) (categorical CG) is now resolved for all types (Theorem **??**); the remaining input for the full four-package MC3 beyond type A is thick generation of $\mathrm{Rep}_{\mathrm{fd}}$ by evaluation modules.

Conjecture 2.2.7 is the *physics completion*: it depends logically on the others. The master package maps to the nine futures (§2.4) as follows: Theorem 2.2.1 → Future 2; Theorem 2.2.2 → Futures 3, 4; Corollary **??** → Future 4 (quintuple → effective Γ-quadruple, proved for standard landscape); Theorem **??** → Future 4 (quadruple universal at generic level via Whitehead–KL; Theorem **??** extends to all non-critical levels for algebraic families; residual Conjecture **??** restricted to non-algebraic-family constructions); Conjecture 2.2.3 → Future 6; Conjecture 2.2.5 → Future 3; Conjecture 2.2.7 → Future 9. Futures 1, 5, 7, 8 are resolved or reformulated independently of the remaining MC3–MC5 dependency chain.

Conjecture 2.2.10 *(Standard-tower MC5 reduction after MC3 and realized MC4;* [conjectured]*).* Consider the standard principal W_∞ tower together with the standard RTT Yangian tower. Assume:

(i) the Yangian side is enlarged beyond the theorematic evaluation-generated core as in Conjecture 2.2.3;

(ii) the standard principal W_∞ MC4 comparison is in place: the target $\mathcal{W}^{\mathrm{ht}}$ satisfies the quotient-system hypotheses of Proposition **??**, and the exact packet identities on $\mathcal{I}_N$ hold. If one wants the first effective stage-4 packet read on the four higher-spin locus rather than the unconditional six-entry locus, assume that $\mathcal{W}^{\mathrm{ht}}$ is stage-4 Ward-normalized in the sense of Definition **??**;

(iii) the standard type-A Yangian realized-MC4 package is in place: the hypotheses of Proposition **??** hold, and the completed categorical step is closed by either Corollary **??** or, equivalently, Corollary **??**;

(iv) the genus-0 algebraic BRST/bar package and the higher-genus modular deformation package are those of Theorem **??** and Theorem 2.2.2.

Then no new infinite-tower algebraic obstruction remains on the standard towers outside the higher-genus BV/BRST/bar identification itself: the remaining MC5 content is exactly the genuswise comparison between the realized H-level targets and the corresponding BV/BRST complexes, compatibly with truncation, Verdier duality, and clutching.

This is a dependency theorem, not a closure theorem for the packet identities themselves. Its hypotheses are exactly the realized MC3/MC4 inputs on the standard principal and RTT towers. The stabilized chain

$$\textit{MC3 enlargement} \Longrightarrow \textit{MC4 realized H-level package} \Longrightarrow \textit{MC5 BV/BRST comparison}.$$

Its scope is only the standard principal and RTT towers; it does not assert the same closure principle for non-principal orbit families or other infinite-generator extensions. Because assumption (i) *imports Conjecture 2.2.3, the reduction principle is conditional on the current record.*

Remark 2.2.11 *(Evidence).* Assumption (ii) says that on the standard principal W_∞ side the MC4 comparison has already been reduced to exact stagewise packet identities on the finite detection system $\mathcal{I}_N$. The additional stage-4 Ward-normalized input of Definition **??**, when imposed, changes only the presentation of the first effective stage-4 block via Corollary **??**; it does not add a further tower-wide algebraic obstruction once the packet identities themselves are assumed.

Assumption (iii) says that on the standard type-A Yangian side the H-level dg-shifted target is already realized by Proposition **??**, and that the remaining completed categorical packet closes by either Corollary **??** or Corollary **??** or, on the spectral vector-line locus, Corollary **??**. Thus, after the MC3 enlargement of assumption (i), no new infinite-tower algebraic input remains on the standard Yangian tower.

Therefore the standard principal W_∞ and standard type-A Yangian towers contribute no remaining infinite-generator algebraic obstruction beyond the realized MC3/MC4 packages. The only further content is the genuswise BV/BRST/bar comparison supplied by assumption (iv), namely the passage from the realized H-level targets to the corresponding BV/BRST complexes in the sense of the genus-0 algebraic BRST/bar package of Theorem **??** together with Theorem 2.2.2, compatibly with truncation, Verdier duality, and clutching. This is exactly the claimed MC5 reduction.

Corollary 2.2.12 *(Standard-tower MC5 closure on the canonical Yangian locus;* [conditional]*).* Consider the standard principal W_∞ tower together with the Yangian

$$\mathcal{A} = \mathcal{Y}(\mathfrak{sl}_M), \qquad \mathfrak{g}_{\mathcal{A}} := T_*\mathfrak{F}_{\mathcal{A}}.$$

Assume:

(i) the Yangian side is enlarged beyond the theorematic evaluation-generated core as in Conjecture 2.2.3;

(ii) the standard principal W_∞ packet identities on $\mathcal{I}_N$ hold; if one wants the first effective stage-4 packet read on the four higher-spin locus rather than the unconditional six-entry locus, assume that the target is stage-4 Ward-normalized in the sense of Definition **??**;

(iii) the Yangian hypotheses of either Corollary **??** or Corollary **??** hold for the canonical H-level target $\mathfrak{g}_{\mathcal{A}}$; and

(iv) the genus-0 algebraic BRST/bar package and the higher-genus modular deformation package are those of Theorem **??** and Theorem 2.2.2.

Then the standard-tower MC5 packet closes.

Proof. Assumption (iii) supplies the standard type-A Yangian realized-MC4 package on the canonical formal-moduli target, via its canonical associative dg model, by either Corollary **??** or Corollary **??**. Therefore

assumptions (i)–(iv) imply the hypotheses of Proposition 2.2.10. That proposition removes every remaining infinite-tower algebraic obstruction beyond the genuswise BV/BRST/bar comparison, and assumption (iv) supplies exactly that comparison. Hence the standard-tower MC5 packet closes. □

Remark 2.2.13 *(The Arakelov–bar transfer problem).* At genus $g \geq 2$ the MC5 problem reduces to proving that the fiberwise curvature d^2_{fib} on the bar complex of a chiral algebra $\mathcal{A}$ on Σ_g equals $\kappa(\mathcal{A}) \cdot \omega_g$. Four a priori constraints force this factorization:

(i) additivity in $\mathcal{A}$ (from the tensor product of curved A_∞-structures);

(ii) antisymmetry under Koszul duality (complementarity, Theorem **??**);

(iii) scalar action on each isotypic component (Schur's lemma for the conformal symmetry);

(iv) $\dim H^{1,1}(\Sigma_g) = 1$ for every $g \geq 1$, which confines the geometric factor to a one-dimensional space.

By Theorem **??**, $\kappa(\mathcal{A})$ is the unique additive duality-constrained invariant with $\hat{A}$-genus generating function, so if d^2 factors as (algebra) $\times$ (geometry), the factors must be $\kappa(\mathcal{A})$ and ω_g. What remains is to *establish the factorization itself*: the Arnold defect on $\Sigma_g^{\times 3}$, which a priori lives in a $\binom{g+1}{2}$-dimensional space (the symmetric part of $H^{1,0}(\Sigma_g)^{\otimes 2}$) of quasi-modular $(1,1)$-forms, must be shown to collapse to a scalar multiple of ω_g after OPE contraction. The required inputs are the Fay trisecant identity (killing the holomorphic component), the Arakelov–Green identity $dd^c G = \delta - \omega_g$ (producing the Arakelov form), OPE convergence in the Arakelov metric, and the cyclic-sum cancellation in the bar formalism.

An alternative approach avoids the direct Arakelov computation: since genus 0 and 1 are proved, non-separating degeneration $\Sigma_g \to \Sigma_{g-1} \cup_{\mathrm{node}}$ reduces each genus to the preceding one plus a local nodal correction controlled by the genus-0 identification. Heuristically, the sewing correction to the scalar genus expansion decreases with g and decays factorially, but the precise asymptotics depend on the Arakelov metric and remain to be established.

Remark 2.2.14 *(Completion problem: resolved).* The completion problem is **resolved** by the strong completion-tower theorem (Theorem **??**). The single axiom $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ (the strong filtration condition, Definition **??**) simultaneously ensures:

(i) *degree cutoff*: on $\mathcal{A}_{\leq N}$, only μ_r for $r \leq N$ survives (Lemma **??**);

(ii) *automatic continuity*: the completed bar differential is quotientwise finite, not an uncontrolled infinite series;

(iii) *automatic Mittag–Leffler*: the surjective tower of finite quotients gives $\varprojlim^1 = 0$;

(iv) *quasi-isomorphism*: the complete filtered comparison lemma passes finite-stage bar-cobar duality to the inverse limit.

The completion closure () carries a quasi-inverse bar-cobar equivalence on the homotopy category (Corollary **??**), stable under MC twisting (Theorem **??**), with completed twisting representability (Theorem **??**). The coefficient-stability criterion (Theorem **??**) reduces verification to finite matrix stabilization, and the uniform PBW bridge (Theorem **??**) connects MC1 to MC4. The remaining example-specific task is H-level target identification (coefficient stabilization on finite reduced-weight windows and comparison with the named completion target).

The completion kinematics (the primitive cumulant quotient $Q(\mathcal{A})$, the primitive generating series $G_{\mathcal{A}}(t)$, the completion Hilbert series $H_{\mathcal{A}}(t)$, and the completion entropy $h_K(\mathcal{A})$) provide numerical invariants of the completed theory that are independent of curvature and OPE data (§1.20 of the concordance).

Remark 2.2.15 *(Local conjecture dependencies).* The individual conjectures in earlier chapters reduce to the master package. Each local conjecture carries a parenthetical annotation identifying its master parent.

MC	Status	Scope
MC1	**Proved** (Thm 2.2.1)	PBW concentration for KM, Vir, principal $\mathcal{W}_N$.
MC2	**Proved** (Thm 2.2.2)	Universal $\Theta_{\mathcal{A}}$, cyclic L_∞, index predictions; ~20 periodicity conjectures adjacent.
MC3	**Proved** (eval-generated core)	All-types prefundamental CG and evaluation-generated-core DK. Type-A shifted-prefundamental generation and pro-Weyl recovery. DK-4/5 (extension beyond evaluation modules) downstream. Compact/completed extension still open (Conj. **??**). Eval-gen. core DK at all types (Cor. **??**). Francis–Gaitsgory completion supplies only the ambient co-completion formalism once the compact-core equivalence is known.
MC4	**Proved** (Conj. 2.2.5)	Strong completion-tower theorem (Thm **??**); $\mathcal{W}_\infty$ unconditional at all stages; Yangian eval-core verified (249 tests); DK-5 categorical packet accessible in type A.
MC5	**Analytic part proved**	HS-sewing / convergence proved; BV/BRST-bar comparison, Feynman, AGT, and holographic identifications remain downstream.
Ind.	n/a	~8 independent problems (periodicity, derived bc-$\beta\gamma$, E_n).

Remark 2.2.16 *(Proof roadmaps).* The proof strategies for the five master conjectures are as follows. *MC1* (PBW concentration): **proved** (Remark **??**; Theorem **??**). *MC2* (cyclic L_∞ and $\Theta_{\mathcal{A}}$): **fully resolved at the level of universal MC existence** (Theorem **??**); the older scalar-trace comparison gauntlet is all-genera only on the proved uniform-weight lane (Theorem **??**). *MC3* (factorization-categorical DK/KL): **proved for all simple types** on the evaluation-generated core. All simple types have the categorical Clebsch–Gordan closure and evaluation-generated-core DK comparison (Theorem **??**, Corollary **??**). Type A proves shifted-prefundamental generation and pro-Weyl recovery. The residual DK-4/5 problem (compact/completed extension, Conjecture **??**) is downstream of MC3. The original type A chromatic strategy begins with: (1) prefundamental Clebsch–Gordan closure $V_n \otimes L^- = \bigoplus_{j=0}^{n} L^-(n-2j)$ (Proposition **??**); (2) K_0 generation via the Baxter TQ system (Theorem **??**); (3) chromatic/conformal-weight filtration reducing to countable finite-dimensional strata; (4) Efimov-type categorical formal completion (pro-Weyl recovery + Mittag-Leffler $R^1\varprojlim = 0$); (5) the remaining DK compact/completion packet isolated by spectral sequence degeneration on weight strata together with the Francis–Gaitsgory completion formalism (Theorem **??**, Remark **??**). Evaluation-generated core proved at all simple types at the H-level (Corollary **??**); on the Yangian module surface, Theorem **??** is now only the type-A thick-generation extension criterion once the ambient extension/comparison package is supplied. Extension to arbitrary simple type (BCD, exceptional) requires

the generalized Hernández–Jimbo prefundamental CG closure for all $L^-_{i,a}$ (Remark **??**). The Yangian equivariance of the Baxter singular vector is proved with spectral constraint $b = a - (\lambda + 1)/2$ (Proposition **??**); naturality on $\mathcal{O}_{\mathrm{poly}}$ follows (Corollary **??**), resolving H4 on this locus. *MC4* (filtered H-level comparison): M-level packages proved for both towers. The $\mathcal{W}_\infty$ stage-4 packet is now **closed** (Theorem **??** and Proposition **??** for explicit stage-4 formulas): the visible diagonal normalization (Proposition **??**), Ward inheritance (Proposition **??**), and Borcherds transport (Proposition **??**) are all resolved via $\mathcal{W}_4$ rigidity (Zamolodchikov–Fateev–Lukyanov), reducing the six-defect block to two primitive square classes matching the DS targets (18 + 121 tests). Promotion through $\mathcal{I}_N$ for $N \geq 5$ follows from $\mathcal{W}_N$ rigidity at each stage. The Yangian degree-2 fundamental-line coefficient is verified (249 tests); the remaining Yangian open problem is the DK-5 categorical packet (downstream of the remaining post-core MC3 completion packet). *MC5* (BV/BRST at all genera): the analytic sewing/convergence component is **proved**, but the genuswise BV/BRST/bar identification remains conjectural. The HS-sewing criterion (Theorem **??**: polynomial OPE growth + subexponential sector growth $\Longrightarrow$ convergence at all genera) and the Heisenberg sewing theorem (Theorem **??**: one-particle Bergman reduction, Fredholm determinant) resolve the analytic convergence at all genera. The genus-0 algebraic BRST/bar comparison is Theorem **??**, while the tree-level moduli-integration/amplitude pairing is only conditional (Corollary **??**); the $\mathcal{W}$-algebra bridge at genus 0 is proved (Volume II), and the $\mathrm{SC}^{\mathrm{ch,top}}$ bar–cobar Quillen equivalence is unconditional (homotopy–Koszulity proved in Volume II); the genus-1 bridge is proved (Volume II, Theorem mc5-genus-one-bridge): the Feynman-defined A_∞ operations on $E_\tau \times \mathbb{R}$ reproduce the curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_1$ via the Arnold defect on the torus. A new approach to $g \geq 2$ via logarithmic FM geometry is now available: Mok's log-smooth degeneration $\overline{\mathrm{FM}}_n(\mathcal{W}/B) \to B$ (proper, flat, with reduced fibres [322, Theorem 5.2.3]) gives an inductive structure where each fibre carries a bar complex and the degeneration formula organizes their compatibility. The Costello renormalization should factor through the log-FM product decomposition on rigid special-fibre components. *Holographic completeness status*: the bar-side algebraic packet is extensive, but the comparison from those algebraic objects to the genuine BV/BRST complexes is still incomplete. The remaining gaps are *analytic* (Costello renormalization at $g \geq 2$, downstream of MC3/MC4) and *physical/comparison-theoretic* (the genuswise BV/BRST/bar identification, including the tree-level integration pairing, together with standing axioms (H1)–(H4) of Volume II). The *graph-first* development (§**??**) derives the full structure of quantum chiral homology from stable graph topology alone: the loop order filtration decomposes the genus-g bar complex by graph Betti number (Theorem **??**), the curvature $\kappa(\mathcal{A})$ is the self-contraction trace $\mathrm{Tr}_{\mathcal{A}} = \sum_a \langle e_a, e^a \rangle$ (Theorem **??**, 27 tests), the Getzler–Kapranov involution $F^2 \simeq \mathrm{id}$ yields bar-cobar inversion as a special case (Corollary **??**), and $\mathfrak{grt}_1$ acts on all-genus bar cohomology through the graph complex weight system (Proposition **??**).

Remark 2.2.17 *(MC4 Yangian residue: low-rank computation target).* The auxiliary-kernel identity $\mathsf{K}^{\mathrm{line}}_{1,2}(N) = \mathsf{K}^{\mathrm{RTT}}_{1,2}(N) = \Lambda^2(V)$ is verified for $N = 2, 3, 4$ (Computation **??**, within 219 tests). The boundary strips $\Delta_{a,0}(4)$ and $\Delta_{a,0}(6)$ vanish on generic tensor lengths 2 and 3 in ranks $2, 3, 4$ (Computations **??** and **??**). The three-layer reduction (residue $\to$ channels $\to$ single line) is verified computationally (Computation **??**, 58 tests).

The remaining Yangian gap is dg-shifted or factorization realization, not a local coefficient computation. Under one-loop exactness, Corollary **??** identifies the local coefficient part of the remaining input as the single degree-2 identity $r^{\mathrm{fund}}_a(z)(e_1 \otimes e_2) = -\hbar(e_2 \otimes e_1)$ on the ordered fundamental line, together with the compact-generator comparison. The factorization side already carries this coefficient (Corollary **??**; Proposition **??**). The full Yangian reduction chain is detailed in §**??**.

Remark 2.2.18 *(MC4 $\mathcal{W}_\infty$ stage-4: concrete computation target).* The principal Drinfeld–Sokolov side is already explicit (Proposition **??**, Computation **??**, 121 tests). The remaining stage-4 task is the six defects

$\mathsf{D}_{s,t;u;m,n}(4) := \mathsf{C}^{\mathrm{res}}_{s,t;u;m,n}(4) - \mathsf{C}^{\mathrm{DS}}_{s,t;u;m,n}(4)$ in the channels $(3,3;4;0,2)$, $(4,4;4;0,4)$, $(3,4;3;0,4)$, $(3,4;4;0,3)$, $(4,4;2;0,6)$, $(3,4;2;0,5)$. Under Ward normalization (Corollary **??**), this contracts to the four higher-spin channels, with the weight-4 self-pairing normalization as the exact remaining scalar input (Proposition **??**; Corollary **??**).

The complete reduction chain $\mathcal{J}_4\,(29) \to \mathcal{J}_4^{\mathrm{top}}\,(7) \to \mathcal{J}_4^{\mathrm{par}}\,(6) = 4+2$ is verified computationally (Computations **??** and **??**, 83 tests). The next downstream reduced stage is $\mathcal{J}_5^{\mathrm{red}}$, an eleven-entry packet (Corollary **??**), with the stage-5 higher-spin packet carrying one effective independent coefficient on the visible pairing locus (Corollary **??**).

Remark 2.2.19 *(Conjecture-by-conjecture attack strategies).* Below we organize the full conjectural landscape into attack fronts, ordered by structural impact and tractability.

Front A : Scalar saturation universality. Conjecture **??**: does $\dim H^2_{\mathrm{cyc}}(\mathcal{A},\mathcal{A}) = 1$ hold at all levels for all modular Koszul chiral algebras with simple Lie symmetry? Proved at generic level (Theorem **??**), on the standard Lie-theoretic landscape (Corollary **??**), and at all non-critical levels for algebraic families with rational OPE coefficients (Theorem **??**). The obstruction is the primitive deformation sector $H^2_{\mathrm{cyc,prim}}(\mathcal{A})$, which must vanish. The algebraic-family theorem subsumes the generic-level result and extends it to admissible parameter values only for the already verified algebraic-family part of the standard landscape; it does not settle the live audit surface of simple admissible quotients; the residual conjecture is restricted to non-algebraic-family constructions.

Front B : Generating function conjectures. Conjectures **??**, **??**, **??**, and **??** are self-consistency checked by 82 tests (recurrence, Koszul dual positivity, growth rates, DS discriminant invariance). The critical verification computations are: $H^5(\overline{B}(\mathcal{W}_3)) \overset{?}{=} 171$, $H^4(\overline{B}(\widehat{\mathfrak{sl}}_3)) \overset{?}{=} 1352$, $H^4(\overline{B}(Y(\mathfrak{sl}_2))) \overset{?}{=} 82$.

Direct computation is feasible for the Yangian case (37M-entry matrix); the Kac–Moody case requires chiral-bar-specific methods (chain groups exceed 786K at degree 5).

Front C : MC3 critical path, CG closure resolved; completion packets remain. The four-step critical path (Conjectures **??**→**??**→**??**→**??**) is no longer treated as globally closed. What is settled uniformly is the categorical Clebsch–Gordan input (Theorem **??**) and the evaluation-generated-core DK comparison (Corollary **??**). Type A proves shifted-prefundamental generation and pro-Weyl recovery, leaving the compact/completed extension conjecture as the remaining packet. The five proof ingredients are therefore:

(i) *Prefundamental Clebsch–Gordan closure* (Proposition **??**): $V_n \otimes L^- \cong \bigoplus_{j=0}^{n} L^-(n-2j)$ for all $n \geq 0$ (**proved**; verified $n = 1,\dots,25$, depth 60). Proves evaluation-stability of L^- and universal character containment (Corollary **??**).

(ii) *K_0 generation via Baxter TQ system*: K_0-level TQ relations verified at 498 tests for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$, with derived-level lifts to distinguished triangles, octahedral axiom for iterated TQ, and $\mathfrak{sl}_3$ Clebsch–Gordan resolution (226 tests at the D^b level). Extended from $\mathcal{O}_{\mathrm{poly}}$ via the chromatic filtration (Theorem **??**). K_0 generation of $\mathcal{O}_Y$ **proved** (Corollary **??**; $31 + 109 = 140$ tests in `test_prefundamental_cg_closure`).

(iii) *Chromatic/conformal-weight filtration*: reduces the infinite-dimensional problem to countably many finite-dimensional strata. Thick generation obstruction analysis (206 tests, Computation **??**): obstruction $\delta(k) = p(k) - 1$ sub-exponential (matches E_1 growth); bidegree-wise spectral-sequence convergence succeeds on each stratum where everything is finite-dimensional.

(iv) *Efimov-type categorical formal completion*: pro-Weyl recovery (344 tests, chain-level Ext, derived inverse limit, Mittag-Leffler $R^1\varprojlim = 0$) assembles the stratum-by-stratum answer. This breaks the circular dependency on L^- compactness by working stratum-by-stratum.

(v) *Remaining compact/completed DK packet*: spectral sequence degeneration on weight strata (101 tests) is part of the type-A evidence for the final packet, but what still remains open is Conjecture **??**: extending

the evaluation-core DK equivalence to the compact shifted-prefundamental core and then comparing that compact-core equivalence with the desired Francis–Gaitsgory/pro-nilpotent completion.

Remaining open problems: the final type-A packet and its propagation beyond type A. Even in type A, the compact/completed DK packet in (v) remains open. Extension to types BCD and the exceptional types further requires the generalized Hernández–Jimbo prefundamental CG closure for all simple Lie algebras, using the prefundamental modules $L^-_{i,a}$ indexed by nodes of the Dynkin diagram. Three structural results established in type A (R1–R3):

(R1) *Prefundamental CG* (Proposition **??**): **proved** for all n in type A.

(R2) *Kernel decomposition*: $K := \mathrm{ch}(L^-) - \mathrm{ch}(M(0))$ has coefficients $p(k) - 1 = \Sigma\, p_2(k)$ (cumulative partitions into parts ≥ 2).

(R3) *Recursive self-similarity*: $K - L^-(\mathrm{hw} = -4)$ recovers partition numbers $p(k)$, connecting MC3 thick generation to classical partition theory.

Conjecture **??** (Strategy IV sectorwise DK extension beyond the evaluation-generated core): E_1 subexponential growth verified for all simple types (301 tests), sectorwise convergence on the evaluation-generated core proved by Theorem **??**, and sectorwise finiteness verified for all simply-laced lattice VOAs (93 tests).

Gap: extend the same sectorwise package and factorization descent to the completed/coderived enlargement.

The lattice route (Theorem **??**) provides an unconditional alternative for simply-laced types at level 1. The Kazhdan–Lusztig N-complex structure (Conjecture **??**) now has a tested M-level certificate on $\mathfrak{u}_q(\mathfrak{sl}_2)$: the tractable roots $N = 2, 3$ are computed directly, the first larger roots $N = 4, 5$ satisfy $d_q^N = 0$ by the quantum-binomial structural certificate, and the reduced degree-2 vanishing is separated from the genuinely non-classical check $d_q^2 \neq 0$ at bar degree ≥ 3 (74 tests).

Explicit computation yields exact KL bar cohomology dimensions at $N = 3$ and $N = 4$: the $N = 3$ degree-4 packet has $\dim H_2^{1,2} = \dim H_3^{2,1} = 3$ (a sparse palindromic weight profile at weights $-3, 0, 3$); the $N = 4$ degree-1 channels all vanish ($H_1^{3,1} = H_1^{2,2} = H_1^{1,3} = 0$); and the sole surviving $N = 4$ degree-2 channel is $\dim H_2^{1,3} = 66$, with palindromic root-weight profile $1, 4, 8, 12, 16, 12, 8, 4, 1$ across weights $-4, \ldots, 4$ (verified exhaustively over all 42,718,284 raw columns; 354,668 prefix signatures; 74 tests).

The $N = 4$ packet is *not* a convolutional shadow of the $N = 3$ data: no exact symmetric transport kernel recovers the staircase profile. The first structured route between them is a finite-window Dirichlet Schrödinger operator with sextic potential. *Gap*: determine whether this S-level operator lifts to an H/M-level KL mechanism or is only a numerical organizing principle.

Genus-1 extension. The sectorwise finiteness of the lattice bar complex persists at genus 1 on E_τ: each lattice sector remains finite-dimensional over $\overline{\mathcal{M}}_{1,1}$. The genus-1 curvature d^2_{fib} concentrates in the Cartan (Heisenberg) sector, where it equals $\mathrm{rank}(\Lambda) \cdot \omega_1$, and *vanishes* in all adjacent-root sectors (where only simple-pole OPEs appear and the genus-1 propagator correction is regular at the pole). In particular:

$$\kappa(V_\Lambda^{N,q}) = \mathrm{rank}(\Lambda), \qquad F_1(V_\Lambda^{N,q}) = \mathrm{rank}(\Lambda)/24,$$

independent of the deformation parameters (N, q). Curvature and braiding are orthogonal: the cocycle $\varepsilon_{N,q}$ modifies only the ordering cycles in the root sectors, while the genus tower is controlled entirely by the Cartan level. This is verified computationally for A_2, A_3, and D_4 at multiple values of N (22 tests).

Front D : MC4 coefficient matching. $\mathcal{W}_\infty$ side: the stage-4 principal DS packet on $\mathcal{I}_4$ is explicit. The four higher-spin channels are given as rational functions of c (Proposition **??**, Computation **??**, 121 tests):

$$c^2_{334} = \frac{42c^2(5c+22)}{(c+24)(7c+68)(3c+46)}, \quad c^2_{444} = \frac{112c^2(2c-1)(3c+46)}{(c+24)(7c+68)(10c+197)(5c+3)},$$

plus inter-coefficient relations $\mathsf{C}^2_{3,4;3;0,4} = \frac{9}{16}\, c^2_{334}$, $\mathsf{C}^2_{3,4;4;0,3} = \frac{5}{7}\, c^2_{334}$ (Corollary **??**). Thus at signless square-class level the four higher-spin principal targets reduce to the primitive pair c^2_{334}, c^2_{444}, with the mixed squares forced by c^2_{334}.

On the residue side, the corresponding four-channel higher-spin packet contracts further to three channels on the additional visible pairing-compatible Ward-normalized locus of Proposition **??** and Corollary **??**: the swap-even channel $\mathsf{C}^{\mathrm{res}}_{3,4;3;0,4}(4)$ is then forced by the $(3,3)$ channel $\mathsf{C}^{\mathrm{res}}_{3,3;4;0,2}(4)$. More sharply, Corollary **??** shows that on this visible pairing locus the entire stage-4 higher-spin comparison reduces at signless square-class level to the primitive-plus-transport triple

$$(3,3;4;0,2), \qquad (4,4;4;0,4), \qquad (3,4;4;0,3),$$

with the swap-even square channel automatic from the first entry.

The visible top-pole Borcherds transport relation of Proposition **??** is now **resolved**: the DS computation gives the relation as an algebraic identity (Corollary **??**), and $\mathcal{W}_4$ uniqueness (Zamolodchikov–Fateev–Lukyanov rigidity) transfers it to the residue calculus (18 tests, `test_w4_borcherds_transport.py`). Under Corollary **??**, this collapses the primitive-plus-transport triple to the two primitive self-coupling square classes. Proposition **??** therefore packages the exact stage-4 local order as the visible weight-4 normalization identity. On the visible pairing locus, the next step is the swap-odd higher-spin transport relation. Under that single additional transport input, Corollary **??** reduces the visible pairing stage-4 higher-spin square problem to the same two primitive self-coupling square classes as on the principal side.

Both principal Virasoro-target identities are verified. The remaining W-side packet is therefore the exact six-defect vanishing problem, equivalently the six identities

$$\mathsf{C}^{\mathrm{res}}_{s,t;u;m,n}(4) = \mathsf{C}^{\mathrm{DS}}_{s,t;u;m,n}(4)$$

on $\mathcal{I}_4$, followed by their promotion through $\mathcal{I}_N$ via the incremental packets $\mathcal{J}_{M+1}$ of Proposition **??**; the additional four-channel contraction follows from Proposition **??** (resolved).

Yangian side: auxiliary-kernel identity $\mathsf{K}^{\mathrm{line}}_{1,2}(N) = \mathsf{K}^{\mathrm{RTT}}_{1,2}(N) = \Lambda^2(V)$ verified for $N = 2,3,4$ (dimensions $1, 3, 6$ = triangular numbers), and the faithful-evaluation boundary-strip packets $\Delta_{a,0}(4)$ and $\Delta_{a,0}(6)$ vanish on generic tensor lengths 2 and 3 in ranks $2,3,4$ (current Yangian residue suite: 219 tests). The theorematic all-rank remainder is the transport of the dg-shifted or factorization one-factor line operator to the standard normalized mixed-tensor residue on the fundamental line; under one-loop exactness, the local coefficient part of this remainder is the single degree-2 fundamental-line coefficient of Corollary **??**.

Front E: KL three-stage (Conjectures ??→??→??). Stage 1: promote the admissible-level N-complex certificate at roots of unity to the periodic/coderived KL object, starting from the first computed $N = 3$ packet

$$\dim H^{1,2}_2 = \dim H^{2,1}_3 = 3$$

and determining whether one flavor or the paired packet carries the KL periodic shadow. Stage 2: coderived equivalence via bar-cobar. Stage 3: braided monoidal upgrade from fusion-tensor compatibility. The lattice route resolves all three stages unconditionally at level 1 for simply-laced types (Theorems **??** and **??**).

Front F: Physics horizon (MC5 children, ~20 conjectures). These include BV/BRST identifications (Conjecture 2.2.7), Feynman diagram/path-integral correspondences (Conjectures **??**, **??**), holographic/AdS-CFT duality (Conjectures 2.3.13, **??**), AGT correspondences (Conjectures **??**, **??**, **??**; now developed in Volume II), and Chern–Simons/topological bridges (Conjectures **??**, **??**).

All are downstream of MC1–MC4; their resolution requires QFT input beyond this manuscript's scope. But the remaining problem is no longer one undifferentiated physics horizon. The exact mathematical route is now:

(i) the finite disk-local packet of Conjecture **??** reduced by Proposition **??** to the binary residue/OPE and ternary residue package, and on the theorematic anomaly-free genus-0 BRST/bar locus reduced further by Corollary **??** to the ternary residue comparison alone;

(ii) the compactification/Stokes upgrade from those local formulas to the FM boundary calculus, using Proposition **??**; after this step, Proposition **??** reduces the compactified ternary packet on $\overline{M}_{0,4}$ to two independent boundary channels, canonically the residues at 0 and 1 in the standard $0, 1, \infty$ chart by Corollary **??**, equivalently the two coefficients of $d \log t$ and $d \log(1-t)$ by Corollary **??**, i.e. the named coefficient identities of Corollary **??**; and

(iii) only then the genuswise clutching and renormalization lift for Conjecture 2.2.7.

In that ordered route, GKW24 supplies the local perturbative bracket package, CDG20 supplies source boundary-chiral examples, and DNP25 supplies line-operator targets that must still be sorted upstream under MC4 whenever they are used to build the Yangian H-level comparison.

Front G : Independent conjectures (~7). Reflected modular periodicity (Conjecture **??**), derived bc-$\beta\gamma$ (Conjecture **??**; Koszul check verified at 57 tests), $\mathcal{W}$-orbit duality (Conjecture **??**; hook-type corridor in type A conditional on hook-wide DS/bar compatibility, Theorem **??**; type-A transport-to-transpose is Conjecture **??**), and the E_n Koszul duality (now Theorem **??**) are not subsumed by any master conjecture. The E_n statement is resolved by chaining Totaro, Lambrechts–Volić, and Knudsen. (The former conjecture on Virasoro $c = 26$ self-duality is now Proposition **??**: NAP gives $\int_X \mathcal{V}_{26} \simeq \mathbb{D}(\int_{-X} \mathcal{V}_0)$; the self-dual point is $c = 13$. Near-rationality of the Virasoro bar sequence is now resolved: Proposition **??** establishes a period-6 Padé matching pattern with excess $O(1)$, subsuming the original $O(\sqrt{q})$ question.)

Remark 2.2.20 *(Structural gaps).* Eight structural gaps separate the proved theorems (Stratum I) from the full package (Stratum II). Three are now closed (H1, H4 on $\mathcal{O}_{\mathrm{poly}}$, H8); the remaining five have clear resolution strategies.

H1 : Coderived Ran-space formalism, *closed*. Foundational theory established: Appendix **??**, §**??** defines curved factorization coalgebras (with CDG-factorization compatibility via the coaugmentation tensor product formula), coacyclic factorization objects, and the coderived/contraderived factorization categories on $\mathrm{Ran}(X)$. Structural theorems: stratified conservative restriction via Positselski's injective characterization (Theorem **??**), provisional embedding (Proposition **??**), bar functor well-definedness (Proposition **??**), and factorization co-contra correspondence for conilpotent coalgebras (Theorem **??**). *Gap closed*: off-Koszul bar-cobar inversion (Theorem **??**; proved via stratified Positselski + conservative assembly).

H2 : $\mathcal{W}_\infty$ H-level realization (MC4). Resolved by Theorem **??**: the factorization target $\mathcal{W}_\infty^{\mathrm{fact}}$ is constructed via E_∞-sectorwise convergence at each finite stage, with DS–KD intertwining (Theorem **??**) identifying $(W_N^k)^! \simeq W_N^{-k-2N}$ and Mittag–Leffler (Corollary **??**) controlling the inverse limit. The residue identities $\mathsf{C}^{\mathrm{res}} = \mathsf{C}^{\mathrm{DS}}$ follow from the chain-level content of the intertwining theorem. The remaining $\mathcal{W}_\infty$ open problem is the large-N coupling identification $\lambda \mapsto 1-\lambda$ (Conjecture **??**).

H3 : Thick generation beyond $\mathrm{Rep}_{\mathrm{fd}}$ (MC3), resolved for type A at all N. Thick generation by evaluation⊕prefundamental modules proved for $Y(\mathfrak{sl}_N)$ at all $N \geq 2$ (Theorem **??**): the compact objects of $\widehat{D}(\mathcal{O}^{\mathrm{sh}}_{\leq 0})$ are the thick closure of $\{V_{\omega_i}(a)\} \cup \{L^-_{i,a}\}$. The **rank-independent Baxter argument** (Step 2) shows: $w_0 = v_{\omega_i} \otimes f_i \cdot v_0$ is annihilated by $\Delta(E_j)$ for all j, unconditionally ($j = i$: $\lambda_i = 0$; $j \neq i$: $[e_j, f_i] = 0$), using only the Chevalley–Serre presentation. Computational verification: 1,690 tests at $\mathfrak{sl}_2$.

H4 : Baxter derived lift (MC3 path (i)), *proved on $\mathcal{O}_{\mathrm{poly}}$; extended to prefundamental locus at $\lambda = 0$.* On $\mathcal{O}_{\mathrm{poly}}$: spectral constraint $b = a - (\lambda+1)/2$ (Proposition **??**, 85 tests), naturality (Corollary **??**), D^b-level octahedral axiom (724 tests). On the prefundamental locus: the $\lambda = 0$ case gives $\Delta(E) \cdot w_0 = 0$

unconditionally, yielding the SES $0 \to L^-(b-1) \to V_1(a) \otimes L^-(b) \to L^-(b+1) \to 0$ for all a, b (13 tests).

H5 : Prefundamental generation (MC3 path (ii)), *resolved for* $Y(\mathfrak{sl}_2)$. The prefundamental Clebsch–Gordan (Proposition **??**): $V_n(a) \otimes L^-(b) \cong \bigoplus_{j=0}^{n} L^-(b+n-2j)$ **proved** for all n (formal series identity; verified $n = 1, \ldots, 20$, depth 60; 109 tests in `test_prefundamental_cg_closure`). This proves evaluation-stability of L^- (Remark **??**), universal character containment for *all* λ (Corollary **??**), and K_0-generation of all Verma modules (Corollary **??**; verified $\lambda = 0, \ldots, 30$). Combined with the unconditional Baxter SES, $L^-(b') \in \mathrm{thick}\langle L^-(b), V_1\rangle$ for any b, b'. Total: 1,800+ tests supporting the four-step proof.

H6 : Pro-Weyl M-level lift (MC3 path (iii)). Shadow-level convergence of the pro-Weyl system is proved: Mittag-Leffler condition holds, $R^1\varprojlim = 0$, and chain-level Ext computations verify the derived inverse limit (344 tests). *Gap*: construct a dg-algebra or A_∞ model for the pro-Weyl limit $M(\Psi) \simeq R\varprojlim W_m$ and prove quasi-isomorphism with the shifted Yangian Verma.

H7 : Holomorphic-to-real propagator extension (MC5). The genus-0 BRST/bar identification (Theorem **??**) uses holomorphic propagators on $\mathbb{P}^1$. Extension to $g \geq 1$ requires passing from holomorphic propagators on Σ_g to real propagators (à la Kontsevich) with Costello's renormalization data. The first missing step is the disk-local packet of Conjecture **??**: identify the local perturbative BRST brackets with the bar residue operations on C_2 and the three codimension-1 strata of $\overline{C}_3$. By Proposition **??**, this is already the full independent local comparison problem. On the anomaly-free genus-0 BRST/bar locus of Corollary **??**, the binary part is already theorematic, so the remaining local check there is the ternary comparison on $\overline{C}_3$. Once that ternary comparison is compactified to $\overline{M}_{0,4} \cong^1$ with logarithmic boundary behavior, Proposition **??** reduces it further to any two of the three boundary channels; on the standard $0, 1, \infty$ chart, Corollary **??** makes this the explicit pair of residues at 0 and 1, equivalently the two scalar coefficients of $d\log t$ and $d\log(1-t)$ by Corollary **??**, i.e. the named coefficient identities $\mathsf{a}_{\mathrm{pert}} = \mathsf{a}_{\mathrm{bar}}$ and $\mathsf{b}_{\mathrm{pert}} = \mathsf{b}_{\mathrm{bar}}$. Once that packet is compactified and matched with the FM Stokes relations, the higher-genus propagator problem becomes an analytic genus-lifting problem rather than a local dictionary problem. This is downstream of MC1–MC4.

H8 : DS transport of T_{br} (spectral branch), *resolved*. Drinfeld–Sokolov reduction preserves the discriminant $\Delta_{\mathcal{A}}(x) = \det(1 - x\,T_{\mathrm{rec}})$ (Theorem **??**). Theorem **??** refines this to a spectral decomposition: the discriminant factors as $(1 - \lambda_{\mathrm{dom}}\,x)\,\Delta^{\mathrm{inv}}(x)$, where DS preserves $\Delta^{\mathrm{inv}}(x)$ and sends the dominant eigenvalue $\dim \mathfrak{g} \mapsto 1$. Full T_{br} similarity holds iff $\mathrm{rank}(\mathfrak{g}) = 1$. Verified at $\mathfrak{sl}_3$ (154 tests).

Homotopy templates for conjectures

The stratification above classifies conjectures by *difficulty*. The following orthogonal classification sorts them by *homotopy type*. Every conjecture in this monograph fits one of seven templates (Convention **??**).

Convention 2.2.21 *(Homotopy templates).* **Type I.** *Existence of a homotopy object.* A filtered cyclic L_∞-algebra, MC element, or factorization (co)algebra is asserted to exist with specified compatibility. *Example:* $\Theta_{\mathcal{A}}$ (Theorem 2.2.2).

Type II. *Equivalence of homotopy categories.* Two ∞-categories (or stable/E_1-monoidal variants) are equivalent; current chain models give a presentation on a dense sublocus. *Examples:* full coderived bar-cobar (Conjecture 2.2.3), full DK, KL from bar-cobar.

Type III. *Shifted-symplectic / Lagrangian enhancement.* A pairing already constructed at the model level upgrades to an n-shifted symplectic structure on a formal moduli problem. *Examples:* complementarity upgrades, moduli of Koszul pairs.

Type IV. *Trace / index statement.* A scalar invariant is the image of a universal deformation object under a trace or pushforward functor. *Examples:* genus generating series ($\hat{A}$-genus), GRR for modular deformations.

Type V. *Periodicity / autoequivalence.* A finite homotopy action, graded autoequivalence, or periodic filtration operator exists on a homotopy object; numerical periods are its shadow. *Examples:* reflected modular periodicity, quantum group periodicity at roots of unity.

Type VI. *Filtered / pro-object comparison.* A theorematic completed model exists stagewise, and the remaining problem is to identify the intended H-level target by compatible finite quotients and finite-detection criteria. *Examples:* $\mathcal{W}_\infty$ duals (Conjecture 2.2.5), Yangian towers.

Type VII. *Physics-dictionary conjecture.* Two deformation problems (one algebraic, one field-theoretic) are controlled by quasi-isomorphic dg Lie or L_∞-algebras. *Examples:* BRST = bar (Conjecture 2.2.7), BV = bar-cobar, path integral = factorization.

Remark 2.2.22 *(Using the templates).* Each individual conjecture in the monograph carries a scope remark identifying its tier (Foundational / Consequence / Physics). The homotopy template provides the complementary structural information: *what kind of homotopy-theoretic statement* the conjecture makes. The resolved PBW theorem and the four remaining master conjectures above are respectively of Types I, II, II, VI, and VII.

2.3 Research directions

Higher-dimensional Koszul duality

Theorem 2.3.1 *(E_n Koszul duality via configuration space integrals;* [proved elsewhere]*).* For an E_n-algebra A on an oriented n-manifold M, the bar-cobar adjunction admits an explicit realization via configuration space integrals on the Fulton–MacPherson compactification $\overline{C}_k(M)$. All four components are proved:

(i) [[proved elsewhere]; Idrissi [232]] The bar differential is given by integration along collision divisors $D_I \subset \overline{C}_k(M)$, with the propagator determined by a Green's form $G \in \Omega^{n-1}(M \times M \setminus \Delta)$. Poincaré duality on configuration spaces ensures $d^2 = 0$ for all n.

(ii) [[proved here]] For $n = 2$ with complex structure (complex curves viewed as real oriented surfaces), this recovers the topological shadow of the chiral bar-cobar duality of Theorem **??**.

(iii) [[proved elsewhere]; Totaro [387], Lambrechts–Volić [288], Knudsen [269]] For $n \geq 3$, the Totaro/graph-complex relations govern the higher-dimensional chain models: the Arnold relations are replaced by Totaro's presentation of $H^*(\overline{C}_k(\mathbb{R}^n))$, and the bar cohomology is governed by E_n-operadic Koszul duality.

(iv) [[proved here]] The Poincaré–Koszul duality of Ayala–Francis ([30], Theorem 7.8) is recovered as the ∞-categorical shadow of this explicit chain-level construction.

Proposition 2.3.2 *($n = 2$ recovery and AF comparison;* [proved here]*).* Parts (ii) and (iv) of Theorem **??** hold:

(a) At $n = 2$, on a complex curve X viewed as a real oriented surface, the FM propagator class admits the holomorphic representative $\eta_{ij} = d\log(z_i - z_j)$ recovers the bar differential of Definition **??**, and the Arnold relations (Appendix **??**) give $d^2 = 0$.

(b) The resulting bar-cobar adjunction is Theorem **??**, which is the chain-level refinement of the ∞-categorical Poincaré–Koszul duality of Ayala–Francis [30] restricted to $n = 2$ (Remark 1.5.1).

Proof. Part (a) is the content of §**??**: the bar differential is $d = \sum_{i<j} \mathrm{Res}_{D_{ij}}(\eta_{ij} \wedge -)$ with η_{ij} the FM propagator on the curve X. Arnold relations ensure $d^2 = 0$ (Corollary **??**). Part (b): by Remark 1.5.1, the AF equivalence $\int_M A \simeq \int_M A^!$ specializes at $n = 2$ to our bar-cobar adjunction, with the factorization homology integral computed via FM configuration space integrals. □

Remark 2.3.3 *(Proof status).* All four parts are proved. Part (i) by Idrissi [232], who establishes that Poincaré duality on configuration spaces ensures $d^2 = 0$ for the propagator-based differential on E_n-algebras for all n. Parts (ii) and (iv) by Proposition 2.3.2 below. Part (iii) follows from a chain of three results: (a) Totaro's presentation [387] of $H^*(\overline{C}_k(\mathbb{R}^n))$; (b) E_n formality (Lambrechts–Volić [288]: the operadic quasi-isomorphism $C_*(\mathsf{E}_n) \xrightarrow{\sim} H_*(\mathsf{E}_n)$ transfers the bar-cobar adjunction to the homological level); (c) Knudsen's E_n Koszul duality [269], which provides PBW and Koszul duality theorems for higher enveloping algebras $U_n(\mathfrak{g})$. Together, these establish that the bar cohomology of any E_n-algebra is governed by E_n-operadic Koszul duality, with the Totaro presentation providing the explicit generators and relations.

What remains as a computational refinement is writing down explicit closed-form expressions for the bar differential in terms of Totaro generators at each $n \geq 3$; the propagator and FM compactification exist, and $d^2 = 0$ is proved, but the individual formulas (analogous to the Arnold-based computations at $n = 2$) have not been tabulated. This is a concrete calculation, not a theoretical gap. (Contributing to Conjecture 2.2.7.)

Remark 2.3.4 *(Proof assembly for part (iii)).* Part (iii) of Theorem **??** is proved by assembling the following known ingredients into the abstract chain-level construction.

Step 1: Formality. The E_n operad is formal over $\mathbb{Q}$ for all $n \geq 2$: Kontsevich [275] and Tamarkin [385] proved this for $n = 2$; Lambrechts–Volić [288] established the general case via configuration space integrals on semi-algebraic models for $\overline{C}_k(\mathbb{R}^n)$. Formality provides a quasi-isomorphism $C_*(\mathsf{E}_n) \xrightarrow{\sim} H_*(\mathsf{E}_n)$ of operads, so the bar-cobar adjunction can be studied at the level of homology.

Step 2: Totaro's presentation. For $n \geq 3$, the cohomology ring $H^*(\mathrm{Conf}_k(\mathbb{R}^n); \mathbb{Q})$ is generated by degree-$(n-1)$ classes ω_{ij} $(1 \leq i < j \leq k)$ subject to: (a) $\omega_{ij} = (-1)^n \omega_{ji}$; (b) $\omega_{ij}^2 = 0$; (c) the generalized Arnold relation $\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0$ [387, 85]. These replace the classical Arnold relations (which govern $n = 1, 2$) with their higher-dimensional analogues; the key difference is that the generators sit in odd degree when n is even and even degree when n is odd.

Step 3: Graph complex connection. Kontsevich's graph complex GC_n computes $H^*(\mathrm{Aut}(\mathsf{E}_n))$, and the bar differential on E_n-algebras factors through GC_n. The passage from $n = 2$ to $n \geq 3$ is governed by the transition from the Lie graph complex ($n = 1$) through the associative graph complex ($n = 2$) to the commutative graph complex ($n \to \infty$); the obstruction theory at each stage is controlled by the parity of the generators.

Step 4: Synthesis. The abstract construction is complete: formality (Step 1) transfers the bar-cobar adjunction to homology; the Totaro presentation (Step 2) provides explicit generators; Knudsen's higher enveloping algebras [269] identify the bar cohomology with E_n-operadic Koszul duality; and Idrissi's Part (i) [232] ensures $d^2 = 0$ at the chain level. What remains as a *computational refinement* is writing down the explicit bar differential formula in Totaro generators at each $n \geq 3$; the natural candidate propagator is the Lambrechts–Volić propagator [288], and the parity of the generators (odd for even n) produces sign patterns that must be tracked through the differential.

Vassiliev invariants from Feynman transform

Conjecture 2.3.5 *(Vassiliev invariants from bar complex;* [conjectured]*).* For $\mathcal{A} = \widehat{\ }_k$, the Feynman transform identification $\overline{B}^{\mathrm{full}}(\mathcal{A}) \simeq \mathrm{FT}_{\mathcal{M}\mathrm{od}}(\mathcal{A})$ (Theorem **??**(ii)) produces universal Vassiliev invariants when specialized to configuration spaces of circles:

(i) The restriction $\overline{C}_n(X) \rightsquigarrow C_n(S^1)$ (from complex curve to real circle) sends the holomorphic propagator $\eta_{ij} = d\log(z_i - z_j)$ to the Kontsevich propagator $\eta^K_{ij} = \frac{1}{2\pi} d\arg(t_i - t_j)$.

(ii) Under this restriction, the bar complex weight systems $W_\Gamma : {}^{\otimes n} \to \mathbb{C}$ match the classical weight systems of Vassiliev theory, with the Lie algebra providing the weight system as in Bar-Natan [33].

(iii) The genus expansion $\sum_g g_s^{2g-2}\overline{B}^{(g)}$ specializes to the loop expansion of the Kontsevich integral.

Proposition 2.3.6 *(Genus-*0 *weight systems from bar complex;* [proved here]*).* At genus 0, the bar complex of $\widehat{\ }_k$ produces classical weight systems:

(i) For each chord diagram Γ with n chords, the degree-n bar differential $d^{(0)} : \overline{B}^n \to \overline{B}^{n-1}$ induces a linear functional $W_\Gamma : {}^{\otimes 2n} \to$ via the residue pairing of the FM propagator on $\overline{C}_{2n}(\mathbb{P}^1)$.

(ii) These coincide with the Bar-Natan weight systems of the Lie algebra with the Killing form at level k: W_Γ assigns to each chord the contraction κ^{ab} and to each vertex the structure constants f^a_{bc}.

Proof. At genus 0, the bar differential is given by residues of the propagator $\eta_{ij} = d\log(z_i - z_j)$ along collision divisors $D_I \subset \overline{C}_n(\mathbb{P}^1)$ (Definition **??**). For a chord diagram Γ with chord (i, j), the residue $\mathrm{Res}_{z_i \to z_j}\eta_{ij}$ extracts the OPE coefficient $J^a(z_i)J^b(z_j) \sim k\kappa^{ab}/(z_i - z_j)^2 + f^{ab}_c J^c/(z_i - z_j)$: the double pole contributes the Killing contraction $k\kappa^{ab}$ and the simple pole the structure constants f^{ab}_c. These are precisely the assignments in Bar-Natan's construction of weight systems from Lie algebras [33]. □

Remark 2.3.7 *(Scope).* Proposition 2.3.6 establishes the genus-0 weight system extraction. The Feynman transform of Com recovers the full graph complex computing Vassiliev invariants (Getzler–Kapranov [199]), and our identification $\overline{B}^{\mathrm{full}} \simeq \mathrm{FT}$ (Theorem **??**) extends this to Koszul chiral algebras. The gap at higher genus is the passage from complex curves to real 1-manifolds: our propagators are holomorphic, while Kontsevich's are real. Comparing requires an analytic continuation argument from holomorphic to topological Feynman transforms. The correct bridge between holomorphic and topological frameworks is Chern–Simons gauge equivalence, not merely restriction from complex curves to S^1: Cattaneo–Mnev [80] establish that configuration space integrals on 3-manifolds produce Vassiliev invariants via BV integration, and the comparison map involves the CS partition function (not a pointwise restriction of propagators).

Homotopy template (Convention **??**): Type VII (physics-dictionary). The conjecture asserts that the dg Lie algebras controlling holomorphic (chiral bar) and topological (Kontsevich integral) deformation problems are quasi-isomorphic. (Contributing to Conjecture 2.2.7.)

Anomaly cancellation and Koszul structure

Theorem 2.3.8 *(Anomaly cancellation as Koszul constraint;* [proved here]*).* For the matter-ghost system $\mathcal{A}_{\mathrm{tot}} = \mathcal{A}_{\mathrm{matter}} \otimes \mathcal{A}_{\mathrm{ghost}}$:

(i) The obstruction coefficient is additive: $\kappa(\mathcal{A}_{\mathrm{tot}}) = \kappa(\mathcal{A}_{\mathrm{matter}}) + \kappa(\mathcal{A}_{\mathrm{ghost}})$ (Corollary **??**).

(ii) The total bar complex is uncurved ($d^2_{\mathrm{bar}} = 0$) if and only if $\kappa(\mathcal{A}_{\mathrm{matter}}) + \kappa(\mathcal{A}_{\mathrm{ghost}}) = 0$.

(iii) For the Virasoro matter-ghost system, $\kappa(\mathrm{Vir}_c) = c/2$ and $\kappa(\mathrm{ghost}) = c_{\mathrm{ghost}}/2 = -13$, so $\kappa_{\mathrm{tot}} = 0$ precisely at $c = 26$.

Proof. Part (i) is Corollary **??**: the singular OPE of the tensor product is the sum of the individual singular OPEs (the fields act on disjoint sectors), so the genus-1 curvature is additive. Part (ii) follows from $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_1 \cdot \mathrm{id}$ (Theorem **??**, Convention **??**): the total bar complex is uncurved if and only if $\kappa_{\mathrm{tot}} = 0$. Part (iii) is the computation $\kappa(\mathrm{Vir}_c) = c/2$ (Computation **??**) combined with $c_{\mathrm{ghost}} = -26$, so $\kappa_{\mathrm{tot}} = c/2 - 13 = 0$ at $c = 26$. □

THEOREM 2.3.9 *(Physical anomaly cancellation, genus 0;* [proved here]*).* The condition $\kappa_{\mathrm{tot}} = 0$ (Theorem 2.3.8) is equivalent, at genus 0, to the BRST anomaly cancellation $c_{\mathrm{matter}} + c_{\mathrm{ghost}} = 0$. The quasi-isomorphism $\Phi\colon (\mathcal{A}_{\mathrm{tot}}^{\mathrm{rel}}, Q_{\mathrm{BRST}}) \xrightarrow{\sim} (\overline{B}_0^{\mathrm{ch}}(\mathcal{A}_{\mathrm{tot}}), d_{\mathrm{bar}})$ of Theorem **??** identifies both anomaly cancellation conditions.

Proof. This is Corollary **??**. □

THEOREM 2.3.10 *(Physical anomaly cancellation for affine Kac–Moody algebras;* [proved here]*).* For affine Kac–Moody algebras $\widehat{\mathfrak{g}}_k$ at $k \neq -h^\vee$, the bar-semi-infinite comparison and anomaly cancellation hold at genus 0 *without restriction on the central charge*:

(i) Semi-infinite (BRST) cohomology is computed via the bar complex: $H^*(\overline{B}(\mathcal{A})) \cong H^{\infty/2+*}(\mathcal{A})$ (Theorem **??**).

(ii) The anomaly duality $\kappa + \kappa' = 0$ holds for affine Kac–Moody algebras under Feigin–Frenkel duality (Corollary **??**). The matter-ghost anomaly cancellation $\kappa_{\mathrm{tot}} = 0$ holds at $c = c_{\mathrm{crit}}$.

(iii) At genus 0, the algebraic Kac–Moody package relevant to a Chern–Simons comparison is in place: classical Maurer–Cartan = flatness, $\overline{B}(\widehat{\mathfrak{g}}_k) \simeq C^{\infty/2+\bullet}(\widehat{\mathfrak{g}}_k, V_k)$, and the dual curvature coefficient satisfies $\kappa + \kappa' = 0$ (Theorem **??**).

Proof. Parts (i) and (ii) are restated from the cited theorems and corollaries, which are proved using the PBW filtration, Arkhipov functor, and Eilenberg–Moore spectral sequence (Theorem **??**). Part (iii) is the content of Theorem **??**: the classical Maurer–Cartan/flatness identification, the affine bar–semi-infinite quasi-isomorphism, and the dual cancellation of the curvature coefficient are all proved. The WZW/Chern–Simons boundary interpretation remains external to the manuscript's proved core. □

Conjecture 2.3.11 *(Physical anomaly cancellation, higher genus;* [conjectured]*).* The genus-0 identifications of Theorem 2.3.10 extend to all genera: the BV-BRST identification $\overline{B}(\mathcal{A}) \simeq C^{\mathrm{BRST}}(\mathcal{A})$ holds as a quasi-isomorphism of *genus-graded* complexes, so that the quantum master equation $\hbar\Delta S + \frac{1}{2}\{S, S\} = 0$ is the chain-level refinement of $D_g^2 = 0$ at every genus (Convention **??**).

Remark 2.3.12 *(Scope).* The genus-0 case is now fully proved for two classes of algebras: (a) arbitrary conformal vertex algebras at $c = 26$ (Theorem **??**); (b) Kac–Moody algebras at any non-critical level (Theorem 2.3.10); (c) for principal W-algebras obtained by DS reduction, the analogous bar/semi-infinite transport remains conditional on Theorem **??**. The remaining gap is the *higher-genus* extension: at genus $g \geq 1$, the path integral measure on $\overline{\mathcal{M}}_g$ introduces counterterms (Costello [100]) that must be matched with the genus-g bar complex. The algebraic infrastructure is in place (curved A_∞ at all genera, Feynman transform identification); what is missing is the analytic comparison between renormalized BRST and the bar differential at genus $g \geq 1$. Note, however, that *scalar* anomaly cancellation extends to all genera unconditionally for pairs with $\kappa + \kappa' = 0$: Theorem D (Theorem **??**) gives $\mathrm{obs}_g(\mathcal{A}) + \mathrm{obs}_g(\mathcal{A}^!) =$

$(\kappa + \kappa') \cdot \lambda_g = 0$ for such pairs (Kac–Moody, free fields, lattice VOAs; see Theorem D for the general formula).

Homotopy template (Convention **??**): Type VII (physics-dictionary). The conjecture asserts quasi-isomorphism of the L_∞ algebras controlling BRST and bar deformation problems, genus by genus. (Contributing to Conjecture 2.2.7.)

$\mathrm{AdS}_3/\mathrm{CFT}_2$ as curved Koszul duality

Conjecture 2.3.13 *(Holographic duality from bar-cobar;* [conjectured]*).* The curved bar complex of a boundary chiral algebra $\mathcal{A}$ provides an algebraic model for the bulk theory on AdS_3:

(i) The bar complex $\overline{B}(\mathcal{A})$ classifies twisting morphisms (couplings between $\mathcal{A}$ and $\mathcal{A}^!$); the *Koszul dual* $\mathcal{A}^!$ models the bulk field content (Costello–Paquette [102]). The bulk *observable* algebra is the derived centre $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, not $\overline{B}(\mathcal{A})$ itself.

(ii) The curvature $m_0 = \kappa \cdot \mathbf{1}$ encodes the cosmological constant $\Lambda \sim -1/R^2_{\mathrm{AdS}}$, with $\kappa \sim c/2 \sim R^2_{\mathrm{AdS}}/\ell_P^2$.

(iii) The genus expansion $\sum_g g_s^{2g-2}\overline{B}^{(g)}$ is the $1/N$ expansion of the dual gauge theory, with $g_s \sim 1/N$.

(iv) Bar-cobar inversion $\Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$ recovers the boundary algebra (Theorem B). The boundary-to-bulk map is the chiral centre construction $\mathcal{A} \mapsto C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$.

Remark 2.3.14 *(Scope).* Costello–Paquette [102] conjecture (footnote 8) that curved Koszul duality controls twisted supergravity on AdS_3. Our framework provides the mathematical structure they require: the curved A_∞ bar complex (Theorem **??**), the genus expansion (Theorem **??**), and the complementarity theorem (Theorem **??**) relating boundary and bulk. The bar–semi-infinite identification for affine Kac–Moody algebras (Theorem **??**) is proved, and the principal W-algebra extension remains conditional on Theorem **??**. Thus the theorematic boundary algebraic package is fully in place on the Kac–Moody side and only conditionally in place on the principal W side. The open problem is the bulk side of the dictionary: a filtered H-level bulk target, if required, belongs upstream to MC4, and the comparison with twisted supergravity is the downstream MC5 step. Concretely, the remaining gap is the identification of the *bulk* observables with the twisted supergravity theory, which requires input from 3d gauge theory (Chern–Simons formulation of gravity on AdS_3, $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ gauge group).

Homotopy template (Convention **??**): Type VII (physics-dictionary). At H-level: an equivalence of factorization categories (boundary chiral vs. bulk twisted SUGRA). The M-level model is the curved bar complex; the S-level shadow is the $1/N$ expansion. (Contributing to Conjecture 2.2.7.)

Remark 2.3.15 *(BRST chain map: existing ingredients).* The genus-0 component of the bar-BRST identification already exists in pieces in the literature:

(i) Arkhipov [23] constructs a chain map from the semi-infinite (BRST) complex to the bar complex for associative algebras at genus 0.

(ii) Markl [308] clarifies the relationship between BV brackets and bar complex structures via higher-order derivations.

(iii) Si Li [295, 296] establishes BV quantization for holomorphic Chern–Simons theory, producing the A_∞ structure from Feynman diagrams; the Feynman graph integrals yield almost modular forms that connect to our genus expansion.

(iv) The Francis–Gaitsgory factorization framework [167] provides the ∞-categorical envelope.

This synthesis is fully achieved for Kac–Moody algebras via Theorem **??**. For principal W-algebras the corresponding genus-0 bar–BRST identification is conditional on Theorem **??**. The higher-genus generalization is proved by the inductive genus determination (Theorem **??**) and the HS-sewing criterion (Theorem **??**).

3d mirror symmetry from E_1-chiral Koszul duality

Conjecture 2.3.16 *(3d mirror symmetry;* [conjectured]*).* For 3d $\mathcal{N} = 4$ gauge theories with Coulomb and Higgs branches, E_1-chiral Koszul duality provides the algebraic mechanism:

(i) The Coulomb branch algebra (in the sense of Braverman–Finkelberg–Nakajima [69]) is the E_1-chiral bar complex of the Higgs branch algebra.

(ii) The curved A_∞ structure on the bar complex encodes mass and FI parameter deformations.

(iii) Symplectic duality (Braden–Licata–Proudfoot–Webster) is the module-level avatar of E_1-chiral Koszul duality (Theorem **??**).

Remark 2.3.17 *(Scope).* This is highly speculative. The mathematical ingredients (E_1-chiral Koszul duality (Theorem **??**), Yangian bar complexes (§**??**), and the Coulomb branch construction of [69]) exist independently; the gauge-theoretic side is developed in Bittleston–Costello–Zeng [109], who derive 3d mirror symmetry from S-duality in the teleparallel formulation. However, connecting them requires: (a) identifying the Coulomb branch ring with a specific E_1-chiral bar complex; (b) showing that symplectic duality (exchange of Coulomb and Higgs branches) corresponds to bar-cobar inversion; (c) relating the mass/FI deformations to the curvature of the bar complex. Each of these steps requires substantial new work combining 3d gauge theory, symplectic geometry, and the E_1-chiral framework.

Homotopy template (Convention **??**): Type II (equivalence of homotopy categories). The conjecture asserts an equivalence of E_1-monoidal ∞-categories (Coulomb branch factorization vs. Higgs branch bar-cobar). (Contributing to Conjecture 2.2.3.)

Noncommutative Hodge theory from genus tower

Definition 2.3.18 *(Chiral Hodge numbers).* For a Koszul chiral algebra $\mathcal{A}$ and genus $g \geq 1$, the *chiral Hodge numbers* are

$$h_g^{p,q}(\mathcal{A}) \;:=\; \dim E_2^{p,q}(\overline{B}^{(g)}(\mathcal{A}))$$

where $E_2^{p,q}$ is the E_2 page of the genus spectral sequence (Spectral Sequence **??**).

Proposition 2.3.19 *(Hodge symmetry from complementarity;* [proved here]*).* For a Koszul chiral algebra $\mathcal{A}$ with dual $\mathcal{A}^!$:

(i) (Hodge symmetry) $h_g^{p,q}(\mathcal{A}) = h_g^{q,p}(\mathcal{A}^!)$ for all g, p, q.

(ii) (Hodge-to-de Rham degeneration) The genus spectral sequence degenerates at E_2 for Koszul algebras, so that the total chiral Hodge number $h_g^n(\mathcal{A}) = \sum_{p+q=n} h_g^{p,q}(\mathcal{A})$ equals the hypercohomology dimension $\dim \mathbb{H}^n(\overline{\mathcal{M}}_g, \overline{B}_\bullet^{(g)}(\mathcal{A}))$.

(iii) (Self-dual Hodge diamond) If $\mathcal{A} \simeq \mathcal{A}^!$ (self-dual), then $h_g^{p,q}(\mathcal{A}) = h_g^{q,p}(\mathcal{A})$.

Proof. Part (i): Theorem **??** provides the complementarity isomorphism $E_2^{p,q}(\mathcal{A}) \cong E_2^{q,p}(\mathcal{A}^!)^\vee$, which gives $h_g^{p,q}(\mathcal{A}) = h_g^{q,p}(\mathcal{A}^!)$. Part (ii): the E_2 collapse for Koszul algebras is Theorem **??**. Part (iii) is immediate from (i). □

Remark 2.3.20 *(Hodge symmetry from universal MC duality).* The Hodge symmetry $h_g^{p,q}(\mathcal{A}) = h_g^{q,p}(\mathcal{A}^!)$ has a transparent origin in the universal Maurer–Cartan class (Theorem **??**): Verdier duality sends $\Theta_{\mathcal{A}} \mapsto -\Theta_{\mathcal{A}^!}$, which exchanges the E_2 pages of $\mathcal{A}$ and $\mathcal{A}^!$ (underlying complexes: $\mathbb{D}(\overline{B}^{(g)}(\mathcal{A})) \simeq (\mathcal{A}^!)_\infty^{(g)}$, factorization *algebra*, not coalgebra). For self-dual algebras ($\mathcal{A} \simeq \mathcal{A}^!$), this gives an internal involution on the deformation package and hence a symmetric chiral Hodge diamond. Self-duality alone does *not* force $\Theta_{\mathcal{A}}$ or $\kappa(\mathcal{A})$ to vanish; it only constrains how the Hodge pages are paired. The strongest uncurved form occurs on the scalar-critical locus $\kappa = 0$, where $\Theta_{\mathcal{A}}^{\min} = 0$ and the bar complex is uncurved.

Remark 2.3.21 *(Computability of the weight filtration).* The weight filtration on $H^*(\overline{C}_n(X))$ that enters the genus spectral sequence $E_2^{p,q}$ is explicitly computable by combinatorial methods. Bibby–Gadish [51] establish effective algorithms for computing the mixed Hodge structure on configuration space cohomology via intersection lattice data. For our chiral Hodge numbers $h_g^{p,q}(\mathcal{A})$, this means the E_2 page can in principle be computed from the combinatorial data of the Orlik–Solomon algebra (for genus 0) and the augmented OS algebra (for genus $g \geq 1$), without needing the full geometric machinery of mixed Hodge modules.

Conjecture 2.3.22 *(Twistor interpretation of genus expansion;* [conjectured]*).* The genus expansion variable g_s in the total bar complex $\overline{B}^{\mathrm{total}}(\mathcal{A}) = \sum_g g_s^{2g-2}\,\overline{B}^{(g)}(\mathcal{A})$ plays the role of the twistor parameter λ in the Kontsevich–Soibelman noncommutative Hodge theory [278]: there exists a $\mathbb{C}^*$-action on the de Rham complex of the genus tower such that the weight-k subspace at genus g has dimension $h_g^{k,n-k}(\mathcal{A})$.

Remark 2.3.23 *(Scope).* The Hodge numbers and their symmetry are proved (Proposition 2.3.19), and the E_2 degeneration is established (Theorem **??**). The remaining conjecture is the identification with the Kontsevich–Soibelman $\mathbb{C}^*$-action, which requires establishing a twistor structure on the genus tower. There is currently no framework in which to formulate this precisely for chiral algebras.

Homotopy template (Convention **??**): Type III (shifted-symplectic/Lagrangian enhancement). The conjecture upgrades the genus spectral sequence to a $\mathbb{C}^*$-equivariant shifted-symplectic structure on the total deformation complex. (Part of the MC2 theorem package; the universal class exists by Theorem **??**, while the older comparison route is Theorem **??**.)

Analytic completion and the sewing envelope

Conjecture 2.3.24 *(Analytic completion programme;* [conjectured]*).* The algebraic bar-cobar engine admits a canonical analytic completion:

(i) *Sewing envelope*: the Hausdorff completion $\mathcal{A}^{\mathrm{sew}}$ for the locally convex topology generated by sewing-amplitude seminorms is a universal analytic completion of the algebraic chiral core (Definition **??**).

(ii) *HS-sewing criterion*: under Hilbert–Schmidt sewing (Definition **??**), closed amplitudes become trace class and partition functions converge (Proposition **??**).

(iii) *Analytic Koszul duality*: the genus-zero bar-cobar adjunction lifts to an equivalence of complete analytic module categories (Definition **??**).

(iv) *Coderived shadows*: at curved genus, the analytic shadow $Q_g^{\mathrm{an}}(\mathcal{A})$ lives in the coderived analytic category (Definition **??**); the shadow partition function (**??**) is a convergent supertrace.

The decisive test case is the Heisenberg VOA, where the sewing envelope should be $\mathrm{Sym}\,A^2(D)$ (Theorem **??**). Recent work: Moriwaki [323, 324] constructs conformally flat factorization homology valued in IndHilb; Adamo–Moriwaki–Tanimoto [3] prove conformal Osterwalder–Schrader axioms for unitary full VOAs.

Remark 2.3.25 *(Scope).* All items are [conjectured]. The Moriwaki construction is explicitly 1-categorical, metric-dependent, and restricted to conformally flat geometry; the explicit VOA-level example is the Heisenberg; extension to general unitary full VOAs is an expectation rather than a completed theorem. The programme is orthogonal to the algebraic proved core (Theorems A–H); it asks whether the formal shadow invariants of §**??** survive analytic completion and become positivity-compatible invariants.

Homotopy template (Convention **??**): Type V (filtered/pro-object comparison). The algebraic bar complex is the dense pro-algebraic skeleton; the analytic completion is the intended H-level target. (Part of the MC5 frontier; genus ≥ 2 requires analytic control.)

The nine directions: overview and interdependence

Programmes I (higher-dimensional Koszul duality, §2.3) and II (Vassiliev invariants, §2.3) share the critical-level bar complex as common foundation, but diverge in application: Programme I computes E_n-operadic Koszul duals via configuration space integrals on n-manifolds, while Programme II specializes from complex curves to real circles, extracting knot invariants via the Kontsevich integral. At $n = 3$, the two programmes converge: the E_3-bar complex on a 3-manifold computes perturbative Chern–Simons invariants, and the restriction to configuration spaces of circles in S^3 produces precisely the Vassiliev weight systems of Programme II (Proposition 2.3.6).

The anomaly cancellation programme (§2.3) is the physical interpretation of the curvature theory: $\kappa_{\mathrm{tot}} = 0$ is both the mathematical condition $d^2_{\mathrm{bar}} = 0$ and the physical condition for consistent quantization (Theorem 2.3.8). This feeds directly into the $\mathrm{AdS}_3/\mathrm{CFT}_2$ programme (§2.3), where the curvature $m_0 = \kappa \cdot \mathbf{1}$ plays the role of the cosmological constant: the bar-cobar adjunction $\overline{B}(\mathcal{A}) \dashv \Omega(\overline{B}(\mathcal{A}))$ models the boundary-to-bulk and bulk-to-boundary maps of holographic duality. The 3d mirror symmetry programme (§2.3) extends this physical circle from 2d to 3d, with E_1-chiral Koszul duality (Theorem **??**) replacing the E_2-chiral theory and Coulomb–Higgs exchange replacing bar-cobar inversion.

The noncommutative Hodge programme (§2.3) reinterprets the entire genus tower as a twistor family: the genus expansion variable g_s plays the role of the Kontsevich–Soibelman twistor parameter, and the chiral Hodge numbers $h^{p,q}_g(\mathcal{A})$ provide the Hodge-theoretic data. This programme depends on the complementarity theorem (Theorem **??**), which supplies the Hodge symmetry $h^{p,q}_g(\mathcal{A}) = h^{q,p}_g(\mathcal{A}^!)$.

Remark 2.3.26 *(Unifying vision: modular Koszul duality for factorization algebras).* All of the programmes above are facets of a single conjecture: for a factorization algebra $\mathcal{F}$ on an oriented n-manifold M, the Feynman transform $\mathrm{FT}(\mathcal{F})$ computes the Koszul dual factorization algebra on M with reversed orientation. The evidence is the following hierarchy of established and conjectural instances:

(i) At $n = 1$ (A_∞-algebras): classical bar-cobar duality (Loday–Vallette [299]). This is the foundational case, with the bar and cobar constructions as mutual inverses on the Koszul locus.

(ii) At $n = 2$ with complex structure (chiral algebras): the Main Theorems A, B, C of this monograph (Theorems **??**, **??**, **??**). The Feynman transform is realized by Verdier duality on FM compactifications, and the genus expansion provides the quantum corrections.

(iii) At $n = 2$ topological (E_2-algebras): Kontsevich formality [273], which identifies the E_2-bar complex with the deformation complex of the associative operad.

(iv) At $n = 3$ (E_3-algebras): perturbative Chern–Simons invariants, with Vassiliev weight systems extracted from the E_3-bar complex (Conjecture 2.3.5).

(v) At $n = \infty$ (E_∞-algebras): factorization homology duality of Ayala–Francis [30], which provides the ∞-categorical framework for Poincaré–Koszul duality on arbitrary manifolds.

This monograph provides case (ii) in full detail; the passage from (ii) to the general conjecture is the content of §1.20. For the status of each programme-level aspiration, see §2.4.

2.4 Programme status assessment

Status: nine futures

The development of this monograph was guided by nine programme-level aspirations. We summarize the status of each, then record the working definition and the concrete modular datum that constitute the realized core.

Future 1 (all-genus parameterized bar-cobar theory): **proved** after reformulation. The correct parameter source is fiber cohomology $H^1(\Sigma_g, \mathbb{C})$, not moduli cohomology $H^1(\mathcal{M}_g)$ (which vanishes for $g \geq 2$ by Harer). The total corrected differential D_g satisfies $D_g^2 = 0$ (Theorem **??**), with fiberwise curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ (Theorem **??**).
Future 2 (modular Koszul programme): **largely realized.** Theorems A–D and H establish the core; MC2 is fully resolved (Theorem 1.20.8). The remaining gap is the coderived Ran-space formalism (§1.20).
Future 3 (modular homotopy theory for factorization algebras on curves): the mature endpoint of the project. The working definition appears below (Definition 2.4.1).

Definition 2.4.1 *(Modular homotopy theory: working definition).* A *modular homotopy theory* on a smooth curve X is the global completion of the nilpotence-periodicity correspondence (Remark **??**) from genus 0 to all genera. Concretely, it consists of:

(i) *The categorical* exp/log *adjunction.* An ∞-categorical bar-cobar adjunction $\overline{B}_X \dashv \Omega_X$ on the coderived category $D^{\mathrm{co}}(\mathrm{Fact}(X))$ of curved factorization algebras on $\mathrm{Ran}(X)$: the categorical logarithm and its inverse, analytically continued to the multi-valued regime via the coderived framework.

(ii) *Functoriality of the logarithm over moduli.* Functoriality of the adjunction over the modular operad $\{\overline{\mathcal{M}}_{g,n}\}_{g,n}$, intertwined with Verdier duality: $\mathbb{D}_{\mathrm{Ran}}\, \overline{B}_X(\mathcal{A}) \simeq \mathcal{A}^!_\infty$ (factorization *algebra*, not coalgebra). The monodromy of the logarithm varies coherently over moduli.

(iii) *The complete logarithm.* A universal Maurer–Cartan class $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}) \,\widehat{\otimes}\, R\Gamma(\overline{\mathcal{M}}_{g,\bullet}, \mathbb{Q}))$ controlling the genus tower as a single deformation: the full logarithmic datum from which the characteristic hierarchy (Remark **??**) and its exponential (the modular automorphy) are derived.

(iv) *The Lagrangian branch structure.* A shifted-symplectic pairing on $R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ for which the deformation and obstruction complexes $Q_g(\mathcal{A}), Q_g(\mathcal{A}^!)$ are complementary Lagrangians: the branch structure of the categorical logarithm, with each branch (algebra vs. Koszul dual) contributing a Lagrangian half of the total monodromy.

Of these four requirements: (ii) and (iv) are proved (Theorems A, C; Proposition **??**), where (ii) is established as functorial compatibility over $\overline{\mathcal{M}}_{g,n}$ rather than as an equivalence of modular operads; (i) is proved at genus 0 and on the Koszul locus at all genera (Theorems A, B); (iii) is resolved by Theorem **??** (see also Theorem 1.20.8); the older comparison route of Theorem **??** is now understood as a scalar-lane refinement.

Theorem 2.4.2 *(Volume I concrete modular datum;* [proved here]*).* Let $\mathcal{A}$ be a modular Koszul chiral algebra on a smooth projective curve X. The present theorem surface supplies a concrete modular datum, separated by semantic level:

(i) *H-level.* Definition **??** provides the homotopy bar object $\mathbf{B}_X(\mathcal{A})$, the dual object $\mathcal{A}^!$, and the center local system $\mathcal{Z}_{\mathcal{A}}$, with Verdier intertwining and Lagrangian complementarity supplied by Theorems **??** and **??**. The cyclic deformation object $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$ and the universal Maurer–Cartan class $\Theta_{\mathcal{A}}$ exist by the bar-intrinsic construction (Theorem **??**), without restriction to simple Lie symmetry.

(ii) *M-level.* Definition **??** provides the dg bar model $\overline{B}_X(\mathcal{A})$ together with the genus-0 differential d_0, the fiberwise curved differential d_{fib}, and the total corrected differential D_g, satisfying $d_0^2 = 0, d_{\mathrm{fib}}^2 = \kappa(\mathcal{A})\,\omega_g$, and $D_g^2 = 0$ by Theorems **??**, **??**, and **??**.

(iii) *S-level.* Definition **??** packages the scalar, spectral, periodic, and genus-family shadows as $C_{\mathcal{A}} = (\Theta_{\mathcal{A}}, \kappa(\mathcal{A}), \Delta_{\mathcal{A}}, \Pi_{\mathcal{A}}, \mathcal{H}_{\mathcal{A}})$. Here $\kappa(\mathcal{A})$ is proved by Theorem **??**, $\Delta_{\mathcal{A}}$ by Theorem 1.20.12, $\Pi_{\mathcal{A}}$ has a proved structural framework by Remark **??**, and $\mathcal{H}_{\mathcal{A}}$ is the genus-graded deformation family induced by $\Theta_{\mathcal{A}}$ when the latter is available.

Proof. The H-level datum is exactly Definition **??**, together with Theorems **??** and **??**; the universal Maurer–Cartan enhancement is supplied intrinsically by Theorem **??**, while Theorem **??** is the scalar-lane comparison refinement on the simple-Lie-symmetry locus. The M-level dg model is Definition **??**, and its three differential identities are the cited theorems on genus-0 nilpotency, genus-g curvature, and corrected nilpotence. The S-level hierarchy is Definition **??**, with the scalar, spectral, and periodic statements supplied by the cited theorem and remark references. □

Remark 2.4.3 *(What remains open).* The remaining open problems concern the outer continuation of this datum: coderived bar-cobar persistence beyond the proved loci, the H-level MC4 coefficient identities for $\mathcal{W}_\infty$ and Yangian towers, non-principal W-algebra duality, and analytic completion.

Future 4 (universal modular characteristic hierarchy): **existence proved.** The full characteristic quintuple $C_{\mathcal{A}} = (\Theta_{\mathcal{A}}, \kappa(\mathcal{A}), \Delta_{\mathcal{A}}, \Pi_{\mathcal{A}}, \mathcal{H}_{\mathcal{A}})$ (Definition **??**) exists unconditionally:

- $\kappa(\mathcal{A})$: proved (Theorem **??**);
- $\Delta_{\mathcal{A}}(x) = \det(1 - x\,T_{\mathrm{rec}})$: proved (Theorem 1.20.12; Definition 1.20.14);
- $\Pi_{\mathcal{A}}$: structural framework proved (Remark **??**; Definition **??**), though specific sharp modular and geometric periods remain conjectural (Future 7). Under the nilpotence-periodicity correspondence (Remark **??**), $\Pi_{\mathcal{A}}$ is the *exponential* of the logarithmic data $(\kappa(\mathcal{A}), \Delta_{\mathcal{A}}, \Theta_{\mathcal{A}})$: the periodicity profile is derived from the nilpotent characteristic via exponentiation, not an independent invariant;
- $\Theta_{\mathcal{A}}$: **proved** (Theorem **??**), resolving Theorem 1.20.8 intrinsically from the genus-completed bar differential. The older explicit completion route (Theorem **??**) now gives the all-genera scalar-trace package only on the proved uniform-weight lane, with genus 1 universal separately;
- $\mathcal{H}_{\mathcal{A}}$: the genus-graded deformation family exists as a formal consequence of $\Theta_{\mathcal{A}}$.

Scalar saturation is proved on the standard landscape (Corollary **??**); sharp modular periods and functoriality under embeddings remain open.

Future 5 (index-theoretic genus expansions): **realized** (Theorem 1.20.16).

Future 6 (derived Drinfeld–Kohno and quantum groups): **substantially realized**, decomposed into a DK ladder. DK-0 through DK-$1\frac{1}{2}$ are proved. DK-2/3 is proved for the evaluation-generated core at all simple types (Corollary **??**); for fd type A, Theorem **??** supplies only the conditional thick-generation extension criterion once the ambient extension/comparison package is constructed; Koszulness uses Molev PBW [326]). DK-4 is reduced to the algebraic identification of the tangent Lie algebra with the dg-shifted Yangian (Conjecture **??**), after the Mittag-Leffler property for the RTT tower (Theorem **??**) and the completed bar-cobar equivalence for Yangians (Corollary **??**). DK-5 remains conjectural (Conjecture **??**).

Remark 2.4.4 *(The category $\mathcal{O}$ generation obstruction).* The generated-core DK comparison surface (traditionally packaged as DK-2/3) is resolved on the evaluation-generated core at all simple types (Corollary **??**). Beyond this core, the natural domain for thick generation is $\mathcal{O}_{\mathrm{poly}}$ (polynomial subcategory); thick generation holds there for $N = 2$ unconditionally and for all N (Theorem **??**). By Lemma **??**, the thick closure of evaluation modules in $D^b(\mathcal{O})$ equals $D^b(\mathcal{O}_{\mathrm{poly}})$ and cannot reach infinite-length modules. For infinite-dimensional modules, the correct framework is localizing generation in a completed/coderived category; what is proved here is the evaluation-generated-core DK surface, while the remaining completion packet beyond that core stays open. Candidate strategies are developed in §**??**.

Future 7 (periodicity and tautological depth): **conceptually corrected.** The geometric contribution is nilpotent (not periodic): $\kappa(\lambda)^{g-1} = 0$ on smooth moduli (Theorem **??**). Quantum periodicity is proved unconditionally; modular bar periodicity remains conjectural (Conjectures **??** and **??**).

Future 8 (spectral discriminant geometry): **proved.** $\Delta_{\mathcal{A}}(x) = \det(1 - xT_{\mathrm{rec}})$ is established (Theorem 1.20.12) with DS invariance (Theorem **??**).

Future 9 (BV/BRST/path-integral completion): genus-0 proved (Theorem **??**); higher genus proved via inductive genus determination (Theorem **??**), two-dimensional convergence (Proposition **??**), and the HS-sewing criterion (Theorem **??**); see also (Remark **??**, Conjecture **??**). Volume II's bar–cobar Quillen equivalence (now unconditional) supplies the operadic identification needed for the disk-local packet; the remaining content is the analytic genus-lifting.

Remark 2.4.5 *(Summary of the nine futures).* Of the nine programme-level aspirations:

#	Future	Status
1	$H^1(\mathcal{M}_g)$-parameterized theory	**Proved**
2	Modular Koszul programme	**Largely realized**
3	Modular homotopy theory	Alive (Def. 2.4.1)
4	Universal char. package	**Core proved with one repaired gap**: standard landscape (Cor. **??**) gives the effective Γ-quadruple, generic cyclic rigidity (Thm **??**) and algebraic-family rigidity at all non-critical levels (Thm **??**) give vanishing primitive tangent space and level-direction concentration, and the repaired equivalence (Prop. **??**) matches that weaker statement; the strong scalar identity $\Theta^{\min}_{\mathcal{A}} = \kappa(\mathcal{A})\eta \otimes \Lambda$ is proved on the uniform-weight lane and *fails* for multi-weight families: the genus-g free energy receives a cross-channel correction (Theorem **??**, resolving Open Problem **??** negatively); functoriality (Prop. **??**) is one-channel generically and scalar only on the proved lane; $\Pi_{\mathcal{A}}$ sharp open
5	Index genus expansions	**Realized** (Thm 1.20.16)
6	Derived Drinfeld–Kohno	DK-0/1/$1\frac{1}{2}$ proved; generated-core DK comparison surface (traditionally packaged as DK-2/3) proved on the evaluation-generated core for all types (uses Molev PBW); $\mathcal{O}_{\mathrm{poly}}$ belongs to the separate type-A thick-generation lane; full $\mathcal{O}$/KL extension conjectural; Positselski/ind-completion/ML theorem; DK-4/5 formalized
7	Periodicity & tautological depth	Corrected: geometric contribution is nilpotent depth (proved, index $\leq g-1$); quantum proved; modular bar periodicity conjectural
8	Spectral discriminant	**Proved** + branch (Def. 1.20.14)
9	BV/BRST completion	Genus 0 proved; genus 1 bridge proved (Feynman curvature = $\kappa \cdot \omega_1$); Quillen equiv. unconditional; $g \geq 2$ analytic

The project is becoming *modular homotopy theory for factorization algebras on curves*: the global completion of the nilpotence-periodicity correspondence (Remark **??**) from genus 0 to all genera. Its structural identity is the principle that *nilpotence is the logarithm of periodicity*: the bar construction is the categorical logarithm whose single-valued regime recovers, via formal-disk restriction, classical Koszul duality over a point (the degree-2 stratum) and whose multi-valued regime is the full chiral theory (already at genus 0, configuration-space topology on curves has more structure than the classical picture over a point; at genus $g \geq 1$, Hodge curvature enters); the curvature $\kappa(\mathcal{A})$ is the infinitesimal generator of the monodromy; the coderived category is the analytic continuation; the characteristic hierarchy (Remark **??**) organizes the logarithmic data (κ, Δ, Θ) from which the periodic data (N, Π, modular automorphy) are derived via exponentiation.

The monograph realizes this identity through a GRR genus theorem (Future 5), a spectral characteristic package (Futures 4, 8), and a staged derived Drinfeld–Kohno ladder beyond the evaluation locus (Future 6). It is no longer becoming a single theorematic theory of all-genus corrections controlled by $H^1(\mathcal{M}_g)$ parameters (Future 1, superseded) or a sharp global periodicity theorem (Future 7, stratified). These older self-images have been replaced by the nilpotence-periodicity framework, which correctly classifies the geometric contribution as nilpotent depth (not period), derives modular periodicity from the scalar characteristic via exponentiation, and identifies the Koszul locus as the convergence domain of the categorical logarithm.

Remark 2.4.6 *(Status notes from the conjecture census).* The current conjecture census forces the following programme-level status notes:

(i) The modular factorization adjunction $U_X^{\mathrm{mod}} \dashv \mathrm{Prim}^{\mathrm{mod}}$ is **proved** (Theorem **??**).

(ii) The all-types categorical Clebsch–Gordan packet of MC3 is now **proved** (Theorem **??**), while the remaining completion packets stay open beyond the evaluation-generated core. The folding approach (Proposition **??**) provides an alternative route for types B_n and C_n.

(iii) Transport-closure of hook vertices equals $\mathrm{Par}(N)$ for all $N \leq 7$ (Theorem **??**).

(iv) The formality route (derived DS = classical DS by Kazhdan formality (more precisely, E_1 degeneration of the Kazhdan filtration spectral sequence)) resolves DS–Koszul intertwining for all abelian-$\mathfrak{n}_+$ orbits (Proposition **??**).

Index of conjectures and related results

The following conjectures and related results are organized by theme. Each entry gives the label, a brief description, and the section where it appears.

Algebraic structure.

- Conjecture **??**: Anomaly cancellation implies uncurved duality (§**??**).
- Conjecture **??**: Heuristic BRST/bar comparison, general genus (§**??**).
- Conjecture **??**: Cobar elements as on-shell propagator templates (§**??**).
- Conjectures **??**, **??**, **??**: Modular periodicity of bar cohomology (conjectured for minimal models, WZW, and general rational chiral algebras; T-matrix periodicity proved unconditionally) (§**??**).
- Conjecture **??**: Reflected periodicity for non-self-dual algebras (§**??**).
- Conjecture **??**: downstream large-N inverse-limit specialization of the standard $\mathcal{W}_\infty$ MC4 package (§**??**).

Representation theory and Langlands.

- Conjectures **??**–**??**: Kazhdan–Lusztig recovery from bar-cobar (three stages: periodic CDG, coderived lift, braided semisimplified target; §**??**).

- Conjecture **??**: Factorization-categorical DK/KL bridge beyond the evaluation locus (§**??**).
- Conjecture **??**: CoHA Koszulness for quiver varieties (§**??**).
- Conjecture **??**: Shifted Yangian and geometric Langlands (§**??**).
- Conjecture **??**: E_1-chiral genus theory for Yangians (§**??**).
- Conjecture **??**: sectorwise DK extension to the completed/coderived MC3 domain for all types (§**??**). E_1 sub-exponential growth is proved (Proposition **??**), and sectorwise convergence on the evaluation-generated core is proved (Theorem **??**); the open step is extension of this package to the completed/coderived enlargement.
- Conjecture **??**: Factorization finiteness criterion (abstract bypass of thick generation; §**??**).

Generating functions and combinatorics.

- Conjecture **??**: $\widehat{\mathfrak{sl}}_3$ bar generating function (§**??**).
- Conjecture **??**: Non-simply-laced discriminant families (§**??**).
- Conjecture **??**: $\mathcal{W}_3$ bar algebraicity (§**??**). Seven independent consistency checks support $H^5 = 171$ (Appendix **??**).
- Conjecture **??**: Yangian bar generating function (§**??**). Seven independent consistency checks support $H^4 = 82$ (Appendix **??**).
- Conjecture **??**: Kodaira–Spencer operator on reduced H^1 (§**??**).

Physics and BV-BRST. *Items marked Heuristic require physics input beyond the algebraic framework.*

- Remark **??**: BV formalism equals bar-cobar at all genera (Section **??**).
- Remark **??**: Complete quantum master equation from bar-cobar (Section **??**).
- Conjecture **??**: Bar-cobar as rigorous path integral (Section **??**).
- Conjecture **??**: Feynman rules dictionary (Section **??**).
- Conjecture **??**: String amplitudes from bar complex (Section **??**).
- Theorem **??**: Modular anomaly equation from bar-cobar (Section **??**).
- Conjecture **??**: Modular anomaly via BRST (Section **??**).
- Theorem **??**: higher A_∞ correction package for quantum $\mathcal{W}$-algebras (Section **??**).
- Conjecture **??**: BPHZ recursion from bar differential (Section **??**).

Holomorphic-topological and dualities.

- Theorem **??**: E_n Koszul duality from configuration spaces (§2.3; proved).
- Conjecture **??**: Goddard–Witten S-duality from Koszul (§**??**).
- Conjecture **??**: Holographic duality from deformation quantization (§**??**).
- Conjecture **??**: Deformation quantization = holographic dictionary (§**??**).
- Conjecture **??**: Toroidal algebra Koszul dual (elliptic/double-affine MC4 extension) (§**??**).
- Conjecture **??**: Extended fermion-ghost duality (derived/super MC4 extension) (§**??**).
- Conjecture **??**: Chern–Simons/Koszul dictionary for Yangian–quantum affine duality (§**??**).
- Remark **??**: Periodicity–exchange duality (§**??**).
- Conjecture **??**: Superstring degeneration from genus tower (§**??**).

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